Open Problems in Geometric Analysis

A reorganized volume from the Yau series

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Editorial Note

This volume is a focused book built from the existing Yau series.[1] It keeps only two chapters from the geometry-and-analysis selection:

1. *Problem Section (With Commentary)*,
2. *Open Problems in Geometry*.

The original mathematical content is preserved by importing each selected article in full. The Zhu–Chen commentary has been folded into the first chapter as working comparison notes for later problem-by-problem revision. The two chapter guides are grouped together before the reproduced articles, so the internal arrangement is easier to follow before the original text begins.

[1] The front matter has been lightly edited for grammar and readability, while the reproduced source articles remain unchanged.

Source References

The selected source articles used in this volume are listed below.[2]

1. *Problem Section (With Commentary)*.
   Source folder: `4_Problem_section`.
2. *Open Problems in Geometry*.
   Source folder: `12_open_problem`.

[2]The source list is intentionally typeset with tighter spacing so that the references read more neatly on the page.

Guide to the Chapters

This page links to the two editorial guide pages gathered at the start of the main text.

1. Guide to Chapter 1: Problem Section (With Commentary)
2. Guide to Chapter 2: Open Problems in Geometry

Guide to Chapter 1: Problem Section (With Commentary)

Editorial roadmap for the chapter structure. Click any line below to jump to the corresponding place in the source chapter.

I. Curvature and the Topology of Manifolds
- **A. Sectional curvature, positive curvature, and topological restrictions** — Problems 1–17
- **B. Ricci curvature, scalar curvature, and Einstein metrics** — Problems 18–32

II. Curvature and Complex Structure
- **A. Kähler curvature, complex structure, and canonical metrics** — Problems 33–49

III. Submanifolds
- **A. Immersions, embeddings, and geometric structures on submanifolds** — Problems 50–66

IV. The Spectrum
- **A. Eigenvalues, eigenfunctions, and spectral geometry** — Problems 67–80

V. Problems Related to Geodesics
- **A. Closed geodesics, variational questions, and rigidity** — Problems 81–87

VI. Minimal Submanifolds
- **A. Minimal surfaces, higher-codimension questions, and variational geometry** — Problems 88–113

VII. General Relativity and the Yang–Mills Equation
- **A. Geometric problems motivated by relativity and gauge theory** — Problems 114–120

Guide to Chapter 2: Open Problems in Geometry

Editorial roadmap for the chapter structure. Click any line below to jump to the corresponding place in the source chapter.

I. Metric Geometry
A. Curvature bounds, rigidity, collapse, and scalar curvature Problems 1–17

II. Classical Euclidean Geometry
A. Convexity, hypersurfaces, singularities, and classical surface theory Problems 18–34

III. Partial Differential Equations
A. Geometric PDE, nonlinear analysis, and curvature equations Problems 35–51

IV. Kähler Geometry
A. Canonical metrics, moduli, and Calabi–Yau questions Problems 52–80

V. Further Kähler Geometry
A. Birational geometry, mirror symmetry, and later complex-geometric questions Problems 81–100

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Theorem[section] [theorem]Lemma [theorem]Corollary [theorem]Definition [theorem]Conjecture [theorem]Remark [theorem]Proposition [theorem]Example [theorem]Fact

[footnotesize]caption2

lastpage

Commentary on

Problem Section

In March 1980, I gave three lectures on open problems in geometry that I considered to be important. My talks aroused much interest in the differential geometry community, and I was rather excited by this group of problems myself.

I still remember that I got the flu during the second lecture. But I insisted on finishing all three lectures, wearing a heavy jacket while giving them. Many people in the audience contributed their opinions. There were very lively conversations indeed. The whole set of problems should be considered a joint effort, although in the final version printed later, I added many more problems to the original set.

In order to encourage Chinese geometers to work on more modern questions, I gave extensive talks on these problems at the 1980 Beijing DD Symposium on differential geometry and differential equations. The conference was organized by Chern in the summer of 1980 and had a rather deep influence on the development of geometry in China.

Many geometers worked on those problems. There were 120 problems in all, and I am pleased that about a quarter of the problems were solved. In 1986, I was asked by Princeton University Press to update this set of problems. My former student Bennett Chow and I spent a month working on this project. But when the manuscript was sent to the Press, it was vetoed by Wu-Chung Hsiang who was then the mathematics editor.

In this commentary, I made some comments myself and also included comments from many friends and colleagues.

Shing-Tung Yau

Problem Section[‡]

Shing-Tung Yau

[‡]Shing-Tung Yau, Problem section, Seminar on Differential Geometry, pp. 669–706, Ann. of Math. Stud., 102, Princeton University Press, Princeton, N.J., 1982.

Contents

During the last part of the geometry year in the Institute, there were a lot of requests for a lecture on open problems. After discussions with the following mathematicians: Bourguignon, Calabi, Cheng, Kazdan, Li, Schoen, Simon, Treibergs and Uhlenbeck, I made up a list of sixty problems and gave two lectures on it. Later I was encouraged to enlarge the problem set somewhat.

I should emphasize that the problem set is by no means exhaustive. The choice of the topics depends a lot on the author's personal taste. Apart from a few exceptions, I do not attempt to mention problems in the other subjects which are closely related to differential geometry or can probably be solved by geometric means.

The difficulty of the problems ranges from "elementary" to "deep." Whereas "deep" problems may be solved by a beginning student within a few months and elementary problems can be open for a long time, I do hope that this problem set will provide a condensed overview of the subject for beginning students.

Most, if not all, of the problems are well known. If a problem has an explicit reference or the problem is explicitly suggested by a single person, it will be mentioned. Otherwise the reader can assume that the problem is well known.

Finally, I would like to thank all the mathematicians who read the original draft and suggested corrections, further discussion and references, including F. J. Almgren, Jr., M. Berger, A. Borel, E. Calabi, J. Cheeger, M. Gromov, and H. B. Lawson. I am also grateful to James Mackraz for his valuable help in organizing the problems and compiling the bibliography, and to Neola Crimmins and Kathy Lunetta for a wonderful job of typing this work.

I. Curvature and the Topology of Manifolds

A. Sectional curvature

Problem 1 (The Hopf Conjecture) — Open as of 2026

Does $\mathbb{S}^2 \times \mathbb{S}^2$ admit a metric with positive sectional curvature?

Status. Open. The product metric on $\mathbb{S}^2 \times \mathbb{S}^2$ has nonnegative sectional curvature and many zero two-planes; the question is whether these zero planes can be removed by some global deformation of the metric. No metric of positive sectional curvature is known, and no known topological, analytic, or variational obstruction rules out all such metrics. The available results are therefore genuinely partial: they exclude positive curvature near the standard product metric, under symmetry, under pinching or stronger curvature hypotheses, or under additional analytic assumptions.

It is useful to separate this product problem from the Euler-characteristic form of Hopf's conjecture. The latter predicts that every compact even-dimensional positively curved manifold has positive Euler characteristic; it gives no obstruction here, since $\chi(\mathbb{S}^2 \times \mathbb{S}^2) = 4$. A solution of this problem would require a finer obstruction, one sensitive to the middle-dimensional product structure or to the indefinite intersection form, or else a construction of a new positively curved four-manifold outside the currently known simply connected examples $\mathbb{S}^4$ and $\mathbb{CP}^2$.

The earliest result in this direction is Berger's local obstruction [Br1], which was strengthened by Bourguignon, Deschamps and Sentenac [BDS]. They proved that, in a neighborhood of the product metric on $\mathbb{S}^2 \times \mathbb{S}^2$, there is no metric with positive sectional curvature. Thus the product metric cannot be perturbed locally to a positively curved metric.

There are also extrinsic obstructions which help explain why the problem is not merely a search for an elementary deformation of the standard product. Hopf's hypersurface theorem rules out a positively curved hypersurface realization in codimension one, and Weinstein's codimension-two theorem [Wei] implies that a positively curved $\mathbb{S}^2 \times \mathbb{S}^2$ could not occur as a submanifold of $\mathbb{R}^6$ with the induced metric. These results do not obstruct an abstract intrinsic metric, but they are part of the classical evidence that the product topology is hard to reconcile with strict sectional positivity. Ziller's survey [Zi] places this product problem among the central unresolved examples in positive sectional curvature and is a useful guide to the surrounding construction and obstruction landscape.

The next decisive progress concerns symmetry. Hsiang and Kleiner [HK] proved that a compact simply connected positively curved four-manifold with an isometric circle action is homeomorphic to $\mathbb{S}^4$ or $\mathbb{CP}^2$. Consequently a hypothetical positively curved metric on $\mathbb{S}^2 \times \mathbb{S}^2$ would have finite isometry group; in particular it cannot carry a nontrivial Killing field. Seaman [S] obtained a different kind of restriction: $\mathbb{S}^2 \times \mathbb{S}^2$ admits no metric whose sectional curvature is pinched between 0.19 and 1. Motivated partly by the role of Killing fields in positive curvature problems, Li and Liu [LL] introduced a geometric heat flow for vector fields on closed manifolds with positive sectional curvature. That flow is not an obstruction to arbitrary positively curved metrics, but it belongs to the analytic program of detecting hidden symmetry, which is exactly what the Hsiang–Kleiner theorem would then rule out on $\mathbb{S}^2 \times \mathbb{S}^2$.

The four-dimensional topology suggests another analytic route. Liu and Wan [LW] proved that a compact oriented Riemannian four-manifold with positive sectional curvature is definite if its harmonic two-forms satisfy a suitable Kato-type inequality. Since $\mathbb{S}^2 \times \mathbb{S}^2$ has indefinite intersection form, this theorem identifies a precise missing estimate: a general proof of the required

inequality for positively curved four-manifolds would imply the Hopf product conjecture. At present this remains a conditional mechanism rather than a theorem in the unrestricted setting.

Stronger curvature hypotheses give complete classification results, but they also show why the original problem is difficult. Hamilton [Ham1] proved that a compact simply connected four-manifold with positive curvature operator is diffeomorphic to $\mathbb{S}^4$. In the setting of positive isotropic curvature, Hamilton [Ham2] initiated the Ricci-flow analysis in dimension four, and Chen, Tang and Zhu [CTZ] later obtained the complete classification of compact four-manifolds with positive isotropic curvature. In higher dimensions, Böhm and Wilking [BW] proved that manifolds with positive curvature operator are space forms. Brendle and Schoen [BS] proved the differentiable sphere theorem for pointwise quarter-pinched curvature. These Ricci-flow and pinching theorems do not rule out an arbitrary positively curved metric on $\mathbb{S}^2 \times \mathbb{S}^2$, but they show that once sectional positivity is strengthened in natural analytic directions, the topology collapses to space-form type behavior.

The higher-dimensional symmetry program gives the broader context. Grove and Searle [GS] classified the maximal symmetry rank case: if a compact simply connected positively curved n-manifold has an effective isometric torus action of maximal possible rank, then the equality cases are the sphere and complex projective space. Wilking [Wil2] extended this direction to large-symmetry-rank manifolds; in particular, for $n \geqslant 10$ and symmetry rank at least $\frac{n}{4} + 1$, the manifold is homeomorphic to $\mathbb{S}^n$ or $\mathbb{HP}^{n/4}$, or is homotopy equivalent to $\mathbb{CP}^{n/2}$. Later work on the Hopf conjecture with symmetry gives further evidence in this direction. Püttmann and Searle [PS] proved the Hopf Euler-characteristic conjecture under low-cohomogeneity or high-symmetry-rank assumptions. Kennard [Ke] proved the Hopf conjecture for positively curved manifolds under a logarithmic lower bound on the symmetry rank. Amann and Kennard [AK] proved that, under a logarithmic symmetry-rank assumption, products $N \times N$ of simply connected closed manifolds do not admit positively curved metrics; they also gave evidence for the conjecture that compact simply connected symmetric spaces of rank greater than one should not admit positive sectional curvature.

Finally, there are examples illustrating the gap between positive curvature everywhere and positivity on a large set. Petersen and Wilhelm [PW] constructed metrics with positive sectional curvature almost everywhere on the unit tangent bundle of $\mathbb{S}^4$ and on a real cohomology $\mathbb{CP}^3$. Wilking [Wil1] constructed further almost-positive examples. These examples do not produce positive curvature everywhere on $\mathbb{S}^2 \times \mathbb{S}^2$, but they show that almost positive curvature is a much weaker condition than positive sectional curvature.

Recent work also clarifies how delicate the curvature assumption is. Kennard, Mouillé and Nienhaus [KMN] studied Hopf-type Euler-characteristic questions under positive second intermediate Ricci curvature. They note that the Hopf conclusion can fail under such relaxations, while proving a positive Euler-characteristic result under additional torus symmetry. This is not a solution of the sectional-curvature Hopf conjecture, but it helps separate which parts of the problem depend specifically on sectional curvature.

Schoen's recent work [Sch] gives a complementary minimal-surface restriction on the topology of positively curved manifolds. He proved that an odd-dimensional closed positively curved manifold cannot contain an incompressible real projective plane, and that in dimension $4k$ the fundamental group acts trivially on π_2. In particular, $\mathbb{RP}^2 \times \mathbb{S}^2$ does not admit a metric of positive sectional curvature. This does not settle the simply connected product $\mathbb{S}^2 \times \mathbb{S}^2$, but it is closely related in method: stable minimal two-spheres detect restrictions on the interaction between low homotopy groups and positive sectional curvature, a theme that any four-dimensional obstruction to the Hopf product conjecture would likely have to confront.

Related open problems.

1. *Original direction.* Is every compact simply connected four-manifold with positive sectional curvature homeomorphic, or even diffeomorphic, to $\mathbb{S}^4$ or $\mathbb{CP}^2$? This is the natural four-dimensional classification problem behind the Hopf product question. It is known under circle symmetry by Hsiang–Kleiner [HK], and related high-symmetry results support the same picture in higher dimensions [GS, Wil2, Ke].

2. *Original direction.* More generally, can any nontrivial product $M_1 \times M_2$ of compact positive-dimensional manifolds admit a metric with positive sectional curvature? The special case $\mathbb{S}^2 \times \mathbb{S}^2$ is Problem 1; the product version with symmetry is discussed by Amann–Kennard [AK].

3. *Reference-driven formulation.* Can a compact simply connected symmetric space of rank greater than one admit a metric with positive sectional curvature? This would include $\mathbb{S}^2 \times \mathbb{S}^2$ as the first rank-two example, and is explicitly discussed as a generalization of Hopf's problem in [AK].

4. *Original direction.* Does every compact simply connected manifold with nonnegative sectional curvature admit some metric with positive sectional curvature? Yau emphasized this as a basic surrounding question in the original problem list [Yau]. The product metric on $\mathbb{S}^2 \times \mathbb{S}^2$ is the first test case, and the failure of all known deformation methods there is one reason the question is difficult.

5. *Original direction.* Does the Hopf Euler-characteristic conjecture hold in all even dimensions: if M^{2n} is compact and has positive sectional curvature, must $\chi(M) > 0$? It is known in several symmetric situations, for example [PS, Ke], but remains open in general.

6. *Reference-driven formulation.* What are the strongest possible restrictions on the action of $\pi_1(M)$ on $\pi_2(M)$ for compact manifolds with positive sectional curvature? Schoen's theorem [Sch] gives strong results in odd dimensions and in dimensions divisible by four, but the full structure of low homotopy groups under positive sectional curvature remains part of the broader topology problem.

7. *Reference-driven formulation.* Can the harmonic-form criterion of Liu–Wan [LW] be proved directly from positive sectional curvature in dimension four, or can one construct a positively curved four-manifold for which the required Kato-type estimate fails? Either outcome would clarify whether the intersection-form approach can reach the Hopf product conjecture.

8. *AI-generated formulation.* Find a curvature or topological obstruction which is strong enough to rule out positive sectional curvature on products, but flexible enough to detect metrics far from the product metric. This question is a synthesis of the local obstruction [Br1, BDS], symmetry obstructions [HK, GS, Wil2, PS, Ke, AK], extrinsic restrictions [Wei], and pinching/curvature-operator classification results [S, Ham1, Ham2, BW, BS, CTZ].

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Problem 2

Is there any metric with positive curvature on exotic spheres?

Status. Needs verification. The problem is claimed affirmative in dimension seven, pending a fully published verification. Petersen and Wilhelm [PW2] announced a metric of positive sectional curvature on the Gromoll–Meyer exotic 7-sphere. If this construction is accepted in the stated form, it answers Yau's original existence question affirmatively. Since the reference used here is an arXiv preprint and not a published account, the entry records the result as a claimed solution rather than as an unconditional settled theorem. In any case, the broader classification problem–which exotic spheres admit positive sectional curvature–remains open.

Milnor's discovery of exotic 7-spheres [Mil] made it natural to ask whether positive sectional curvature can occur on a manifold homeomorphic but not diffeomorphic to a standard sphere. This is a much sharper question than positive scalar curvature. Hitchin [H1] showed that many exotic spheres, detected by the α-invariant, do not admit metrics of positive scalar curvature; hence they cannot admit positive sectional curvature. These index-theoretic obstructions rule out large families in certain dimensions, but they do not rule out all exotic spheres and do not decide the Gromoll–Meyer sphere.

The first major positive construction was the Gromoll–Meyer sphere. Gromoll and Meyer [GM1] realized a Milnor exotic 7-sphere as a biquotient of $\mathrm{Sp}(2)$ and constructed on it a metric of nonnegative sectional curvature. Their metric has zero-curvature planes, but the biquotient presentation became the basic model for all later attempts to produce strict positivity. Grove and Ziller [GZ] later constructed nonnegatively curved metrics on many Milnor spheres. This shows that nonnegative sectional curvature is far more flexible than positive sectional curvature in the exotic-sphere setting, just as in the Hopf product problem.

The next stage was almost positive curvature. Petersen and Wilhelm [PW1] constructed examples with positive curvature almost everywhere. Eschenburg and Kerin [EK] then showed that the Gromoll–Meyer sphere admits metrics whose sectional curvature is positive outside four explicit subvarieties of codimension at least one. These results did not solve the original problem, but they reduced it to understanding whether the remaining zero-curvature planes could be removed.

Petersen and Wilhelm [PW2] subsequently announced a deformation of the Gromoll–Meyer construction with positive sectional curvature. Mathematically, this is the decisive point in the problem: the announced result would supply the first exotic sphere with positive sectional curvature and would answer Yau's original question. For the purposes of this commentary, however, the status should keep the publication caveat visible. A later attempted explicit construction by Ge and Tang [GT] was withdrawn; the authors state that the metrics constructed there have positive sectional curvature almost everywhere, but the claimed everywhere-positive curvature statement should not be used as a proof.

The proof architecture in [PW2] is substantially more involved than a single perturbation. Starting from an almost positively curved metric, the authors combine Cheeger deformations with a redistribution perturbation, scale the fibers over $\mathbb{S}^4$ to obtain positive integrated curvature on the former flat tori, and use a partial conformal change to turn that gain into pointwise positivity. The main technical burden is then to control a uniform neighborhood of every old zero plane in the Grassmannian. This is encoded in two-variable curvature polynomials for planes of the form $\mathrm{span}\{\zeta + \sigma z, W + \tau V\}$ and completed by further Cheeger deformations. The arXiv record for v3 notes that a referee found an error in the original Theorem 5.3; the corrected result is Lemma 5.3. Since the current manuscript remains a preprint, the status above records the claim and its method without treating it as a fully verified published solution.

Recent work has also returned to the geometry of the Gromoll–Meyer sphere from a more explicit metric viewpoint. Berman, Cederwall and Schettini Gherardini [BCSG] study Kaluza–Klein type metrics on the Gromoll–Meyer sphere using $SU(2)$ instanton data on the base $\mathbb{S}^4$. Their construction gives an explicit family in which the Ricci tensor can be computed and positive Ricci curvature can be ensured for a range of the scale parameter. It also permits concrete sectional-curvature calculations. This is useful recent progress for understanding the metric geometry of the exotic 7-sphere, but it should not be confused with a new proof of positive sectional curvature everywhere.

The differentiable sphere theorem of Brendle and Schoen [BS] gives an important complementary restriction: a complete manifold with pointwise $1/4$-pinched sectional curvature is diffeomorphic to a spherical space form. Therefore no exotic sphere can admit a pointwise $1/4$-pinched metric, even though the Gromoll–Meyer sphere would admit positive sectional curvature if the Petersen–Wilhelm construction is accepted in the announced form.

Related open problems.

1. *Original direction.* Which exotic spheres admit positive sectional curvature? The Petersen–Wilhelm preprint claims an affirmative example on the Gromoll–Meyer sphere [PW2], while Hitchin's obstruction rules out many exotic spheres even for positive scalar curvature [H1]. A classification remains open.

2. *Original direction.* Which Milnor spheres admit positive sectional curvature? Grove and Ziller proved nonnegative curvature for many Milnor spheres [GZ], but the passage from nonnegative to positive curvature is still poorly understood.

3. *Reference-driven formulation.* Can the Petersen–Wilhelm construction [PW2] be written in a fully explicit and independently checkable form, for example by giving curvature formulas that make the removal of all zero-curvature planes transparent? This is motivated by the withdrawal of the attempted explicit construction [GT] and by the need for concrete models.

4. *Reference-driven formulation.* Which compact manifolds with quasipositive or almost positive sectional curvature can be deformed to metrics with positive sectional curvature? The Gromoll–Meyer sphere is the motivating positive case [GM1, EK, PW2], while Wilking's examples [Wil1] show that deformation questions in positive curvature are delicate.

5. *Reference-driven formulation.* Can the instanton or Kaluza–Klein ansatz for the Gromoll–Meyer sphere be pushed beyond positive Ricci curvature to produce new explicit metrics with nonnegative, almost positive, or positive sectional curvature? The recent computations in [BCSG] give a concrete testing ground for this question.

6. *AI-generated formulation.* Develop a computer-verifiable curvature package for explicit metrics on exotic spheres: write the curvature operator in invariant coordinates, isolate the zero-plane locus, and test whether Cheeger deformation, partial conformal change, or fiber scaling removes the zero planes without creating new negative directions.

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Problem 3

Let M be an N-dimensional manifold with nonnegative sectional curvature. Is it true that the ith Betti number of M is not greater than the ith Betti number of $\mathbb{T}^N$, the N-torus?

Status. Open in this sharp form. The statement is usually read for closed manifolds, or for complete manifolds after using the Cheeger–Gromoll soul theorem [CG] to reduce the homotopy type to a compact totally convex soul. Gromov proved that the total Betti number of a complete N-manifold with nonnegative sectional curvature is bounded above by a constant depending only on N [Gr], and Abresch improved the quantitative form of this estimate [Ab]. This is far weaker than the conjectural sharp torus bound $b_i(M) \leqslant b_i(\mathbb{T}^N) = \binom{N}{i}$.

The first Betti number case is classical. By the Bochner formula, nonnegative Ricci curvature already implies $b_1(M) \leqslant N$, with equality only in the flat torus case after passing to the appropriate cover. Thus Problem 3 begins in earnest in the middle Betti numbers. Recent work on almost nonnegative Ricci curvature gives a related structural refinement of this first-Betti-number picture. Huang, Huang, Wang and Zhu [HHWZ] prove fibration theorems over the $b_1(M)$-torus under almost nonnegative Ricci curvature together with a lower sectional-curvature bound, removing an upper sectional curvature hypothesis from earlier Yamaguchi-type fibration theorems. Their proof passes to the torsion-free homology cover, takes an equivariant Gromov–Hausdorff limit with an Abelian action, constructs an equivariant almost submetry to a torus, and regularizes it by a center-of-mass procedure before descending the resulting fibration. This is not a solution of the sharp sectional-curvature Betti-number bound, but it shows that the first Betti number is tied to torus-fibration structure in a robust collapsed or almost nonnegative setting.

Gromov's theorem [Gr] remains the main unconditional theorem in the direction of the problem: for every field F,

$$\sum_i b_i(M;F) \leqslant C(N)$$

for all complete N-manifolds with nonnegative sectional curvature. For closed manifolds, Gromov's proof treats the distance function as a nonsmooth Morse function: Toponogov comparison controls its critical points, an isotopy lemma removes critical-point-free annuli, and a quantitative covering argument turns local information into a global homology bound. Abresch [Ab] reorganized this mechanism using ranks of homology maps between nested metric sets, a sharper packing lemma, and metric annuli; this also bounds the topology and number of ends under an integrable asymptotic lower-curvature condition. These results imply, for example, that connected sums of sufficiently many copies of $\mathbb{CP}^2$ cannot carry nonnegative sectional curvature. They do not, however, identify the optimal bound, nor do they explain why the flat torus should be extremal for each individual Betti number.

The conjectural structural mechanism behind the sharp bound is the Bott–Grove–Halperin rational ellipticity conjecture: a compact simply connected manifold with nonnegative sectional curvature should be rationally elliptic [FHT, FOT]. For elliptic spaces, the formal dimension controls the rational Betti numbers: in particular, Félix–Halperin–Thomas, Theorem 33.2 [FHT], gives $b_i(M;\mathbb{Q}) \leqslant \binom{N}{i}$. Thus rational ellipticity would give a conceptual explanation for the expected rational Betti-number restrictions in the simply connected part of the problem. It would not by itself settle every integral or finite-field formulation, and it leaves the fundamental-group contribution to be separated by splitting and covering arguments.

The rational homotopy picture is subtle. Totaro [To] showed that closed simply connected nonnegatively curved manifolds do not fall into finitely many rational homotopy types in each dimension; already in dimension six there are infinitely many rational homotopy types. Thus one cannot hope for a simple finiteness theorem for rational homotopy types to imply Problem 3, even though Gromov's theorem gives a uniform bound on homology ranks.

There is also a noncompact analytic analogue involving harmonic functions of polynomial growth. Colding and Minicozzi [CM] proved finite dimensionality for spaces of polynomial growth harmonic functions on complete manifolds with nonnegative Ricci curvature, and Li [Li] gave a short proof. In the Kähler setting with nonnegative holomorphic bisectional curvature, Ni [Ni] proved sharp dimension estimates under maximal volume growth, and Chen–Fu–Yin–Zhu [CFYZ] proved the sharp dimension estimate with rigidity for holomorphic functions of polynomial growth. These results do not solve the compact Betti-number problem, but they show that Yau's "torus or Euclidean model gives the sharp dimension" philosophy has successful analogues in function theory.

Related open problems.

1. *Reference-driven formulation.* Prove the Bott–Grove–Halperin conjecture: every compact simply connected nonnegatively curved manifold is rationally elliptic [FOT]. This would give a conceptual explanation for the expected rational Betti-number restrictions.

2. *Original direction.* Determine the optimal constant in Gromov's Betti-number theorem. The conjectural optimal total bound is 2^N, the total Betti number of $\mathbb{T}^N$ [Gr, Ab, FHT].

3. *AI-generated formulation.* Classify the equality case in Problem 3: if $b_i(M) = b_i(\mathbb{T}^N)$ for some nontrivial range of i, must M be closely related to a flat torus? This is modeled on the rigidity phenomena in Bochner's theorem and in polynomial-growth function theory [CM, Li, CFYZ].

4. *Reference-driven formulation.* Separate the compact simply connected part of the problem from the fundamental-group part: after passing to the soul in the complete noncompact case [CG], and after using the Cheeger–Gromoll splitting theorem for the universal cover, determine how much of the torus bound reduces to the rationally elliptic factor.

5. *Reference-driven formulation.* Extend the recent first-Betti-number fibration theorems for almost nonnegative Ricci curvature [HHWZ] toward middle-dimensional cohomology: is there a collapsed-geometry mechanism which explains why the torus should remain extremal for all Betti numbers under nonnegative sectional curvature?

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Problem 4

Let M be a compact manifold with positive curvature. Does M admit a smooth, effective $\mathbb{S}^1$ action?

Status. Open. The question is about the smooth topology of the underlying manifold, not only about the isometry group of a particular positively curved metric. All known compact manifolds with positive sectional curvature admit nontrivial smooth circle actions, and in fact the known metrics usually come with much larger isometry groups. No theorem is known, however, which forces a compact positively curved manifold to admit any effective $\mathbb{S}^1$-action, and no positively curved example is known whose smooth manifold is asymmetric in this sense.

The motivation is that all classical and currently known compact examples of positive sectional curvature are highly symmetric. Rank one symmetric spaces, Berger spaces, Wallach spaces, Aloff–Wallach spaces, Eschenburg spaces, Bazaikin spaces and the Gromoll–Meyer exotic sphere all arise from homogeneous, biquotient, or related quotient constructions; see the surveys of Ziller and Wilking [Zi, WilS]. These constructions produce not only a positive curvature metric, but also a visible group action. Yau's question asks whether this is an accident of construction or a genuine topological consequence of positive sectional curvature.

The symmetry assumption has produced many strong classification theorems. Grove and Searle [GS] classified the maximal symmetry-rank case. Wilking [Wil2] obtained strong restrictions for high symmetry rank, including connectedness and periodicity phenomena. Püttmann– Searle [PS] and Kennard [Ke] proved Hopf-type Euler-characteristic results under symmetry hypotheses. These results show that symmetry is a powerful organizing principle for positive curvature. They also make clear the gap in Problem 4: the theorems begin once a torus action is given, while Yau asks whether such an action must exist at all.

Recent work has refined the high-symmetry picture rather than settling the existence of a circle action. Kennard, Wiemeler and Wilking [KWW] found a link between torus representations occurring in positive-curvature arguments and regular matroids. At a torus fixed point, the relevant isotropy representation is encoded by a totally unimodular matrix. Seymour's classification reduces its difficult part to weighted graphs; graph systole and matroid cogirth estimates then produce a circle with a large fixed-point component, to which connectedness and cohomological periodicity can be applied. This leads to new obstructions to positive sectional curvature in the presence of torus symmetry. Kennard, Khalili Samani and Searle [KKS] developed parallel restrictions for discrete elementary abelian symmetry: coding-theoretic estimates select involutions with controlled fixed-set codimension, followed by connectedness, periodicity, and induction. These papers are important for the modern reporting of the problem: they show that once symmetry is assumed, the topology is increasingly rigid and computable. They do not, however, supply the missing first step of proving that a positively curved manifold must have any nontrivial smooth circle action. In particular, finite $(\mathbb{Z}/2)^r$ symmetry does not imply continuous circle symmetry.

There are two possible routes to the problem. A topological route would show that any manifold supporting positive sectional curvature must lie in a class of smooth manifolds admitting effective circle actions. An analytic route would try to produce an isometric circle action from the metric itself, equivalently a nonzero Killing field after changing neither the manifold nor the metric. Li and Liu [LL] proposed an approach of the latter kind: they introduced a geometric heat flow for vector fields on closed manifolds with positive sectional curvature, with the goal of detecting or producing Killing fields. The flow gives criteria for the existence of Killing fields, but it has not yet resolved the general circle-action problem.

Related open problems.

1. *Original direction.* Does there exist a compact positively curved manifold which admits no effective smooth $\mathbb{S}^1$-action? This is the direct negation of Problem 4 and is stronger than asking for a particular positive-curvature metric with finite isometry group.

2. *Original direction.* Does every compact positively curved manifold admit some positive-curvature metric with a nontrivial isometric circle action? This is stronger than Yau's smooth-action question but closer to the structure of all known examples.

3. *Reference-driven formulation.* Can the Li–Liu heat flow be strengthened to force convergence to a nonzero Killing field under positive sectional curvature [LL]? This is an analytic version of the circle-action problem.

4. *Reference-driven formulation.* Classify compact positively curved manifolds with small isometry group. The high-symmetry case is much better understood [GS, Wil2, PS, Ke, KWW], so the low-symmetry regime is the natural frontier.

5. *Reference-driven formulation.* Determine which restrictions coming from torus symmetry have analogues for discrete abelian symmetry. The results of Kennard–Khalili Samani–Searle [KKS] suggest that discrete symmetry can sometimes substitute for continuous torus symmetry, but it is not a replacement for an effective circle action on the manifold.

6. *AI-generated formulation.* Develop topological tests, using characteristic classes, fixed-point data, and rational homotopy, which could decide whether a proposed positive-curvature manifold admits any effective circle action before one tries to construct a symmetric metric.

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Problem 5

Is there an example of a compact, simply-connected manifold with nonnegative Ricci curvature that does not admit a metric with nonnegative sectional curvature?

Status. Solved affirmatively by Sha and Yang. Their 1989 paper already constructed compact simply connected manifolds with positive Ricci curvature, hence with nonnegative Ricci curvature, whose Betti numbers are too large to allow any metric with nonnegative sectional curvature [SY1]. Their 1991 theorem supplied the especially transparent connected-sum family used below [SY2]. Thus the problem is not merely solved by weakening sectional curvature to Ricci curvature; it is solved by examples showing that the two conditions have sharply different topological consequences.

The point is that positive Ricci curvature is much more flexible than nonnegative sectional curvature. Sha and Yang first constructed examples of manifolds with positive Ricci curvature [SY1], and then proved that connected sums of products $\mathbb{S}^n \times \mathbb{S}^m$ admit positive Ricci curvature [SY2], in the relevant simply connected range $n, m \geqslant 2$. For a fixed dimension $n+m$, taking more connected summands increases the middle Betti numbers while preserving positive Ricci curvature. In particular, connected sums of many copies of $\mathbb{S}^2 \times \mathbb{S}^2$ give simply connected four-dimensional examples with positive Ricci curvature.

On the other hand, Gromov's Betti-number theorem [Gr] gives a bound, depending only on the dimension, for the total Betti number of a closed manifold with nonnegative sectional curvature. The Betti numbers of the Sha–Yang connected sums grow with the number of summands, while the dimension is fixed. Hence, for sufficiently many summands, these manifolds cannot admit any metric with nonnegative sectional curvature. This is the clean mechanism resolving Yau's question: Ricci positivity survives connected-sum constructions that sectional nonnegativity cannot tolerate because of Gromov's theorem.

Later work made the positive-Ricci side still more flexible. Burdick [Bu] developed core-metric techniques, based on Perelman's connected-sum ideas, and constructed Ricci-positive metrics on large classes of connected sums of projective spaces. The mechanism builds round convex boundaries on punctured summands, attaches them to a Ricci-positive "docking station," and uses Perelman's second-fundamental-form gluing criterion. Reiser [Re] extended the core-metric method to broad classes of fibre and projective bundles. Reiser and Wraith [RW] proved that in dimension $d \geqslant 5$ a Gromov-type Betti-number bound fails for $\mathrm{Ric}_k > 0$ whenever $k \geqslant \lfloor d/2 \rfloor + 2$, using doubly warped surgery metrics, an algebraic positivity criterion, and a smoothing theorem. These developments sharpen the lesson of Problem 5: the transition from sectional curvature to Ricci-type positivity is not a small technical weakening, but a substantial change in the topology allowed by curvature.

The newest Ricci-positive constructions strengthen this conclusion. Reiser [Re] proves that core metrics can be lifted along broad classes of fibre bundles and applies this to represent all classes in the torsion-free oriented bordism ring by connected manifolds with positive Ricci curvature. In a related paper, Reiser [ReT] proves that positive Ricci curvature can be lifted through twisted suspensions, with applications to simply connected manifolds with large symmetry rank and prescribed homology. Reiser and Wraith [RW2] extend the connected-sum philosophy to positive intermediate Ricci curvature via k-core metrics. None of these results weakens the Sha–Yang solution of Yau's question; rather, they show that the positive-Ricci side has become an active construction theory, while the nonnegative-sectional side remains constrained by comparison topology.

Related open problems.

1. *Original direction.* Classify which simply connected manifolds with positive Ricci curvature can also carry nonnegative sectional curvature. Problem 5 shows that positive Ricci curvature alone is not enough [SY2, Gr].

2. *Reference-driven formulation.* Find sharper topological obstructions to nonnegative sectional curvature beyond Gromov's total Betti-number bound [Gr]. This is closely related to Problem 3.

3. *Reference-driven formulation.* Determine the sharp threshold among intermediate Ricci curvature conditions at which Gromov-type Betti-number bounds begin or fail. Reiser–Wraith show failure in a range of positive intermediate Ricci curvatures [RW], while sectional nonnegativity retains Gromov's bound [Gr].

4. *Reference-driven formulation.* Which positive-Ricci connected-sum, fibre-bundle, twisted-suspension, or bordism constructions can be upgraded to positive intermediate Ricci curvature? Recent work of Reiser and Reiser–Wraith [Re, ReT, RW2] gives a concrete framework for this question.

5. *AI-generated formulation.* Determine whether large families produced by modern Ricci-positive connected-sum constructions admit any curvature positivity stronger than Ricci but still weak enough to allow unbounded Betti numbers [Bu, Re, RW, RW2].

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Problem 6

Do all vector bundles over a manifold with positive curvature admit a complete metric with nonnegative sectional curvature?

Status. Open for the original positive-curvature base question. The problem asks for a converse to the soul theorem in the special case where the soul itself has positive sectional curvature. There are important positive results for bundles over spheres and compact rank-one symmetric spaces, but there are also counterexamples and necessary conditions for natural extensions of the question when the base is merely nonnegatively curved, such as a flat torus.

The problem is rooted in the soul theorem of Cheeger and Gromoll [CG2]. Their theorem says that a complete noncompact manifold with nonnegative sectional curvature is diffeomorphic to the normal bundle of a compact totally convex soul. Problem 6 asks for a converse in a favorable case: if the base already admits positive sectional curvature, must every vector bundle over it occur as the total space of a complete nonnegatively curved metric? The positivity of the base is essential; without it the converse is known to fail in natural families.

The Ricci-curvature analogue is much more flexible. Nash [Na] studied positive Ricci curvature on fibre bundles, and later work on positive Ricci curvature shows that many bundle and connected-sum constructions are possible. Sectional curvature is more rigid because O'Neill's formulas, horizontal zero-curvature planes, and the choice of connection on the bundle all enter the curvature calculation. Thus the vector-bundle question is not just a topological existence problem; it is also a problem about constructing connection metrics or more flexible complete metrics with enough horizontal curvature.

Rigas [Ri] proved the stable sphere theorem: after adding a sufficiently large trivial summand, every real vector bundle over any sphere admits nonnegative sectional curvature. Grove and Ziller [GZ] obtained the decisive unstable low-dimensional results. Every vector bundle and every sphere bundle over $\mathbb{S}^4$ or $\mathbb{S}^5$ admits a complete metric with nonnegative sectional curvature. Their construction encodes a cohomogeneity-one manifold by the two disk bundles of its singular orbits, builds an invariant nonnegatively curved metric when those orbits have codimension two, and passes to associated bundles by Riemannian submersion. They also constructed such metrics for every three-dimensional and 88 of the 144 four-dimensional vector bundles over $\mathbb{S}^7$, together with the corresponding sphere bundles.

The broader nonnegative-base version is false. For flat torus bases, there are counterexamples, and Tapp [Ta] formulated differential conditions involving the normal connection curvature, the horizontal sectional curvature, and the Hessian of the vertical curvature which any vector bundle admitting a complete nonnegatively curved metric must satisfy. The inequality comes from the Hessian of the sectional curvature function at a mixed zero-curvature plane along the soul. A strict form is sufficient for positive curvature on the associated sphere bundle and, in the connection-metric case, for complete nonnegative curvature on the vector bundle. Thus the positivity of the base in the original question is not a harmless hypothesis. González-Álvaro [GA] proved a stable positive result: after adding a sufficiently large trivial bundle, every real or complex vector bundle over a compact rank-one symmetric space admits a complete metric with nonnegative sectional curvature. The proof identifies stable classes with reduced K-theory, realizes them by homogeneous bundles through representation rings, and equips those representatives with quotient metrics. This stable theorem is one of the closest modern approximations to Yau's original question.

Thus the present frontier has two different forms. The unstable problem asks whether the original bundle itself carries a complete nonnegatively curved metric. The stable problem asks for the minimal rank of a trivial summand after which such a metric exists. The González-Álvaro theorem answers the second question for compact rank-one symmetric spaces, but leaves the first question open even in several natural cases where the base has positive sectional curvature.

Related open problems.

1. *Original direction.* Does every vector bundle over a sphere admit a complete metric with nonnegative sectional curvature? The cases $\mathbb{S}^4$ and $\mathbb{S}^5$, and many bundles over $\mathbb{S}^7$, are known [Ri, GZ].

2. *Original direction.* Does every vector bundle over every compact rank-one symmetric space admit such a metric without adding a trivial bundle? González-Álvaro proves this after stabilization [GA].

3. *Reference-driven formulation.* Find intrinsic topological obstructions for a vector bundle over a positively curved base to admit complete nonnegative sectional curvature. Tapp's conditions [Ta] are an important step, but a classification is not known.

4. *Reference-driven formulation.* Compare connection metrics with more general complete metrics on the total space: are Tapp-type differential conditions also close to sufficient after allowing deformations away from connection metrics [Ta]?

5. *Reference-driven formulation.* Determine the minimal stable rank in González-Álvaro's theorem [GA]: for a given real or complex bundle over a compact rank-one symmetric space, how many trivial summands are actually necessary before a complete nonnegatively curved metric exists?

6. *AI-generated formulation.* Develop computable characteristic-class tests predicting the minimal rank of a trivial summand needed before a given vector bundle admits nonnegative sectional curvature, extending the stable estimates in [GA].

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Problem 7

(Chern). Let M be a compact, positively curved manifold. Is it true that every abelian subgroup of $\pi_1(M)$ is cyclic?

Status. False. Shankar gave counterexamples in 1998 [S]: compact positively curved manifolds can have noncyclic abelian subgroups in their fundamental groups. Thus Chern's proposed analogue of the negative-curvature cyclicity theorem fails, although the broader problem of classifying fundamental groups in positive sectional curvature remains open.

Chern's question was motivated by two classical comparison pictures. For compact negatively curved manifolds, Preissmann's theorem [P] implies that every nontrivial abelian subgroup of the fundamental group is infinite cyclic. For constant positive curvature, the spherical space-form problem restricts the possible finite groups acting freely on spheres [W]. Since Myers' theorem already implies that the fundamental group of a compact positively curved manifold is finite, Chern asked whether positive sectional curvature might force the abelian subgroups of this finite group to be cyclic, as happens for spherical space-form groups in many familiar cases.

The answer is negative. Shankar [S] constructed positively curved examples whose finite fundamental groups contain noncyclic abelian subgroups. The construction is related to finite quotients of positively curved homogeneous spaces, including Aloff–Wallach type examples. Grove, Shankar and Ziller [GSZ] later produced further counterexamples through Eschenburg spaces and biquotient constructions, showing that the failure of Chern's proposed cyclicity condition is not isolated and is compatible with the symmetry-rich world of known positive-curvature examples. More precisely, for every finite subgroup $\Gamma \subset \mathrm{SO}(3)$ they construct infinitely many positively curved Eschenburg spaces on which Γ acts freely and isometrically. The quotients therefore realize Γ as fundamental group; taking $\Gamma = \mathbb{Z}_2 \oplus \mathbb{Z}_2$ gives explicit Abelian counterexamples.

Nevertheless, the broader classification problem for fundamental groups of positively curved manifolds remains very much open. Synge's theorem gives strong restrictions in even dimensions, and Myers' theorem gives finiteness of the fundamental group in all dimensions. Under additional torus symmetry, Kennard [Ke] obtained substantial classification results for fundamental groups of positively curved manifolds. Thus the correct post-Shankar problem is no longer cyclicity of abelian subgroups, but the classification of finite groups which can occur under positive sectional curvature, possibly with extra hypotheses such as dimension, orientability, symmetry rank, or cohomological restrictions. One concrete frontier is already visible in the known examples: $\mathbb{Z}_3 \oplus \mathbb{Z}_3$ can occur, whereas it remains unknown whether $\mathbb{Z}_p \oplus \mathbb{Z}_p$ can occur for any prime $p \geqslant 5$ [Ke].

Related open problems.

1. *Original direction.* Classify the finite groups which occur as fundamental groups of compact positively curved manifolds in each dimension. Shankar's counterexamples [S] show that the answer is broader than Chern's original conjecture, while Myers' theorem still forces finiteness.

2. *Reference-driven formulation.* Classify fundamental groups of positively curved manifolds under symmetry assumptions. Kennard's results [Ke] give a systematic picture in the torus-symmetry setting, but the general case remains open.

3. *Reference-driven formulation.* Determine which spherical space-form group restrictions survive for positively curved manifolds that are not space forms, and which fail in the homogeneous and biquotient examples [W, S, GSZ].

4. *AI-generated formulation.* Find the best replacement for Chern's cyclicity conjecture: what universal restrictions do positive sectional curvature, dimension, orientability, and symmetry rank impose on abelian subgroups of $\pi_1(M)$? This is based on [S, GSZ, Ke].

5. *Reference-driven formulation.* Decide whether a positively curved manifold can have fundamental group containing $\mathbb{Z}_p \oplus \mathbb{Z}_p$ for a prime $p \geqslant 5$. The cases visible in the current constructions stop at $p = 3$ [GSZ, Ke].

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Problem 8

For an even-dimensional compact manifold with positive sectional curvature, prove that the Euler number is positive.

Status. Open as of 2026 in every even dimension at least six. The conjecture is classical in dimensions two and four, and it is known under a number of additional hypotheses, especially large symmetry, suitable fixed-point data, and rational-homotopy restrictions. The main lesson of the subject is that the Euler characteristic should be controlled by the global topology forced by positive curvature, not merely by the sign of a local curvature polynomial: Geroch showed that in dimensions at least six positive sectional curvature does not force the Gauss–Bonnet–Chern integrand to be positive [Ge].

The low-dimensional cases explain both the attraction and the limitation of the Gauss–Bonnet approach. For surfaces the assertion is exactly Gauss–Bonnet. In dimension four there is a very short topological proof. If M^4 is orientable and has positive sectional curvature, Synge's theorem implies that M is simply connected. Thus $b_1(M) = b_3(M) = 0$, and the Euler–Poincaré formula gives

$$\chi(M) = 1 - b_1(M) + b_2(M) - b_3(M) + b_4(M) = 2 + b_2(M) > 0.$$

If M is nonorientable, the orientable double cover has positive sectional curvature and positive Euler characteristic, and $\chi(\widetilde{M}) = 2\chi(M)$, so again $\chi(M) > 0$. There is also a local four-dimensional proof: Milnor's unpublished observation, recorded by Chern, shows that the algebraic Hopf statement for the Gauss–Bonnet integrand is valid in dimension four [Ch2]. These two proofs are complementary. Synge's proof uses global topology; the Chern–Milnor proof uses the special four-dimensional curvature decomposition. Neither mechanism extends directly to arbitrary dimension.

Historically, Hopf's work on total curvature of hypersurfaces and Chern's intrinsic Gauss–Bonnet theorem made it natural to ask whether positive sectional curvature should force positivity of the Euler characteristic [Ho], [Ch1]. The strongest local version would assert that every algebraic curvature tensor with positive, or nonnegative, sectional curvature has positive, respectively nonnegative, Pfaffian. That statement is true in dimension four, but Geroch disproved it in every even dimension at least six [Ge]. Hence the remaining conjecture is genuinely global: the topology of a closed positively curved manifold may still force $\chi > 0$, but this cannot be proved by pointwise positivity of the Gauss–Bonnet integrand alone.

There are important positive results outside the intrinsic low-dimensional case. Hopf's original extrinsic theorem gives positivity for closed positively curved hypersurfaces in Euclidean space, and Weinstein proved a codimension-two analogue for positively curved submanifolds of $\mathbb{R}^{n+2}$ [Ho], [We]. These results show that the conjectural sign is compatible with strong ambient realization hypotheses, although they do not address arbitrary intrinsic metrics.

The most productive intrinsic progress has come from symmetry. Hopf and Samelson proved that for compact homogeneous spaces G/H the Euler characteristic is nonnegative, and is positive precisely when $\operatorname{rank} G = \operatorname{rank} H$ [HS]. Wallach's classification of compact homogeneous spaces with strictly positive sectional curvature then implies the Hopf conclusion for homogeneous positive-curvature examples [Wa]. Later work extended this philosophy beyond homogeneous spaces. Puttmann and Searle proved the conjecture for compact even-dimensional positively curved manifolds admitting an isometric compact Lie group action with principal isotropy H and cohomogeneity k satisfying

$$k - \big(\operatorname{rank} G - \operatorname{rank} H\big) \leqslant 5,$$

and also for certain high-symmetry torus actions [PS]. Kennard proved a logarithmic symmetry-rank version using Steenrod operations, and subsequent work with Amann obtained further restrictions on Euler characteristics under large torus symmetry [Ke], [AK].

A more recent development reduces part of the symmetry problem to representation theory. Kennard, Wiemeler and Wilking proved that the Hopf Euler-characteristic conjecture holds for positively curved even-dimensional manifolds whose isometry group has rank at least five [KWW1]. Their argument uses a splitting theorem for torus representations whose isotropy groups are connected, together with the connectedness and periodicity techniques that come from positive curvature. Nienhaus subsequently improved the conclusion from T^5 to T^4 symmetry [Ni]. His new four-periodicity theorem exploits the natural complex structure on the normal bundle of a circle fixed set: the periodicity-inducing Euler class becomes a Chern class, so characteristic classes and Steenrod operations force rational four-periodicity under a weaker codimension bound. A classification of the relevant T^4 weight systems then gives CROSS-type rational cohomology for every torus fixed-set component, and Berger's fixed-point theorem and Lefschetz yield $\chi(M) > 0$. In later work, Kennard–Wiemeler–Wilking linked positive curvature with torus symmetry to regular matroids, obtaining new obstructions to positively curved metrics with torus symmetry [KWW2]. These results do not settle the general conjecture, but they mark a shift: the symmetry approach is no longer only a high-rank classification program; it now gives effective low-rank structural tools.

There are also recent results for discrete symmetry and for curvature conditions adjacent to positive sectional curvature. Bosgraaf, Escher and Searle lowered the rank assumptions in fixed-point-set theorems for $\mathbb{Z}_2$-tori acting on positively curved

manifolds, obtaining cohomological classification conclusions for fixed-point components [BES]. Kennard, Mouillé and Nienhaus studied Hopf-type Euler-characteristic questions under positive second intermediate Ricci curvature; they emphasize that the Hopf conclusion can fail under such relaxations, but prove a positive Euler-characteristic theorem in dimensions divisible by four under sufficiently large torus symmetry [KMN]. These papers are not solutions of the sectional curvature Hopf conjecture. They are useful because they show which parts of the symmetry method survive when one changes either the symmetry group or the curvature condition.

A complementary 2025 preprint of Huang and Wang [HW] treats curvature subclasses rather than symmetry. By decomposing the curvature operator into its Schouten–Ricci and Weyl parts, they prove the Hopf sign conclusions for compact locally conformally flat manifolds and under an explicit pointwise Weyl-pinching condition relative to a strict sectional-curvature bound. Integral Weyl bounds give further harmonic-form vanishing. Positive sectional curvature alone does not imply these Weyl smallness assumptions, so this is a new partial result rather than a solution of the general conjecture.

A third line connects the conjecture with the Bott–Grove–Halperin rational-ellipticity philosophy. The Bott–Grove–Halperin conjecture predicts that simply connected closed manifolds with nonnegative sectional curvature are rationally elliptic. Rational ellipticity gives strong constraints on Betti numbers and would provide a conceptual path toward the nonnegative Euler-characteristic form of Hopf's conjecture [GrH], [FHT]. Amann and Kennard proved versions of Hopf's conjecture under the joint assumptions of rational ellipticity and torus symmetry, including positivity of $\chi(M)$ in broad logarithmic-symmetry regimes [AK2]. The missing bridge is still the curvature-to-rational-homotopy step: positive sectional curvature itself is not yet known to imply enough rational-homotopy structure to force the Euler sign.

The analogy with algebraic geometry is useful but should not be confused with the Riemannian problem. Bloch and Gieseker proved positivity of Chern classes for ample vector bundles [BG]. This supports the general theme that positive geometric curvature should force positivity of characteristic numbers, but it does not settle the Riemannian sectional-curvature conjecture.

Related open problems.

1. *Original direction.* Prove the positive-curvature Hopf conjecture in the first open case, real dimension six. A proof in dimension six would have to bypass Geroch's obstruction to pointwise Pfaffian positivity and identify a genuinely global positive-curvature mechanism [Ge].

2. *Original direction.* Prove the nonnegative version: if a compact even-dimensional manifold has nonnegative sectional curvature, then $\chi(M) \geqslant 0$. This is the natural weak-curvature companion of Hopf's conjecture and is tied to the Bott–Grove–Halperin rational-ellipticity conjecture [GrH], [FHT].

3. *Reference-driven formulation.* Reduce the symmetry hypotheses in the known theorems. In particular, determine whether the Hopf conclusion follows from rank below four, from a single isometric circle action, or from structural restrictions on the fixed point set [PS], [Ke], [AK], [KWW1], [Ni], [BES].

4. *Reference-driven formulation.* Extend the Kennard–Wiemeler–Wilking representation-theoretic and matroid methods to torus actions with disconnected isotropy, lower rank, or weaker fixed-point data [KWW1, KWW2].

5. *Reference-driven formulation.* Determine whether Hopf-type Euler-characteristic positivity under positive intermediate Ricci curvature is possible only with substantial symmetry, and identify the exact point at which sectional-curvature methods break down [KMN].

6. *Reference-driven formulation.* Determine whether the conjecture is true for all positively curved manifolds whose rational cohomology is concentrated in even degrees, and find geometric hypotheses which force that cohomological conclusion [GrH], [FHT], [AK2].

7. *AI-generated formulation.* Identify curvature cones stronger than positive sectional curvature but weaker than positive curvature operator for which the Gauss–Bonnet–Chern integrand is pointwise nonnegative in all even dimensions. Geroch's examples show that ordinary positive sectional curvature is insufficient [Ge].

8. *Reference-driven formulation.* Remove or weaken the Weyl smallness assumptions in the curvature-operator approach, especially in dimension six, where the conjecture first remains open [HW].

9. *AI-generated formulation.* Develop a rational-homotopy criterion, usable directly from positive sectional curvature, which rules out odd rational cohomology and hence forces positive Euler characteristic in the rationally elliptic case [GrH], [FHT], [AK2].

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Problem 9

Characterize the groups which can appear as the fundamental group of some compact manifold with negative curvature.

Status. Open. There is no known intrinsic group-theoretic characterization of the fundamental groups of closed negatively curved manifolds. If M^n is closed and negatively curved, then $\pi_1(M)$ is a torsion-free word-hyperbolic Poincaré duality group of dimension n, and its Gromov boundary is homeomorphic to $\mathbb{S}^{n-1}$. These are powerful necessary conditions, but they are not a characterization: the converse realization problem for hyperbolic groups with sphere boundary is still open even in closely related forms.

The first restrictions are classical. By the Cartan–Hadamard theorem the universal cover is contractible, so a compact negatively curved manifold is a $K(\pi, 1)$ and its fundamental group is torsion-free. Preissmann proved in 1943 that every nontrivial abelian subgroup is infinite cyclic [P]. Yau strengthened this type of restriction by showing that subgroups containing subnormal abelian subgroups are cyclic; in particular, solvable subgroups are severely constrained [Y]. Milnor proved that the fundamental group has exponential growth [M1].

The next stage is geometric and dynamical. Margulis' thin-thick theory gives quantitative restrictions on small displacement subgroups and leads to strong information about conjugacy classes [Ma], [BGS]. Eberlein showed that such groups contain nontrivial free subgroups [E]. These results already show that the group must be large and nonamenable, while its amenable subgroups are small. In the later language of geometric group theory, the universal cover is a Gromov-hyperbolic space and $\pi_1(M)$ is a torsion-free word-hyperbolic group [G], [BH]. Consequently, every amenable subgroup is virtually cyclic; in the torsion-free closed negatively curved case, it is cyclic.

Gromov's 1987 theory of hyperbolic groups changed the form of the problem. It gives a broad necessary framework, but it does not solve the realization question: not every torsion-free hyperbolic group is known, or expected, to be the fundamental group of a smooth closed manifold with a negatively curved metric. The boundary at infinity sharpens the obstruction. Since the universal cover of a closed negatively curved n-manifold is diffeomorphic to $\mathbb{R}^n$ and has visual boundary $\mathbb{S}^{n-1}$, the Gromov boundary of the group is a sphere. Bestvina–Mess related the cohomology of a hyperbolic group to the topology of its boundary; in particular, Poincaré duality for the group is reflected in homological properties of the boundary [BM]. Thus Yau's problem can be rephrased as a realization problem inside the much smaller class of torsion-free hyperbolic Poincaré duality groups with spherical boundary.

This boundary formulation also explains why the problem is close to major open questions in geometric topology. In dimension three, geometrization gives a sharp answer for genuine 3-manifold groups: closed hyperbolic 3-manifold groups are exactly the fundamental groups of closed 3-manifolds carrying hyperbolic geometry. But the purely group-theoretic recognition problem remains subtler. Cannon's conjecture asks whether a word-hyperbolic group with boundary $\mathbb{S}^2$ acts geometrically on hyperbolic 3-space; Markovic reduced this to the existence of enough quasiconvex surface subgroups [CS], [Mar]. In higher dimensions, spherical boundary by itself is still not a known Riemannian negative-curvature realization criterion.

Recent work has clarified how much extra structure must be added to a spherical-boundary hyperbolic group before one can hope for a manifold realization theorem. Bregman and Incerti-Medici [BIM] proved a higher dimensional Cannon-type result for torsion-free hyperbolic groups with boundary $\mathbb{S}^{n-1}$, for $n \neq 5$, assuming the existence of sufficiently many quasiconvex codimension-one subgroups whose locally flat spherical limit sets separate boundary points. Their proof uses this system to cubulate the group, passes to malnormal finite-index data, builds an equivariant boundary cell complex, and invokes generalized Schoenflies and topological rigidity. Their conclusion is a virtual closed orientable aspherical manifold realization, not a smooth Riemannian metric of negative sectional curvature. This distinction is important for Yau's problem: current geometric group theory can sometimes turn boundary and cubulation data into manifold topology, but the additional step from aspherical manifold group to negatively curved Riemannian manifold remains a separate and much stronger requirement.

Locally symmetric examples provide important test cases. For irreducible locally symmetric spaces of noncompact type and dimension greater than two, Borel's theorem and Mostow rigidity imply strong restrictions, for example finiteness of the outer automorphism group in the locally symmetric case [B]. However, Mostow and Siu constructed compact Kähler surfaces of negative sectional curvature not homotopy equivalent to locally symmetric spaces [MS]. Thus locally symmetric groups do not exhaust the negatively curved Kähler or Riemannian possibilities. Hyperbolic arithmetic constructions of Millson and Vinberg show that first Betti number phenomena also vary substantially among constant-curvature examples [Mil], [Vi].

The construction side has also expanded far beyond locally symmetric spaces. Charney–Davis strict hyperbolization produces many locally $\mathrm{CAT}(-1)$ aspherical spaces and manifolds [CD]. Ontaneda's Riemannian hyperbolization then showed that, for smooth input in the appropriate dimensions, one can smooth the strict hyperbolization construction to obtain closed smooth manifolds with sectional curvatures pinched in $[-1-\epsilon, -1]$ [O]. The construction chooses hyperbolization pieces with large normal widths, uses compatible normal smooth structures on links, and combines hyperbolic extension, warp forcing, and two-variable deformation to obtain uniform C^2 curvature control. It also produces a degree-one map to the input manifold which injects cohomology and preserves rational Pontryagin classes. These results give large new families of negatively curved

manifold groups and show that locally symmetric lattices are only a small part of the landscape. They do not, however, provide an intrinsic test telling whether a given hyperbolic group is realized by a smooth closed negatively curved manifold.

The Kähler version remains especially delicate. If a compact negatively curved manifold is also Kähler, one can ask whether some finite cover must have nonzero first Betti number. This is not known in general. Results on Kähler groups and negatively pinched groups give obstructions in special settings, but they do not yet amount to a characterization [JY].

Related open problems.

1. *Original direction.* Characterize which torsion-free word-hyperbolic Poincaré duality groups with boundary $\mathbb{S}^{n-1}$ are realized by closed smooth manifolds with negative sectional curvature [G], [BH], [BM].

2. *Reference-driven formulation.* Determine whether every compact negatively curved Kähler manifold has a finite cover with $b_1 > 0$, or produce a geometric obstruction to this expectation [MS], [JY].

3. *Reference-driven formulation.* Decide whether finiteness of $\mathrm{Out}(\pi_1 M)$ holds for all closed negatively curved manifolds in the same generality as in the irreducible locally symmetric case [B], [MS].

4. *Reference-driven formulation.* In dimension three, relate the group-recognition problem for negatively curved manifolds to Cannon's conjecture and to the existence of quasiconvex surface subgroups [CS], [Mar].

5. *Reference-driven formulation.* In higher dimensions, determine whether the cubulated spherical-boundary realization results of Bregman–Incerti-Medici [BIM] can be strengthened from virtual aspherical manifold realization to smooth closed manifolds admitting negatively curved metrics, or whether new smooth/geometric obstructions appear.

6. *Reference-driven formulation.* Resolve the missing $n = 5$ case in the higher-dimensional Cannon-type manifold-realization theorem, and determine whether local flatness of the separating limit spheres follows from more intrinsic cubulation or boundary hypotheses [BIM].

7. *AI-generated formulation.* Find a purely group-theoretic criterion, stable under quasi-isometry, which distinguishes fundamental groups of smooth closed negatively curved manifolds from general torsion-free word-hyperbolic groups with sphere boundary [G], [BH], [O].

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Problem 10

As a continuation of the previous problem, let M^{2N} be a compact manifold with negative curvature. Is it true that $(-1)^N\chi(M) > 0$?

Status. Open as of 2026 in general, with the first open strictly negative-curvature case in real dimension six. The conjecture is known in real dimensions two and four, for locally symmetric examples, and in several Kähler or pinched settings. It is the negative-curvature half of the Hopf sign problem: positive sectional curvature should force $\chi > 0$, while negative sectional curvature on a closed $2N$-manifold should force $(-1)^N\chi > 0$.

The two-dimensional case is exactly Gauss–Bonnet. In real dimension four the Chern–Gauss–Bonnet integrand has a special sign structure under nonpositive sectional curvature; consequently a closed nonpositively curved four-manifold satisfies $\chi \geqslant 0$, and the strictly negative case gives $\chi > 0$ [Ch]. This is a genuinely low-dimensional phenomenon. Geroch showed that in every even dimension at least six the Gauss–Bonnet–Chern integrand is not controlled pointwise by the sign of sectional curvature [Ge]. Thus, as in the positive-curvature Hopf problem, a proof in higher dimensions must use global geometry or topology rather than only a local Pfaffian sign.

Locally symmetric examples form the basic model class. For compact locally symmetric spaces of noncompact type, Hirzebruch proportionality relates characteristic numbers to those of the compact dual, and gives the expected Euler sign [Hi]. Products of negatively curved surfaces give the simplest non-locally-irreducible examples: the sign follows immediately from multiplicativity of Euler characteristic. These model cases suggest the right answer, but they also hide the main difficulty: variable negative curvature has far less algebraic structure than local symmetry.

The analytic route is the Singer L^2-cohomology philosophy. Since a closed negatively curved manifold is aspherical by Cartan–Hadamard, its Euler characteristic can be studied through the universal cover. Singer's conjecture predicts that for a closed aspherical manifold,

$$b_p^{(2)}(\widetilde{M}) = 0 \qquad \text{if } 2p \neq \dim M.$$

For a closed negatively curved manifold one expects, in addition, positivity of the middle L^2-Betti number in even dimensions. If this vanishing and middle-dimensional positivity hold, then Atiyah's L^2 index theorem for coverings gives

$$\chi(M) = \sum_p (-1)^p b_p^{(2)}(\widetilde{M}),$$

and the desired sign follows [At].

The Kähler case is the most complete analytic success story. Gromov introduced Kähler-hyperbolic manifolds and proved that a compact Kähler-hyperbolic manifold of complex dimension n satisfies $(-1)^n\chi(M) > 0$ [Gr]. Compact Kähler manifolds with negative sectional curvature are Kähler-hyperbolic, so the conjecture is true in that case. Cao–Xavier and Jost–Zuo extended the L^2-Hodge method to prove $(-1)^n\chi(M) \geqslant 0$ for closed nonpositively curved Kähler manifolds under their hypotheses [CX], [JZ]. In algebraic geometry, Bloch–Gieseker's positivity theorem for ample vector bundles gives a parallel positivity phenomenon for characteristic classes [BG]. The open Riemannian problem is therefore not a failure of the L^2 idea, but the lack of a Kähler package: without type decomposition, Lefschetz operators, and complex positivity, the middle-degree concentration is much harder to force.

There are also partial Riemannian vanishing theorems. Under suitable pinching or curvature-decay hypotheses, Jost–Xin and Ballmann–Brüning proved vanishing results for L^2 harmonic forms outside the middle degree [JX], [BB]. These results support Singer's route, but they do not yet cover arbitrary closed negatively curved manifolds. Recent work on L^2-Betti numbers for groups with hierarchies and for Dehn-filling constructions gives additional evidence for the Hopf–Singer viewpoint in large topological classes, but these methods still do not give a general sectional-curvature theorem [OS], [PSu]. Petrosyan and Sun [PSu] construct a classifying space for group-theoretic Dehn fillings and combine quantitative L"uck approximation, the Cohen–Lyndon property, special-quotient theorems, and Atiyah-conjecture discreteness to prove stability of L^2-Betti numbers for broad deep fillings. As an application they obtain Anderson-type closed Einstein manifolds satisfying the Singer conjecture. These special Einstein examples need not have negative sectional curvature, so they test the L^2 mechanism without settling Yau's geometric case.

The topological analogue is the aspherical Euler-sign problem, often called the Hopf–Thurston sign conjecture: if M^{2N} is closed and aspherical, should $(-1)^N\chi(M) \geqslant 0$ hold? Kan–Thurston's theorem shows that aspherical models are extremely flexible homologically [KT], while the Charney–Davis conjecture gives a combinatorial nonpositive-curvature version for flag triangulations and piecewise Euclidean manifolds [CD]. These topological forms are related to, but distinct from, the strict Riemannian negative-curvature problem. A negative-curvature proof would be stronger than a purely topological theorem for that geometric class, but it might use analytic information invisible to the fundamental group alone.

Recent work also clarifies the equality case in low-dimensional nonpositive-curvature geometry. Connell–Ruan–Wang proved that if a closed nonpositively curved Riemannian 4-manifold has $\chi = 0$, then the Ricci tensor must degenerate somewhere, and

in the analytic case the universal cover has a nontrivial Euclidean de Rham factor [CRW]. Their proof uses pointwise Pfaffian vanishing to isolate a distinguished Ricci eigenline, varies the metric in that direction to force a totally geodesic flat orthogonal foliation, and finishes with a divergence argument and analytic rank rigidity. Kim–Wan subsequently proved that a closed nonpositively curved four-manifold with nonzero Euler characteristic obeys the quantitative inequality

$$\|M\| \geqslant \frac{1}{11}|\chi(M)|,$$

and in particular obtained a four-dimensional positivity result under negative Ricci curvature [KW]. These theorems do not settle the higher-dimensional negative-curvature problem, but they sharpen the boundary between strict negativity, flat directions, Euler characteristic, and simplicial volume. Their proof straightens a fundamental cycle to geodesic simplices and applies the Allendoerfer–Weil Gauss–Bonnet formula simplex by simplex, bounding all face and angle contributions by the ℓ^1-norm.

Related open problems.

1. *Original direction.* Prove the conjecture in real dimension six. This is the first dimension in which the four-dimensional Chern–Gauss–Bonnet sign argument and all pointwise Pfaffian strategies fail [Ch], [Ge].

2. *Reference-driven formulation.* Prove Singer's L^2-Betti-number vanishing conjecture for all closed negatively curved manifolds, and prove positivity of the middle L^2-Betti number in even dimensions [At], [Lu].

3. *Reference-driven formulation.* Find a higher-dimensional replacement for the four-dimensional Chern–Gauss–Bonnet sign argument, or prove that no pointwise curvature-polynomial refinement can exist beyond the known low-dimensional cases [Ch], [Ge].

4. *Reference-driven formulation.* Extend the Kähler-hyperbolic argument beyond the Kähler setting: find a geometric condition implied by negative sectional curvature that forces middle-dimensional L^2 cohomology and the Hopf sign [Gr], [CX], [JZ].

5. *Original direction.* Determine whether the Hopf–Thurston Euler-sign conjecture holds for all closed aspherical manifolds, or at least for all closed smooth manifolds admitting a nonpositively curved metric [KT], [CD].

6. *Reference-driven formulation.* In real dimension four, understand whether the Connell–Ruan–Wang rigidity picture for $\chi = 0$ has smooth, nonanalytic, or higher-dimensional analogues; in particular, decide when Euler-characteristic equality forces a Euclidean de Rham factor [CRW], [KW].

7. *Reference-driven formulation.* Find a six-dimensional analogue of the Kim–Wan geodesic-simplex estimate which controls the signed Euler form, rather than only an absolute simplicial-volume contribution [KW].

8. *Reference-driven formulation.* Determine whether the L^2-Betti stability methods for deep Dehn fillings can be combined with pinched negatively curved constructions, producing new geometric test families for the Hopf sign [PSu].

9. *AI-generated formulation.* Develop a pinching-free Bochner or coarse-index method which proves the L^2 vanishing outside the middle degree for universal covers of closed negatively curved manifolds [JX], [BB].

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Problem 11

The Cohn–Vossen inequality says that the total curvature of a complete surface is dominated by its Euler number. Finn [Fi] and Huber [Hu] studied the difference of these numbers in terms of geometric quantities. The question is how to generalize this inequality to higher dimensions. If M is a complete manifold with finite volume and bounded curvature, when is the Euler number equal to the Gauss–Bonnet–Chern integral? If the manifold is locally symmetric, this is true and due to Harder [Ha]. It can be proved that the assertion is true if the curvature of M is bounded between two negative constants. In a private conversation, Gromov claimed that the Gauss–Bonnet–Chern integral is an integer and is the Euler number if one merely assumes that the curvature is nonpositive and the metric is real analytic.

If M is complete and has nonnegative curvature, Poor [Po], R. Walter [Wa] and Greene and Wu (Theorem 9 of [GW]) proved that the Cohn–Vossen inequality holds true if $\dim M = 4$. What is the geometric constraint on M if the equality holds? What happens when $\dim M = 2n$ with $n > 2$?

Status. Open in the general form. The problem should be separated into two related but different questions: a finite-volume Gauss–Bonnet–Chern equality problem, and a nonnegative-curvature Cohn–Vossen inequality/equality problem. Both are understood in several important model geometries, but there is still no general end-geometry criterion in all even dimensions.

In dimension two, Cohn–Vossen's inequality says that for a complete noncompact surface of finite total curvature the curvature integral is bounded above by the Euler characteristic [CV]. Finn and Huber refined the missing term: the defect between the Euler number and the curvature integral is measured by the geometry of the ends [Fi], [Hu]. The higher-dimensional problem is therefore not merely to integrate the Pfaffian curvature form. One must also identify the boundary transgression terms produced by an exhaustion and prove that their limits are intrinsic invariants of the ends.

For complete finite-volume manifolds with bounded curvature, Cheeger–Gromov gave a powerful open-manifold Gauss–Bonnet theorem in the language of L^2-cohomology and von Neumann dimension [CG]. This explains why Yau's question is close to, but not identical with, the compact Gauss–Bonnet–Chern theorem: the natural analytic invariant at infinity is an L^2 Euler characteristic, and the comparison with the ordinary Euler number requires topological and geometric control of the ends. The technical heart of this comparison is the construction of good exhaustions with controlled second fundamental form and curvature transgression terms; the Schoen–Yau exhaustion construction and Dafermos' account are useful references for this point [SY], [D]. Finite-volume locally symmetric spaces form the cleanest positive class, by Harder's arithmetic Gauss–Bonnet formula [Ha].

The negatively curved and nonpositively curved cases should also be kept distinct. Under pinched negative curvature and finite volume, the ends have enough controlled asymptotic geometry for the expected Gauss–Bonnet–Chern identity to hold. Gromov's private real-analytic nonpositive-curvature claim, as quoted by Yau, is stronger and should still be recorded as a problem unless one imposes hypotheses which force sufficiently rigid ends. In the smooth nonpositive-curvature category there are many possible cusp and collapse phenomena, so the integer-valuedness of the Pfaffian integral is itself part of the difficulty.

The nonnegative-curvature direction is a different Cohn–Vossen problem. Poor, Walter, and Greene–Wu proved four-dimensional inequalities under nonnegative curvature hypotheses [Po], [Wa], [GW]. Equality should mean that all boundary defect terms at infinity vanish. In dimension two this is a strong asymptotic statement about the ends; in dimension four and higher its geometric meaning is not yet classified. A natural expectation is that equality forces strong splitting, cone, or flatness behavior at infinity, but there is no general theorem in dimensions $2n > 4$.

Several structured theories now give the best evidence for the correct formulation. Chang–Qing–Yang obtained four-dimensional conformally flat end formulae in which the Chern–Gauss–Bonnet defect is explicitly expressed through asymptotic quantities [CQY1], [CQY2]. Fang proved a conformal Gauss–Bonnet–Chern inequality for complete locally conformally flat manifolds with integrable Q-curvature or Pfaffian curvature hypotheses [Fa]. Buzano–Nguyen extended this circle to noncompact four-manifolds with conformally flat ends and isolated singularities, where the error terms are isoperimetric deficits, and then to all even dimensions in the conformally flat singular-end setting [BN], [BN2]. These results strongly support the view that the right high-dimensional replacement of the Finn–Huber defect is an explicit end invariant.

There is also recent progress on scalar-curvature versions of the Cohn–Vossen theme. Deng proved Cohn–Vossen-type scalar-curvature inequalities for complete noncompact manifolds with nonnegative Ricci curvature under additional topological, decay, locally conformally flat, or Bakry–Emery hypotheses [De]. Ma proved a locally conformally flat theory for complete noncompact manifolds with nonnegative Ricci curvature, verifying the scalar-curvature form of Yau's Cohn–Vossen conjecture in that class and proving a sharp gap theorem [Ma]. The mechanisms are particularly instructive. In Deng's topological case, a subgroup $\mathbb{Z}^{n-2} \subset \pi_1(M)$ forces the universal cover to split an $\mathbb{R}^{n-2}$ factor, reducing the estimate to the classical surface Cohn–Vossen theorem. In Ma's conformally flat case, a refined singularity theorem for n-superharmonic functions recovers the logarithmic mass of the conformal factor and identifies it with the asymptotic volume ratio ν. For $n > 3$ this gives the exact

formula

$$\lim_{r\to\infty} r^{2-n}\int_{B_r(p)} R_g\, d\mu_g = (n-1)\big(\nu^{\frac{n-3}{n-1}} - \nu\big)|S^{n-1}|.$$

These results are quite close in spirit to the original question, but they are scalar or conformally flat substitutes for the full Gauss–Bonnet–Chern problem. The remaining task is to turn such estimates into a Chern–Weil formula with computable boundary terms in arbitrary even dimension.

Related open problems.

1. *Original direction.* Give necessary and sufficient end-geometry conditions under which a complete finite-volume bounded-curvature $2n$-manifold satisfies
$$\chi(M) = \int_M e(TM, \nabla),$$
where $e(TM, \nabla)$ is the Gauss–Bonnet–Chern form [CG], [D], [SY].

2. *Original direction.* Classify equality cases in the known four-dimensional Cohn–Vossen inequalities under nonnegative curvature [Po], [Wa], [GW].

3. *Original direction.* Decide whether equality in the nonnegative-curvature Cohn–Vossen inequality forces splitting, asymptotic flatness, asymptotic conical structure, or vanishing of an explicitly defined end defect.

4. *Reference-driven formulation.* Extend the Chang–Qing–Yang conformally flat four-dimensional end-defect formula to higher even dimensions or to broader curvature classes [CQY1], [CQY2], [Fa], [BN], [BN2], [Ma].

5. *Reference-driven formulation.* Determine whether the recent scalar-curvature Cohn–Vossen inequalities under Ricci, locally conformally flat, or Bakry–Emery hypotheses can be lifted to full Chern–Weil boundary formulae in even dimensions [De], [Ma].

6. *Reference-driven formulation.* Clarify the relation between Cheeger–Gromov's L^2 Gauss–Bonnet theorem and the ordinary Euler characteristic for finite-volume bounded-curvature manifolds with collapsed or nonstandard ends [CG].

7. *AI-generated formulation.* Formulate an intrinsic "isoperimetric deficit at infinity" in dimension $2n$ whose sum over ends is exactly the difference between $\chi(M)$ and the Gauss–Bonnet–Chern integral, reducing to Finn's formula in dimension two [Fi], [CQY1], [BN], [BN2].

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Problem 12

Let M_1 and M_2 each have negative curvature. If $\pi_1(M_1) = \pi_1(M_2)$, prove that M_1 is diffeomorphic to M_2.

Status. False in the differentiable category in high dimensions. The correct positive statement is topological rigidity: under suitable dimension hypotheses, the fundamental group determines the manifold up to homeomorphism, but not up to diffeomorphism or PL homeomorphism.

The original motivation is Mostow rigidity. If the negatively curved manifolds are locally symmetric of rank one and dimension at least three, an isomorphism of fundamental groups is induced by an isometry. For general variable negative curvature this is too strong. Early progress of Cheeger–Gromov and Farrell–Hsiang showed that $\pi_1(M)$ determines various stabilized or associated objects; for example, Farrell–Hsiang proved stable topological information under nonpositive-curvature hypotheses [FH].

The decisive high-dimensional picture was developed by Farrell and Jones. On the positive side, they proved topological rigidity for compact nonpositively curved manifolds: in dimensions different from 3 and 4, if one manifold is closed nonpositively curved and the other is aspherical, then a homotopy equivalence is homotopic to a homeomorphism [FJ1], [FJ3]. Since closed negatively curved manifolds are aspherical by Cartan–Hadamard, this gives the expected homeomorphism conclusion in the relevant high-dimensional cases. In modern language this is part of the Borel-conjecture and Farrell–Jones assembly-map story. Bartels–Lück prove the required algebraic K- and L-theory assembly isomorphisms by controlled methods on hyperbolic and CAT(0) flow spaces; the surgery exact sequence then collapses the topological structure set. In particular, they proved the Borel conjecture for word-hyperbolic groups and for CAT(0) groups [BL], so the homeomorphism-level rigidity expected from the fundamental group is now known for closed negatively curved manifolds in the standard high-dimensional topological category. In dimension three, geometrization and Mostow rigidity settle the closed hyperbolic case and give the corresponding topological rigidity picture for negatively curved 3-manifolds. Dimension four remains the exceptional dimension; even the homeomorphism form is tied to the difficult four-dimensional Borel-conjecture landscape.

On the negative side, Farrell and Jones constructed negatively curved manifolds with exotic smooth structures [FJ2]. Thus two closed negatively curved manifolds can be homeomorphic, and hence have the same fundamental group, without being diffeomorphic. Farrell–Jones–Ontaneda strengthened this phenomenon by constructing examples in dimensions greater than five which are not even PL-homeomorphic, while still carrying negatively curved metrics [FJO]. These examples show that the original differentiable statement of the problem cannot be true.

Farrell–Jones's Theorem 1.1 gives more than a qualitative counterexample. Starting from a closed real-hyperbolic manifold of dimension at least five, they pass to a finite cover and form connected sums with representatives of the relevant homotopy-sphere classes. Smoothing theory distinguishes the resulting differentiable structures, while a warping deformation inserts the exotic transition into a sufficiently long hyperbolic annulus. Making the annulus long makes every curvature error arbitrarily small, so the exotic smoothings carry metrics with sectional curvature in $(-1-\delta, -1+\delta)$ for any prescribed $\delta > 0$ [FJ2]. This explains why neither strong pinching nor the fundamental group detects the smooth structure.

This problem should therefore be separated into three levels: homeomorphic rigidity, PL rigidity, and smooth rigidity. The homeomorphic version is a special case of the Borel conjecture for aspherical manifolds and is positive for the hyperbolic and CAT(0) group classes relevant here [BL]. The smooth and PL versions fail in high dimensions because exotic smooth and triangulation phenomena are compatible with negative curvature [FJ2], [FJO]. Related differentiable rigidity questions under stronger geometric assumptions, such as entropy or locally symmetric hypotheses, are treated by Besson–Courtois–Gallot and related work [BCG].

Later work shows that the failure of smooth rigidity is not just a bare existence phenomenon. Farrell–Ontaneda proved that the space of nonpositively curved metrics on a negatively curved manifold can be highly disconnected, so the moduli of curvature-constrained metrics remembers more than the fundamental group [FO]. Bustamante–Tshishiku studied symmetries of exotic negatively curved manifolds homeomorphic but not diffeomorphic to hyperbolic manifolds, producing examples where the hyperbolic isometry group acts smoothly and examples where large subgroups cannot act smoothly [BT]. Thus the modern refinement of Yau's question asks not only whether π_1 determines the underlying topological manifold, but how much differentiable and symmetry information can survive under negative curvature.

Related open problems.

1. *Reference-driven formulation.* Prove the homeomorphism version in dimension four for closed negatively curved manifolds, or identify the precise four-dimensional obstruction. This is the remaining low-dimensional edge of the Borel-rigidity picture [FJ3], [BL].

2. *Reference-driven formulation.* Classify which exotic smooth structures on a given closed negatively curved topological manifold admit negatively curved metrics [FJ2], [FJO].

3. *Original direction.* Determine geometric hypotheses stronger than negative curvature, but weaker than local symmetry, under which an isomorphism of fundamental groups forces diffeomorphism or isometry rather than only homeomorphism [BCG], [BL].

4. *Reference-driven formulation.* Study the moduli space of negatively curved metrics on a fixed topological manifold and determine whether different smooth structures can lie in naturally comparable geometric components [FJ2], [FJO], [FO].

5. *Reference-driven formulation.* Determine which finite isometry groups or finite group actions on a hyperbolic manifold can be realized smoothly on exotic negatively curved smoothings, and which exotic smoothings force symmetry breaking [BT].

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Problem 13

Let M_1 and M_2 be compact Einstein manifolds with negative curvature. Suppose $\pi_1(M_1) \cong \pi_1(M_2)$ and $\dim M_1 \geqslant 3$. Is M_1 isometric to M_2?

Status. False as literally stated, because no scale normalization is imposed. If g is a negatively curved Einstein metric, then cg is again negatively curved and Einstein for every $c > 0$, while (M, g) and (M, cg) are not isometric in general. The meaningful normalized version—fix the Einstein constant or the volume and ask for uniqueness up to diffeomorphism—is open in general. It is true for important locally symmetric models and locally in the moduli space of negatively curved Einstein metrics, but no general theorem says that the fundamental group determines the normalized Einstein metric.

The locally symmetric case is the model. If both manifolds are compact locally symmetric spaces of negative curvature and dimension at least three, Mostow's strong rigidity theorem shows that an isomorphism of fundamental groups is induced by an isometry after the natural normalization of scale [Mo]. This is stronger than topological rigidity and uses the symmetric-space structure, not merely the Einstein condition.

The harmonic-map approach gives another important class of rigidity theorems. Eells–Sampson proved existence of harmonic representatives for maps into nonpositively curved targets [ES]. Sampson and Siu developed Bochner-type rigidity results for harmonic maps; in particular, for compact Kähler manifolds with sufficiently negative curvature, a homotopy equivalence is forced to be holomorphic or anti-holomorphic, and hence becomes much closer to an isometry after imposing the relevant canonical metric condition [Sa], [Si]. This explains the older comment that strongly negative Kähler curvature is a setting where harmonic maps can prove biholomorphic rigidity.

Entropy rigidity provides a third line of partial progress. Besson, Courtois, and Gallot proved minimal entropy rigidity for negatively curved locally symmetric spaces, giving a powerful extension and new proof of Mostow-type rigidity [BCG]. These results are extremely strong when one of the manifolds is locally symmetric, but they do not settle the general case of two arbitrary negatively curved Einstein manifolds.

There are also four-dimensional and local-rigidity results which show both the strength and the limits of the Einstein condition. LeBrun used Seiberg–Witten theory to prove uniqueness results for Einstein metrics on compact quotients of irreducible four-dimensional symmetric spaces of noncompact type, giving a Riemannian version of Miyaoka–Yau type rigidity [Le]. Koiso's local rigidity theorem shows that compact Einstein metrics with negative sectional curvature have no nontrivial nearby Einstein deformations modulo diffeomorphism and scaling; Fine–Krasnov–Singer later proved a chiral four-dimensional generalization under negativity of the self-dual curvature operator [FKS]. Their proof recasts an oriented Einstein four-metric as a critical point of a pure-connection functional and proves positivity of its Hessian after removing gauge directions. These are local or special-model theorems, not a global classification of all negatively curved Einstein metrics with a fixed fundamental group.

The problem has become more subtle because negatively curved Einstein manifolds need not be locally homogeneous. Fine and Premoselli constructed compact Einstein four-manifolds with negative sectional curvature which are not locally homogeneous [FP]. These examples do not by themselves disprove the problem, but they show that the desired rigidity, if true, would have to apply well beyond the locally symmetric world. Their construction first interpolates between a smooth Einstein model near a codimension-two branch locus and a pulled-back hyperbolic metric, then uses uniform L^2 coercivity and a parameter-dependent inverse-function theorem to correct the small Einstein defect.

Very recent work strengthens this point in all dimensions at least four. Hamenstädt and Jäckel constructed infinitely many mutually non-homotopic closed n-manifolds, for every $n \geqslant 4$, which admit negatively curved Einstein metrics but admit no locally symmetric metric [HJ]. This does not produce two Einstein metrics with the same fundamental group, so it is not a counterexample to the normalized version of Yau's question. It does, however, show that the class of negatively curved Einstein manifolds is much larger than the locally symmetric class even at the level of homotopy types. Any solution of the problem must therefore combine topological rigidity with a genuinely Einstein-specific uniqueness mechanism.

Related open problems.

1. *Original direction.* Decide whether a compact negatively curved Einstein manifold is determined up to isometry by its fundamental group after fixing the Einstein constant or volume, under additional pinching assumptions [BCG], [FP].

2. *Original direction.* Prove or disprove uniqueness of negatively curved Einstein metrics on a fixed closed aspherical smooth manifold, up to scaling and diffeomorphism [FP].

3. *Reference-driven formulation.* Extend harmonic-map rigidity from strongly negative Kähler curvature to broader Einstein negative-curvature settings [ES], [Sa], [Si].

4. *Reference-driven formulation.* Relate local Einstein rigidity under negative sectional curvature to global uniqueness in a fixed aspherical homotopy type [FKS], [Le].

5. *Reference-driven formulation.* Determine whether the Fine–Premoselli negatively curved Einstein four-manifolds admit any nontrivial Einstein deformations or any distinct Einstein metrics with the same fundamental group [FP], [FKS].

6. *Reference-driven formulation.* For the Hamenstädt–Jäckel Gromov–Thurston Einstein manifolds, determine whether the negatively curved Einstein metric is unique up to scaling and diffeomorphism, and whether their fundamental groups can occur for more than one Einstein smooth model [HJ].

7. *Reference-driven formulation.* Decide whether every Einstein metric on a closed real-hyperbolic manifold is homothetic to the hyperbolic metric, as asked explicitly by Hamenstädt–Jäckel [HJ].

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Problem 14

Let M be a compact manifold of dimension N. Can one find a positive constant δ_N depending only on N so that M is diffeomorphic to a manifold with constant negative curvature whenever the curvature of M lies between $-1-\delta_N$ and -1?

Let M be a compact manifold whose sectional curvature lies between -4 and -1. If M is not diffeomorphic to a manifold with constant negative curvature, is M locally symmetric?

Status. False in every dimension $N \geqslant 4$. Gromov–Thurston disproved the proposed dimension-only rigidity constant by constructing nonhyperbolic manifolds with curvature arbitrarily close to -1. Hamenstädt–Jäckel now give, in every dimension at least four, near-hyperbolic manifolds which are not even homeomorphic to a closed locally symmetric space and which nevertheless carry negatively curved Einstein metrics [HJ]. The low dimensions should be kept separate: surfaces are governed by uniformization, and in dimension three geometrization gives a much more rigid topological picture for closed negatively curved manifolds.

The first question asks whether sufficiently strong pinching near constant curvature forces a compact negatively curved manifold to be diffeomorphic to a hyperbolic manifold. Gromov and Thurston constructed, in dimensions at least four, closed manifolds with sectional curvatures arbitrarily close to -1 which are not diffeomorphic to any hyperbolic manifold [GT]. Thus there is no dimension-only pinching constant δ_N with the proposed conclusion once $N \geqslant 4$. This distinguishes the problem sharply from the two- and three-dimensional settings, where uniformization and geometrization leave far less room for exotic smooth or PL behavior.

The smooth and PL failures are even stronger. Farrell, Jones, and Ontaneda constructed hyperbolic manifolds with negatively curved exotic triangulations in dimensions greater than five [FJO]. This is closely related to the exotic smooth-structure phenomena in Problem 12: high dimensional negative curvature is compatible with topology and smooth structures that constant-curvature locally symmetric metrics cannot carry.

The second question is also false. Ontaneda's pinched smooth hyperbolization, later developed as Riemannian hyperbolization, constructs closed Riemannian manifolds with curvatures in intervals arbitrarily close to $[-1,-1]$ which are not homeomorphic to locally symmetric spaces in the relevant high-dimensional range [O1], [O2]. Thus pinching alone does not force local symmetry.

Hamenstädt–Jäckel give a particularly sharp modern form of this failure. For every $n \geqslant 4$ and every $\varepsilon > 0$, their Theorem 1 constructs infinitely many pairwise nonhomeomorphic closed n-manifolds which admit a metric with sectional curvature in $[-1-\varepsilon, -1+\varepsilon]$ and also an Einstein metric of negative sectional curvature, but are not homeomorphic to any closed locally symmetric space [HJ]. The proof combines arithmetic subgroup separability and codimension-two branched covers with a uniformly invertible gauge-fixed Einstein operator and an implicit-function argument. Thus even adding the Einstein condition does not rescue the second proposed rigidity statement.

The hyperbolization viewpoint has also produced flexible noncompact extensions. Belegradek showed that hyperbolization pulls back regular neighborhoods of PL submanifolds and, using Ontaneda's Riemannian hyperbolization, obtained open complete manifolds of pinched negative curvature which are homotopy equivalent to closed smooth manifolds but have no smooth spines, as well as examples related to nonsmoothable closed manifolds [Be]. These examples are not compact counterexamples to the first sentence of the problem, but they illustrate how little pinched negative curvature alone controls smooth or PL topology once one leaves the locally symmetric category.

There are, however, important positive results once extra structure is imposed. In the Kähler setting, quarter-pinching is far more rigid: Hernandez and Yau–Zheng proved complex hyperbolic rigidity results for negatively quarter-pinched Kähler manifolds [He], [YZ]. Recent work on almost quarter-pinched Kähler examples shows that the exact pinching threshold is a meaningful boundary, not merely a technical artifact [Mi].

Related open problems.

1. *Reference-driven formulation.* Identify additional geometric structures, such as Kähler, locally symmetric, or curvature-operator assumptions, under which pinching near -1 does imply rigidity [He], [YZ].

2. *Reference-driven formulation.* Classify which smooth or PL exotic structures on hyperbolic manifolds admit metrics pinched arbitrarily close to -1 [FJO], [O1].

3. *Original direction.* Determine sharp pinching thresholds for rigidity in the Kähler and complex-hyperbolic settings [He], [YZ], [Mi].

4. *AI-generated formulation.* Study whether Ontaneda's Riemannian hyperbolization examples can be distinguished from locally symmetric manifolds by quantitative invariants beyond topology, such as entropy, marked length spectrum, or bounded cohomology [O2].

5. *Reference-driven formulation.* Determine which compact phenomena in the Gromov–Thurston and Ontaneda examples persist for complete open pinched negatively curved manifolds with controlled topology at infinity, and whether "no smooth spine" behavior has a compact analogue [Be], [O2].

6. *Reference-driven formulation.* Determine which Gromov–Thurston and smooth-hyperbolization manifolds admit negatively curved Einstein metrics, extending the Hamenstädt–Jäckel construction [HJ], [O1], [O2].

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Problem 15

Develop a useful notion of curvature for PL manifolds so that one could obtain appropriate pinching theorems and formulas for characteristic classes.

Status. Open. There are powerful curvature theories for polyhedral, piecewise-flat, Alexandrov, CAT, and more general singular spaces, but none yet supplies the full package asked for here: a PL notion of sectional curvature strong enough for pinching theorems and, at the same time, a Chern–Weil theory producing characteristic classes. The target is not merely a discretization of one scalar invariant. It is a theory in which PL approximations of positively curved smooth manifolds carry a natural positive-curvature condition, and in which curvature data still remembers Euler and Pontrjagin classes.

The existing theory has developed along several complementary axes. Banchoff proved a polyhedral Gauss–Bonnet theorem for embedded polyhedra [Ba]. Regge introduced a piecewise-flat scalar curvature concentrated on codimension-two faces, now the basic curvature quantity in Regge calculus [R]. Cheeger developed analytic and Hodge-theoretic tools for spaces with cone-like singularities and Riemannian pseudomanifolds [C1], [C2]. Cheeger–Müller– Schrader constructed curvature measures for piecewise-flat spaces and proved convergence results under suitable approximation hypotheses [CMS]. This is one of the closest achievements to the characteristic-class side of Yau's question, but curvature measures are not yet a substitute for a PL sectional curvature theory with sphere, pinching, and rigidity theorems.

A recent advance makes the measure-valued Riemann tensor itself more intrinsic. Gawlik–McKee use compatible moving frames on blown-up polytopes to derive a distributional curvature satisfying the weak Cartan structure equations and the correct gauge-transformation law [GM]. Their formula has three strata: ordinary Riemann curvature in the interiors, jumps of the second fundamental form across codimension-one faces, and angle-defect rotations on codimension-two faces. They also prove a Gauss–Bonnet theorem for compact two-dimensional Regge manifolds, including boundary and corner terms. This solves an important covariance problem, but it does not yet define nonlinear invariant polynomials of singular curvature in higher dimensions; products of measure-valued curvature are precisely where the Euler and Pontrjagin forms become delicate.

Alexandrov and CAT geometry give a different kind of answer. The Burago–Gromov–Perelman theory of Alexandrov spaces with curvature bounded below and the $\mathrm{CAT}(\kappa)$ theory described in Bridson–Haefliger are now indispensable frameworks for singular curvature [BGP], [BH]. They provide compactness, stability, splitting, and comparison theorems, and they are well suited to Gromov–Hausdorff limits. Their curvature bounds, however, are metric-comparison conditions rather than local Chern–Weil representatives of Pontrjagin or Euler classes. This is the essential gap between synthetic curvature and the characteristic-class part of the problem.

More recent work clarifies parts of the differential-geometric dictionary in discrete settings. Glickenstein studied discrete conformal variations and scalar curvature on two- and three-dimensional piecewise-flat manifolds [Gl]. McDonald–Miller–Alsing–Gu–Wang–Yau developed an exterior calculus and curvature formalism for piecewise-flat manifolds [McY]. These tools are valuable for discrete differential geometry and numerical geometry, but the original problem remains broader: one needs compatibility among subdivision, smoothing, convergence to smooth curvature, pinching rigidity, and topological characteristic classes.

There is also a computationally effective piecewise-flat curvature program. Conboye and Miller constructed discrete scalar, sectional, and Ricci curvatures on three-dimensional simplicial piecewise-flat triangulations, using dual volumes and parallel transport around hinges, and reported convergence to smooth curvatures and Ricci flow in tested examples [CM]. Conboye later developed piecewise-flat approximations of local extrinsic mean and directed curvatures for non-Euclidean embeddings [Co]. These results are valuable for approximation and numerical geometry. They should nevertheless be distinguished from Yau's requested PL theory: convergence of averaged curvature quantities does not yet supply a subdivision-stable sectional-curvature condition, a PL pinching theorem, or Chern–Weil representatives of Pontrjagin classes.

Related open problems.

1. *Original direction.* Formulate a PL sectional curvature condition which is stable under subdivision and converges to smooth sectional curvature under controlled triangulations.

2. *Original direction.* Develop a Chern–Weil theory for piecewise-flat manifolds whose characteristic representatives converge to the smooth Pontrjagin and Euler forms [CMS].

3. *Original direction.* Determine whether every sufficiently fine PL approximation of a positively curved smooth manifold can be assigned a natural positive PL curvature in such a theory [BGP], [BH].

4. *Reference-driven formulation.* Compare curvature-measure approaches with Alexandrov lower-curvature and CAT upper-curvature conditions: determine which implications survive under subdivision, Gromov–Hausdorff limits, and smoothing [BGP], [BH], [CMS].

5. *Reference-driven formulation.* Test whether the dual-volume discrete scalar, sectional, and Ricci curvatures used in piecewise-flat Ricci-flow computations can be upgraded from numerical approximations to subdivision-stable hypotheses in comparison or pinching theorems [CM], [Co], [Gl].

6. *Reference-driven formulation.* Construct Chern–Weil invariant polynomials for the gauge-covariant Gawlik–McKee curvature measure without undefined products of singular measures, and prove that they represent the PL Euler and Pontrjagin classes [GM], [CMS].

7. *AI-generated formulation.* Prove a PL sphere theorem under a natural pointwise pinching condition and identify exactly which additional regularity assumptions on the triangulation are necessary.

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Problem 16

What can one say about the Pontrjagin classes and the Stiefel–Whitney classes of a compact manifold with negative curvature? For example, is it true that a finite covering of such a manifold is spin?

Status. Open in general. Negative sectional curvature does not force the rational Pontrjagin classes to vanish, and no general classification of Pontrjagin or Stiefel–Whitney classes of closed negatively curved manifolds is known. The virtual spin question is also open in this generality: after passing to an orientable finite cover, one asks whether a further finite cover can kill w_2.

Several distinctions are essential. First, one must distinguish characteristic classes from characteristic numbers. A class may be nonzero even when all corresponding top-dimensional characteristic numbers vanish, and finite covers can change integral and mod–two information in different ways. Second, locally symmetric negative curvature and variable negative curvature have very different flexibility. Classical rigidity gives strong restrictions in locally symmetric cases, but hyperbolization constructions show that variable negative curvature allows much wider topology. Long–Reid used characteristic classes to study geometric boundaries of hyperbolic manifolds [LR]. Lafont–Roy gave conceptual proofs of vanishing and nonvanishing results for compact nonpositively curved locally symmetric spaces, and also proved vanishing results for Stiefel–Whitney numbers of finite covers of the Gromov–Thurston negatively curved examples [LRo]. Thus the picture is neither pure vanishing nor complete freedom.

Ontaneda's hyperbolization program gives decisive evidence against overly optimistic vanishing conjectures. Pinched smooth and Riemannian hyperbolization produce negatively curved closed manifolds in dimensions greater than three with nonzero first rational Pontrjagin class [O1], [O2]. Thus the right question is not whether characteristic classes vanish, but which characteristic-class patterns are compatible with negative curvature, which are stable under finite covers, and which are ruled out by additional pinching or local symmetry.

For the spin question, after passing to the orientation double cover, a finite spin cover exists precisely when $w_2(M)$ pulls back to zero on some finite-index subgroup. Since a closed negatively curved manifold is aspherical, this becomes a virtual vanishing problem in the cohomology of $\pi_1(M)$. No known mechanism from variable negative curvature alone forces this restriction map to vanish.

The real-hyperbolic case is nevertheless solved affirmatively. A theorem of Deligne–Sullivan says that every finite-volume real-hyperbolic manifold has a finite cover which is stably parallelizable, hence spin. Long–Reid gave a direct spin-group proof [LRS]: lift the lattice to the central double cover $\mathrm{Spin}^+(n,1)$, use linearity and residual finiteness to separate the central element -1 in a finite quotient, and pass to the corresponding finite-index subgroup. Thus Yau's virtual-spin question has a positive answer for constant curvature, while its variable-curvature form remains open.

The flat case is a useful warning: Hamrick–Royster proved that flat manifolds bound, hence all their Stiefel–Whitney numbers vanish [HR], but numerical vanishing is weaker than vanishing of the class w_2 itself. A virtual spin theorem would therefore be substantially stronger than any statement only about Stiefel–Whitney numbers.

Passing to a cover is essential even in constant curvature. Martelli–Riolo– Slavich constructed compact orientable hyperbolic manifolds without spin structures in every dimension at least four [MRS]. Chen then constructed, in every dimension at least five, infinitely many commensurability classes of closed orientable hyperbolic manifolds without spin^c structures, and more generally examples with nonzero w_{4k-1} in every dimension at least $4k+1$ [Chn]. The mechanism recursively embeds arithmetic hyperbolic manifolds as geodesic hypersurfaces with controlled normal class and propagates Stiefel–Whitney nonvanishing by the Whitney product formula.

In the cusped category, Reid–Sell showed that every commensurability class of cusped arithmetic hyperbolic manifolds of simplest type in dimensions $2n+2 \geqslant 6$ contains examples with nonzero even Stiefel–Whitney classes and separate examples without spin^c structures [RS]. None of these results contradicts virtual spin: their characteristic-class obstructions disappear on some finite cover. They instead show how much information can be present before that cover is taken.

Related open problems.

1. *Original direction.* Characterize the possible rational Pontrjagin classes of closed negatively curved manifolds in each dimension.

2. *Original direction.* Decide whether every closed negatively curved manifold has a finite spin cover, equivalently whether w_2 becomes zero on some finite-index subgroup after orienting.

3. *Reference-driven formulation.* Determine whether strict pinching, for example near -1, imposes stronger restrictions on Stiefel–Whitney classes or numbers than arbitrary negative curvature [LRo], [O1], [O2].

4. *Reference-driven formulation.* Separate the possible Pontrjagin classes from the possible Pontrjagin numbers for negatively curved manifolds, especially in the locally symmetric, Gromov–Thurston, and Ontaneda-hyperbolized families [LRo], [O2].

5. *Reference-driven formulation.* Compare closed negatively curved manifolds with finite-volume cusped hyperbolic manifolds: determine which nonvanishing Stiefel–Whitney phenomena in the arithmetic cusped setting can occur in closed negatively curved manifolds or in their finite cover towers [RS], [Chn].

6. *Author question.* Construct closed orientable non-arithmetic hyperbolic manifolds without spin^c structures and decide whether they occur in infinitely many commensurability classes [Chn].

7. *Author question.* Decide whether a closed orientable hyperbolic manifold can fail to be spin^c while $w_3 = 0$, and study analogous obstructions to higher spin structures [Chn].

8. *AI-generated formulation.* Find computable obstructions, expressed in the cohomology of the fundamental group, to a closed negatively curved manifold admitting a finite spin cover.

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Problem 17

Prove that the Stiefel–Whitney numbers of a flat manifold are zero.

Status. Solved. Hamrick and Royster proved in 1982 that every compact flat Riemannian manifold is a boundary [HR]. By Thom's description of unoriented cobordism, this is equivalent to the vanishing of all Stiefel–Whitney numbers [Th]. At the time of Yau's list, the problem belonged to the topology of Bieberbach manifolds and compact solvmanifolds; Auslander–Szczarba had already studied characteristic classes in that setting [AS].

The point of the Hamrick–Royster theorem is that it proves a stronger cobordism statement, not merely a numerical cancellation. A compact flat manifold represents zero in the unoriented cobordism ring; Thom's theorem then identifies this with the vanishing of all Stiefel–Whitney numbers [Th]. Thus the original flat problem is completely settled.

The natural continuation is the almost-flat case. By the Gromov–Ruh theory, almost flat manifolds are infranilmanifolds [Gr], [Ru]. For oriented almost flat manifolds, Chern–Weil theory forces rational Pontrjagin numbers to vanish: the curvature can be made arbitrarily small while the diameter is kept bounded. Stiefel–Whitney numbers are subtler because they are mod–two cobordism invariants rather than Chern–Weil integrals. Their vanishing is equivalent to unoriented null-cobordism, and the general almost-flat bounding problem remains open.

There are important positive results in this broader direction. Farrell–Zdravkovska proved bounding theorems for almost flat manifolds under several holonomy hypotheses [FZ]. Davis–Fang later showed that an almost flat manifold bounds when the 2-Sylow subgroup of its holonomy group is cyclic or generalized quaternionic; in particular, cyclic holonomy is covered [DF]. Han–Huang–Zhang proved that every almost-flat spin^c manifold bounds an orientable manifold and, in odd dimensions, every chosen spin^c structure bounds spin^c [HHZ, Theorem 1.1]. Their integral spin^c Wu classes, together with Atiyah–Singer divisibility, force the relevant Stiefel–Whitney numbers to vanish. They also prove bounding when the Schur multiplier of the holonomy group has odd order and settle the orientable and spin bordism versions in dimension four. Thus the unresolved part of the almost-flat conjecture is concentrated in non-spin^c holonomy extensions.

Spin and spin^c structures are not automatic even in the almost-flat world. Gasior–Petrosyan–Szczepański gave necessary and sufficient conditions for almost-flat manifolds with cyclic holonomy to admit spin structures and classified the four-dimensional orientable cyclic-holonomy examples without spin structures [GPS]. Together with the spin^c bounding theorem of Han–Huang–Zhang, this makes the remaining problem sharper: the unoriented bounding conjecture for all almost-flat manifolds is not merely a spin-theoretic question, but a question about the full holonomy extension and its mod–two characteristic classes.

There is also a geometric filling version. Ontaneda proved that if an almost flat manifold bounds, then the bounding manifold can be chosen so that its interior carries a complete finite-volume metric with sectional curvatures pinched in $[-1-a,-1]$ for any $a>0$ [O]. Thus the topological bounding problem controls which infranilmanifolds can occur as cusp cross-sections in this negatively curved setting. Long–Reid showed that one cannot generally upgrade such fillings to constant curvature: eta-invariant obstructions prevent some flat manifolds from occurring as totally geodesic boundaries of compact hyperbolic manifolds [LR].

Related open problems.

1. *Reference-driven formulation.* Prove or disprove that every almost flat manifold bounds, equivalently that all of its Stiefel–Whitney numbers vanish.

2. *Reference-driven formulation.* Extend the known Farrell–Zdravkovska and Davis–Fang holonomy cases to arbitrary finite holonomy groups of infranilmanifolds.

3. *Reference-driven formulation.* Determine whether the spin^c bounding theorem for almost-flat manifolds can be upgraded to unoriented bounding without spin^c hypotheses, or whether the remaining obstructions are genuinely non-spin^c holonomy phenomena [GPS], [HHZ].

4. *Reference-driven formulation.* Classify which flat or almost flat manifolds occur as cusp cross-sections of complete finite-volume negatively pinched manifolds.

5. *Reference-driven formulation.* Classify which flat manifolds occur as totally geodesic boundary components of compact hyperbolic manifolds.

6. *Reference-driven formulation.* Determine the optimal pinching possible for negatively curved fillings of a given flat or almost flat manifold.

7. *AI-generated formulation.* Express the best possible filling pinching constant purely in terms of the Bieberbach group of the flat manifold.

8. *AI-generated formulation.* Describe the almost-flat bounding obstruction as an explicit group-cohomological invariant of the infranil holonomy extension.

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Problem 18

Given M a complete, noncompact manifold with sectional curvature $K \geqslant 0$; if, at some point $x, K(x) > 0$, prove that M is diffeomorphic to $\mathbb{R}^N$.

Status. Solved. Perelman proved in 1994 that if a complete open manifold has nonnegative sectional curvature and has positive sectional curvature at one point, then its soul is a point [Pe]. Hence the manifold is diffeomorphic to $\mathbb{R}^N$, exactly as Yau asked.

The background is the Cheeger–Gromoll soul theorem: every complete open manifold with nonnegative sectional curvature is diffeomorphic to the normal bundle of a compact totally geodesic soul [CG]. If the soul is a point, the manifold is $\mathbb{R}^N$; if the soul has positive dimension, the topology is controlled by its normal bundle. Gromoll–Meyer had already proved the point-soul conclusion under the stronger assumption of positive sectional curvature everywhere [GM]. The Cheeger–Gromoll soul conjecture asked whether positivity at a single point is enough.

Perelman's proof established exactly this one-point version [Pe]. The key geometric background is Sharafutdinov's distance nonincreasing retraction from the open manifold to a soul [Sh]. Perelman proved the additional regularity and submersion property needed to rule out a positive-dimensional soul when the ambient sectional curvature is positive at one point. Thus the topological part of the problem is now a theorem.

A separate nearby direction asks whether zero-curvature directions in a nonnegatively curved metric can be removed while preserving completeness and the underlying manifold. This is not an unresolved clause of Yau's stated Problem 18; it is a local-to-global perturbation problem sensitive to the soul, the Sharafutdinov fibration, and the normal-bundle geometry.

Modern work on the converse soul problem shows how subtle this broader program remains. González-Àlvaro and Zibrowius extended known existence results to simply connected positively curved souls: they constructed pairs of simply connected positively curved manifolds which are tangentially homotopy equivalent but not homeomorphic, and used stable converse-soul results to obtain open manifolds admitting nonhomeomorphic positively curved souls [GZ]. This does not affect Perelman's point-soul theorem under positive curvature at one point. It does show that, once the soul has positive dimension or one asks for deformations and moduli of nonnegatively curved metrics, the topology of possible souls is flexible and not determined only by a coarse homotopy type.

Related open problems.

1. *Original direction.* Classify complete open manifolds which admit positive sectional curvature.
2. *Reference-driven formulation.* Determine when a nonnegatively curved open manifold whose curvature is positive off a small set admits a deformation to positive sectional curvature.
3. *Reference-driven formulation.* Study whether the Sharafutdinov fibration and normal-bundle structure give effective obstructions to removing zero-curvature strata from nonnegatively curved open metrics.
4. *Reference-driven formulation.* Determine when a vector bundle over a positively curved closed manifold admits a complete metric of nonnegative sectional curvature for which the zero section is a soul, and how many nonhomeomorphic positively curved souls can occur on one open manifold [GZ].
5. *AI-generated formulation.* Find quantitative lower-curvature, volume-growth, or convexity assumptions which force the soul to be a point without assuming strict positivity at a point.

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B. Ricci curvature

Problem 19

Let Ω be a closed $4k$-form defined on a compact manifold M which represents some Pontrjagin class of M. Can one find a metric on M so that Ω is represented by the curvature form according to Chern [Ch]? If there are other topological obstructions, what are they? It is rather clear that such obstructions should exist. The problem is to find a sufficient condition.

One can ask the same question for Chern classes of Kähler manifolds. For the first Chern class, this was conjectured by Calabi and solved in [Y].

A resolution of this question will give a deep understanding of the curvature tensor.

Status. Open. For the first Chern class in the Kähler case, the corresponding prescription of the Ricci form is settled by Yau's solution of the Calabi conjecture [Y]. For higher Pontrjagin forms of a real Riemannian metric, no satisfactory necessary and sufficient condition is known.

Chern–Weil theory associates closed differential forms to invariant polynomials in the curvature of a connection; for the Levi-Civita connection these forms represent the Pontrjagin classes of the tangent bundle [Ch]. The problem asks for a converse prescription theorem: given a closed representative of a Pontrjagin class, can it be realized as the Chern–Weil form of some Riemannian metric? The case of scalar curvature is much more flexible and was developed by Kazdan–Warner [KW], while prescribed Ricci curvature has a local theory due to DeTurck [D]. These results show that curvature prescription problems can have very different answers depending on which curvature contraction is prescribed. The Pontrjagin-form problem remains substantially more rigid because the forms are nonlinear in the full curvature tensor and must satisfy Bianchi-type identities.

It is important to distinguish this question from the corresponding problem for arbitrary connections on a fixed vector bundle. The space of connections is affine and much larger than the space of Levi-Civita connections. Universal-connection theory of Narasimhan–Ramanan [NR] and later work on Chern–Weil representatives show that characteristic forms can often be studied flexibly once the connection is allowed to vary freely. Yau's question asks for a representative coming from the Riemannian metric itself; this imposes torsion-freeness, metric compatibility, and the algebraic symmetries of the Riemann curvature tensor. These extra constraints are the source of the missing "sufficient condition" in the problem.

In complex geometry, the first Chern form is the Ricci form of a Kähler metric; Calabi's conjecture, proved by Yau, gives a precise prescription theorem in a fixed Kähler class [Y]. Higher Chern forms and real Pontrjagin forms are not presently governed by an analogous general theorem. Recent work on representability of Chern–Weil forms for Hermitian metrics on vector bundles gives useful models and examples, but it still does not provide a general Levi-Civita prescription theorem for Pontrjagin forms [P].

A new higher-form Monge–Ampère model gives another useful comparison. George introduced a volume form associated to a positive (p, p)-form and proved existence and uniqueness for its prescribed-volume equation on a compact Kähler manifold [Ge]. The associated cohomology-preserving flow extends Kähler–Ricci flow and converges in the fixed positive-form setting. The decisive estimate splits multi-indices relative to the largest Hessian eigenvalue and uses a Kronecker-product structure to control otherwise unmanageable third-order terms. This is a genuine higher-form prescription theorem, but its "Ricci form" is defined from the induced volume of a positive (p, p)-form; it is not yet a theorem that a prescribed higher Chern–Weil form comes from the curvature of a Kähler or Levi-Civita connection.

There are also newer scalar analogues in Hermitian geometry. Recent work on prescribed Chern scalar curvature, including flow approaches on compact Hermitian manifolds of negative Gauduchon degree and complete noncompact extensions, shows that Chern-connection prescription problems can sometimes be attacked by conformal or parabolic methods [Yu1], [Yu2]. These results concern scalar contractions of Chern curvature, not full Chern–Weil forms and not Levi-Civita Pontrjagin forms. They are nevertheless useful comparisons: they illustrate how extra complex structure and a chosen connection can convert a curvature-prescription question into a tractable PDE, whereas Yau's Pontrjagin-form problem still lacks such a canonical elliptic formulation.

Related open problems.

1. *Original direction.* Find local obstructions, beyond de Rham cohomology, for a closed $4k$-form to be a Pontrjagin form of a Levi-Civita connection.

2. *Reference-driven formulation.* Develop a prescribed Chern–Weil form theory for Kähler metrics and compare it with complex Monge–Ampère equations.

3. *Reference-driven formulation.* Compare the flexible prescription problem for arbitrary connections with the constrained problem for Levi-Civita connections, and isolate the first metric-compatibility obstruction.

4. *Reference-driven formulation.* Determine whether the Hermitian prescribed Chern scalar curvature methods can be lifted from scalar contractions to Chern–Weil representatives in fixed Bott–Chern or de Rham cohomology classes [P], [Yu1], [Yu2].

5. *Reference-driven formulation.* Find geometric conditions under which George's positive (p, p)-form flow preserves the subclass of forms arising as higher Chern forms of a Kähler metric [Ge].

6. *AI-generated formulation.* Determine whether every sufficiently small exact perturbation of a Pontrjagin form of a generic metric can again be realized as a Pontrjagin form after perturbing the metric.

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B. Ricci curvature

Problem 20

Find necessary and sufficient conditions on a symmetric tensor T_{ij} on a compact manifold so that one can find a metric g_{ij} to satisfy

$$R_{ij} - R/2 g_{ij} = T_{ij}$$

whence R_{ij} is the Ricci tensor and R is the scalar curvature of g_{ij}. If g_{ij} is the Lorentz metric on a four-dimensional manifold, this is simply the Einstein field equation. If M has boundary, what are suitable boundary conditions to impose?

Status. Open as a global prescription problem. The local prescribed Ricci curvature theory gives important partial results, but a necessary and sufficient global theorem for prescribing the Einstein tensor on a compact manifold is not known.

The contracted Bianchi identity gives an immediate necessary condition: the tensor T_{ij} must be divergence-free with respect to the unknown metric g. This makes the problem different from prescribing a fixed tensor with respect to a fixed background metric. In the Riemannian setting, DeTurck proved a local existence theorem for prescribed Ricci curvature [D]. Since the Einstein tensor determines the Ricci tensor algebraically when the dimension is not two, this is closely related to the local form of the present problem, after accounting for gauge and trace issues. Indeed, if $G_g = T$, then

$$\mathrm{Ric}_g = T - \frac{\mathrm{tr}_g T}{n-2}\, g, \qquad n > 2,$$

but the trace on the right is taken with respect to the unknown metric. This is why the problem is not merely DeTurck's prescribed-Ricci problem in different notation. Globally, topology, elliptic gauge fixing, and the dependence of the divergence condition on g remain serious obstructions.

In Lorentzian signature the equation is the Einstein field equation, and the tensor T is normally constrained by matter equations and a variational or hyperbolic evolution system. Thus the geometric prescription problem and the physical Cauchy-boundary problem should be separated. The Cauchy problem, beginning with Choquet-Bruhat's work and its later global formulation with Geroch, treats Einstein's equations as a hyperbolic system after gauge choice and requires the initial data to satisfy the Einstein constraint equations [CBG, CBY]. Boundary versions are even more delicate: they must include boundary geometry, gauge, compatibility conditions, and constraints, not only Dirichlet data for the metric.

For Riemannian Einstein metrics with boundary, modern work shows that the naive Dirichlet and Neumann boundary maps are generally not Fredholm. Anderson identified natural mixed boundary data, involving the conformal class of the boundary metric together with mean-curvature-type data, which lead to Fredholm boundary-value problems in the Einstein case [A]. This gives a model for what the boundary part of Yau's problem should look like, but it is still far from a general Einstein-tensor prescription theorem with arbitrary source tensor T.

Recent work has pushed this boundary perspective directly to the prescribed Einstein tensor. Delay proved local invertibility near a compact Einstein manifold with boundary for a map recording Ricci curvature, the induced boundary conformal class, and boundary mean curvature, under explicit Hodge- and Lichnerowicz-spectrum nondegeneracy [Del, Theorem 1.1]. His Theorem 6.2 treats an Einstein-type operator whose parameter choice $\kappa = -1/2$ is the Einstein tensor itself. Thus small Einstein-tensor data together with nearby mixed boundary data $([\gamma], H)$ determine a unique nearby metric. The proof adds a Bianchi-gauge correction, splits trace and trace-free tensors, uses the spectral exclusions to invert the linearized mixed problem, and finally removes the gauge term using the Bianchi identity. This is perturbative near a nondegenerate Einstein background, not a global characterization of all tensors T.

Related open problems.

1. *Original direction.* Classify divergence-free symmetric tensors which occur as Einstein tensors of Riemannian metrics on a fixed compact manifold.

2. *Original direction.* Formulate well-posed boundary data for the Riemannian Einstein-tensor prescription problem.

3. *Reference-driven formulation.* Determine whether Anderson-type mixed boundary data extend from Einstein metrics to the full prescribed Einstein-tensor equation.

4. *Reference-driven formulation.* Separate the Riemannian elliptic prescription problem from the Lorentzian initial-boundary value problem with matter fields and constraint propagation.

5. *Reference-driven formulation.* Extend Delay's local Einstein-tensor inversion near a nondegenerate Einstein metric to a global range or degree theorem under geometric compactness hypotheses [A], [Del].

6. *AI-generated formulation.* Determine whether the Einstein-tensor map is locally surjective, modulo diffeomorphisms and scaling, at a generic compact Einstein metric.

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Problem 21

Let M be a complete manifold with positive Ricci curvature. Can M be deformed to a compact manifold with boundary?

Status. False in general. Complete manifolds with positive Ricci curvature need not have finite topological type, hence need not be diffeomorphic to the interior of a compact manifold with boundary.

The question is motivated by the soul theorem for nonnegative sectional curvature, where an open manifold has finite topological type. Ricci curvature is much weaker. Sha–Yang constructed complete metrics with positive Ricci curvature on infinite connected sums, including infinite connected sums of $\mathbb{S}^2 \times \mathbb{S}^2$, giving explicit examples of infinite topological type [SY]. Thus the direct analogue of the soul theorem fails.

Positive results require additional hypotheses. Shen proved finite topological consequences under nonnegative Ricci curvature together with large volume growth assumptions [S]. Later work developed this theme using asymptotic Ricci lower bounds, curvature decay, rewinding volume, and volume-growth hypotheses [H, Z]. These theorems show that finite topology can be recovered when the geometry at infinity is controlled strongly enough.

The boundary of the theory is sharp. Even quite strong asymptotic conditions do not automatically restore finite topological type: Jiang–Yang constructed complete manifolds of positive Ricci curvature with quadratically asymptotically nonnegative sectional curvature and infinite topological type in high dimensions [JY]. Thus the modern form of the problem is not simply "positive Ricci plus mild control at infinity," but rather to identify the exact package of volume, curvature-decay, and local noncollapsing assumptions that prevents topological complexity from escaping to infinity.

The dimensions and topology in these counterexamples are now quite precise. Jiang–Yang first produced a six-dimensional example [JY] and then constructed examples in every dimension at least five with specified infinite intermediate Betti numbers [JY2]. Their mechanism is an infinite sequence of Perelman-type surgeries inside a doubly warped end: punctured $\mathbb{CP}^2$-type cores are attached through positive-Ricci necks, while the warping functions balance Ricci positivity against quadratic sectional-curvature decay.

The negative side has become even sharper. Jiang recently constructed in every dimension at least four complete open manifolds with positive Ricci curvature, sectional curvature bounded from below, and infinite intermediate Betti numbers, with volume growth $O(r^{2+\gamma})$ for arbitrarily small $\gamma > 0$ [Ji]. In a related but nonnegative-Ricci direction, Brue–Naber–Semola constructed smooth complete examples with $\mathrm{Ric} \geqslant 0$ and infinitely generated fundamental group, first in dimension seven and then in dimension six [BNS1], [BNS2]. These latter examples answer Milnor's finite-generation conjecture rather than Yau's finite-topological-type question directly, but they reinforce the same lesson: Ricci lower bounds alone allow topological complexity at infinity that has no analogue under nonnegative sectional curvature.

Related open problems.

1. *Original direction.* Find sharp geometric conditions, weaker than sectional curvature bounds, which force finite topological type under positive Ricci curvature.

2. *Original direction.* Classify which infinite connected sums admit complete positive Ricci curvature.

3. *Reference-driven formulation.* Determine the optimal combination of large volume growth, curvature decay, and local rewinding-volume control that implies finite topological type under nonnegative or positive Ricci curvature.

4. *Reference-driven formulation.* Decide in which dimensions positive Ricci curvature with quadratically asymptotically nonnegative sectional curvature can still have infinite topological type.

5. *Reference-driven formulation.* Decide whether the almost quadratic volume-growth examples with positive Ricci curvature and infinite Betti numbers can be improved to exactly quadratic volume growth, or whether quadratic growth is a genuine threshold for finite topology under a lower sectional curvature bound [Ji].

6. *Reference-driven formulation.* Compare finite topological type with finite generation of the fundamental group under Ricci lower bounds: determine which features of the Brue–Naber–Semola counterexamples can coexist with positive Ricci curvature or with stronger volume-growth assumptions [BNS1], [BNS2].

7. *AI-generated formulation.* Determine whether positive Ricci curvature together with Euclidean volume growth and a quantitative lower injectivity-radius condition forces finite topological type.

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Problem 22

Characterize the fundamental group of a complete manifold with positive Ricci curvature.

If the manifold is compact, the splitting theorem of Cheeger–Gromoll [CG] provides a rather satisfactory answer.

The case of noncompact manifolds is more complicated. Recently, P. Nabonnand [N], under the direction of Bérard–Bergery, provided an example of a noncompact, complete manifold with positive Ricci curvature whose fundamental group is infinite cyclic. This example has dimension $\geqslant 4$.

In the case of three dimensions, Schoen and Yau [SY] proved that such manifolds are diffeomorphic to $\mathbb{R}^3$. It may be possible that the fundamental group of the manifold is always a finite extension of a polycyclic group.

Status. Open in its full positive Ricci form. The compact case is well understood up to the crystallographic extension given by the Cheeger–Gromoll splitting theorem [CG]. The noncompact case is much subtler: many nilpotent groups occur, dimension three is understood, and the analogous finite-generation expectation for nonnegative Ricci curvature is now known to be false in high dimensions. No audited counterexample upgrades the latter construction to strictly positive Ricci curvature, so that distinction is part of the remaining problem rather than a technicality.

In the compact nonnegative Ricci case, Cheeger and Gromoll proved that the universal cover splits isometrically as $\mathbb{R}^d \times N$ with N compact [CG]. Hence the fundamental group is virtually crystallographic. Conversely, Wilking proved that the expected abstract groups occur as fundamental groups of compact manifolds with nonnegative sectional curvature [W].

In dimension three, the picture is much more rigid. Schoen and Yau proved that a complete noncompact three-manifold with positive Ricci curvature is diffeomorphic to $\mathbb{R}^3$ [SY]. Hamilton's Ricci-flow classification treats the compact nonnegative Ricci case [Ham]. Liu later completed the noncompact nonnegative Ricci classification in dimension three: either the manifold is diffeomorphic to $\mathbb{R}^3$ or its universal cover splits isometrically as $N^2 \times \mathbb{R}$ [Liu]. Pan gave another proof of Milnor's finite-generation conjecture in dimension three using Cheeger–Colding theory [Pan].

In higher-dimensional noncompact geometry, the original expectation was strongly influenced by Milnor's theorem that every finitely generated subgroup of the fundamental group of a complete manifold with nonnegative Ricci curvature has polynomial growth [Mil]. If the whole group were finitely generated, Gromov's theorem on groups of polynomial growth would imply that it is virtually nilpotent [G]. Several partial results supported this picture: Li proved finiteness under Euclidean volume growth [Li], Anderson proved finite generation under additional volume growth assumptions [A], and Sormani proved finite generation under small linear diameter growth [So]. Kapovitch and Wilking proved a Ricci-lower-bound Margulis lemma, showing that local fundamental groups contain nilpotent subgroups of uniformly bounded index [KW].

The situation changed substantially with the work of Brue–Naber– Semola, who constructed complete manifolds with nonnegative Ricci curvature and infinitely generated fundamental group, disproving Milnor's conjecture in high dimensions [BNS]. This does not by itself classify the positive Ricci case, but it shows that the noncompact Ricci-lower-bound world is broader than the earlier virtually nilpotent picture suggested.

The counterexample side has also been sharpened dimensionally. The Brue–Naber–Semola construction now gives a smooth complete six-dimensional manifold with $\mathrm{Ric} \geqslant 0$ and infinitely generated fundamental group exactly $\mathbb{Q}/\mathbb{Z}$ [BNS2]. Its universal cover is diffeomorphic to $\mathbb{S}^3 \times \mathbb{R}^3$; the curvature is bounded and decays polynomially, and the example is uniformly locally noncollapsed. The topology is assembled from nested finite cyclic groups by recursively twisted gluing. In particular, the Milnor finite-generation conjecture is false in all dimensions at least six, while the four- and five-dimensional cases remain natural borderline questions. This is important for Yau's positive-Ricci problem because any attempted classification must now distinguish carefully between positive Ricci curvature, nonnegative Ricci curvature, and the strength of the asymptotic hypotheses imposed at infinity.

After these counterexamples, a more refined positive theory has begun to emerge under volume-growth hypotheses. Navarro–Pan–Zhu proved that for complete manifolds with nonnegative Ricci curvature and linear volume growth, the fundamental group is virtually free abelian, and if the Ricci curvature is positive everywhere then the fundamental group is finite [NPZ1]. More precisely, the nonnegative-Ricci conclusion is that $\pi_1(M)$ contains $\mathbb{Z}^k$ with finite index. Their subsequent theorem uses the relative volume function $RV(s)$: under sublinear end diameter, $RV(s) = o(s^2)$ forces virtual abelianness, while a stronger $RV(s) = o(s^{1+\delta})$ hypothesis and Ricci positivity at one point force finiteness [NPZ2]. Absolute subquadratic volume growth is not known to give the same conclusion. Huang–Huang further proved that in the linear- growth case the universal cover splits as $\mathbb{R}^k \times N$, with a finite cover fibering over $\mathbb{T}^k$, and positivity at one point already forces finiteness [HH]. These results suggest that the replacement for Milnor's conjecture should be formulated in terms of asymptotic volume geometry rather than Ricci positivity alone.

There are also positive realization results. Nabonnand constructed a complete noncompact positive Ricci example with fundamental group $\mathbb{Z}$ [N]. Wei proved that every finitely generated torsion-free nilpotent group occurs as the fundamental group of a complete manifold with positive Ricci curvature [Wei]. These examples show that even under positive Ricci curvature, the class of possible noncompact fundamental groups is large.

Related open problems.

1. *Original direction.* Characterize the groups which occur as fundamental groups of complete noncompact manifolds with positive Ricci curvature.

2. *Original direction.* Decide whether every finitely generated virtually nilpotent group occurs as the fundamental group of a complete positive Ricci manifold, allowing torsion.

3. *Reference-driven formulation.* Determine whether the high-dimensional infinitely generated examples with nonnegative Ricci curvature can be modified to have positive Ricci curvature everywhere [BNS], [BNS2].

4. *Reference-driven formulation.* Determine whether positive Ricci curvature plus linear or slow relative volume growth is the optimal regime in which the fundamental group must be finite or virtually abelian [NPZ1], [NPZ2], [HH].

5. *Reference-driven formulation.* Decide the Milnor finite-generation conjecture in dimensions four and five, or find a geometric reason why the six-dimensional Brue–Naber–Semola mechanism cannot occur there [BNS2].

6. *Original direction.* Find the exact restrictions imposed by positive Ricci curvature on abelian subgroups of the fundamental group in the noncompact case.

7. *Reference-driven formulation.* Decide whether absolute subquadratic volume growth, without a sublinear end-diameter hypothesis, forces finite generation or virtual abelianness [NPZ1], [NPZ2].

8. *AI-generated formulation.* Formulate a replacement for Milnor's finite-generation conjecture which is true for positive Ricci curvature and stable under measured Gromov–Hausdorff limits.

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Problem 23

Construct an explicit metric with zero Ricci curvature on the $K3$ surface. The existence of such a metric was proved in Yau [Y]. Is there any four-dimensional manifold with zero Ricci curvature which is not covered by a torus or a $K3$ surface? A simpler unsolved question is whether such a manifold can be diffeomorphic to $\mathbb{S}^4$ or $\mathbb{S}^2 \times \mathbb{S}^2$.

Status. Open for an explicit closed-form metric on a compact $K3$ surface. Yau's theorem gives existence and uniqueness of the Ricci-flat Kähler metric in each Kähler class [Y], but no fully explicit formula is known for a compact $K3$ surface. The broader four-dimensional Ricci-flat classification question remains open as a Riemannian problem, although the compact Kähler surface case is controlled by the Enriques–Kodaira classification and Yau's Calabi–Yau theorem.

Yau's solution of the Calabi conjecture proves that every complex $K3$ surface carries a Ricci-flat Kähler metric in each Kähler class [Y]. The proof is analytic and solves a complex Monge–Ampère equation; it does not produce an elementary explicit formula. This is why $K3$ remains a central test case for numerical and asymptotic approaches to Calabi–Yau metrics. Gross and Wilson analyzed collapsing Ricci-flat metrics on elliptic $K3$ surfaces near the large complex structure limit [GW]. Donaldson proposed and developed numerical approximation methods for Calabi–Yau metrics [D]. Yu-Shen Lin's thesis belongs to this numerical and string- theoretic direction [L].

More recent work gives detailed asymptotic and gluing models for families of $K3$ Ricci-flat metrics. Foscolo constructed collapsing hyperkähler metrics on $K3$ using ALF gravitational instantons [F]. Hein–Sun–Viaclovsky–Zhang exhibited collapsing families with nilpotent structures and bubbling models [HSVZ]. Chen–Viaclovsky– Zhang constructed collapsing Ricci-flat Kähler metrics on elliptic $K3$ surfaces with precise descriptions near singular fibers [CVZ]. These works give powerful models of the metric geometry, but they do not give a single elementary explicit compact $K3$ metric.

The Kummer construction continues to be one of the closest rigorous approaches to an explicit model. Thomas Jiang gave new analytic proofs of the Kummer construction of Calabi–Yau and hyperkähler metrics on $K3$, including improved estimates and a description of large families of noncollapsed hyperkähler orbifold limits [J]. A recent preprint of Shackleton revisits the Kummer surface using Donaldson's gluing strategy and ordinary compact Hölder and Sobolev theory, rather than weighted spaces or conformal changes to model spaces [Sh]. It identifies the Eguchi–Hanson and Gibbons–Hawking descriptions, builds an approximate metric on the resolution of $\mathbb{T}^4/\{\pm 1\}$, and solves for the correction by a Banach fixed-point argument. Thus it makes the local ingredients and the analytic perturbation more explicit, but the corrected compact metric is still defined implicitly as a PDE solution, not by a closed formula.

There is also a fast-growing computational direction. Machine-learning and balanced-metric methods approximate Calabi–Yau metrics very accurately, and recent symbolic-approximation work tries to distill neural approximations into compact analytic expressions in highly symmetric Calabi–Yau examples [MM]. This is directly relevant to the "explicit metric" aspect of Yau's question: it suggests a possible route from numerical approximations to human-readable formulae, while also emphasizing that no exact compact $K3$ formula is presently known.

For the topological question, compact examples of Ricci-flat four-manifolds include flat quotients of tori and Ricci-flat Kähler metrics on $K3$ surfaces and their finite quotients. In the complex surface category, Kodaira dimension zero gives complex tori, $K3$ surfaces, Enriques surfaces, and bi-elliptic or hyperelliptic surfaces; after passing to a finite cover, these are covered by a torus or a $K3$ surface. Yau's theorem supplies the Ricci-flat Kähler metrics in this setting.

The Riemannian question is different. A compact Ricci-flat four-manifold is Einstein, so it satisfies the Hitchin–Thorpe inequality [HT]. Equality is very rigid, but the inequality alone does not rule out $\mathbb{S}^4$ or $\mathbb{S}^2 \times \mathbb{S}^2$. No Ricci-flat metric is known on either of these manifolds. The remaining issue is whether a compact Ricci-flat four-manifold can have generic Riemannian holonomy and topology not accounted for by the torus/$K3$ finite-cover picture.

Related open problems.

1. *Original direction.* Construct a closed-form Ricci-flat Kähler metric on a compact $K3$ surface, even for a highly symmetric algebraic $K3$ such as the Fermat quartic.

2. *Original direction.* Decide whether $\mathbb{S}^4$ or $\mathbb{S}^2 \times \mathbb{S}^2$ admits a Ricci-flat metric.

3. *Original direction.* Classify compact Ricci-flat four-manifolds not finitely covered by a torus or a $K3$ surface, or prove that none exist.

4. *Reference-driven formulation.* Determine whether the Hitchin–Thorpe inequality, Seiberg–Witten theory, or collapse theory can rule out Ricci-flat metrics on $\mathbb{S}^4$ and $\mathbb{S}^2 \times \mathbb{S}^2$.

5. *Reference-driven formulation.* Give effective error estimates for Kummer-type gluing and collapsing approximations to Ricci-flat $K3$ metrics [J], [Sh].

6. *Reference-driven formulation.* Determine whether symbolic or machine-learned approximations to highly symmetric Calabi–Yau metrics can be converted into certified approximate solutions with explicit residual bounds for the complex Monge–Ampère equation [D], [MM].

7. *AI-generated formulation.* Give a rigorous convergence theorem comparing numerical balanced-metric approximations of a $K3$ Calabi–Yau metric with the analytic estimates in Yau's proof.

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Problem 24

Can every manifold with dimension $\geqslant 3$ admit a metric with negative Ricci curvature?

It is very hard to suggest what the right answer to this question is. One does not know if $\mathbb{S}^3$ or $\mathbb{T}^3$ admit such a metric. However, there are many examples of simply connected Kähler manifolds with negative Ricci curvature [Y].

Perhaps a compact manifold with nonpositive Ricci curvature does not admit an $\mathrm{SU}(2)$ action. This conjecture is partly motivated by Bochner's theorem that $\mathrm{SU}(2)$ cannot act isometrically and partly motivated by the theorem of Lawson and Yau [LY] stating that a manifold with effective $\mathrm{SU}(2)$ action admits a metric of positive scalar curvature.

If this last conjecture is true, then one can prove that the complex structures over $\mathbb{S}^2 \times \mathbb{S}^2$ are given by the standard Hirzebruch surfaces.

Status. Solved affirmatively as an existence problem. Gao and Yau proved the compact three-dimensional case [GY], and Lohkamp proved that every smooth manifold of dimension at least three admits a Riemannian metric with negative Ricci curvature; in the noncompact case the metric can be chosen complete [L1, L2]. In fact, Lohkamp obtains dimensional constants $a(n) > b(n) > 0$ and a complete metric satisfying $-a(n) < \mathrm{Ric} < -b(n)$ [L1].

The result shows a sharp contrast between Ricci curvature and sectional curvature. In dimension two, negative Ricci curvature is the same as negative Gaussian curvature and is strongly constrained by topology. In dimensions at least three, Lohkamp's theorem says that negative Ricci curvature alone imposes no topological restriction on the underlying smooth manifold [L2]. Thus even $\mathbb{S}^3$ and $\mathbb{T}^3$ admit metrics of negative Ricci curvature.

The symmetry issue has an exact answer in the negative-Ricci category. On a compact manifold the Bochner principle kills every Killing field, so the isometry group must be finite. Conversely, Lohkamp proved that every finite subgroup $G < \mathrm{Diff}(M)$ is the full isometry group of some negative-Ricci metric [L1, Theorem 2]. Thus the finite- group equivariant problem is solved, even though a naive averaging argument would not preserve the Ricci inequality. One must also distinguish an isometric action from a smooth action on the underlying manifold: $\mathrm{SU}(2)$ cannot act effectively by isometries of a compact negative-Ricci metric, but the existence of such a metric does not prevent the manifold from carrying an unrelated smooth $\mathrm{SU}(2)$ action. In that latter interpretation, Yau's suggested smooth-action obstruction is false.

In the Kähler category, negative Ricci curvature is much more rigid. If the first Chern class is negative, Aubin and Yau produce a Kähler–Einstein metric with negative Ricci curvature [Au, Y]. For general compact complex manifolds, the existence of a Kähler metric with negative Ricci form is governed by positivity of the canonical class rather than by the smooth topology alone. For manifolds with large symmetry, Lawson and Yau proved that many nonabelian group actions imply the existence of positive scalar curvature metrics [LY]; this motivates the question of how negative Ricci curvature interacts with effective group actions once one imposes equivariance, Kählerness, conformal flatness, or sectional-curvature bounds.

Recent work in special geometric categories reinforces this distinction. For example, Payne proved that no closed manifold admits a closed G_2-structure with negative Ricci curvature, and obtained further noncompact restrictions under pinching assumptions [P]. This does not contradict Lohkamp's theorem, because a closed G_2-structure is substantial extra structure. It is nevertheless a useful modern model for how Yau's solved smooth existence question can be refined into structured negative-Ricci questions where topology, torsion, and special holonomy data again become visible.

Related open problems.

1. *Original direction.* Classify compact Kähler manifolds admitting Kähler metrics with negative Ricci curvature, in terms of the canonical class and algebro-geometric positivity.
2. *Original direction.* Determine which closed manifolds admit metrics with negative Ricci curvature together with additional restrictions such as conformal flatness, local symmetry, or sectional-curvature bounds.
3. *Original direction.* Study whether effective nonabelian group actions impose meaningful restrictions on negative Ricci curvature when combined with stronger curvature assumptions.
4. *Reference-driven formulation.* Develop a parametric version of Lohkamp's construction for families of metrics and for quotients by diffeomorphism groups.
5. *Reference-driven formulation.* Strengthen Lohkamp's exact finite-isometry-group realization theorem by prescribing local geometry, curvature pinching, volume, or diameter while retaining the given group action [L1].
6. *Reference-driven formulation.* Determine which special geometric structures, such as closed G_2-structures, symplectic structures, or calibrated structures, obstruct negative Ricci curvature even though the underlying smooth manifold admits a negative-Ricci metric [P].

7. *AI-generated formulation.* Find quantitative versions of Lohkamp's construction which control volume, diameter, or a mixed sectional-Ricci curvature bound.

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Problem 25

Classify four-dimensional compact Einstein manifolds with negative Ricci curvature. Can $\mathbb{S}^4$ admit such a metric?

The Thorpe-Hitchin inequality [H] gives some relation on the Euler number and the index of these manifolds.

Status. Open. The classification of compact four-dimensional Einstein manifolds with negative Einstein constant is far out of reach. The special question whether $\mathbb{S}^4$ admits an Einstein metric with negative Ricci curvature is still open.

The first general restriction is the Hitchin–Thorpe inequality, which relates the Euler characteristic and signature of an oriented compact Einstein four-manifold [H]. This is only a necessary condition and is far from a classification. Even in the Kähler case, the problem contains the diffeomorphism classification of minimal complex surfaces of general type. In practice, one asks geography questions, concerning possible pairs of characteristic numbers, and botany questions, concerning the number of smooth structures for fixed characteristic numbers.

There is a large constructive Kähler source. If a compact complex surface has ample canonical bundle, equivalently negative first Chern class, the Aubin–Yau theorem produces a Kähler–Einstein metric with negative Ricci curvature [Au, Y]. This supplies many examples among surfaces of general type, but it does not classify the underlying smooth four-manifolds and does not address non-Kähler–Einstein metrics.

There are strong rigidity theorems in locally symmetric cases. Besson–Courtois–Gallot proved minimal-entropy rigidity, implying in particular that on a compact real-hyperbolic four-manifold the standard hyperbolic metric is rigid among Einstein metrics up to scale and diffeomorphism [BCG]. In the complex-hyperbolic case, LeBrun used Seiberg–Witten theory to obtain rigidity results for Einstein metrics on complex-hyperbolic surfaces [L1].

Seiberg–Witten theory gives many obstructions to Einstein metrics on four-manifolds. LeBrun derived sharp curvature estimates and nonexistence results for Einstein metrics on manifolds with nontrivial Seiberg–Witten invariants [L2]. These results show that the Hitchin–Thorpe inequality is not sufficient. On the constructive side, Fine and Premoselli recently produced compact negatively curved Einstein four-manifolds which are not locally homogeneous, using branched covers of hyperbolic four-manifolds and perturbation of approximate Einstein metrics [FP]. Thus the class is larger than the locally symmetric examples, but still very far from classified.

There is also progress on rigidity within curvature-sign subclasses. Fine–Krasnov–Singer proved local rigidity for compact oriented Einstein four-manifolds satisfying a chiral curvature condition, namely negativity of the self-dual part of the curvature operator [FKS]. This generalizes the familiar local rigidity of negatively sectionally curved Einstein metrics in a four-dimensional direction. It does not classify negative Einstein four-manifolds, but it gives a useful boundary between flexible construction problems and rigid moduli problems.

The sphere question remains one of the cleanest test cases. The four-sphere satisfies the Hitchin–Thorpe inequality, so the basic topological obstruction gives no information. A negative Einstein metric on $\mathbb{S}^4$ would be neither Kähler nor locally symmetric in any standard way; conversely, no known gauge-theoretic or minimal-surface obstruction rules it out.

Related open problems.

1. *Original direction.* Decide whether $\mathbb{S}^4$ admits an Einstein metric with negative Einstein constant.
2. *Original direction.* Determine the geography region of pairs $(\chi(M), \tau(M))$ realized by compact Einstein four-manifolds with negative Einstein constant.
3. *Reference-driven formulation.* Determine which simply connected smooth four-manifolds in the Hitchin–Thorpe region admit negative Einstein metrics, and compare the answer with complex-surface geography [Au], [H], [L2], [Y].
4. *Original direction.* Classify compact negatively curved Einstein four-manifolds up to diffeomorphism in any natural subclass beyond the locally symmetric case.
5. *Reference-driven formulation.* Understand whether the Fine–Premoselli branched-cover construction can be varied to realize new geography regions or prescribed fundamental groups [FP].
6. *Reference-driven formulation.* Determine whether the Fine–Krasnov–Singer chiral-curvature rigidity hypothesis can be converted into a global compactness or finiteness theorem for negative Einstein four-manifolds [FKS].
7. *AI-generated formulation.* Find effective Seiberg–Witten or monopole-type obstructions which separate the topological Hitchin–Thorpe region into realizable and non-realizable Einstein geography regions.

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Problem 26

Find for each N constants $c_N < C_N$ so that if the Ricci curvature of a compact manifold satisfies $c_N\delta_{ij} \leqslant R_{ij} \leqslant C_N\delta_{ij}$ then the manifold admits an Einstein metric.

Status. Open in the stated generality. No known universal pair of constants depending only on N gives a theorem of this form from two-sided Ricci pinching alone.

The problem asks whether sufficiently strong Ricci pinching forces the underlying compact manifold to admit some Einstein metric, not necessarily the original one. This is weaker in conclusion than proving that the given metric evolves or deforms to an Einstein metric, but stronger in topology because the hypothesis controls only the Ricci tensor. In dimension three, Hamilton's Ricci flow theorem shows that compact three-manifolds with positive Ricci curvature are spherical space forms and hence admit constant-curvature Einstein metrics [Ham]. This gives a positive result in a special dimension with a stronger sign condition. Catino–Djadli obtained a different three-dimensional partial answer: an explicit integral pinching condition between the L^2 norms of the Ricci and scalar curvatures, together with positive scalar curvature, forces the manifold to be a spherical space form and hence to admit a positive Einstein metric [CD].

In higher dimensions, the strongest general convergence and classification theorems usually assume pinching of the full curvature operator, isotropic curvature, or sectional curvature, rather than only Ricci curvature. Hamilton's four-dimensional work on positive curvature operator [Ham4], Böhm–Wilking's theorem for positive curvature operator [BW], and Brendle–Schoen's differentiable sphere theorem under pointwise quarter-pinching [BS] show how powerful curvature pinching can be when it controls more than the Ricci tensor. These results are evidence for the philosophy of the problem, but do not settle the Ricci-pinching statement.

There are also analytic stability results near known Einstein metrics or in noncompact settings, where normalized Ricci flow can converge to an Einstein metric under nondegeneracy and sufficiently strong pinching assumptions [Kr, QSW]. Tian–Wang's work on almost-Einstein manifolds shows that, under noncollapsing and flow-type hypotheses, limits retain much of the structure known for Einstein limits [TW]. Such theorems are local, compactness-theoretic, or require additional structure; they do not give the universal compact theorem asked for here.

Anderson proved a precise "almost Einstein implies Einstein" theorem, but only inside a precompact geometric class [An, Corollary B(4)]. In dimension four his hypotheses control $|\operatorname{Ric}|$, a scaled $L^{q/2}$ norm of $D^2\operatorname{Ric}$ for some $q > 4$, the shortest inessential geodesic loop, volume, diameter, and b_2; if in addition $\int_M |\operatorname{Ric} - \lambda g|$ is sufficiently small, the underlying manifold admits a metric with $\operatorname{Ric} = \lambda g$. In general dimension the b_2 bound is replaced by an $L^{n/2}$ bound on the full curvature. This is the closest classical theorem to Yau's formulation, but its extra compactness and derivative hypotheses are essential to the proof and are not consequences of two-sided Ricci pinching alone. Recent pinching results in submanifold geometry similarly show strong topological rigidity from Ricci lower bounds when extrinsic curvature data are present [GTZ]. They are useful analogies, not solutions of Yau's intrinsic problem.

The main difficulty is that two-sided Ricci control does not control the Weyl tensor. A metric may have Ricci tensor close to a constant multiple of the metric while its full curvature tensor, topology, or collapse behavior remains complicated. Thus a plausible modern version of the problem probably needs either a noncollapsing normalization, a curvature-operator hypothesis, or a Ricci-flow stability assumption in addition to pointwise Ricci pinching.

Related open problems.

1. *Original direction.* Find a topological obstruction to the existence of universal Ricci-pinching constants $c_N < C_N$ in this problem.

2. *Reference-driven formulation.* Determine which of the derivative, loop-length, volume, diameter, and curvature-energy hypotheses in Anderson's almost-Einstein theorem can be removed when one assumes pointwise two-sided Ricci pinching [An].

3. *Reference-driven formulation.* Determine whether two-sided Ricci pinching plus noncollapsing and a uniform bound on $\|\operatorname{Rm}\|_{L^{n/2}}$ suffices, without an a priori $D^2\operatorname{Ric}$ bound, to force existence of an Einstein metric [An], [TW].

4. *Original direction.* Prove or disprove an analogous theorem for compact manifolds with positive Ricci curvature operator pinched in L^∞ around a constant multiple of the metric.

5. *Reference-driven formulation.* Decide whether Ricci-pinching theorems for compact submanifolds have an intrinsic analogue after replacing extrinsic curvature control by harmonic-radius, Weyl-energy, or Sobolev-constant bounds [GTZ].

6. *AI-generated formulation.* Find a Ricci-flow invariant pinching condition involving only the traceless Ricci tensor and the scalar curvature which forces convergence to an Einstein metric.

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C. Scalar curvature

Scalar curvature is the weakest of the classical pointwise curvature conditions, but it is also the one most tightly connected with topology, conformal geometry, and general relativity. Since Yau's 1980 list, three themes have transformed the subject: Ricci flow and geometrization in dimension three, surgery and index theory for positive scalar curvature in high dimensions, and minimal hypersurface or minimal-slicing methods which give non-spin obstructions. The problems below should therefore be read not only as classification questions, but also as requests for the right interface between local scalar lower bounds, global topology, and large-scale geometry.

Problem 27

Classify complete, three-dimensional manifolds with nonnegative scalar curvature.

This is considerably interesting for general relativity because the "universe" tends to have such metrics. In fact, under physically reasonable assumptions, Schoen and Yau [SY2] did prove that such metrics always exist on the universe.

Schoen–Yau [SY3] also proved that the fundamental group of such a manifold does not contain a subgroup which is isomorphic to the fundamental group of a compact surface of genus $\geqslant 1$. In the case of a compact manifold, this was proven in [SY4]. For dimension exceeding three, the problem was considered in Schoen–Yau [SY5] and Gromov–Lawson [GL].

Status. Partially resolved. Several sharp three-dimensional regimes are classified, with major recent progress on the open case, but there is not yet an unrestricted classification of all complete noncompact three-manifolds with scalar curvature $R \geqslant 0$. The closed orientable positive-scalar-curvature case is classified by geometrization. The complete open case with uniformly positive scalar curvature is now topologically classified, while the borderline nonnegative case remains more delicate because flat regions, splitting, and degeneration at infinity can occur.

For compact orientable three-manifolds, the positive scalar curvature classification says that the manifold is a connected sum of spherical space forms and copies of $\mathbb{S}^2 \times \mathbb{S}^1$; the nonnegative case also allows flat manifolds. Schoen–Yau's minimal surface arguments supplied early topological restrictions, excluding incompressible higher-genus surface subgroups in the relevant settings [SY3], [SY4]. Perelman's Ricci-flow proof of geometrization then gives the full compact topological classification.

For complete noncompact three-manifolds, the first decisive classification theorem was obtained under both bounded geometry and uniform positivity. Bessières–Besson–Maillot proved that an orientable open three-manifold admits a complete bounded-geometry metric with uniformly positive scalar curvature if and only if it is a possibly infinite connected sum of $\mathbb{S}^2\times\mathbb{S}^1$'s and spherical space forms from a finite list [BBM]. This is a genuine noncompact analogue of the compact geometrization classification. Bessières–Besson–Maillot– Marques later proved path connectedness of the moduli space of such bounded-geometry uniformly positive scalar curvature metrics [BBMM].

Recent work has pushed the classification well beyond the original bounded-geometry framework. Jian Wang proved that every complete orientable three-manifold with uniformly positive scalar curvature is homeomorphic to an infinite connected sum of spherical three-manifolds and copies of $\mathbb{S}^1 \times \mathbb{S}^2$ [W3]. His proof uses μ-bubbles to produce a prime decomposition and minimal-surface arguments to identify the prime factors. This removes the bounded-geometry hypothesis from the topological classification in the uniformly positive case and should now be regarded as one of the closest modern answers to Yau's original three-dimensional question.

Other recent results attack the gap between uniform positivity and mere nonnegativity. In a series of papers, Jian Wang studied complete contractible three-manifolds with positive scalar curvature: the Whitehead manifold, and more generally contractible genus-one three-manifolds, admit no complete positive-scalar-curvature metric; moreover, a complete contractible three-manifold with positive scalar curvature and trivial fundamental group at infinity is homeomorphic to $\mathbb{R}^3$ [W1], [W2]. Chodosh–Lai–Xu proved that a complete contractible three-manifold with positive scalar curvature and bounded geometry must be $\mathbb{R}^3$, and that an open handlebody of genus larger than one admits no complete nonnegative-scalar-curvature metric with bounded geometry [CLX]. Their argument uses inverse mean curvature flow in a form suited to noncompact topology. Yan–Zhu recently obtained borderline nonnegative-scalar-curvature results under additional harmonic function regularity assumptions. More precisely, they assume a global lower Ricci bound and, in the nonparabolic case, decay to zero of the minimal positive Green function at infinity; under these hypotheses contractible examples are forced to be $\mathbb{R}^3$, and open handlebodies are forced to have genus at most one [YZ]. The proof replaces inverse mean curvature flow by exhaustions through level sets of proper positive harmonic functions and controls their genus by refined average-gradient estimates.

There are also new obstruction results for open manifolds in the same circle of ideas. Shi–Wang–Wu–Zhu introduced open Schoen–Yau–Schick type conditions and proved nonexistence of complete positive-scalar-curvature metrics for such open manifolds [SWWZ]. Together these papers show a clear modern strategy: combine μ-bubbles, stable minimal surfaces, inverse

mean curvature flow, harmonic functions, and three-dimensional topology to isolate which ends and handlebody structures are compatible with scalar curvature lower bounds.

Two further results clarify how topology at infinity enters. Zhu proved a scalar-curvature analogue of the Calabi–Yau volume theorem for manifolds of dimension at most four which are aspherical at infinity, and showed that a complete contractible three-manifold with positive scalar curvature and sublinear volume growth is $\mathbb{R}^3$ [Z]. Chen–Chu–Zhu proved in dimensions three, four, and five that a connected sum of a closed aspherical manifold with an arbitrary noncompact manifold admits no complete metric of positive scalar curvature; their nonnegative version yields a flat rigidity alternative [CCZ].

In the asymptotically flat setting, Haslhofer introduced a mass-decreasing flow built from Ricci flow with surgery and conformal rescaling [Ha]. These results show that scalar curvature classification is accessible when the geometry at infinity is controlled, and increasingly also under topological or analytic conditions at infinity; the unresolved part is what remains true for complete metrics with only $R \geqslant 0$ and no bounded-geometry, uniform positivity, harmonic-function, or asymptotic-flatness assumption.

In higher dimensions, positive scalar curvature behaves quite differently. Schoen–Yau and Gromov–Lawson developed the surgery and minimal-hypersurface theory of positive scalar curvature [SY5], [GL]. The high-dimensional simply connected classification involves spin, index-theoretic, and surgery-theoretic obstructions, rather than a three-dimensional prime-decomposition picture.

Related open problems.

1. *Original direction.* Classify complete orientable three-manifolds with $R \geqslant 0$ without assuming bounded geometry, uniform positivity, or asymptotic flatness.

2. *Reference-driven formulation.* Determine whether Jian Wang's infinite-connected-sum classification for uniformly positive scalar curvature admits a borderline analogue for $R \geqslant 0$, possibly after separating flat ends and splitting regions [W3].

3. *Reference-driven formulation.* Determine the weakest geometric or analytic hypotheses under which the Bessières–Besson–Maillot and Wang connected-sum classification remains valid [BBM], [BBMM], [W3].

4. *Reference-driven formulation.* Develop a borderline theory for complete three-manifolds with $R \geqslant 0$ which separates flat, splitting, asymptotically flat, and uniformly positive scalar curvature ends [Ha], [YZ], [Z].

5. *Original direction.* Classify asymptotically flat three-manifolds with nonnegative scalar curvature and minimal boundary in terms of mass, topology, and Ricci-flow surgery.

6. *Reference-driven formulation.* Decide whether every complete contractible three-manifold with positive scalar curvature is $\mathbb{R}^3$, and identify the minimal hypotheses at infinity needed to rule out Whitehead-type behavior [CLX], [W1], [W2], [YZ].

7. *Reference-driven formulation.* Classify open handlebodies admitting complete metrics with $R > 0$ or $R \geqslant 0$ under bounded geometry, harmonic-function regularity, or other natural control at infinity [CLX], [YZ].

8. *Reference-driven formulation.* Decide whether a complete three-manifold with $R \geqslant 0$ and controlled topology at infinity admits a canonical Ricci-flow or inverse-mean-curvature-flow decomposition [CLX], [Ha].

9. *Reference-driven formulation.* Extend the open Schoen–Yau–Schick obstruction framework to borderline nonnegative scalar curvature and to weakly controlled ends [CCZ], [SWWZ].

10. *AI-generated formulation.* Formulate a coarse topological invariant of a complete noncompact three-manifold which detects whether it can carry a complete nonnegative scalar curvature metric with controlled geometry at infinity.

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Problem 28

Classify all compact, four-dimensional, Einstein manifolds with positive scalar curvature.

Status. Open in general. There is no classification of compact four-dimensional Einstein manifolds with positive scalar curvature without additional hypotheses. Several important subclasses are understood.

The basic topological restriction is the Hitchin–Thorpe inequality: if (M^4, g) is compact, oriented and Einstein, then $2\chi(M) \geqslant 3|\tau(M)|$, with equality only in the flat or Ricci-flat Kähler cases and their quotients [H]. This gives a first geography constraint, but it is far from sufficient.

Under extra complex or symplectic structure, much more is known. In the Kähler case, positive Einstein constant is the Fano Kähler–Einstein problem. The modern Yau–Tian–Donaldson theorem identifies the existence of Kähler–Einstein metrics on smooth Fano manifolds with algebro-geometric stability [CDS, Ti]. In complex dimension two this places the positive Kähler–Einstein part inside the well-developed classification of del Pezzo surfaces, while still leaving the purely Riemannian Einstein problem much broader. Chen–LeBrun–Weber classified the conformally Kähler–Einstein case arising in this setting [CLW]. LeBrun then gave a complete diffeomorphism classification for compact four-manifolds which admit both a complex or symplectic structure and an Einstein metric with nonnegative Einstein constant [L1]. The resulting list is governed by del Pezzo surfaces in the positive case, together with the Ricci-flat possibilities in the nonnegative case.

The moduli viewpoint has also become much sharper in the Kähler subclass. Odaka–Spotti–Sun proved that the Gromov–Hausdorff compactification of moduli spaces of Kähler–Einstein del Pezzo surfaces agrees with an algebro-geometric compactification by singular log Fano surfaces [OSS]. This does not classify all positive Einstein four-manifolds, but it gives a model for what a successful classification may need to include: not only diffeomorphism types, but also moduli components, degenerations, and singular compactifications.

There are also rigidity theorems under stronger curvature hypotheses. Gursky and LeBrun proved that a compact oriented Einstein four-manifold with nonnegative sectional curvature and positive intersection form is, up to scaling and isometry, the Fubini–Study metric on $\mathbb{CP}^2$ [GL]. LeBrun's Seiberg–Witten estimates give stronger curvature inequalities and nonexistence results in many cases [L2]. More recently, LeBrun classified compact oriented Einstein four-manifolds with $\det(W^+) > 0$: there are exactly fifteen oriented diffeomorphism types, and on each the metrics in this curvature class form exactly one connected component of the Einstein moduli space [L3]. Dameno obtained a complementary twistorial rigidity theorem: if the scalar curvature of the Riemannian twistor space is constant along every twistor fiber, then the base Einstein four-manifold is half conformally flat; in the compact positive case this leaves only the standard $\mathbb{S}^4$ and $\mathbb{CP}^2$ [D]. These results show substantial progress, but they still fall well short of the unrestricted classification problem.

The known examples already show that the classification is not merely the list of locally symmetric spaces. Besides the round sphere, $\mathbb{CP}^2$, and products such as $\mathbb{S}^2 \times \mathbb{S}^2$, there are inhomogeneous positive Einstein metrics such as the Page metric and the Chen–LeBrun–Weber metric. The central difficulty is to understand whether these constructions exhaust natural topological classes or whether there are many undetected positive Einstein smooth four-manifolds.

Even the variational behavior of the known examples contains useful structure. Biquard–Ozuch proved that every compact conformally Kähler Einstein four-manifold with nonnegative scalar curvature which is not half conformally flat is unstable for the Hilbert–Einstein functional; in particular this applies to the Chen–LeBrun–Weber metric [BO]. Thus existence, moduli, degeneration, and dynamical or variational stability are genuinely distinct layers of the classification problem.

Related open problems.

1. *Original direction.* Classify compact simply connected Einstein four-manifolds with positive Einstein constant.

2. *Original direction.* Decide whether the round metric is the unique positive Einstein metric on $\mathbb{S}^4$, up to scaling and diffeomorphism, under progressively weaker curvature assumptions.

3. *Original direction.* Decide whether every compact positive Einstein four-manifold with positive intersection form must be $\mathbb{CP}^2$ under weaker curvature assumptions than nonnegative sectional curvature.

4. *Original direction.* Determine the geography of compact positive Einstein four-manifolds in terms of (χ, τ) and smooth structure.

5. *Reference-driven formulation.* Determine whether the moduli space of positive Einstein metrics on a fixed compact four-manifold has finitely many components under natural curvature, volume, or diameter bounds [L3], [OSS].

6. *Reference-driven formulation.* Determine whether the twistor-fiber scalar-curvature condition in Dameno's theorem can be replaced by an integral pinching or small-oscillation hypothesis [D].

7. *Reference-driven formulation.* Classify the stable critical points of the Hilbert–Einstein functional among positive Einstein four-manifolds, beyond the conformally Kähler class [BO].

8. *Reference-driven formulation.* Relate the Fano Kähler–Einstein classification in complex dimension two to the larger Riemannian positive Einstein geography problem [CDS], [OSS], [Ti].

9. *AI-generated formulation.* Find a curvature condition strictly weaker than $\det(W^+) > 0$ but strong enough to yield a finite diffeomorphism classification of positive Einstein four-manifolds.

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Problem 29

Prove that a compact $K(\pi, 1)$ manifold with nonnegative scalar curvature is flat.

Status. Partially resolved. The statement is known in dimensions at most five; the new cases 4 and 5 were proved by Chodosh–Li using generalized soap bubbles [CL]. Gromov gave an independent closely related proof in dimension 5 through stable μ-bubbles and torical symmetrization [Gr2]. In these dimensions a smooth closed aspherical manifold admits no metric of positive scalar curvature, and every metric of nonnegative scalar curvature is flat. The problem remains open in full generality for arbitrary closed aspherical manifolds in higher dimensions, although it is known for important classes, including tori, enlargeable manifolds, many spin manifolds detected by higher index theory, and several classes accessible by minimal-hypersurface methods.

The statement is a rigidity form of the positive scalar curvature problem. If an aspherical manifold admits a metric with scalar curvature nonnegative and positive somewhere, then a conformal deformation often converts it to a positive scalar curvature problem. Thus obstructions to positive scalar curvature become obstructions to the nonnegative case, except for the rigid scalar-flat possibility. Bourguignon's deformation theorem shows that a scalar-flat metric which is not Ricci-flat can be deformed to one with positive scalar curvature [B]. Hence, once positive scalar curvature is excluded, any nonnegative-scalar-curvature metric must be Ricci-flat. On a closed aspherical manifold, the Cheeger–Gromoll splitting theorem then forces the universal cover of a Ricci-flat metric to split as Euclidean space; equivalently, the compact quotient is a flat Bieberbach manifold [CG]. This explains why the main difficulty is the exclusion of positive scalar curvature.

The problem originates in the Schoen–Yau program relating stable minimal hypersurfaces to scalar curvature. Schoen–Yau proved strong low-dimensional results, including scalar-flat rigidity phenomena for tori in the dimensions where the regularity theory is classical [SY]. Their strategy is to find a stable hypersurface in a suitable homology class, use the stability inequality and the Gauss equation to descend the scalar-curvature condition by one dimension, and iterate until a topological contradiction appears. Their later minimal-slicing program extends this obstruction mechanism beyond the classical regularity range and is designed precisely to handle singularities of stable hypersurfaces [SY2].

Chodosh–Li completed the Schoen–Yau outline for closed aspherical 4-manifolds and developed the additional slice-and-dice and μ-bubble arguments needed in dimension 5 [CL]. The technical issue in dimension 5 is not merely the existence of a stable hypersurface, but the need to localize the argument and control the topology and diameter of the lower-dimensional pieces produced by the descent. Gromov obtained a closely related dimension-5 theorem through stable μ-bubbles and torical symmetrization, proving in particular the nonexistence of positive scalar curvature on closed aspherical manifolds of dimension $\leqslant 5$ [Gr2]. Gromov–Lawson introduced enlargeability and proved that enlargeable spin manifolds, including tori, do not admit positive scalar curvature [GL1], [GL2]. Together with rigidity arguments, this gives the expected flat conclusion for many classical examples. Rosenberg's index-theoretic formulation and the Gromov–Lawson–Rosenberg program connect the general aspherical case with higher index theory and the Novikov/Baum– Connes circle of ideas [R]. Hanke–Schick strengthened the enlargeability obstruction by proving nontriviality of the universal index class for enlargeable spin manifolds [HS].

There is also nearby recent progress for open four-manifolds with uniformly positive scalar curvature. Chodosh–Maximo–Mukherjee proved topological obstructions for complete noncompact four-manifolds, including restrictions for interiors of compact contractible four-manifolds and for exotic $\mathbb{R}^4$'s [CMM]. These results do not solve the closed $K(\pi, 1)$ problem, but they show that the soap-bubble and low-dimensional scalar-curvature methods are now probing genuinely four-dimensional smooth phenomena beyond the closed aspherical setting.

The low-dimensional obstruction has also acquired noncompact and splitting forms. Chen–Chu–Zhu proved that, for dimensions three through five, the connected sum of a closed aspherical manifold with an arbitrary noncompact manifold carries no complete positive-scalar- curvature metric, and their nonnegative theorem has a flat rigidity alternative [CCZ]. He–Zhu proved that a complete orientable aspherical manifold of dimension at most five with two ends and nonnegative scalar curvature splits as a product of a closed flat manifold with $\mathbb{R}$ [HZ]. These results do not settle the closed problem above dimension five, but they identify quantitative filling and homology-vanishing statements which a higher-dimensional proof would have to supply.

A very recent extension concerns symplectic rather than homotopy asphericity. Di Cerbo–Dranishnikov–Jauhari proved that a closed c-symplectically aspherical manifold with a spin finite cover admits no Riemannian metric of positive scalar curvature, proved the unconditional result in real dimensions two and four, and showed that smooth projective varieties mapping birationally to symplectically aspherical projective varieties admit no positive-scalar-curvature Kähler metric [DDJ]. They formulate the broader conjecture that every closed symplectically aspherical manifold is obstructed from positive scalar curvature. This is not a solution of Yau's $K(\pi, 1)$ problem, but it enlarges the class of "aspherical" structures for which index and gauge-theoretic mechanisms can be tested.

The modern perspective separates three layers. First, positive scalar curvature on aspherical manifolds is obstructed by stable minimal hypersurfaces, enlargeability, and higher indices. Second, the scalar-flat borderline requires a rigidity theorem showing that a nonnegative-scalar-curvature metric which cannot be deformed to positive scalar curvature must be Ricci-flat, and then

flat in the aspherical setting. Third, Gromov's macroscopic scalar-curvature program asks for quantitative versions in which lower scalar curvature bounds force restrictions on Urysohn width, macroscopic dimension, filling radius, or enlargeability at large scale [Gr]. The full statement for all compact $K(\pi, 1)$ manifolds remains a major open direction.

Related open problems.

1. *Original direction.* Prove the nonnegative scalar curvature rigidity statement for all closed aspherical manifolds of dimension at least 6.

2. *Reference-driven formulation.* Quantify the Bourguignon–Cheeger–Gromoll rigidity mechanism: under which noncollapsing and diameter hypotheses does a closed aspherical metric with $R \geqslant -\epsilon$ and small negative part converge to a flat metric as $\epsilon \to 0$ [B], [CG]?

3. *Reference-driven formulation.* Relate this rigidity problem to the strong Novikov conjecture and the Baum–Connes assembly map for the fundamental group.

4. *Reference-driven formulation.* Determine whether minimal-slicing methods prove the aspherical positive-scalar-curvature obstruction for all closed non-spin $K(\pi, 1)$ manifolds in dimensions at least 6 [SY2].

5. *Reference-driven formulation.* Determine whether the Chodosh–Li μ-bubble and slice-and-dice argument admits a higher-dimensional replacement for the controlled-diameter decomposition used in dimension 5 [CL], [Gr2].

6. *Reference-driven formulation.* Extend scalar-flat rigidity from tori and enlargeable spin manifolds to non-spin aspherical manifolds detected by minimal-slicing or macroscopic scalar-curvature methods.

7. *Reference-driven formulation.* Compare the closed aspherical rigidity problem with the new obstruction theory for open four-manifolds and exotic $\mathbb{R}^4$'s with uniformly positive scalar curvature and with noncompact aspherical splitting theorems [CCZ], [CMM], [HZ].

8. *Reference-driven formulation.* Decide whether the spin-cover and low-dimensional hypotheses in the symplectically aspherical obstruction can be removed [DDJ].

9. *AI-generated formulation.* Establish a quantitative version: if a closed aspherical manifold has scalar curvature $R \geqslant -\epsilon$ together with noncollapsing and diameter or width control, must it be close in a geometric or coarse sense to a flat, enlargeable, or lower-dimensional model when ϵ is small?

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Problem 30

Prove that a compact, simply-connected, three-dimensional manifold with positive scalar curvature is homeomorphic to the sphere. It is proved in Meeks–Simon–Yau [MSY] that the connected sum of two fake three-spheres does not admit a metric with positive Ricci curvature.

Status. Solved. The conclusion follows from Perelman's proof of the Poincaré conjecture, which is a consequence of the geometrization theorem and finite-extinction argument [Pe], [Pe2]. The scalar curvature assumption is no longer needed.

Before Perelman, the problem was connected with the interaction between scalar curvature, Ricci curvature, and minimal surface methods in dimension three. Meeks–Simon–Yau proved that the connected sum of two fake three-spheres does not admit a metric of positive Ricci curvature [MSY]. Perelman's Ricci-flow proof shows that a compact simply connected three-manifold is homeomorphic to $\mathbb{S}^3$, so the original scalar-curvature version is now a topological corollary.

For the simply connected case, Perelman's finite-extinction theorem provides a particularly direct analytic bridge: on a closed oriented three-manifold whose prime decomposition has no aspherical factor, every Ricci flow with surgery becomes extinct in finite time [Pe2, Theorem 1.1]. Applied together with the surgery and canonical-neighborhood theory, this gives elliptization and hence the Poincaré conclusion.

The modern refinement concerns not only existence but the topology of the space of metrics. Bamler–Kleiner proved that, for any closed three-manifold, the space of positive-scalar-curvature metrics is either empty or contractible [BK]. In particular, on $\mathbb{S}^3$ the moduli-level analogue of the solved topological problem is also extremely rigid: once positive-scalar-curvature metrics exist, their space has no higher homotopical ambiguity before quotienting by diffeomorphisms. Their proof evolves continuous families by canonical singular Ricci flows and uses "partial homotopies" to pass through neck singularities while preserving positive scalar curvature. Their Ricci-flow-through-singularities techniques also prove the generalized Smale conjecture for spherical space forms, linking scalar curvature, Ricci flow, and diffeomorphism groups.

Related open problems.

1. *Original direction.* Understand which geometric curvature conditions give direct analytic proofs of the three-dimensional sphere theorem without using the full geometrization theorem.

2. *Reference-driven formulation.* Classify moduli spaces of positive scalar curvature metrics on $\mathbb{S}^3$ and on connected sums of spherical space forms and $\mathbb{S}^2 \times \mathbb{S}^1$ [BK].

3. *Reference-driven formulation.* Determine whether canonical paths produced by Ricci flow with surgery give natural deformation retractions of spaces of positive scalar curvature metrics in dimension three [BK], [Pe].

4. *AI-generated formulation.* Find a min-max or stable minimal surface proof of the scalar-curvature sphere conclusion which recovers the Poincaré conclusion under an explicit geometric normalization.

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Problem 31

Classify compact hypersurfaces in $\mathbb{R}^{N+1}$ which have constant scalar curvature. Are they isometric to $\mathbb{S}^N$? If they are convex, then the answer is yes and was proved by Cheng–Yau [CY].

Status. Solved for embedded compact hypersurfaces. Ros proved that a compact embedded hypersurface in Euclidean space with constant scalar curvature is a round sphere [R]. Thus the answer to the original compact embedded problem is yes.

Cheng and Yau proved earlier that a complete oriented hypersurface in a real space form, with constant scalar curvature and nonnegative sectional curvature, is totally umbilical, a product of two totally umbilical factors, or flat; in Euclidean space their complete noncompact classification gives the generalized cylinders $\mathbb{S}^{N-k} \times \mathbb{R}^k$ [CY]. In the compact convex case this gives the round sphere. Their main analytic device is the self-adjoint second-order operator associated with the first Newton transformation of the shape operator. Ros later removed the convexity assumption in the compact embedded case by combining a Reilly identity on the bounded domain with the Newton inequalities [R], [R2].

The distinction between embedded and immersed hypersurfaces is essential. Alexandrov-type reflection and Reilly identities use the global embedded domain in an important way. For embedded compact hypersurfaces, the constant scalar curvature result fits into a wider family of rigidity theorems for constant higher-order mean curvatures: Ros proved corresponding Alexandrov-type theorems for higher mean curvatures [R2]. The immersed case is more flexible and is not covered by the same argument.

Recent work around adjacent curvature hypotheses clarifies how special the constant-scalar-curvature condition is. Ge–Zhao proved that, for hypersurfaces in real space forms, having constant Ricci eigenvalues is equivalent to curvature homogeneity, and hence falls under the known classification of curvature-homogeneous hypersurfaces [GZ]. In particular, a complete hypersurface with constant Ricci eigenvalues is isoparametric in a sphere and, in Euclidean space, whenever it is nonflat; the hyperbolic statement has explicit exceptional cases. This is a stronger condition than constant scalar curvature alone, but it gives a nearby rigidity model: the Gauss equation turns the fixed Ricci spectrum into algebraic constraints on the principal curvatures, and real-analytic continuation fixes their multiplicities. Freitas– Guimarães recently extended the Reilly-formula congruence mechanism to compact hypersurfaces in a hemisphere [FG]. In a different direction, Fu–Hong–Yang–Zhan studied biconservative hypersurfaces with constant scalar curvature in space forms and obtained low-dimensional classification results [FHYZ]. These developments do not change Ros's solution in the compact embedded Euclidean case, but they identify natural strengthened and generalized settings in which classification remains active.

For complete noncompact hypersurfaces, stability hypotheses lead to different rigidity and nonexistence questions. Alencar–Santos–Zhou obtained nonexistence results for stable constant-scalar-curvature hypersurfaces in Euclidean spaces under curvature and finite total-curvature assumptions [ASZ].

Related open problems.

1. *Original direction.* Classify immersed, rather than embedded, compact hypersurfaces in Euclidean space with constant scalar curvature.

2. *Reference-driven formulation.* Study analogous rigidity for hypersurfaces with constant higher order mean curvature in space forms [R2].

3. *Reference-driven formulation.* Classify compact hypersurfaces with constant scalar curvature in hyperbolic space or the sphere under the weakest embeddedness, star-shapedness, or curvature-sign assumptions, comparing the result with the stronger constant-Ricci-eigenvalue classification [GZ].

4. *Reference-driven formulation.* Prove quantitative stability versions of the Cheng–Yau and Ros sphere theorems under small oscillation of scalar curvature or higher mean curvature [CY], [R], [R2].

5. *Reference-driven formulation.* Determine how much of the Ge–Zhao constant-Ricci-eigenvalue classification survives when one assumes only constant scalar curvature plus one additional curvature pinching or multiplicity condition [GZ].

6. *Reference-driven formulation.* Extend the spherical hemisphere congruence theorem from comparison of mean-curvature vectors to a natural scalar-curvature comparison [FG].

7. *AI-generated formulation.* Determine whether a quantitative almost-constant scalar curvature condition forces a compact embedded hypersurface to be close to a round sphere, with an explicit rate in terms of the scalar-curvature oscillation and a noncollapsing constant.

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Problem 32

(Yamabe) Prove that any metric on a compact manifold can be conformally deformed to a metric of constant scalar curvature. Yamabe published a proof [Ya], but N. Trudinger [Tr] found a gap in the work after Yamabe's death. Nonetheless, Yamabe's original proof can be pushed to cover a large class of metrics, as was made clear by Trudinger (see also Eliasson [E]).

Pushing further, Aubin [Au] solved the problem for an even broader class, in dimension $s \geqslant 6$

However, even for surfaces of genus zero it is nontrivial to find a proof without use of a complex analysis.

One can formulate a similar conjecture in the class of complete noncompact manifolds. Progress has been made recently by W. M. Ni [Ni].

Status. Solved for compact manifolds. The Yamabe problem was completed by the combined work of Yamabe, Trudinger, Aubin, and Schoen, with Schoen's 1984 paper closing the remaining cases [Ya], [Tr], [Au], [S].

Yamabe introduced the variational problem and claimed a proof [Ya]. Trudinger found a gap and established important partial results [Tr]. Aubin developed the sharp Sobolev and test-function estimates which settled many cases, especially in high dimensions or when the Weyl tensor is nonzero somewhere [Au]. Schoen completed the proof by using the positive mass theorem to handle the locally conformally flat and remaining low-dimensional cases [S]. Thus every conformal class on a compact smooth manifold contains a metric of constant scalar curvature.

After existence, the next questions concern uniqueness, compactness, flow convergence, and the structure of the solution set. If the Yamabe constant is nonpositive, the Yamabe metric is unique up to scaling. In the positive case uniqueness can fail; Pollack proved nonuniqueness results [P]. Schoen's compactness conjecture predicted compactness of the set of Yamabe solutions away from the conformal sphere. This is true in several ranges and under additional hypotheses, but false in high dimensions: Brendle constructed counterexamples in dimensions $n \geqslant 52$ [B], and Brendle–Marques extended counterexamples to $25 \leqslant n \leqslant 51$ [BM]. Khuri–Marques– Schoen proved compactness of the full solution set for all smooth compact manifolds of dimension at most 24, away from the conformal round sphere [KMS]. No spin hypothesis occurs in the statement of their main compactness theorem. Their proof combines sharp blow-up expansions, Weyl vanishing through a dimension-dependent quadratic form, and positive mass; the same quadratic form acquires negative directions beginning in dimension 25, which drive the counterexamples [B], [BM], [BMS]. Thus the solved existence theorem opened a finer geometric analysis problem: describe precisely when bubbling occurs and how the conformal geometry detects it.

For complete noncompact manifolds the direct analogue is false in general. Jin constructed counterexamples to the complete noncompact Yamabe problem [J]. The noncompact problem remains active under extra hypotheses; negative scalar curvature is better understood, while the nonnegative and positive cases are more delicate.

Several modern variants show that the Yamabe problem has become a prototype for conformally covariant geometric flows and nonlocal operators. Brendle first proved arbitrary-initial-energy convergence when $3 \leqslant n \leqslant 5$ or the initial metric is locally conformally flat [Br1]. His higher-dimensional paper uses a Weyl-tensor condition and the positive mass theorem; in its original form the corresponding spin or curvature hypotheses must be retained, while the modern all-dimensional positive mass theorem supplies the missing global input [Br2]. More recent work studies weighted and fractional analogues: Yan proved convergence of a weighted Yamabe flow on smooth metric measure spaces [Yan], while An–Chan–Ho proved convergence of the fractional Yamabe flow for arbitrary initial energy under the validity of the corresponding fractional positive mass conjecture [ACH]. These are not new proofs of the classical compact Yamabe theorem, but they show how the Yamabe mechanism continues to organize conformal geometry beyond the original local scalar-curvature equation. In CR geometry, Afeltra–Pinamonti–Ho obtained new compactness and noncompactness results for the CR Yamabe problem [APH], reinforcing the lesson that compactness is a separate and subtler question from existence.

Related open problems.

1. *Original direction.* Give a complete classification of conformal classes with unique Yamabe metric up to scaling.
2. *Reference-driven formulation.* In dimensions $n \geqslant 25$, classify the conformal classes for which the full Yamabe solution set is nevertheless compact, and isolate geometric conditions that exclude the negative quadratic-form directions used in the counterexamples [KMS], [B], [BM].
3. *Reference-driven formulation.* Describe the bubbling set and blow-up profiles for Yamabe metrics in all noncompactness regimes, including the role of Weyl jets and mass-type invariants [B], [BM].
4. *Original direction.* Classify complete noncompact manifolds whose conformal class contains a complete constant scalar curvature metric.

5. *Reference-driven formulation.* Determine convergence and singularity formation for the Yamabe flow in conformal classes where the Yamabe solution set is noncompact or nonunique [Br1], [Br2].

6. *Reference-driven formulation.* Develop relative, boundary, CR, weighted, and fractional Yamabe theories compatible with positive mass, minimal boundary, asymptotically flat geometry, and fractional positive mass hypotheses [Ni], [APH], [Yan], [ACH].

7. *AI-generated formulation.* Develop a compactification theory for noncompact Yamabe solutions which detects bubbling and loss of completeness by geometric invariants at infinity, and compare the resulting boundary data with the compactness theory for the compact Yamabe equation.

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II. Curvature and Complex Structure

Problem 33

Let M be a compact Kähler manifold with nonnegative bisectional curvature. Prove that M is biholomorphic to a locally symmetric Kähler manifold, at least when the Ricci curvature is positive.

If the bisectional curvature is strictly positive, the manifold is in fact biholomorphic to $\mathbb{CP}^N$, as was conjectured by Frankel and proved in Mori [Mo] and Siu-Yau [SiY].

Status. Solved. Mok proved the generalized Frankel conjecture: a compact Kähler manifold with nonnegative holomorphic bisectional curvature has universal cover biholomorphic and isometric to a product of complex Euclidean space, complex projective spaces, and irreducible compact Hermitian symmetric spaces [Mok]. More precisely, the projective-space factors may carry nonnegative-bisectional Kähler metrics rather than only their canonical symmetric metrics, whereas the irreducible rank-at-least-two Hermitian symmetric factors carry their canonical metrics. In the compact case with positive Ricci curvature there is no flat factor, giving the biholomorphic locally symmetric conclusion asked for here.

The strictly positive bisectional curvature case was the original Frankel conjecture. Mori proved the algebraic version via the Hartshorne conjecture on ample tangent bundles [Mo], and Siu–Yau gave the differential-geometric proof that a compact Kähler manifold with positive bisectional curvature is biholomorphic to $\mathbb{CP}^N$ [SiY]. Bando obtained an important three-dimensional classification under nonnegative bisectional curvature [B]. Mok's uniformization theorem completed the general nonnegative case [Mok].

Gu later gave a short transcendental proof of the irreducible core of Mok's theorem [Gu]. Under Kähler–Ricci flow, nonnegative bisectional curvature is preserved and the strong maximum principle transports its zero set by holonomy. In the nonsymmetric case full $U(n)$ holonomy then forces strict positivity and hence the $\mathbb{CP}^n$ conclusion; otherwise Berger holonomy leads to the locally symmetric alternative. The Howard–Smyth–Wu splitting theorem then restores the full universal-cover decomposition.

Later developments study weaker curvature conditions. Nonnegative quadratic bisectional curvature and orthogonal bisectional curvature were introduced and developed as possible extensions of the same classification philosophy [WYZ], [CT], [Z]. These conditions are weaker than full nonnegative bisectional curvature, so the corresponding classification problems remain subtler. Li–Wu–Zheng made this separation concrete: the seven-dimensional Kähler C-space (B_3, α_2) has nonnegative quadratic bisectional curvature but admits no Kähler metric with nonnegative orthogonal bisectional curvature [LWZ].

More recent work continues to search for curvature cones weaker than nonnegative bisectional curvature but still strong enough to retain Frankel-type consequences. Ni–Zheng introduced and studied cross quadratic bisectional curvature and its dual; among other results, they showed that positivity or nonnegativity of the first cross-quadratic condition gives Fano-type conclusions under suitable hypotheses and is well behaved with respect to Kähler-Ricci-flow deformation [NZ]. Li has recently classified manifolds admitting a new "2-positive bisectional curvature" condition by using Yang–Mills flow on rational curves [Li]. In complex dimension two the class is exactly the del Pezzo surfaces; in dimensions at least three it consists precisely of $\mathbb{CP}^n$ and the smooth complex hyperquadric. The proof turns the desired classification into a lower bound for minimal anticanonical degrees and derives that bound from a Hamilton-type maximum principle for Yang–Mills flow on $\mathbb{CP}^1$. These results should be read as extensions of the solved Mok–Siu–Yau picture, not replacements for it: the main open issue is which weakened curvature cones still force a rational homogeneous or locally symmetric classification.

Related open problems.

1. *Original direction.* Classify compact simply connected locally irreducible Kähler manifolds with nonnegative quadratic bisectional curvature.

2. *Reference-driven formulation.* Determine the sharp classification under nonnegative orthogonal holomorphic bisectional curvature [CT], [GuZ].

3. *Reference-driven formulation.* Extend Mok-type uniformization results to natural Hermitian, Sasaki, or orbifold settings, while preserving a curvature condition that is stable under a geometric flow [Mok], [Gu].

4. *Reference-driven formulation.* Determine whether cross quadratic bisectional curvature or its dual gives a classification of compact Kähler manifolds beyond the Fano conclusion known under first-order positivity assumptions [NZ].

5. *Reference-driven formulation.* Test whether Li's "2-positive bisectional curvature" condition is preserved by natural flows and whether it forces rational homogeneity under additional simply connectedness or positivity assumptions [Li].

6. *AI-generated formulation.* Find a curvature cone strictly between nonnegative orthogonal bisectional curvature and nonnegative bisectional curvature which is preserved by Kähler-Ricci flow and still forces a locally symmetric classification.

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Problem 34

Let M be a complete, noncompact, Kähler manifold with positive bisectional curvature. Prove that M is biholomorphic to $\mathbb{C}^N$.

Status. Open in full generality. The strongest known results prove the conclusion under additional hypotheses such as maximal volume growth, bounded curvature, curvature decay, or other quantitative control at infinity.

The first obstruction is already analytic: from positive bisectional curvature alone it is still not known in general that M is Stein. Greene and Wu proved that positive sectional curvature gives strictly plurisubharmonic exhaustion functions in important cases, hence Steinness in that stronger Riemannian curvature setting [GW]. Siu and Yau developed related uniformization criteria for complete Kähler manifolds under curvature and decay assumptions [SiY].

The next stage concerns maximum volume growth and curvature decay. Mok–Siu–Yau used Poincaré–Lelong type techniques on complete Kähler manifolds to obtain uniformization results under strong curvature decay and volume growth assumptions [MSY]. Mok then proved an embedding theorem: a complete Kähler manifold with positive bisectional curvature, quadratic curvature decay, and maximal volume growth is biholomorphic to an affine algebraic variety; in complex dimension two, with positive sectional curvature, this gives $\mathbb{C}^2$ [Mok]. Shi's work on Ricci flow for complete noncompact Kähler manifolds gave a major analytic tool for later progress [Shi].

In complex dimension two, the conclusion was strengthened under bounded positive bisectional curvature, Euclidean volume growth, and average scalar curvature decay: the surface is biholomorphic to $\mathbb{C}^2$ [CTZ]. Related work also showed that bounded positive sectional curvature together with finite analytic first Chern number forces a complete noncompact Kähler surface to be biholomorphic to $\mathbb{C}^2$ [CZ]. In all dimensions, Kähler-Ricci flow methods then yielded the theorem that bounded nonnegative holomorphic bisectional curvature plus maximal volume growth implies biholomorphism to $\mathbb{C}^n$ [CT1]. A different partial result shows that, under bounded curvature and uniform linear average scalar curvature decay, the manifold is holomorphically covered by a pseudoconvex domain in $\mathbb{C}^n$ homeomorphic to $\mathbb{R}^{2n}$; in particular the simply connected case is Stein [CT2].

A later decisive development is Gang Liu's proof of Yau's uniformization conjecture under maximal volume growth: every complete noncompact Kähler manifold with nonnegative bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$ [Liu]. Thus the remaining core of the problem is the case without maximal volume growth or comparable control at infinity. The nonnegative bisectional curvature "complex soul" picture also remains an important related direction: one expects a complete noncompact Kähler manifold with nonnegative bisectional curvature to admit a holomorphic structure analogous to a vector bundle over a compact nonnegatively curved Kähler manifold; partial results are known under bounded curvature assumptions [CCZ], and plurisubharmonic-function methods give structure and splitting theorems [NT].

More recent work has clarified the algebraic and analytic mechanisms behind the maximal-volume-growth case. Liu proved finite generation of the ring of polynomial-growth holomorphic functions on complete noncompact Kähler manifolds with nonnegative bisectional curvature, and showed that in the maximal-volume-growth case the manifold is an affine algebraic variety before proving the full biholomorphism to $\mathbb{C}^n$ [Liu2], [Liu]. Lee–Tam gave a Ricci-flow route to the same maximal-volume-growth uniformization by first deforming suitable noncollapsed complete metrics to bounded nonnegative bisectional curvature [LT]. Chen–Zhu's volume-growth and curvature-decay estimates, and Ogaja's work on nonmaximal volume growth, emphasize that the main remaining obstacle is to produce enough holomorphic functions or a Stein exhaustion without assuming the strongest asymptotic volume condition [CZ2], [O].

Shi has now made the maximal-growth mechanism quantitative [Shi2]. Assuming nonnegative bisectional curvature and no flat splitting of the universal cover, maximal volume growth is equivalent to the existence of a nonconstant polynomial-growth holomorphic function and to finite average scalar-curvature decay. Refined minimal degrees of global holomorphic coordinates and of canonical forms determine the asymptotic volume ratio and scalar-curvature decay, and also describe Lyapunov exponents of the Kähler–Ricci flow. This unifies the Liu and flow-based proofs, but it also exposes their limitation: in the nonmaximal-growth case these polynomial-growth functions do not exist.

A major 2026 advance removes every hypothesis at infinity in a stronger-curvature surface case. Datar–Pingali–Seshadri proved that every complete noncompact Kähler surface with positive real sectional curvature is biholomorphic to $\mathbb{C}^2$ [DPS]. Their method replaces polynomial-growth coordinates by uniformly Lipschitz plurisubharmonic weights with finite Monge–Ampère mass, weighted L^p holomorphic functions, and new Bézout and multiplicity estimates; finite generation and a two-stage algebraic embedding then allow Ramanujam's surface criterion to finish the proof. Positive real sectional curvature is stronger than positive holomorphic bisectional curvature. Indeed, their bisectional-curvature theorem still assumes strong Steinness and contractibility, followed by simple connectivity at infinity to obtain $\mathbb{C}^2$. Thus this breakthrough does not settle Yau's full conjecture, but it provides a genuinely new mechanism which does not start from maximal volume growth.

Related open problems.

1. *Original direction.* Prove Steinness for complete noncompact Kähler manifolds with positive holomorphic bisectional curvature, without assuming bounded curvature, maximal volume growth, or scalar-curvature decay.

2. *Original direction.* Remove maximal volume growth from the known $\mathbb{C}^n$ uniformization theorems for nonnegative or positive bisectional curvature.

3. *Reference-driven formulation.* Develop a precise "complex soul theorem" for complete noncompact Kähler manifolds with nonnegative bisectional curvature, identifying when the manifold is biholomorphic to a holomorphic vector bundle over a compact Kähler manifold with nonnegative bisectional curvature [CCZ], [NT].

4. *Reference-driven formulation.* Determine whether polynomial-growth holomorphic functions, heat-flow deformation of plurisubharmonic functions, or Kähler-Ricci flow asymptotics can replace volume-growth assumptions in the uniformization problem [Liu2], [LT], [Shi2].

5. *Reference-driven formulation.* Relate lower bounds on volume growth and average curvature decay to the existence of enough holomorphic functions to prove Steinness [CZ2], [O].

6. *Reference-driven formulation.* Extend the finite Monge–Ampère-mass weight and Bézout method from positive sectional curvature to positive bisectional curvature, first for complex surfaces and then in higher dimension [DPS].

7. *Reference-driven formulation.* In complex dimension two, decide whether positive bisectional curvature forces contractibility and simple connectivity at infinity, thereby supplying the missing topological hypotheses in the Datar–Pingali–Seshadri criterion [DPS].

8. *AI-generated formulation.* Find a curvature-decay condition strictly weaker than maximal volume growth but still strong enough to force Steinness and eventually biholomorphism to $\mathbb{C}^n$, perhaps by producing a canonical algebra of polynomial-growth holomorphic functions.

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Problem 35

Let M be a complete, simply-connected, Kähler manifold with negative bisectional curvature. Prove that M is Stein.

Status. False as stated. Mohsen constructed positive-dimensional compact simply connected Kähler complete intersections carrying metrics of negative holomorphic bisectional curvature [Moh]; his dimension estimate includes compact simply connected complex surfaces. A compact Kähler metric is complete, so these examples satisfy the completeness and simple-connectedness hypotheses in Yau's question. However, a positive-dimensional compact complex manifold cannot be Stein: every holomorphic function on it is constant by the maximum principle. Thus negative holomorphic bisectional curvature alone does not force noncompactness, and the Stein conclusion cannot be true without an additional hypothesis.

This correction is conceptually important. In complex dimension one, a simply connected complete surface with negative curvature is modeled by the disk in the classical uniformization theory, and is therefore Stein. One might hope for a higher-dimensional analogue by analogy with Cartan–Hadamard geometry. That analogy is misleading here: negative holomorphic bisectional curvature is a condition on complex two-plane directions, not a statement that all real sectional curvatures are nonpositive. Mohsen's examples show that the missing real-sectional-curvature input is not a technical gap but a genuine obstruction to the proposed proof strategy.

The compact examples fit naturally with the modern algebraic picture. Negative holomorphic bisectional curvature implies negative holomorphic sectional curvature. By the Wu–Yau theorem, a compact Kähler manifold with negative holomorphic sectional curvature has ample canonical bundle [WY], with extensions to the quasi-negative setting due to Tosatti–Yang and Diverio–Trapani [TY, DT]. These results do not exclude compact examples; rather, they place them among canonically polarized projective manifolds. Mohsen's Donaldson–Auroux complete-intersection construction realizes compact simply connected examples in precisely this general-type world.

The right surviving problem is therefore a noncompact Stein problem. If compact counterexamples are excluded explicitly, one can still ask whether a complete noncompact simply connected Kähler manifold with negative holomorphic bisectional curvature admits a strictly plurisubharmonic exhaustion. That is the analytic heart of the Stein question. Noncompact examples with complete negatively curved Kähler metrics on vector-bundle total spaces show that the noncompact theory is substantial [WY2]. Work on real bisectional curvature and mixed curvature also indicates that related curvature cones can force positivity properties of the canonical bundle, but such algebraic positivity is weaker than the existence of a global Stein exhaustion [YZ, CLT].

One should also separate the compact classification problem from the noncompact analytic problem. Mohsen's theorem gives compact simply connected examples in every complex dimension at least two: take a simply connected projective ambient manifold of dimension at least $4d-1$ and a sufficiently high-degree complete intersection of dimension d; Lefschetz gives simple connectivity [Moh]. Thus compact simply connected surfaces already occur, and complex dimension two is the smallest possible dimension. What remains is to classify their topology and deformation types and to find more explicit or lower-codimension constructions. Conversely, a positive answer to a suitably formulated noncompact Stein theorem would not contradict Mohsen's examples, because those examples are compact.

Recent work also sharpens the two sides of the corrected question. On the positive side, Lee proved that Stein manifolds with negatively pinched holomorphic sectional curvature admit complete negative Kähler–Einstein metrics and that normalized Kähler-Ricci flow can deform such metrics to the complete Kähler–Einstein metric [Lee]. This is a converse-type result: it starts from Steinness and produces canonical complete negative curvature, rather than deriving Steinness from bisectional negativity. On the construction side, Wan produced negative holomorphic sectional and bisectional curvature metrics on compact relative Kähler fibrations under curvature negativity conditions on the base and fibers [Wan]. Such examples reinforce the need to separate compact projective negative-curvature constructions from the noncompact Stein-exhaustion problem.

Bakkacha provides a useful converse model on bounded domains [Bak]. If a bounded domain carries a complete negatively pinched bisectional-curvature metric outside a compact set, the metric can be modified to have the same pinching everywhere. In particular this applies to strictly pseudoconvex bounded domains with C^2 boundary and to bounded domains whose squeezing function tends to one at every boundary point. The construction adds a large multiple of a ball Bergman metric and controls curvature by a spherical-comparison inequality. Since the domain is already Stein, this constructs curvature from Stein geometry rather than deriving Steinness from curvature.

Related open problems.

1. *Reference-driven formulation.* Classify the topology and deformation types of Mohsen's compact simply connected negatively bisectional Kähler surfaces, and find explicit low-degree or lower-codimension models [Moh].

2. *Original direction.* Prove or disprove the noncompact replacement: if M is complete, noncompact, simply connected, and Kähler with negative holomorphic bisectional curvature, must M be Stein?

3. *Reference-driven formulation.* Find geometric hypotheses at infinity, such as bounded geometry, curvature pinching, finite topological type, or controlled Kobayashi geometry, which force a strictly plurisubharmonic exhaustion in the noncompact case [Lee], [WY2], [Bak].

4. *Reference-driven formulation.* Clarify the relation between algebraic negativity of the tangent bundle, positivity of the cotangent bundle, positivity of the canonical bundle, and the existence of a Kähler metric with negative bisectional curvature [WY], [TY], [DT], [Moh].

5. *AI-generated formulation.* Find a curvature condition between negative holomorphic sectional curvature and negative bisectional curvature which is strong enough to imply Steinness for complete simply connected noncompact manifolds and still flexible enough to include natural bounded-domain examples.

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Problem 36

If M is complete, Kähler, of finite volume, and has bounded curvature, is M a Zariski open set of some projective manifold? If M has negative bisectional curvature, does M have a finite automorphism group?

Status. The compactification assertion is false as stated, even if noncompactness is added. The automorphism-group assertion remains open under the stated pointwise negative bisectional-curvature hypothesis, although it is known under stronger uniform pinching and end-control assumptions.

Here is a direct counterexample to the first assertion. Let T be a nonprojective compact complex torus and let $\Sigma = \mathbb{P}^1\setminus\{0, 1, \infty\}$ carry its complete finite-area hyperbolic metric. The product $T \times \Sigma$, with the product of a flat Kähler metric and the hyperbolic metric, is complete, noncompact, of finite volume, and has bounded curvature. It cannot be quasi-projective: if it were a Zariski open subset of a projective variety, the compact analytic slice $T \times \{p\}$ would be projective by Chow's theorem, contradicting the choice of T. Thus finite volume and bounded curvature alone do not detect an algebraic complex structure.

After excluding this elementary nonalgebraic factor obstruction, one can ask which additional geometric hypotheses force a quasi-projective algebraic origin. Siu and Yau proved an affirmative result when the sectional curvature is pinched between two negative constants [SiY]. This places the manifold in a projective compactification framework and is a prototype for the later theory of finite-volume negatively curved Kähler manifolds.

Mok then proved compactification theorems for complete Kähler surfaces of finite volume under curvature assumptions [Mok]. Mok and Zhong extended the method to complete Kähler-Einstein manifolds of finite topological type and bounded curvature [MZ]. These results show that the desired Zariski-open conclusion is available when the geometry at infinity is sufficiently controlled; the unresolved part is whether finite volume plus bounded curvature alone is enough to supply that control.

For the automorphism group, negative Ricci curvature tends to destroy continuous holomorphic automorphisms by Bochner-type arguments; in compact Kähler-Einstein settings this goes back to Matsushima-type structure results [Mat]. In the complete finite-volume setting, Lawson–Yau and Avérous–Kobayashi studied automorphisms and isometries of nonpositively curved manifolds of finite volume [LY, Ko]. Yeung later developed related compactness and bounded-domain techniques, including results under uniform squeezing-type hypotheses [Y]. More recent negatively pinched complete Kähler work relates the background metric, Kähler-Einstein metric, Kobayashi metric, and Bergman metric, giving additional control of automorphisms in strongly negatively curved situations [WY].

Wu–Yau give a precise strong case [WY]. If the holomorphic sectional curvature satisfies $-B \leqslant H \leqslant -A < 0$, the manifold has a unique complete negative Kähler–Einstein metric, uniformly equivalent to the original metric, and its curvature tensor has bounded covariant derivatives. Every biholomorphism preserves this canonical metric. In conjunction with finite volume and the finite-volume automorphism results, this yields finiteness in the uniformly pinched setting. The gap in Yau's original formulation is genuine: bounded curvature and strict pointwise negativity do not imply a uniform estimate $H \leqslant -A < 0$ at infinity.

Recent quasi-projective work is useful even though it often assumes the compactification rather than proving it. Rochon–Zhang described the precise asymptotic behavior of natural complete finite-volume Kähler metrics on complements of normal-crossing divisors and showed how these asymptotics behave under Ricci flow [RZ]. Guenancia proved that quasi-projective manifolds carrying Kähler metrics with negative holomorphic sectional curvature are of log general type [Gue]. These results suggest a two-step refinement of Yau's question: first prove a quasi-projective compactification from finite volume and bounded curvature, then identify which curvature hypotheses force the boundary divisor and log canonical class to have the expected positivity. Cap–Gover's c-projective compactification supplies another useful intermediate notion: suitable complete Kähler ends acquire a smooth CR boundary and a precise asymptotic metric form [CG]. What is not known is how to promote such differential-geometric boundary data to a projective algebraic compactification.

Related open problems.

1. *Corrected original direction.* Classify the additional hypotheses which eliminate the nonprojective-torus product counterexamples and force a complete finite-volume bounded-curvature Kähler manifold to be quasi-projective.

2. *Reference-driven formulation.* Remove finite-topological-type assumptions from the Mok–Zhong compactification theorem [MZ].

3. *Reference-driven formulation.* Prove finite generation of the fundamental group, or finite topological type, from finite volume and bounded Kähler curvature under natural negative-curvature assumptions [SiY], [Mok], [MZ].

4. *Original direction.* Determine whether negative holomorphic bisectional curvature and finite volume imply finiteness of the full holomorphic automorphism group.

5. *Reference-driven formulation.* Starting from a quasi-projective compactification, characterize the curvature asymptotics at a normal-crossing boundary which are stable under Ricci flow and compatible with finite volume [RZ].

6. *Reference-driven formulation.* Determine whether negative holomorphic sectional or bisectional curvature on the finite-volume manifold forces the eventual compactification to be of log general type [Gue].

7. *Reference-driven formulation.* Characterize when a c-projective compactification with CR boundary is induced by a normal-crossing algebraic compactification [CG].

8. *AI-generated formulation.* Find an intrinsic curvature-and-volume criterion at the cusps which is equivalent to the existence of a normal-crossing projective compactification and strong enough to imply discreteness, or finiteness, of the automorphism group.

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Problem 37

Let M be a compact Kähler manifold with negative sectional curvature. Prove that if $\dim_{\mathbb{C}} M > 1$, then M is rigid, i.e., there is only one complex structure on M.

Status. Solved in complex dimension two; open in complex dimension at least three. The natural conclusion is strong rigidity up to complex conjugation: a homotopy-equivalent compact complex manifold should be biholomorphic or anti-biholomorphic to M.

The locally symmetric case is the model. If M is a quotient of the complex ball of complex dimension at least two, Mostow strong rigidity fixes the locally symmetric structure from the fundamental group. Yau's use of Kähler-Einstein metrics connects this Riemannian rigidity with the complex structure of the ball quotient [Y, Mos].

The main analytic method is the harmonic-map method. Sampson and Yau developed the foundational curvature identities for harmonic maps from Kähler manifolds, and Siu proved that harmonic maps are pluriharmonic or holomorphic under suitable strong negativity assumptions on the target curvature [Sam, Si]. This gives strong rigidity theorems for compact Kähler manifolds whose curvature tensor satisfies the stronger complex negativity conditions needed in the Bochner formula.

The difficulty is that negative Riemannian sectional curvature is not the same as Siu's strong negativity condition on the complexified curvature tensor. Thus the known harmonic-map argument does not directly apply to an arbitrary compact Kähler metric with negative sectional curvature. Later work of Jost–Yau and others refined harmonic-map techniques, but the full deformation rigidity of the complex structure in this curvature class remains unsettled [JY].

In complex dimension two, Zheng found the missing curvature bridge [Z]. For a Kähler surface, negative (respectively nonpositive) real sectional curvature is equivalent to negativity (respectively nonpositivity) of the complex curvature operator. Consequently the first Pontrjagin form $C_1^2 - 2C_2$ is nonnegative. If its integral vanishes, Zheng's equality analysis forces a bidisc universal cover with a product metric; this is incompatible with strictly negative real sectional curvature because the product has zero mixed sectional planes. Hence $c_1^2 > 2c_2$, and Zheng's Proposition 3 uses the unique null direction of the complex curvature operator together with the pluriharmonic part of Siu's Bochner–Kodaira identity to prove strong rigidity. Thus Yau's conjecture is a theorem for Kähler surfaces.

There has also been progress on the algebro-geometric side. Since negative sectional curvature implies negative holomorphic sectional curvature, the canonical bundle is ample by the Wu–Yau theorem [WY]. Chen–Yang proved that compact Kähler manifolds homotopic to negatively curved Riemannian manifolds admit Kähler-Einstein metrics of general type and satisfy hyperbolicity-type conclusions [CY]. These results confirm that the complex structures lie in a very restrictive projective general-type region, but they do not yet give uniqueness of the complex structure on the underlying smooth manifold.

Recent work around nearby curvature cones underlines why Yau's rigidity question remains delicate. Lee–Streets studied compact complex manifolds with negative Chern curvature operator and obtained a dichotomy: either a positive $(1,1)$ current of dd^c-exact type appears, or the manifold is Kähler with ample canonical bundle; in complex dimension two they obtained a complete classification [LS]. Broder–Gaussier have recently constructed complete Kähler metrics with negative bisectional curvature on new hyperbolic complex manifolds, and also produced projective Kobayashi hyperbolic surfaces with prescribed Chern slope behavior [BG]. These results are not rigidity theorems for compact Kähler manifolds of negative Riemannian sectional curvature, but they show that negative holomorphic and Chern curvature conditions allow a much wider geography than the locally symmetric model. The remaining problem is therefore not only to prove hyperbolicity or ampleness, but to turn the real sectional-curvature hypothesis into a deformation-rigidity statement for the complex structure.

Related open problems.

1. *Higher-dimensional original direction.* Prove that negative sectional curvature implies the complex curvature negativity needed in Siu's harmonic-map rigidity, perhaps after passing to the Kähler-Einstein metric in the same canonical class.

2. *Original direction.* Determine whether a compact smooth manifold can carry two non-biholomorphic Kähler complex structures, both admitting negative sectional curvature metrics.

3. *Reference-driven formulation.* Establish local deformation rigidity for compact Kähler manifolds with negative sectional curvature, even before global uniqueness of complex structure [Si], [JY].

4. *Reference-driven formulation.* Relate the Teichmüller space of complex structures on a negatively curved Kähler manifold to harmonic-map energy and Kähler-Einstein metrics [Sam], [Si], [CY].

5. *Reference-driven formulation.* Compare negative Riemannian sectional curvature with negative Chern curvature operator, real bisectional curvature, and negative bisectional curvature as possible curvature cones for rigidity [LS], [BG].

6. *Reference-driven formulation.* Extend Zheng's Pontrjagin-form and null-direction argument from complex surfaces to higher-dimensional curvature blocks or Chern-number inequalities [Z].

7. *AI-generated formulation.* Find a curvature pinching condition strictly weaker than Siu strong negativity but stronger than negative sectional curvature which still forces complex structure rigidity.

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Problem 38

Given M a simply connected, complete Kähler manifold with sectional curvature less than or equal to -1, prove there exists a nonconstant bounded holomorphic function on M.

One would like to even prove that there is a branched immersion of M onto a bounded domain in $\mathbb{C}^N$.

Status. Open, including under two-sided negative sectional-curvature pinching in complex dimension at least two. A frequently cited preprint claiming the pinched result in complex dimension at least three was withdrawn after an error was found in its boundary analysis [CS]. The sole upper bound $K \leqslant -1$ and the branched-immersion statement are therefore still further out of reach.

The Riemannian background is the asymptotic Dirichlet problem on Cartan–Hadamard manifolds. Sullivan and Anderson constructed many bounded harmonic functions on negatively curved simply connected manifolds under pinched negative curvature assumptions [Su, An]. These results show that the ideal boundary carries rich potential theory, but harmonic functions are not automatically real parts of holomorphic functions in several complex variables.

The Kähler problem asks for a holomorphic replacement of this potential theory. Schoen and Yau emphasized the role of harmonic functions, L^2 estimates, and complex analytic methods in the negative-curvature setting [SY]. Cao and Shaw proposed an approach for a simply connected complete Kähler manifold whose sectional curvature satisfies

$$-b^2 \leqslant K_M \leqslant -a^2 < 0,$$

in complex dimension at least three [CS]. Their proposed proof used bounded plurisubharmonic exhaustion functions, the $\overline{\partial}$ equation, CR functions on geodesic spheres, and uniform Hölder estimates as the spheres tend to infinity. The authors withdrew the paper on February 6, 2016, after G. Székelyhidi pointed out an error in Chapter 3. No replacement proof was located in this audit. The failed uniform boundary estimate should be recorded as a concrete gap, not as a theorem.

The stronger desired conclusion would be a supply of enough bounded holomorphic functions to give a branched holomorphic map, or immersion, into a bounded domain of some $\mathbb{C}^N$. This is substantially stronger than producing one bounded holomorphic function and would amount to a much more precise complex compactification of the ideal boundary.

There is also a CR-geometric version of the problem at infinity. Cao–Chang constructed a bounded contact form on a simply connected complete Kähler manifold with sectional curvature $\leqslant -1$ and showed that the sphere at infinity carries a bounded contact structure and a bounded Tanaka–Webster pseudo-Hermitian structure [CC]. This does not by itself solve the bounded holomorphic function problem, but it supplies the kind of boundary structure one would need in order to imitate the unit ball: prescribe CR data at infinity, solve an analytic extension problem, and then use the resulting bounded holomorphic functions to separate points or directions. Recent work on hyperbolic complex manifolds and negatively curved Kähler metrics also suggests that the existence of negative curvature alone is not enough; the boundary regularity, CR extension, and separation properties have to be tracked as separate analytic inputs [BG].

Wu–Yau nevertheless obtained a strong invariant-metric theorem under two-sided sectional pinching [WY]. On such a simply connected manifold the Bergman, Kobayashi–Royden, and negative Kähler–Einstein metrics exist and are uniformly equivalent to the background metric. This does not solve Problem 38: the Bergman metric is built from square-integrable canonical forms, whereas the Carathéodory metric and Yau's requested map require bounded scalar holomorphic functions. The missing passage from nondegenerate Bergman geometry to bounded scalar holomorphic separation is one precise formulation of the obstruction.

Related open problems.

1. *Original direction.* Prove the existence of nonconstant bounded holomorphic functions under the sole assumption $K \leqslant -1$, without a lower sectional-curvature bound.

2. *Reference-driven formulation.* Repair or replace the failed uniform CR-boundary estimate in the withdrawn Cao–Shaw approach, and settle first the complex dimension two pinched case [CS].

3. *Reference-driven formulation.* Construct enough bounded holomorphic functions to separate points and tangent directions on negatively curved complete simply connected Kähler manifolds [CS].

4. *Original direction.* Prove or disprove the existence of a branched holomorphic immersion of such M into a bounded domain in $\mathbb{C}^N$.

5. *Reference-driven formulation.* Develop the Cao–Chang CR boundary at infinity into an extension theorem: identify which CR functions on $S_\infty(M)$ extend to bounded holomorphic functions in the interior [CC].

6. *Reference-driven formulation.* Use the uniformly nondegenerate Bergman geometry of Wu–Yau to produce bounded scalar holomorphic functions, or identify a new obstruction separating the Bergman and Carathéodory theories [WY].

7. *AI-generated formulation.* Build a boundary Poisson transform for Kähler Hadamard manifolds with $K \leqslant -1$ which sends regular CR boundary data to bounded holomorphic functions and gives explicit separation criteria.

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Problem 39

Let M be a compact Kähler manifold with positive first Chern class. Suppose M admits no holomorphic vector field. Prove that M admits a Kähler-Einstein metric. This was conjectured by Calabi [Ca].

Status. False as stated. The corrected existence theorem is now known: a smooth Fano manifold admits a Kähler-Einstein metric if and only if it is K-polystable.

Calabi's problem asked for canonical metrics in a fixed Kähler class [Ca]. In the Fano case $c_1(M) > 0$, the absence of holomorphic vector fields is not enough. Classical necessary conditions already came from the automorphism group and Futaki's character: Matsushima's theorem restricts the Lie algebra of holomorphic vector fields, and Futaki constructed an obstruction to Kähler-Einstein metrics [Mat, Fu].

Tian supplied an explicit smooth counterexample [Ti]. For a generic three-plane $P_a \subset \Lambda^2\mathbb{C}^7$, he considered the smooth Fano threefold

$$X_{P_a} = \{U \in G(4,7) : P_a \text{ projects to zero in } \Lambda^2(\mathbb{C}^7/U)\}.$$

For generic a, this threefold has no nontrivial holomorphic vector field, but a special degeneration has a generalized Futaki invariant which violates Tian's necessary weak K-stability condition. Hence X_{P_a} admits no Kähler–Einstein metric. This is the exact status-changing example, rather than only a general appeal to the later stability theory.

The decisive correction was to add an algebro-geometric stability condition. Tian introduced an analytic form of K-stability and proved existence under suitable stability assumptions; his 1997 work also made clear that the naive "no holomorphic vector field" condition is insufficient [Ti]. Donaldson then gave the influential algebraic formulation using test configurations and related scalar curvature to stability of polarized varieties [Do].

The smooth Fano Yau–Tian–Donaldson theorem was proved by Chen–Donaldson–Sun: a smooth Fano manifold admits a Kähler-Einstein metric exactly when it is K-polystable [CDS1, CDS2, CDS3]. Tian also gave a proof of the same correspondence [Ti2]. Berman proved the necessity of K-polystability in a broad possibly singular setting [Ber]. Later variational, continuity method, and Kähler-Ricci flow approaches gave further proofs and extensions [BBJ, DS, CSW].

Thus the original problem should be recorded historically as a false formulation whose corrected version became the Yau–Tian–Donaldson correspondence. The active frontier has moved to singular Fano varieties, log Fano pairs, effective criteria for K-stability, and moduli spaces of K-polystable Fano varieties.

The most active recent developments are algebraic and moduli-theoretic. Valuative criteria of Fujita–Li type, and the related stability threshold or delta invariant, have made K-stability much more computable; Dervan–Legendre's valuative framework for polarized varieties records how far this philosophy extends beyond the Fano case [DL]. Xu's survey gives an algebro-geometric account of the modern Fano theory, including valuations, special test configurations, and the role of singularities [Xu]. On the moduli side, the construction and projectivity of K-moduli spaces have become central: Blum–Halpern-Leistner–Liu–Xu related properness to optimal destabilizing degenerations [BHLX], while Xu–Zhuang proved positivity results for the CM line bundle on K-moduli spaces [XZ]. Recent explicit work, such as Liu–Zhao's K-moduli analysis of a family of Fano threefolds, shows that the corrected problem has become a classification program rather than a single existence theorem [LZ]. DeVleming's 2026 survey is a useful current entry point for the K-stability and K-moduli viewpoint [DeV].

The moduli statement can now be made unconditionally. The finite-generation theorem for valuations computing the stability threshold supplies the optimal destabilizations required in [BHLX]. Consequently the stack of K-semistable Fano varieties of fixed dimension and volume is proper, its good moduli space of K-polystable objects is projective, and the CM line bundle gives the natural polarization [BHLX, XZ, DeV].

Related open problems.

1. *Reference-driven formulation.* Develop effective and computable K-stability criteria for explicit Fano manifolds and singular Fano varieties, especially through valuations, beta invariants, and delta invariants [Xu], [DL].

2. *Original direction.* Extend the Yau–Tian–Donaldson picture as far as possible from Kähler-Einstein Fano metrics to constant scalar curvature Kähler metrics on general polarized varieties.

3. *Reference-driven formulation.* Describe the compactified moduli spaces of K-polystable Fano varieties, including boundary singularities, CM-line positivity, and degeneration phenomena [BHLX], [XZ], [DeV].

4. *Reference-driven formulation.* Relate analytic flows, such as the Kähler-Ricci flow, to optimal destabilizing degenerations in unstable Fano cases [CSW], [BHLX].

5. *Reference-driven formulation.* Carry out explicit K-moduli computations for Fano threefolds and higher-dimensional families, following the model of recent concrete K-moduli calculations [LZ].

6. *AI-generated formulation.* Produce an explicit "stability atlas" for low-dimensional Fano manifolds: for each deformation family, record the exact K-stable, strictly semistable, and unstable loci together with the corresponding Gromov–Hausdorff or K-moduli boundary points.

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Problem 40

Let M be a complete Kähler manifold with zero Ricci curvature. Prove that M is a Zariski open set of some compact Kähler manifold. If this is true, we shall have algebraic means to classify these manifolds.

Status. Open in this generality. Many complete Ricci-flat Kähler metrics are known on quasi-projective or otherwise compactifiable complex manifolds, but the converse statement–that an arbitrary complete Ricci-flat Kähler manifold must come from deleting an analytic subset of a compact Kähler manifold–is not known.

The compact case is governed by Yau's solution of the Calabi conjecture: a compact Kähler manifold with vanishing first Chern class carries a unique Ricci-flat Kähler metric in each Kähler class [Y]. The noncompact problem is much less rigid, because completeness allows infinite volume and many possible geometries at infinity.

One strong positive direction starts from algebraic geometry and constructs complete Ricci-flat metrics. Tian and Yau constructed complete Ricci-flat Kähler metrics on complements of suitable divisors in compact Kähler or projective manifolds [TY1, TY2]. Bando–Kobayashi and later van Coevering extended this circle of ideas to further quasi-projective and asymptotically conical settings [BK, vC]. Conlon and Hein developed a systematic existence theory for asymptotically conical Calabi–Yau metrics on quasi-projective varieties, giving sharp analytic control at infinity in many divisor-complement situations [CH]. Santoro studied asymptotic expansions of Tian–Yau type complete Ricci-flat metrics on quasi-projective manifolds [Sa1, Sa2]. These results show that Zariski-open Calabi–Yau manifolds provide a large and natural supply of examples, but they start from an algebraic compactification rather than deriving one from the complete metric.

The reverse direction is harder. Finite-volume negatively curved Kähler manifolds admit compactification results under strong curvature and topology assumptions, as in Problem 36, but Ricci-flat complete manifolds usually have infinite volume and their ends need not resemble finite-volume cusps. In complex dimension two, the theory of gravitational instantons gives detailed classifications in several asymptotic regimes; many examples admit algebraic or elliptic-surface compactifications, but these classifications depend on strong curvature decay and volume-growth hypotheses rather than Ricci-flatness alone [HSVZ]. The L^2 Hodge theory of gravitational instantons also shows that the asymptotic type controls analytic invariants in a way that should enter any compactification theorem [HHM].

There is also a separate line related to positive bisectional curvature. Chen and Zhu proved compactification-type results for complete noncompact Kähler manifolds under positive bisectional curvature hypotheses [CZ]. Such results are important but do not settle the Ricci-flat case, where the curvature has no definite sign.

Newer constructions further emphasize that the metric-to-algebraic direction is not a formal consequence of Ricci-flatness. Conlon–Rochon constructed warped quasi-asymptotically conical Calabi–Yau metrics on smoothings of products of Calabi–Yau cones and described their asymptotics using compactifications by manifolds with corners and weighted blow-ups [CR]. Their examples show that even the tangent cone at infinity may fail to determine the complete Ricci-flat Kähler metric uniquely. Liang–Zhang constructed complete Calabi–Yau metrics on noncompact abelian-fibered threefolds and supplied compactifications in that setting [LZ]. Liang has also constructed complete Calabi–Yau metrics on noncompact K3-fibered threefolds whose sectional curvature is unbounded at infinity [Liang]. These recent examples are compactifiable or built from algebro-geometric input, but they warn that any converse compactification theorem must specify which asymptotic, curvature, or fibration data are being recovered from the metric.

There is, however, a direct converse theorem under strong asymptotic control. Liu proved that a complete noncompact Ricci-flat Kähler manifold with maximal volume growth and quadratic curvature decay is a crepant resolution of a normal affine algebraic variety; that affine variety degenerates in two steps to the unique metric tangent cone at infinity [Liu]. The proof constructs and finitely generates polynomial-growth holomorphic functions using three-circle estimates, plurisubharmonic approximations to the squared distance, and multiplicity bounds. This is the closest theorem in the present audit to the direction asked by Yau, but both maximal volume growth and quadratic decay are essential hypotheses in the argument.

Liu's later tangent-cone rigidity shows that, under Euclidean volume growth and quadratic curvature or scalar-curvature control, the Ricci-flat Kähler condition can force uniqueness of the tangent cone from much weaker initial information [Liu2]. This helps remove one input in some algebraization arguments, but it still does not treat collapsed ends or arbitrary complete Ricci-flat metrics.

Related open problems.

1. *Original direction.* Find natural metric hypotheses at infinity, weaker than known asymptotic models, which imply that a complete Ricci-flat Kähler manifold is Zariski open in a compact Kähler manifold.

2. *Reference-driven formulation.* Classify complete Ricci-flat Kähler surfaces under minimal curvature-decay and volume-growth assumptions, and determine exactly which ones admit algebraic compactifications [HSVZ, HHM].

3. *Reference-driven formulation.* Determine whether bounded curvature, Euclidean or polynomial volume growth, and finite topological type are sufficient for compactifiability in the Ricci-flat Kähler setting [CH, CR, Liu].

4. *Reference-driven formulation.* Develop an analytic compactification theory using harmonic functions, holomorphic volume forms, fibration structures, and tangent cones at infinity [CR, LZ, Liang].

5. *Reference-driven formulation.* Extend Liu's finite generation and crepant-resolution theorem from maximal volume growth to collapsed ALG, ALH, and fibered ends [Liu, Liu2, LZ, Liang].

6. *AI-generated formulation.* Define a "Calabi–Yau end of algebraic type" purely from metric data and prove that it is equivalent to being a punctured neighborhood of a divisor in a compact Kähler manifold.

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Problem 41

Classify all compact, two-dimensional Kähler surfaces with zero scalar curvature. (See [Y].)

Status. Partially resolved. The possible birational types are known: a compact scalar-flat Kähler surface is either Ricci-flat, hence of Kodaira dimension zero, or is rational or ruled, hence of Kodaira dimension $-\infty$ [KLP, LS]. What remains open is the exact classification of the rational and ruled surfaces, their blow-up configurations, and the Kähler classes which carry scalar-flat metrics.

For a compact Kähler surface (X, ω), scalar-flat Kähler means that the scalar curvature is identically zero. The integrated condition is

$$\int_X s_\omega \, \omega^2 = 8\pi \, c_1(X) \cdot [\omega] = 0,$$

so the Kähler class must lie on the wall $c_1(X) \cdot [\omega] = 0$. The Chern–Weil identities sharpen this wall condition. They imply $c_1^2(X) \leqslant 0$, with equality precisely in the Ricci-flat case. Combining this with the Enriques–Kodaira classification, Kim–LeBrun– Pontecorvo and LeBrun–Singer show that the non-Ricci-flat case must be rational or ruled [KLP, LS]. In particular, surfaces of Kodaira dimension one or two do not admit scalar-flat Kähler metrics. When $c_1(X) = 0$ in real cohomology, Yau's theorem supplies the Ricci-flat branch: complex tori, K3 surfaces, Enriques surfaces, and bielliptic surfaces [Y2].

The scalar-flat condition is nevertheless broader than Ricci-flatness. In real dimension four it is equivalent to anti-self-duality for the complex orientation. The deformation theory of LeBrun–Singer turns an anti-self-dual conformal construction into scalar-flat Kähler metrics after verifying a cohomological obstruction and then solving the nonlinear deformation problem [LS]. Kim–LeBrun– Pontecorvo prove that every ruled surface becomes scalar-flat after sufficiently many blow-ups unless the metric lies in the Ricci-flat branch [KLP]. Rollin–Singer use parabolic stability, orbifold metrics, and gluing to construct further examples; in particular, $\mathbb{CP}^2$ blown up at ten suitable points admits a scalar-flat Kähler metric, and ten is forced by $c_1^2 \leqslant 0$ [RS].

For a fixed polarized surface, scalar-flat Kähler existence is also a special case of the cscK problem. Arezzo–Pacard show that a cscK metric, including a scalar-flat one, persists under sufficiently small blow-ups when the relevant automorphism obstruction vanishes [AP]. Chen–Cheng identify cscK existence with properness of the Mabuchi energy in the analytic framework [CC1, CC2], while Stoppa shows that cscK existence with discrete automorphism group implies K-stability [St]. These results explain why the remaining classification depends on the polarization and the position of the exceptional curves, not only on the differentiable surface.

Brown proves that Kähler blow-ups admit metrics whose scalar curvature is uniformly close to that of the original metric; as an application, a compact complex surface admits a positive-scalar- curvature Kähler metric exactly when it has Kodaira dimension $-\infty$ [Brown]. This settles the adjacent positive chamber, but does not decide which points on the zero-average wall are represented by scalar-flat metrics. The remaining problem is thus a class-level existence and moduli problem inside the already-known rational/ruled birational types.

Related open problems.

1. *Original direction.* Within the known rational/ruled branch, determine exactly which complex surfaces admit scalar-flat Kähler metrics [KLP, LS, RS].

2. *Original direction.* Determine, for each complex surface, exactly which Kähler classes contain scalar-flat Kähler metrics.

3. *Reference-driven formulation.* Relate scalar-flat Kähler existence on surfaces to explicit K-stability or uniform K-stability conditions for the corresponding polarized surface [CC1, CC2, St].

4. *Reference-driven formulation.* Classify scalar-flat Kähler metrics on blow-ups of ruled and rational surfaces, including sharp bounds on the number and position of blow-up points [KLP, LS, AP, RS].

5. *Reference-driven formulation.* Determine whether parabolic stability gives a necessary as well as sufficient criterion for the ruled-surface constructions of Rollin–Singer [RS].

6. *AI-generated formulation.* Build a wall-crossing description for scalar-flat Kähler surfaces as the Kähler class crosses the hyperplane $c_1(X) \cdot [\omega] = 0$.

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Problem 42

Let M be a compact simply-connected symplectic manifold. Does M admit a Kähler structure? M. Berger says that Serre indicated a counterexample to him in 1955, where $\pi_1(M) \neq 0$. See [Bs].

For any symplectic structure on a manifold, one can define an almost-complex structure. Conversely, it may be true that for any almost-complex structure, one can also find an associated symplectic structure. Is it true that the almost-complex structure determines the symplectic structure up to conjugation by a diffeomorphism? One does not know the answer of this last question even for $\mathbb{CP}^N$. However, Moser [Mo] has proved that all elements of a one-parameter family of symplectic structures are mutually conjugate by diffeomorphisms.

Status. False in each universal formulation. There are compact simply connected symplectic manifolds with no Kähler structure; an almost-complex structure need not admit any compatible symplectic form; and neither cohomology nor the induced canonical class determines a symplectic form up to diffeomorphism. The classification on $\mathbb{CP}^N$ remains open beyond low-dimensional and special cases.

The early non-Kähler symplectic examples of Kodaira and Thurston were not simply connected [Th]. McDuff gave the first simply connected example: a symplectic blow-up construction whose third Betti number is odd, hence impossible for a compact Kähler manifold by Hodge theory [Mc]. Gompf's symplectic sums realize every finitely presented group as a symplectic fundamental group and give, in particular, simply connected examples not homotopy equivalent to Kähler manifolds [G]. Babenko–Taimanov constructed simply connected nonformal symplectic manifolds in every even dimension at least ten; formality of compact Kähler manifolds gives a separate rational-homotopy obstruction [BT].

The converse from almost-complex to symplectic is false without extra hypotheses. The simplest obstruction is cohomological: a compact symplectic $2n$-manifold must contain a class $[\omega]$ with $[\omega]^n \neq 0$, so an almost-complex manifold such as $\mathbb{S}^6$ cannot be symplectic because $H^2(\mathbb{S}^6) = 0$. In real dimension four the failure is much more refined. Taubes showed that a symplectic 4-manifold has nontrivial Seiberg–Witten invariants in the canonical spin^c class [T1, T2], while Witten's framework implies strong finiteness constraints for basic classes [W]. Thus many orientation-compatible almost-complex structures on a fixed smooth 4-manifold cannot be compatible with any symplectic form.

The uniqueness question is also delicate. Moser's theorem says that a smooth path of cohomologous symplectic forms is induced by an isotopy of diffeomorphisms [Mo]. This is a local-in-moduli statement and does not classify disconnected components of the symplectic cone. For $\mathbb{CP}^2$, Gromov's theory of pseudoholomorphic curves gives strong uniqueness of symplectic forms up to deformation and diffeomorphism [Gr], with further four-dimensional refinements by McDuff and others [McS]. In higher dimensional $\mathbb{CP}^N$ and on general manifolds, the analogous global classification remains open.

Recent examples show how far the uniqueness question is from the Moser-path situation. McMullen–Taubes constructed a closed simply connected four-manifold with symplectic forms whose canonical classes lie in distinct diffeomorphism orbits [MT]. Ning constructed closed manifolds in every even dimension at least six carrying cohomologous symplectic forms with different Gromov widths and first Chern classes [Ning]. These symplectic invariants show that the cohomology class alone cannot classify symplectic forms. For almost-complex-to-symplectic questions, Lejmi proved that the almost complex structure underlying a strict nearly Kähler six-manifold, including the standard almost complex structure on $\mathbb{S}^6$, is not compatible with any symplectic form even locally [Lej]. In dimension four, Draghici–Li–Zhang's decomposition of cohomology associated to an almost complex structure gives a useful framework for the question of which classes are represented by J-compatible or J-tamed symplectic forms [DLZ].

Related open problems.

1. *Original direction.* Classify symplectic structures on $\mathbb{CP}^N$ up to diffeomorphism and deformation for $N \geqslant 3$.

2. *Reference-driven formulation.* Determine which almost-complex structures in dimension $2n \geqslant 6$ are homotopic to ones tamed or compatible with a symplectic form [Lej, DLZ].

3. *Reference-driven formulation.* Develop higher-dimensional analogues of Taubes' Seiberg–Witten obstruction, possibly using Gromov–Witten or Floer-theoretic invariants [T1, T2, W].

4. *Reference-driven formulation.* Classify simply connected symplectic non-Kähler manifolds within fixed homeomorphism or diffeomorphism types [Mc, G].

5. *Reference-driven formulation.* Understand when cohomologous symplectic forms are distinguished by capacities, Gromov–Witten invariants, or first Chern class [Ning].

6. *AI-generated formulation.* Define a computable "symplectic realizability cone" for almost-complex Chern data and compare it with the full topological cone of almost-complex structures.

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Problem 43

Let Ω be a bounded, pseudoconvex domain in $\mathbb{C}^N$. Cheng and Yau [CY] have established the existence of a canonical Kähler-Einstein metric on Ω. Under general conditions, e.g. $\partial\Omega$ is C^2, that metric is complete. Is the metric always complete?*

Status. Solved. Mok and Yau proved the completion of this problem: a bounded domain admits a complete Kähler-Einstein metric of negative Ricci curvature if and only if it is pseudoconvex [MY].

Cheng and Yau first established the existence and regularity theory for complete Kähler-Einstein metrics on broad classes of noncompact complex manifolds and bounded strongly pseudoconvex domains [CY]. Their work is based on the complex Monge–Ampère equation and Fefferman's approximate solution near the boundary. For smooth strongly pseudoconvex domains, the Kähler-Einstein metric is complete and has precise boundary asymptotics.

The remaining issue in the original question was whether pseudoconvexity alone forces completeness of the canonical Kähler-Einstein metric on a bounded domain. Mok–Yau proved this affirmatively and also obtained the converse characterization of domains of holomorphy by curvature conditions [MY]. Thus the existence-completeness question is settled, but the detailed boundary behavior remains an active analytic theme, especially for weakly pseudoconvex or nonsmooth domains.

Recent work has shifted the problem toward local boundary models, Bergman-kernel localization, and rigidity of nearby canonical metrics. Gontard proved local boundary regularity and curvature asymptotics for the complete Kähler-Einstein metric near strictly pseudoconvex boundary points and obtained estimates for tube domains near weakly pseudoconvex points [Go1, Go2]. Hsiao–Huang–Li recently established Bergman-kernel localization near finite-type boundary points for unbounded pseudoconvex domains, a tool designed for problems where the local scaling limit is not itself bounded [HHL]. Sha studied Kähler-Ricci solitons on bounded pseudoconvex domains and proved, under suitable hypotheses, that such solitons reduce to Kähler-Einstein metrics [Sha1]. These results do not change the solved status of the completeness question, but they make the residual problem more precise: understand how the complete Kähler-Einstein metric degenerates, localizes, and compares with the Bergman geometry at weak or singular boundary strata.

Related open problems.

1. *Original direction.* Describe sharp boundary asymptotics of the Cheng–Yau Kähler-Einstein metric on weakly pseudoconvex finite-type domains.

2. *Reference-driven formulation.* Determine the optimal boundary regularity under which the Kähler-Einstein potential has a full polyhomogeneous expansion near strictly pseudoconvex or finite-type points [CY, Go1, HHL].

3. *Reference-driven formulation.* Relate the curvature asymptotics of the complete Kähler-Einstein metric to CR invariants of the boundary and to the model geometry of weakly pseudoconvex points [Go1, Go2].

4. *Reference-driven formulation.* Study stability of the complete Kähler-Einstein metric under domain degeneration, especially for families of pseudoconvex domains with singular limits [MY, HHL].

5. *Reference-driven formulation.* Extend rigidity results for Kähler-Ricci solitons and cscK perturbations on bounded pseudoconvex domains to weaker boundary regularity [Sha1, Sha2].

6. *AI-generated formulation.* Develop a boundary stratification theory for the Cheng–Yau metric on arbitrary bounded pseudoconvex domains, analogous to metric tangent cones in Riemannian convergence theory.

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*Mok–Yau have recently shown that a bounded domain admits a complete Kähler-Einstein metric iff the domain is pseudoconvex. However, it is still desirable to study the boundary behavior of this metric.

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Problem 44

Describe the Kähler-Einstein metric constructed by Cheng–Yau [CY] on the Teichmüller space. What is its relation to the Bergman metric? In general, if a domain is not biholomorphic to a product domain and covers a compact Kähler manifold, is the Kähler-Einstein metric equal to the Bergman metric?

Status. Partially resolved. The Teichmüller/moduli-space part has been solved at the level of metric equivalence and bounded geometry by Liu–Sun–Yau. The broader question asking for exact equality on a non-product domain covering a compact Kähler manifold remains open in the stated generality. Exact equality is much more rigid than the established quasi-isometry results.

Cheng and Yau constructed complete Kähler-Einstein metrics on bounded pseudoconvex domains and on Teichmüller space in this framework [CY]. Teichmüller space also carries several canonical metrics: the Teichmüller/Kobayashi metric, the Carathéodory metric, the Bergman metric, the Weil–Petersson metric, McMullen's metric, and metrics derived from Ricci curvature.

Liu–Sun–Yau introduced the Ricci and perturbed Ricci metrics and used them to compare these canonical metrics on Teichmüller space and on the moduli space of Riemann surfaces [LSY1, LSY2]. They proved that the Bergman and Carathéodory metrics are uniformly equivalent to the Kobayashi metric, and hence to the Kähler–Einstein, McMullen, Ricci, and perturbed Ricci metrics [LSY1, LSY2]. Here "equivalent" is defined to mean quasi-isometric: there is a uniform $C > 1$ with $C^{-1}g_1 \leqslant g_2 \leqslant Cg_1$. Their proof uses Bers embeddings between fixed Euclidean balls, monotonicity of intrinsic metrics, and upper and lower estimates for Bergman peak sections. They also obtained bounded geometry and detailed boundary control for the Kähler–Einstein metric [LSY2]. Yeung independently proved quasi-isometry results in this setting [Ye].

Equivalence should be distinguished from equality. Equality between the Bergman metric and a complete Kähler-Einstein metric is characteristic of highly symmetric situations, such as bounded symmetric domains with their invariant metrics after normalization. The available comparison theorems do not prove or disprove equality on Teichmüller space or on every universal cover in Yau's question. The useful modern question is therefore to identify conditions under which the Bergman, Kobayashi, and Kähler-Einstein metrics are equivalent, have comparable curvature, or share the same boundary asymptotics.

Recent Bergman-Einstein rigidity results sharpen the equality side of the problem. Huang–Xiao proved Cheng's conjecture for smoothly bounded strongly pseudoconvex domains: if the Bergman metric is Kähler-Einstein, then the domain is biholomorphic to the ball [HX]. Savale–Xiao proved the corresponding ball characterization for two-dimensional finite-type pseudoconvex domains [SX]. Hsiao–Huang–Li proved that every bounded real-analytic pseudoconvex domain with Einstein Bergman metric is biholomorphic to the ball, using localization near finite-type boundary points and CR extension across the boundary [HHL]. Sha's rigidity theorem for complete Kähler-Einstein metrics under cscK perturbations gives another nearby test: on a bounded strictly pseudoconvex domain, a Bergman metric with constant scalar curvature is forced to be Kähler-Einstein, and hence falls into the same ball-rigidity circle after the Huang–Xiao theorem [Sha2]. These results support the guiding principle that equality with the Bergman metric is a uniformization phenomenon, whereas quasi-isometry is the flexible Teichmüller/moduli-space statement.

Related open problems.

1. *Original direction.* Characterize bounded domains for which the Bergman metric is Kähler-Einstein.
2. *Original direction.* Determine whether the normalized Kähler–Einstein and Bergman metrics can be exactly equal on a non-product universal cover of a compact Kähler manifold, and classify all equality cases.
3. *Reference-driven formulation.* Refine the boundary asymptotics of the Kähler-Einstein and Bergman metrics near the Deligne–Mumford boundary of moduli space [LSY1, LSY2, Chan].
4. *Reference-driven formulation.* Extend the Liu–Sun–Yau metric comparison framework to moduli spaces of higher-dimensional varieties [LSY1, LSY2].
5. *Reference-driven formulation.* Extend Bergman-Einstein ball-rigidity from strongly pseudoconvex and real-analytic finite-type domains to broader smooth weakly pseudoconvex classes [HX, SX, HHL].
6. *Reference-driven formulation.* Determine when cscK or Kähler-Ricci soliton assumptions on the Bergman metric already imply the Kähler-Einstein condition [Sha1, Sha2].
7. *AI-generated formulation.* Find a curvature or automorphism-group criterion which detects when metric equivalence between the Bergman and Kähler-Einstein metrics improves to equality after scaling.

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Problem 45

Let M be a compact Kähler-Einstein manifold of complex dimension N, with negative scalar curvature. Yau [Y] proved then that

$$(-1)^N \frac{2(N+1)}{N} C_1^{N-2} C_2 \geqslant (-1)^N C_1^N.$$

Are there any other inequalities of this sort among the Chern numbers of M? When $N = 4$, Bourguignon asked whether or not $C_4(M)$ is positive.

Status. Partially resolved. The fundamental Miyaoka–Yau inequality is proved, including its equality case in the smooth ample-canonical situation. The broader problem of finding all universal Chern-number inequalities for negatively curved Kähler-Einstein manifolds remains open. Bourguignon's positivity question for C_4 in complex dimension four should therefore still be treated as an open problem unless additional hypotheses are imposed.

Historically, the two-dimensional case is the Bogomolov–Miyaoka–Yau inequality. Yau's proof via the Kähler-Einstein metric gave

$$\big(2(N+1)c_2(M) - Nc_1(M)^2\big)(-c_1(M))^{N-2} \geqslant 0$$

for compact Kähler-Einstein manifolds with negative scalar curvature, and equality characterizes quotients of the unit ball. Miyaoka's algebraic proof in the surface case put the inequality into the birational classification of surfaces of general type. In higher dimension, the same inequality is naturally interpreted through the tangent bundle as a Hermite–Einstein bundle and through Bogomolov–Gieseker–Lübke type inequalities.

Greb–Kebekus–Peternell–Taji extended the Miyaoka–Yau inequality and the ball-quotient equality case to klt minimal models of general type, replacing ordinary Chern classes by orbifold Chern classes [GKPT]. Their mechanism is algebro-geometric: stability of a natural Higgs sheaf, a Bogomolov–Gieseker inequality after restriction to surfaces, and a quasi-étale uniformizing cover in the equality case. This is important for moduli and degeneration questions, but it does not by itself give a complete list of all numerical inequalities among the Chern numbers of smooth compact Kähler-Einstein manifolds with $c_1 < 0$.

Several modern results give related inequalities under additional positivity assumptions. Li–Zheng obtain reverse Miyaoka–Yau-type and Schur-polynomial inequalities for very ample polarizations and nef vector bundles [LZ]. Ping Li obtains $\lfloor N/2 \rfloor + 1$ sharp Chern-number inequalities for Kähler-hyperbolic manifolds by combining L^2 Hodge theory, Atiyah's L^2 index theorem, and the -1 expansion of the Hirzebruch χ_y-genus [Li]. These hypotheses are stronger than the existence of a negative Kähler–Einstein metric. The results suggest that the right answer may not be a short finite list of Chern-number inequalities depending only on N. The original problem is therefore best organized as two questions: first, whether there are genuinely new universal inequalities beyond the Miyaoka–Yau inequality; second, whether more refined positivity hypotheses force stronger Chern number positivity, including positivity of C_4 in complex dimension four.

Recent progress has clarified the curvature meaning of the original Miyaoka–Yau expression. Du wrote the quantity

$$-c_1^N(M) + \frac{2(N+1)}{N} c_2(M) c_1(M)^{N-2}$$

as an integral involving the holomorphic sectional curvature and its pointwise average, and derived reverse Yau-type inequalities and improved pinching theorems for compact Kähler-Einstein manifolds [D]. Since $c_4(M)[M] = \chi(M)$, Bourguignon's question is an Euler-sign problem. Negative Ricci curvature alone does not supply the sectional- or bisectional-curvature sign used in known Hopf-type results. Du's formula does not settle this question, but it gives a sharper analytic framework for testing which Chern number inequalities are really consequences of curvature pinching, which are consequences of the Hermite–Einstein tangent bundle, and which require new global input.

Related open problems.

1. *Original direction.* Determine whether every universal linear Chern-number inequality for compact Kähler-Einstein manifolds with $c_1 < 0$ is generated by the Miyaoka–Yau inequality, the obvious sign of K_M^N, and standard topological constraints.

2. *Original direction.* Decide Bourguignon's question: if M is a compact Kähler-Einstein fourfold with negative scalar curvature, must $c_4(M) = \chi(M)$ be positive?

3. *Reference-driven formulation.* Classify the equality and near-equality cases in the Miyaoka–Yau inequality beyond the smooth ball quotient case, especially in degenerating families of canonically polarized manifolds [GKPT].

4. *Reference-driven formulation.* Find sharp Chern-number inequalities under additional curvature assumptions, such as pinched negative holomorphic sectional curvature or negative bisectional curvature [D, Li].

5. *Reference-driven formulation.* Determine whether reverse Yau-type inequalities can be combined with Bogomolov–Gieseker–Lübke inequalities to produce new independent constraints in dimensions $N \geqslant 4$ [B, Lu, D].

6. *AI-generated formulation.* Formulate a cone of "Kähler-Einstein Chern numbers" in each complex dimension N and decide whether its extremal rays are represented by locally symmetric examples, complete intersections, or degenerations of canonical models.

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Problem 46

(Calabi). Let u be a real-valued function defined on $\mathbb{C}^N$ so that $\det \frac{\partial^2 u}{\partial z_i \partial \overline{z}_j} = 1$ and $\sum_{i,j} \frac{\partial^2 u}{\partial z_i \partial \overline{z}_j} dz^i d\overline{z}^j$ defines a Kähler metric. Prove that this metric is flat (see [Ca2]).

The difficulty lies in the fact that the automorphism group of $\mathbb{C}^N$ is very large.

Status. False in the natural complete-metric formulation. LeBrun pointed out that the Taub–NUT metric gives a complete non-flat Ricci-flat Kähler metric on $\mathbb{C}^2$. The surviving problem is the classification of complete Ricci-flat Kähler metrics on $\mathbb{C}^N$, especially under specified asymptotic, symmetry, or compactification hypotheses.

Calabi's problem is a complex Monge–Ampère analogue of Bernstein and Liouville-type rigidity questions. If the metric is assumed to be the standard Euclidean metric up to sufficiently strong global control, one expects rigidity; without such asymptotic assumptions, LeBrun's observation shows that completeness alone is not enough. The Taub–NUT example is hyperkähler, complete, Ricci-flat, and defined on the smooth manifold $\mathbb{C}^2$, but it is not flat.

The post-LeBrun viewpoint is constructive as well as classificatory. There are now several families of complete non-flat Calabi–Yau metrics on affine space or on spaces very close to affine space: Taub–NUT type models in complex dimension two, higher-dimensional extensions in complex dimension three and above, and explicit Hamiltonian-two-form constructions. These examples make clear that the correct rigidity statement must include geometric conditions at infinity.

One natural way to organize the classification is by asymptotic geometry: ALE, ALF, ALG, ALH, asymptotically conical, or more collapsed Taub–NUT type behavior. Another is by algebraic compactification: determine whether the metric is produced from a compactification of $\mathbb{C}^N$ by a divisor, and how the divisor and its normal geometry determine the Ricci-flat metric.

Recent constructions show that the classification problem has at least two genuinely different branches. Székelyhidi and Conlon–Rochon constructed complete Calabi–Yau metrics on $\mathbb{C}^n$, $n \geqslant 3$, with maximal volume growth and singular tangent cones at infinity [Sz, CR]. Firester extended this smoothing picture for Calabi–Yau complete intersections, producing complete metrics whose tangent cone at infinity is $\mathbb{C} \times V_0$ for a Calabi–Yau cone V_0 [Fi]. On the collapsed side, Ma recently constructed complete non-flat Taub–NUT-type Calabi–Yau metrics on $\mathbb{C}^{N+1}$ for every $N \geqslant 3$, with tangent cone at infinity $\mathbb{R}^{N+2}$, non-maximal volume growth, bounded curvature, and torus-fibration asymptotics controlled by a discriminant locus [Ma]. Thus the failure of Calabi's flatness prediction is not isolated: there are both maximal-growth and non-maximal-growth families, and any remaining rigidity theorem must explicitly exclude one or both asymptotic mechanisms.

Related open problems.

1. *Original direction.* Classify complete Ricci-flat Kähler metrics on $\mathbb{C}^N$ under natural volume growth and curvature decay assumptions.

2. *Reference-driven formulation.* Determine which asymptotic models at infinity can occur for complete Calabi–Yau metrics on $\mathbb{C}^N$, including singular Calabi–Yau cones and collapsed Taub–NUT-type fibrations [Sz, CR, Li, Ma].

3. *Reference-driven formulation.* Decide whether every complete Ricci-flat Kähler metric on $\mathbb{C}^N$ with Euclidean volume growth and sufficiently fast curvature decay must be flat, or whether the known maximal-growth examples exhaust the non-flat possibilities [CH, CR, Sz, Fi].

4. *Reference-driven formulation.* Relate complete Ricci-flat Kähler metrics on affine space to algebraic compactifications by divisors, in the spirit of the Tian–Yau and Conlon–Hein theories.

5. *Reference-driven formulation.* Classify complete Calabi–Yau metrics on $\mathbb{C}^N$ with non-maximal volume growth and identify which torus-fibration discriminant loci can occur [AC, Li, Ma].

6. *AI-generated formulation.* Build a moduli theory for complete Ricci-flat Kähler metrics on $\mathbb{C}^N$ with fixed asymptotic fibration type, and identify the Taub–NUT type metrics as boundary or extremal points of that moduli space.

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Problem 47

Let M be a compact Kähler manifold with positive holomorphic sectional curvature or positive Ricci curvature. Prove that M is rationally connected, i.e. any two points of M can be joined by a chain of rational curves.

Status. Solved. The positive Ricci curvature case follows from the Fano case: Campana and Kollár–Miyaoka–Mori proved that smooth Fano manifolds are rationally connected. The positive holomorphic sectional curvature case was solved later by Yang using RC-positivity; in particular a compact Kähler manifold with positive holomorphic sectional curvature is projective and rationally connected.

The historical point is that positive Ricci curvature and positive holomorphic sectional curvature are different conditions. Positive Ricci form gives positivity of $c_1(M)$, hence places the manifold in the Fano world. The algebraic geometry of rational curves then applies. This is the Campana and Kollár–Miyaoka–Mori solution.

The holomorphic sectional curvature direction required a different mechanism. Yang proves that positive holomorphic sectional curvature makes every $\Lambda^p T_M$ RC-positive [Y, Theorem 1.7]. The resulting vanishing theorem excludes holomorphic tensor powers; it also gives projectivity, after which the Campana–Demailly–Peternell criterion implies rational connectedness. Heier–Wong obtained the projective case through total-scalar-curvature and uniruledness methods [HW1, HW2]. Yang's argument removes the projectivity assumption and separates the problem from the Fano route, since positive holomorphic sectional curvature need not imply positive Ricci curvature.

The present frontier is the semipositive and quasi-positive boundary. Recent work of Zhang–Zhang and Matsumura gives structure theorems for compact Kähler manifolds with nonnegative holomorphic sectional curvature: roughly, rationally connected fibers appear over a torus-type base, and quasi-positive or no-truly-flat hypotheses force projectivity and rational connectedness.

There is also a complementary existence problem: which rationally connected or rational manifolds actually carry positive holomorphic sectional curvature? In complex dimension two, Hitchin proved that a compact complex surface with positive holomorphic sectional curvature is rational and constructed examples on Hirzebruch surfaces. Zhang recently proved the converse for rational surfaces: every rational surface admits a Kähler metric with positive holomorphic sectional curvature [SZ]. More generally, Zhang's construction covers projective manifolds obtained from projective toric manifolds by iterated point blow-ups. This turns the solved direction of Yau's problem into a sharper classification question: among rationally connected manifolds, characterize the ones admitting positive holomorphic sectional curvature.

Related open problems.

1. *Reference-driven formulation.* Classify compact Kähler manifolds with nonnegative holomorphic sectional curvature, including the precise role of flat tangent directions [ZZ, Ma].

2. *Original direction.* Determine optimal curvature positivity conditions between positive holomorphic sectional curvature and positive Ricci curvature which still imply rational connectedness.

3. *Reference-driven formulation.* Characterize rationally connected manifolds, beyond rational surfaces and iterated point blow-ups of toric manifolds, which admit positive holomorphic sectional curvature [SZ].

4. *Reference-driven formulation.* Study deformation and moduli behavior of compact Kähler manifolds admitting positive holomorphic sectional curvature, especially under blow-ups and degenerations [SZ].

5. *Reference-driven formulation.* Decide whether natural weaker notions, such as real bisectional curvature positivity or RC-positivity of tensor bundles, give sharp birational conclusions [Y].

6. *AI-generated formulation.* Develop a curvature "MRC-fibration theorem" which reads off the dimension of the rationally connected fiber directly from the null directions of the holomorphic sectional curvature.

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Problem 48

Let M be a compact Kähler manifold with negative sectional curvature. Prove that M is covered by a bounded domain of $\mathbb{C}^N$. One might prove a weaker assertion that the universal cover of M has an abundance of bounded holomorphic functions. (See the example of Mostow–Siu [MS].)

Status. Open both for the bounded-domain statement and, at this level of generality, for the weaker bounded-holomorphic-function direction. Cao and Shaw's arXiv preprint claimed that a simply connected complete Kähler manifold of complex dimension at least three with pinched negative sectional curvature admits a nonconstant bounded holomorphic function, but the authors withdrew the paper after an error was found in Chapter 3 [CS]. Its first version is therefore retained here only as a failed-proof audit and cannot be cited as progress on the problem.

The problem sits between Riemannian negative curvature and several complex-analytic uniformization questions. By Cartan–Hadamard the universal cover is diffeomorphic to Euclidean space, and by classical work of Wu and Greene–Wu one expects strong Stein and convexity properties under nonpositive or negative curvature. However, producing enough bounded holomorphic functions to embed the universal cover as a bounded domain is much harder than proving the existence of some holomorphic functions.

Mostow and Siu constructed compact Kähler surfaces of negative curvature which are not ball quotients. Their examples show that one cannot expect the universal cover to be the unit ball; a successful uniformization theorem must allow more general bounded domains. Jost–Yau and Bland obtained partial results toward bounded holomorphic functions under additional analytic hypotheses. The withdrawn Cao–Shaw argument proposed to combine bounded plurisubharmonic exhaustions, CR analysis on geodesic spheres, and estimates for the $\overline{\partial}$-equation. Because its crucial Chapter 3 step is invalid, this remains a proposed proof architecture rather than an established theorem.

There is also important metric progress. Wu and Yau proved that on complete Kähler manifolds with negatively pinched holomorphic curvature there is a complete Kähler-Einstein metric, and in the simply connected case with negatively pinched sectional curvature the Kähler-Einstein, Kobayashi–Royden, Bergman, and background metrics are uniformly equivalent. This gives strong invariant-metric evidence for the bounded-domain picture, but does not by itself produce a biholomorphic realization of the universal cover as a bounded domain in $\mathbb{C}^N$.

Recent function-theoretic work gives additional tests for any bounded-domain strategy. Bracci–Gaussier–Zimmer studied domains admitting negatively pinched Kähler metrics and proved strong restrictions on convex boundaries: finite type follows from smooth boundary hypotheses, and sufficiently tight negative holomorphic sectional pinching characterizes strong pseudoconvexity among convex domains [BGZ]. Huang–Li proved that for a Stein manifold with well-defined Bergman metric of nonpositive constant holomorphic sectional curvature, the Bergman space separates points; in the negative constant case this leads to a ball-type uniformization outside a possible pluripolar set [HL]. These results are not a solution of Yau's bounded-domain problem, because the universal cover is not initially given as a Euclidean domain with controlled boundary. They do, however, indicate what additional boundary, Bergman-space, or convexity conclusions would be needed after one constructs sufficiently many bounded holomorphic functions.

Related open problems.

1. *Original direction.* Prove or disprove that the universal cover of every compact Kähler manifold with negative sectional curvature is biholomorphic to a bounded domain in $\mathbb{C}^N$.

2. *Original direction.* Settle the bounded-holomorphic-function problem in every complex dimension, including the claimed dimension-at-least-three case whose published preprint was withdrawn.

3. *Reference-driven formulation.* Determine whether the Bergman metric equivalence results imply finite-dimensional bounded-domain embeddings under additional regularity or boundary assumptions [WY, HL].

4. *Reference-driven formulation.* Classify the possible bounded domains which can occur as universal covers of compact negatively curved Kähler manifolds, and test whether convexity, finite type, or strong pseudoconvexity can be forced [MS, BGZ].

5. *Reference-driven formulation.* Repair or replace the failed Chapter 3 argument of Cao–Shaw, then use L^2 $\overline{\partial}$ methods and Bergman-space separation to construct enough bounded functions for a proper holomorphic embedding [CS, HL].

6. *AI-generated formulation.* Find a curvature-pinning or rank condition which forces the universal cover to be a bounded convex domain, and compare it with the Mostow–Siu examples.

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Problem 49

Let M_t be a holomorphic family of Kähler manifolds. Let ds_t^2 be the canonical Kähler–Einstein metric on M_t. What is the behavior of ds_t^2 as the family M_t degenerates?

Status. Partially resolved. Major special regimes have a well-developed theory, but the fully general degeneration theory of canonical Kähler-Einstein metrics is still open. The best developed cases are Fano Kähler-Einstein metrics and their Gromov–Hausdorff limits, Ricci-flat Calabi–Yau degenerations, and negatively curved canonically polarized degenerations.

In the Fano case, the modern picture is quite complete at the level of metric and algebraic compactness. Donaldson–Sun proved that non-collapsed Gromov–Hausdorff limits of Kähler manifolds with two-sided Ricci control are normal algebraic varieties. For Kähler-Einstein Fano manifolds this identifies the limit as a $\mathbb{Q}$-Fano variety carrying a weak Kähler-Einstein metric. Chen–Donaldson–Sun developed the conic continuity method and treated limits with cone angles; this became one of the analytic inputs in the Yau–Tian–Donaldson theorem. The algebraic meaning of the limit is now expressed through K-stability and K-polystability.

For Ricci-flat compact Kähler manifolds, the behavior depends on whether the degeneration is non-collapsing or collapsing. If the Kähler class approaches the boundary of the Kähler cone with positive limiting volume, Tosatti proved smooth convergence away from the subvarieties whose volume tends to zero, and Rong–Zhang identified the metric completion with the global Gromov–Hausdorff limit. If the limiting class has zero volume, the metrics collapse. For holomorphic fibrations, Tosatti proved adiabatic convergence to a metric on the base whose Ricci curvature is controlled by the Weil–Petersson form. Gross–Tosatti–Zhang proved refined collapse results for abelian-fibered Calabi–Yau manifolds, extending the Gross–Wilson analysis of large complex structure limits of $K3$ surfaces.

If the complex structure also varies, the singularities of the central fiber become decisive. When the limit has canonical singularities, one expects and in many settings obtains non-collapsing convergence to a singular Ricci-flat metric on the limit variety, smooth away from the singular set. At large complex structure limits the SYZ picture predicts collapse, after diameter normalization, to a real Monge–Ampère space of half real dimension, with singular set of real codimension at least two. This picture is proved in detail for $K3$ surfaces and in important fibration cases, but is far from complete in higher dimensions.

For canonically polarized manifolds and varieties of general type, Aubin–Yau existence gives the smooth negative Ricci Kähler-Einstein metrics. In degenerating families, Song–Tian and Eyssidieux–Guedj–Zeriahi developed the weak Kähler-Einstein metric theory on singular canonical models. The remaining issues include quantitative asymptotics near the central fiber, metric completion, diameter and curvature control, and compatibility with the algebraic compactification of moduli.

Recent work has made the collapsed Calabi–Yau side much more topological and metric-measure theoretic. Hein–Tosatti proved smooth asymptotic expansions, to all orders away from the singular fibers, for collapsing Calabi–Yau metrics in semiample fibration classes [HT]. Székelyhidi proved that for collapsing Calabi–Yau fibrations the Gromov–Hausdorff limit is homeomorphic to the base and that the discriminant locus has Hausdorff codimension at least two, resolving conjectures of Tosatti in this setting [Sz]. This work also places singular Kähler metrics and canonical metrics naturally in the language of RCD spaces [Sz2]. In complex dimension three, Fu–Guo–Song proved RCD and homeomorphism results for singular Kähler spaces with Ricci lower bounds and bounded Nash entropy [FGS]. For conifold transitions, Benabida obtained polyhomogeneous asymptotics for resolutions and smoothings of Calabi–Yau conifolds [Ben]. These developments make the modern target sharper: one wants not only convergence, but a topological identification of the metric completion, quantitative codimension bounds for the discriminant, and asymptotic expansions near the forming singularities.

Related open problems.

1. *Original direction.* Give a unified metric compactification theorem for degenerating families of canonical Kähler-Einstein metrics in the Fano, Calabi–Yau, and general type cases.

2. *Reference-driven formulation.* Describe the precise Gromov–Hausdorff limits of Ricci-flat Calabi–Yau metrics at higher-dimensional large complex structure limits, including the affine Monge–Ampère structure and singular set [GW, GTZ2, SZh].

3. *Reference-driven formulation.* Establish sharp curvature, diameter, and local non-collapsing estimates near singular fibers in degenerating families [HT, Ben].

4. *Reference-driven formulation.* Relate metric tangent cones and singularities of Kähler-Einstein limits to the birational type and stability of the central fiber [DS, CDS1, CDS2, EGZ].

5. *Reference-driven formulation.* Determine when singular Kähler-Einstein metric completions are RCD spaces homeomorphic to the underlying algebraic variety [Sz2, FGS].

6. *AI-generated formulation.* Construct a single "degeneration dictionary" which takes an algebraic degeneration and predicts the metric limit, collapsed dimension, singular set, and canonical measure.

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III. Submanifolds

Problem 50

Prove that a compact surface in $\mathbb{R}^3$ is rigid, i.e. one cannot find a continuous family of surfaces in $\mathbb{R}^3$ which are isometric to each other and are not obtained from each other by a rigid motion.

This is a very long-standing problem. If we consider polyhedral surfaces, there is a counterexample due to R. Connelly [Co]. Sullivan asked if the (signed) volume enclosed by the surfaces is invariant under isometric deformation.

For the smooth case, Cohn–Vossen proved the rigidity for convex surfaces. In an attempt to generalize Cohn–Vossen's result, L. Nirenberg [Ni] studied the surfaces with $\int K^+ = 4\pi$. He generalized Cohn–Vossen's result assuming the nonexistence of more than one closed asymptotic line. The real analytic case was first established by A. D. Alexandrov [Al].

Status. Open. The smooth rigidity problem remains open in this generality. Important classical cases are known: convex closed surfaces are rigid by Cohn–Vossen and Pogorelov-type global rigidity theory, and real analytic closed surfaces satisfy rigidity theorems going back to Alexandrov. The polyhedral analogue is false: Connelly constructed flexible polyhedral spheres. Sullivan's volume-invariance question for flexible polyhedra is solved by the Bellows theorem of Sabitov and Connelly–Sabitov–Walz.

The historical development separates three levels of regularity. For convex smooth or convex polyhedral surfaces, the Cauchy–Alexandrov and Cohn–Vossen traditions give strong rigidity. For real analytic closed surfaces, analytic continuation and the geometry of asymptotic lines give rigidity in settings where the smooth problem is still not understood. Nirenberg's work on surfaces with $\int K^+ = 4\pi$ is one attempt to push convex-surface methods into the nonconvex smooth setting by controlling asymptotic-line behavior.

Polyhedral geometry shows that rigidity is delicate rather than purely topological. Bricard found self-intersecting flexible octahedra; Connelly later constructed flexible polyhedral spheres, including embedded examples. These examples disprove the naive polyhedral rigidity conjecture, but they led to a new invariant problem. The Bellows theorem says that the volume enclosed by an orientable flexible polyhedron is constant during a flex. More recently, the strong Bellows theorem proves invariance of the Dehn invariant, giving scissors-congruence consequences for flexible polyhedra.

Recent discrete-differential examples sharpen the boundary between rigid and flexible behavior. Izmestiev–Rasoulzadeh–Tervooren studied T-hedra, flexible quadrilateral polyhedral surfaces including Miura folds and discrete surfaces of revolution, and also developed their smooth analogues, T-surfaces, with explicit isometric deformation formulas [IRT]. These examples are not counterexamples to the closed smooth rigidity problem as stated, but they are useful test cases: they show how special curvature-line or grid-like structures can carry actual isometric deformations, and hence what sort of exceptional geometry a smooth rigidity theorem must exclude.

The smooth problem is therefore still a central open question in classical surface theory. One should distinguish global rigidity from infinitesimal rigidity, embedded from immersed surfaces, and smooth from C^1 regularity. Nash–Kuiper type C^1 flexibility shows that low regularity behaves very differently, so the intended problem is a C^2, smooth, or analytic rigidity question. In Lorentzian geometry, related rigidity questions for spacelike two-surfaces in Minkowski or static reference spaces enter the Wang–Yau quasi-local mass theory: vanishing or minimizing of the mass is tied to uniqueness and rigidity of isometric embeddings.

Related open problems.

1. *Original direction.* Prove or disprove global rigidity for smooth closed embedded surfaces in $\mathbb{R}^3$ without convexity assumptions.

2. *Original direction.* Classify smooth closed surfaces in $\mathbb{R}^3$ which admit nontrivial infinitesimal bendings, and determine when such bendings integrate to actual isometric deformations.

3. *Reference-driven formulation.* Develop a smooth analogue of the Bellows theorem: identify which enclosed-volume, total-mean-curvature, or Dehn-type quantities remain invariant under smooth isometric deformations [CSW, GI].

4. *Reference-driven formulation.* Study rigidity of spacelike two-surfaces isometrically embedded in Minkowski space, static spacetimes, or cylindrical Lorentzian three-manifolds, with applications to quasi-local mass [CWY].

5. *Reference-driven formulation.* Use flexible T-surfaces and related grid-like models as local normal forms for possible smooth flexing mechanisms [IRT].

6. *AI-generated formulation.* Find curvature-line or asymptotic-line conditions weaker than convexity and analyticity which still force global rigidity of a smooth closed surface in $\mathbb{R}^3$.

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Problem 51

Let M be the space of immersions of a fixed compact surface into $\mathbb{R}^3$. Prove that the subspace of M which consists of infinitesimally rigid immersions is "generic" in M. How can we describe its complement? Study the same problem in the category of surfaces of rotation.

Status. Open in this smooth infinite-dimensional form. There are strong supporting results in special classes, including convex surfaces and surfaces of revolution, and there is a well-developed finite-dimensional generic rigidity theory for frameworks. A full Baire-generic theorem for smooth immersions of a fixed compact surface in $\mathbb{R}^3$, together with a useful description of the exceptional set, does not appear to be available.

An infinitesimal bending of an immersion $F : \Sigma \to \mathbb{R}^3$ is a vector field V along F such that the first variation of the induced metric vanishes:

$$\langle dF(X), dV(Y)\rangle + \langle dF(Y), dV(X)\rangle = 0.$$

The trivial solutions are restrictions of ambient Killing fields. Thus infinitesimal rigidity means that the kernel of this first-order linear system is exactly the six-dimensional space of Euclidean motions. The problem asks whether this minimal-kernel condition is generic among all immersions.

Classically, smooth closed surfaces of positive Gauss curvature are infinitesimally rigid; a modern proof uses the second variation of the Hilbert–Einstein functional [Iz]. Surfaces of revolution give a more tractable reduction of the bending equations. Minagawa–Rado developed this reduction, and Sabitov later obtained local existence and uniqueness results at low regularity and explicit rigidity criteria for compact analytic surfaces of revolution whose poles have prescribed orders of flattening [MR, Sab]. Sabitov also constructed nonanalytic nonrigid examples with any prescribed pole-flattening order. Thus even in the rotationally symmetric class the exceptional set depends on analyticity and fine jet data, rather than on a single curvature-sign condition. For general smooth immersions, the bending equations may also interact with asymptotic lines, umbilics, and global topology in a much less transparent way.

The finite-framework theory gives a useful analogy but not a direct solution. Asimow–Roth showed that generic rigidity and generic infinitesimal rigidity are closely related for finite bar-and-joint frameworks; there the relevant rank conditions are finite-dimensional algebraic conditions. In the surface problem, the corresponding operator is a differential operator whose kernel can change through analytic and geometric degenerations. Describing the complement should therefore involve both PDE Fredholm theory and the classical geometry of lines of curvature and asymptotic directions.

The recent theory of discrete and smooth T-surfaces provides a concrete testing ground for this complement. In the discrete setting, generic quadrilateral surfaces are rigid, while T-hedra form explicit flexible families; in the smooth setting, Izmestiev–Rasoulzadeh–Tervooren give analytic descriptions and parametrizations of deformations of T-surfaces [IRT]. These models suggest that the exceptional set in the smooth immersion space may not be a single simple codimension-one condition, but a union of geometric strata reflecting special curvature-line coordinates, separation of variables in the bending equations, or hidden integrable-system structure.

Related open problems.

1. *Original direction.* Prove a Baire-generic infinitesimal rigidity theorem for smooth immersions of a fixed compact surface into $\mathbb{R}^3$.

2. *Original direction.* Describe the exceptional set of non-infinitesimally-rigid immersions by geometric conditions on curvature, umbilics, asymptotic-line dynamics, or solvability of the bending equations.

3. *Reference-driven formulation.* Complete the classification of infinitesimally flexible surfaces of revolution and determine whether their complement is open and dense in the natural function space of profiles [MR, Sab].

4. *Reference-driven formulation.* Relate infinitesimal rigidity of smooth surfaces to polyhedral generic rigidity through triangulation, approximation, and convergence of bending fields [AR1, AR2, Sa].

5. *Reference-driven formulation.* Analyze T-surfaces and other explicit flexible smooth families as model strata in the complement of the generic rigid set [IRT].

6. *AI-generated formulation.* Develop a Fredholm package for the infinitesimal bending operator modulo Euclidean Killing fields and identify the geometric strata where the Fredholm index or kernel jumps.

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Problem 52

The Nash embedding theorem insures that every manifold can be isometrically embedded into some Euclidean space, but it does not give us geometric properties of the embedding. For example, one hopes to show that a complete manifold with bounded Ricci curvature and positive injectivity radius can be embedded with bounded mean curvature in a higher dimensional Euclidean space.

Status. Open in the stated form. Nash's smooth embedding theorem and Greene's noncompact extension give high-codimension isometric embeddings, but the construction does not give a mean-curvature bound depending only on n, a Ricci bound, and a positive injectivity-radius lower bound. Known quantitative embedding theorems usually require stronger bounded-geometry hypotheses, such as bounds on the full curvature tensor and its derivatives, or they give bi-Lipschitz rather than isometric embeddings.

The geometric issue is that an isometric embedding solves the metric equations

$$g_{ij} = \sum_{\alpha=1}^{N} \partial_i F^\alpha \, \partial_j F^\alpha,$$

while the mean curvature involves second derivatives of F. Nash's iteration produces smooth embeddings but does not directly control those second derivatives by the intrinsic Ricci curvature and injectivity radius. The Gauss equation also shows that extrinsic curvature and intrinsic curvature are coupled only after the full second fundamental form is known; a bounded mean curvature estimate is weaker than a bounded second fundamental form estimate, but it is still an extrinsic regularity statement not provided by the general embedding theorem.

There are several nearby positive results. For compact manifolds, smooth Nash embeddings exist in dimensions depending only on n. For noncompact manifolds, Greene proved smooth isometric embedding theorems into Euclidean or pseudo-Euclidean spaces. Under stronger bounded geometry assumptions one can construct controlled coordinate systems and obtain quantitative local or bi-Lipschitz embeddings. There are also compactness theorems for immersions whose second fundamental forms are bounded in L^p or uniformly bounded. These results are important, but they do not answer the precise question with only $|\mathrm{Ric}| \leqslant C_0$ and $\mathrm{inj}(M, g) \geqslant i_0$.

One must also distinguish bounded mean curvature from bounded image or bounded ambient radius. Bessa and Montenegro studied restrictions on the mean curvature of Nash-type embeddings into Euclidean balls, showing that intrinsic geometry imposes lower bounds in some settings. Such results indicate that the desired embedding, if it exists, cannot be obtained by arbitrary compression of the Nash embedding without paying extrinsic-curvature cost.

A direct modern comparison is Petrunin's uniform form of Nash's construction [Pe]. If a complete manifold has positive injectivity radius and uniform bounds on the full curvature tensor and all of its covariant derivatives, he constructs a uniformly smooth isometric immersion, hence one with bounded normal curvatures. With an additional uniform polynomial-growth hypothesis he upgrades this to a tubed isometric embedding. The paper explains that finitely many full curvature-derivative bounds suffice for corresponding finite regularity, but it leaves the borderline low-regularity regime open. This is substantially stronger intrinsic input than a Ricci bound and injectivity-radius lower bound; in dimensions at least four a Ricci bound does not even control the full curvature tensor.

Recent work has also developed compactness theories in the reverse direction. Breuning's theorem for immersions with L^p-bounded or uniformly bounded second fundamental form gives a robust language for controlled extrinsic geometry [Br]. Martino–Rivière treat weak immersions whose second fundamental form lies in a critical Sobolev space and obtain induced differential structures and weak sequential closure results [MRi]. These results assume extrinsic curvature control instead of producing it from Ricci and injectivity-radius bounds. The remaining challenge is therefore to lower the intrinsic regularity required by the uniform Nash construction while retaining control of the mean-curvature vector.

Related open problems.

1. *Original direction.* Prove or disprove the stated estimate: for each n, C_0, i_0, find $N(n)$ and $C(n, C_0, i_0)$ so that every complete (M^n, g) with $|\mathrm{Ric}| \leqslant C_0$ and $\mathrm{inj}(M, g) \geqslant i_0$ has an isometric embedding $F : M \to \mathbb{R}^{N(n)}$ with $|H_F| \leqslant C(n, C_0, i_0)$.

2. *Original direction.* Determine whether the same hypotheses can give an isometric embedding with bounded second fundamental form, or whether full curvature bounds are necessary.

3. *Reference-driven formulation.* Establish local versions with uniform mean-curvature control on geodesic balls under Ricci and injectivity-radius bounds.

4. *Reference-driven formulation.* Compare isometric embeddings with controlled mean curvature to quantitative bi-Lipschitz embeddings of bounded-geometry manifolds.

5. *Reference-driven formulation.* Determine whether compactness theories for immersions with bounded or Sobolev-controlled second fundamental form can be reversed or combined with Nash–Moser iteration to produce mean-curvature control [Br, MRi].

6. *Reference-driven formulation.* Decide how far Petrunin's uniformly smooth immersion theorem can be extended from full bounded geometry to two-sided Ricci bounds and positive injectivity radius [Pe].

7. *AI-generated formulation.* Develop a Nash iteration with extrinsic curvature bookkeeping, tracking how each smoothing and correction step changes the mean curvature vector.

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Problem 53

Can one generalize Weyl's embedding problem to higher dimensions?

This would be to prove that a compact, N-dimensional manifold with positive sectional curvature can be isometrically immersed into the Euclidean space of $\frac{N(N+1)}{2}$ dimensions.

One difficulty comes from the lack of understanding of the nonuniqueness of the immersion. P. Griffiths has recently obtained some new insight into this problem.

Status. Open in the stated smooth compact global form. The analytic local isometric embedding problem is settled by the Janet–Cartan theorem, and Nash gives global smooth embeddings in higher codimension. There is also strong smooth local progress at the determined dimension: every smooth three-dimensional metric is locally isometrically embeddable in $\mathbb{R}^6$ at a point where its Riemann tensor is nonzero. What remains open here is the global positive-sectional-curvature problem in the Weyl dimension; in particular, no theorem is known which embeds every compact positively curved three-manifold isometrically in $\mathbb{R}^6$.

The classical Weyl problem asks for an isometric embedding of a positively curved metric on $\mathbb{S}^2$ into $\mathbb{R}^3$. It was solved by Lewy, Nirenberg, Pogorelov, and others through elliptic Monge–Ampère methods and convex-surface theory. In dimension two, positive Gauss curvature gives a scalar elliptic equation for the support function or embedding. In higher dimensions, the isometric embedding equations form a much larger nonlinear system:

$$g_{ij} = \sum_{\alpha=1}^{N(N+1)/2} \partial_i F^\alpha \, \partial_j F^\alpha.$$

At the determined dimension the system has exactly as many unknown functions as equations, so one cannot simply use high-codimension flexibility.

Janet and Cartan proved local analytic embedding into $\mathbb{R}^{N(N+1)/2}$. Bryant, Griffiths, and Yang studied the smooth determined system using exterior differential systems and characteristic varieties. In dimension three they obtained local existence under a curvature condition; Nakamura–Maeda and Goodman–Yang extended this to every point where the Riemann tensor is nonzero. Chen–Clelland–Slemrod–Wang–Yang later gave a simpler proof by converting the linearized equations into a strongly symmetric positive system and applying Nash–Moser iteration [CCSWY]. This is genuine progress on the smooth local problem, but it neither crosses a zero-curvature point in full generality nor globalizes over a compact manifold.

In dimensions at least five, Han and Khuri proved the precise Bryant–Griffiths–Yang conjecture for sufficiently small generic parameters: the characteristic variety of the reduced linearized system is not smooth [HK]. Their proof first reduces the large linearized embedding system to an equivalent N-by-N first-order system and then locates the singular locus algebraically. The smallness and genericity qualifications are essential; the result is a structural warning about higher-dimensional PDE analysis, not a global nonembedding theorem.

There is related progress in codimension one under additional assumptions. Li and Weinstein obtained a priori estimates for codimension-one isometric embeddings of metrics with nonnegative sectional curvature and positive scalar curvature on $\mathbb{S}^n$, generalizing estimates in the two-dimensional Weyl problem. These results are valuable but address a different target dimension and require local embeddability or curvature hypotheses tailored to the hypersurface equation.

There is also a new computational and weak-formulation direction around the classical two-dimensional Weyl problem. Gao–Hu–Li–Zhang recently proposed finite element methods for isometric embedding of Riemannian manifolds, beginning with Weyl's problem for positive Gaussian curvature surfaces in $\mathbb{R}^3$, and proved well-posedness, convergence, and error estimates for their numerical scheme [GHLZ]. This does not settle the higher-dimensional Weyl problem, but it suggests a practical auxiliary program: develop numerical diagnostics for the determined isometric embedding system, especially near characteristic degeneration and nonuniqueness.

Related open problems.

1. *Original direction.* Prove or disprove that every compact N-manifold with positive sectional curvature admits a smooth isometric immersion into $\mathbb{R}^{N(N+1)/2}$.

2. *Original direction.* Settle the first open case: smooth compact positively curved three-manifolds into $\mathbb{R}^6$.

3. *Reference-driven formulation.* Develop an elliptic or mixed-type analytic framework for the determined isometric embedding system in positive sectional curvature.

4. *Reference-driven formulation.* Determine whether additional topological, convexity, or curvature-pinching hypotheses force global embeddability in the Weyl dimension.

5. *Reference-driven formulation.* Use characteristic-variety analysis to decide whether the Bryant–Griffiths–Yang degeneracy found by Han–Khuri can be converted into either an obstruction or a solvability condition [BGY, HK].

6. *Reference-driven formulation.* Build reliable numerical tests for the higher-dimensional determined system, using the finite element Weyl program as a two-dimensional model [GHLZ].

7. *AI-generated formulation.* Identify a geometric replacement for convexity in dimension $N \geqslant 3$ which turns the determined isometric embedding system into a globally solvable elliptic system.

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Problem 54

Given a smooth metric in a neighborhood of a point p in a 2-dimensional manifold, can one find a neighborhood of p that embeds isometrically into $\mathbb{R}^3$?

The cases where the metric is either C^ω or of strictly positive or negative curvature are well known. See Pogorelov [Pg] for a possible counterexample in the smooth category.

Status. Open for arbitrary smooth metrics. The analytic case is solved by the Janet–Cartan theorem, and the strictly elliptic or hyperbolic cases $K(p) > 0$ and $K(p) < 0$ are classical. C. S. Lin proved local embedding for sufficiently smooth metrics with nonnegative Gaussian curvature and for clean sign changes. Later work extended the positive results to several finite-order mixed-type configurations. Published counterexamples of Pogorelov and Nadirashvili–Yuan concern $C^{2,1}$ metrics and failure of a local C^2 realization; they do not by themselves disprove the C^∞-metric question. A 2002 Nadirashvili–Yuan preprint claims smooth nonpositively curved metrics with no local C^3 embedding, but it remains unpublished and is recorded here as a claim, not as a settled status theorem [NYp].

The local isometric embedding equations for a surface in $\mathbb{R}^3$ are equivalent, after a choice of coordinates, to a Monge–Ampère type equation. The sign of the Gaussian curvature determines the analytic character of the equation: positive curvature is elliptic, negative curvature is hyperbolic, and curvature vanishing or changing sign produces degenerate or mixed type equations. This explains why the analytic and strictly signed cases behave differently from the general smooth case.

Lin's nonnegative-curvature theorem was a major step because it treated degeneracy at $K = 0$. His later work on clean sign changes opened the mixed-type direction. Khuri proved local existence when the Gaussian curvature changes sign to finite order along a smooth curve; Han–Khuri treated transverse intersections of zero curves; and Tsung-Yin Lin treated tangent zero curves forming cusp domains. These results show that many geometrically natural degeneracies are still embeddable.

The negative results are equally important for formulating the frontier. Pogorelov's $C^{2,1}$ construction, later supplied with detailed corrections by Holland, builds successively smaller bad metric disks accumulating at the base point [Ho, Pg]. Nadirashvili–Yuan modified this construction to obtain nonnegative Gauss curvature while retaining the same low metric regularity [NY]. Their separate smooth preprint arranges infinitely many localized bad regions and uses the ruling structure forced on the hypothetical C^3 image to derive incompatible length estimates [NYp]. Because that smooth claim has not become a published theorem, the original universal smooth assertion and the classification of its possible obstructions should both remain on the open list.

The current frontier is therefore a stratification problem for degenerating Monge–Ampère equations. Clean sign change, transverse zero curves, and cusp domains are positive cases, while published low-regularity examples show that uncontrolled degeneration can fail. The new finite-element work on Weyl's problem for $K > 0$ surfaces [GHLZ] is adjacent rather than decisive here, but it suggests computable diagnostics near curvature degeneracy.

Related open problems.

1. *Original direction.* Settle the original question: prove that every smooth two-dimensional metric is locally smoothly realizable in $\mathbb{R}^3$, or produce a published and independently verifiable smooth counterexample.

2. *Original direction.* Classify the curvature-zero-set configurations for which a smooth two-dimensional metric admits a smooth local isometric embedding into $\mathbb{R}^3$.

3. *Original direction.* Determine sharp regularity thresholds for local embeddability near points where Gaussian curvature vanishes to high or infinite order.

4. *Reference-driven formulation.* Develop a unified Monge–Ampère theory for the mixed elliptic- hyperbolic equations arising from sign-changing Gaussian curvature.

5. *Reference-driven formulation.* Verify, repair, or refute the smooth obstruction mechanism claimed in the Nadirashvili–Yuan preprint, and express it in intrinsic geometric terms [NYp].

6. *Reference-driven formulation.* Extend the positive finite-order zero-set cases from clean curves, transverse intersections, and cusp domains to more general singular or infinitely flat zero sets [Kh, HK, L3].

7. *Reference-driven formulation.* Develop numerical or weak-formulation tests for local embeddability near $K = 0$, using the positive-curvature finite element Weyl theory as a starting model [GHLZ].

8. *AI-generated formulation.* Formulate a local "curvature phase portrait" criterion which decides embeddability from the Taylor expansion and zero-set geometry of K near p.

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Problem 55

Suppose one defines an isometric embedding of a manifold into $\mathbb{R}^N$ to be elliptic if the second fundamental form corresponding to each normal has at least two nonzero eigenvalues of the same sign (see Tanaka [Ta]). Suppose, then, that we have two elliptic isometric embeddings of a fixed compact manifold. Are they congruent to each other? What is the correct generalization of the Cohn–Vossen rigidity theorem to higher dimensions? If M is a complete immersed surface in $\mathbb{R}^3$ with finite area and if K is bounded and nonpositive, then is M rigid?

Status. Open in the full elliptic-embedding generality. There is substantial progress for convex hypersurfaces and for rigidity statements closer to the classical Cohn–Vossen setting, but a general congruence theorem for elliptic isometric embeddings in higher codimension is not known. The finite-area nonpositive-curvature surface question also remains part of a broader open rigidity theory for complete noncompact surfaces in $\mathbb{R}^3$.

Tanaka introduced a notion of elliptic isometric embedding for which the infinitesimal deformation operator becomes elliptic [Ta]. His Theorem 2.4 is a local rigidity theorem in the space of embeddings: if a compact reference embedding is elliptic and infinitesimally rigid, then two sufficiently C^3-near embeddings inducing the same metric are congruent. The infinitesimal-rigidity and neighborhood hypotheses are essential, so this theorem does not answer the original question about an arbitrary pair of elliptic embeddings. Its mechanism is upper semicontinuity of the kernel of a family of elliptic operators, followed by identification of the minimal kernel with ambient Euclidean Killing fields.

For hypersurfaces, the high-dimensional Cohn–Vossen picture is much better understood. Guan–Shen prove congruence for isometric compact star-shaped hypersurfaces in a space form when the normalized scalar curvature is strictly larger than the ambient curvature [GS]. Borisenko proves intrinsic-metric rigidity for arbitrary compact convex hypersurfaces in E^{n+1}, $n \geqslant 3$, without a smoothness assumption, and obtains corresponding constant-curvature results [Bo]. He also proves local congruence near corresponding strictly convex points in ambient dimension at least four [Bo2]. These are decisive answers in the convex hypersurface branch, but not in higher codimension, where several normal directions interact and Tanaka's ellipticity condition is not ordinary convexity.

The final surface question is related to the global theory of negatively curved or nonpositively curved complete surfaces in $\mathbb{R}^3$. Hilbert and Efimov give nonexistence when negative curvature is bounded away from zero everywhere. Gálvez–Martínez–Teruel prove that if this uniform negativity holds only outside a compact set, then every complete immersion has finite total curvature, finite area, is proper, and has ends asymptotic to half-lines [GMT]. Independently, Mendonça proves that such an end with compact boundary cannot have infinite total absolute curvature [Me]. These results isolate a very restricted finite-total-curvature end geometry, but they do not prove rigidity. They also do not settle Yau's full finite-area problem, where K may tend to zero.

Related open problems.

1. *Original direction.* Prove or disprove that two elliptic isometric embeddings of a fixed compact manifold into Euclidean space must be congruent.

2. *Original direction.* Identify the correct convexity or ellipticity hypothesis in higher codimension which plays the role of positive Gauss curvature in the Cohn–Vossen theorem.

3. *Reference-driven formulation.* Extend high-dimensional Cohn–Vossen rigidity beyond hypersurfaces in space forms to higher-codimension elliptic immersions.

4. *Original direction.* Determine whether complete immersed finite-area surfaces in $\mathbb{R}^3$ with bounded nonpositive Gaussian curvature are rigid.

5. *Reference-driven formulation.* Determine how much of Borisenko's global and local convex hypersurface rigidity survives under weaker strict-convexity, low-regularity, or space-form hypotheses [Bo, Bo2].

6. *Reference-driven formulation.* Use the end-geometry restrictions of Gálvez–Martínez–Teruel and Mendonça to isolate the possible finite-area nonpositive-curvature surfaces which could flex [GMT, Me].

7. *Reference-driven formulation.* Analyze the borderline finite-area ends for which $K \to 0$, precisely the regime not excluded by the Efimov-type end theorems [GMT, Me].

8. *AI-generated formulation.* Formulate a normal-bundle positivity condition, invariant under rotations of the normal frame, which makes the higher-codimension isometric embedding system elliptic and globally rigid.

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Problem 56

The famous Efimov theorem [Ef] states that no complete surface with curvature $\leqslant -1$ exists in $\mathbb{R}^3$. One may ask whether or not a complete hypersurface with Ricci curvature less than -1 can exist in $\mathbb{R}^N$. This was asked in [Y3] and [R].

One may also try to generalize Hilbert's theorem and ask if the hyperbolic space form of dimension N can be isometrically embedded in $\mathbb{R}^{2N-1}$.

Another problem is the nature of the singularities of a surface with $K = -1$ in $\mathbb{R}^3$. (See Hopf [Ho].) Can one give a good definition of weak solution for the $K = -1$ embedding equations, analogous to minimal currents for the zero mean curvature equation? Possibly it would be useful to consider objects in the frame bundle.

Status. Partially resolved, with important remaining analytic and geometric questions. Efimov's theorem gives the classical two-dimensional nonexistence result. Smyth–Xavier completely exclude a uniform negative Ricci bound for three-dimensional Euclidean hypersurfaces and prove the corresponding statement in higher dimension provided the sectional curvature omits at least one real value; the unrestricted higher-dimensional case remains open. The low-codimension hyperbolic-space question has recently been settled in the expected negative direction: Moore proved in 2025 that $\mathbb{H}^N$ admits no isometric immersion in $\mathbb{R}^{2N-1}$. The last part, asking for a good weak theory and a useful description of singularities for $K = -1$ surfaces in $\mathbb{R}^3$, remains open in the geometric form stated here, although there is substantial PDE progress through the sine-Gordon equation and weak Gauss–Codazzi compactness.

The historical starting point is Hilbert's theorem for the complete hyperbolic plane in $\mathbb{R}^3$, followed by Efimov's much stronger theorem excluding complete immersed surfaces whose Gaussian curvature is bounded above by a negative constant. These theorems show that the local theory of negatively curved surfaces and the global complete theory behave very differently.

The first higher-dimensional question asks whether an analogous obstruction follows from a negative upper bound on Ricci curvature. In codimension one, the Gauss equation expresses the intrinsic Ricci curvature in terms of the principal curvatures, so the problem becomes a global question about their possible asymptotic behavior and sign pattern. Smyth–Xavier proved that a complete immersed hypersurface $M^3 \subset \mathbb{R}^4$ with negative Ricci curvature satisfies $\inf_M |\operatorname{Ric}| = 0$. Thus the proposed bound $\operatorname{Ric} \leqslant -c < 0$ is impossible in dimension three. For $n > 3$ they proved the same conclusion if the sectional curvature does not take every real value. The remaining exceptional regime—where the sectional curvature is forced to range through all of $\mathbb{R}$—is why the unrestricted higher-dimensional formulation is still open [SX].

The second question is a precise higher-dimensional analogue of Hilbert's theorem: can the complete hyperbolic space $\mathbb{H}^N$ be isometrically immersed in the Euclidean space of the critical dimension $2N - 1$? The answer is now no. Moore's 2025 proof uses the generalized Gauss–Bonnet formula for Riemannian polyhedra of Allendoerfer–Weil and Chern to rule out such an isometric immersion. This result should be recorded separately from the Ricci-curvature question, because Nash's theorem still gives smooth isometric embeddings in sufficiently high codimension, and Fontenele–Xavier show that even in arbitrary codimension isometric embeddings of hyperbolic space must have complicated asymptotic behavior.

The third question concerns the correct weak replacement for smooth pseudospherical surfaces. In asymptotic coordinates, surfaces with $K = -1$ are governed by the sine-Gordon equation, so the problem is linked to integrable systems and to singularities of pseudospherical surfaces. However, the analogy with minimal currents is not yet a complete geometric theory: one would like an intrinsic weak object, a compactness theorem, and a selection principle for singular limits. Chen–Slemrod–Wang developed a compensated-compactness framework for weak solutions of the Gauss–Codazzi–Ricci system, and later work on negatively curved surfaces has produced weak or low-regularity immersions under additional curvature hypotheses. These results give a serious analytic framework, but they do not yet supply a canonical current-like theory for complete $K = -1$ surfaces with singularities.

Related open problems.

1. *Original direction.* For $n > 3$, remove the sectional-curvature-range hypothesis from Smyth–Xavier and exclude complete hypersurfaces $M^n \subset \mathbb{R}^{n+1}$ under only $\operatorname{Ric} \leqslant -c < 0$, or construct an example in the exceptional regime where the sectional curvature takes every real value.

2. *Reference-driven formulation.* Classify the additional pinching, finite-total-curvature, or finite-index assumptions under which the Smyth–Xavier type principal-curvature obstruction becomes a full Ricci-curvature obstruction.

3. *Reference-driven formulation.* Develop a current-like or varifold-like weak theory for pseudospherical surfaces, with a compactness theorem, a boundary operator, and a regularity theory for the singular set.

4. *Reference-driven formulation.* Relate the frame-bundle formulation of $K = -1$ surfaces to weak solutions of the sine-Gordon equation and to weak Gauss–Codazzi data.

5. *Reference-driven formulation.* Extend Moore's nonimmersion theorem for $\mathbb{H}^N \subset \mathbb{R}^{2N-1}$ to broader complete constant-negative-curvature manifolds, or identify the sharp role of the ambient dimension [Mo].

6. *AI-generated formulation.* Find a geometric entropy or defect measure which records the loss of smoothness in a sequence of smooth $K = -1$ immersions and whose vanishing characterizes classical pseudospherical surfaces.

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Problem 57

Find nontrivial sufficient conditions for a complete, negatively curved surface to embed isometrically in $\mathbb{R}^3$. Such a condition might be a rate of decay of the curvature. Related to this is the Dirichlet problem for prescribed Gauss curvature.

Status. Open in its broad embedded form, but there is substantial partial progress for isometric immersions. The central obstruction is the Hilbert–Efimov phenomenon: a complete surface whose negative curvature is bounded away from zero cannot be smoothly immersed in $\mathbb{R}^3$. Therefore any positive result must use decay, finite total curvature, monotonicity, oscillation control, or appropriate boundary/initial data.

Historically, the problem was first clarified by the contrast between local solvability and global obstruction. Locally the isometric embedding equations for negative curvature are hyperbolic and may be written through the Gauss–Codazzi system. Globally, Hilbert's theorem and Efimov's theorem show that constant negative curvature, or curvature bounded above by a negative constant, is impossible for complete smooth immersions in $\mathbb{R}^3$.

Hong gave an important positive result: under suitable hypotheses, including sufficiently fast decay of the negative curvature at infinity, one can solve the global isometric embedding problem. In the formulation recorded in the later comparison notes, faster-than-quadratic curvature decay is one natural sufficient condition. This should be understood as a model theorem: it does not give a final intrinsic characterization, but it shows that decay at infinity can overcome the Hilbert–Efimov obstruction.

The modern PDE approach treats the problem through the Gauss–Codazzi equations as a nonlinear hyperbolic system. Chen–Slemrod–Wang gave a compensated-compactness framework for weak Gauss–Codazzi–Ricci data. Cao–Huang–Wang obtained large L^∞ solutions by a Lax–Friedrichs scheme, giving $C^{1,1}$ isometric immersions for a class of negatively curved surfaces. Cao–Huang–Wang then proved global smooth isometric immersion under an integrability condition allowing slowly decaying negative curvature [CHW2]. More recently, Cao–Han–Huang–Wang proved smooth isometric immersion results for complete simply connected negatively curved surfaces with finite total curvature and controlled oscillation. Their proof rewrites the Gauss–Codazzi equations as a symmetric hyperbolic system whose selected combinations of Riemann invariants have different decay rates and a partial damping mechanism. These results enlarge the class far beyond monotone rapid decay, but they do not by themselves prove injectivity or proper embeddedness and still leave open the desired sharp intrinsic criterion.

The Dirichlet problem for prescribed Gauss curvature is closely related but should be kept conceptually separate. It asks for embedded or immersed surfaces realizing prescribed curvature with boundary data, and its elliptic or hyperbolic character depends on the sign of the curvature and the chosen formulation. For negative curvature, boundary value problems remain delicate because characteristic directions and possible loss of regularity interact with global completeness.

Related open problems.

1. *Original direction.* Find a sharp intrinsic sufficient condition, stated only in terms of the metric and its curvature at infinity, for a complete negatively curved surface to admit a smooth isometric immersion in $\mathbb{R}^3$.

2. *Reference-driven formulation.* Determine whether finite total curvature plus a natural quantitative oscillation condition is close to optimal for global smooth isometric immersion.

3. *Original direction.* Classify the decay rates of negative curvature which separate Hilbert–Efimov nonexistence from global embeddability.

4. *Reference-driven formulation.* Develop a regularity theory which upgrades weak or $C^{1,1}$ Gauss–Codazzi solutions to smooth immersions under checkable geometric hypotheses.

5. *Reference-driven formulation.* Compare rapid monotone decay, slow integrable decay, finite total curvature, and curvature oscillation hypotheses in the Hong and Cao–Huang–Wang programs [H, CHW2, CHHW].

6. *Reference-driven formulation.* Determine which immersion theorems under decay or finite-total-curvature hypotheses can be upgraded to proper isometric embeddings, and identify intrinsic conditions preventing self-intersection.

7. *AI-generated formulation.* Formulate a boundary-at-infinity Dirichlet problem for complete negatively curved metrics whose solvability is equivalent to global isometric immersion in $\mathbb{R}^3$.

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Problem 58

Recall that a Weingarten surface is a surface where the mean curvature H and the Gaussian curvature K satisfy a suitable functional relation of the form $\phi(K, H) = 0$ where ϕ is a nonsingular function defined on the plane. It would be interesting to know if the ellipsoid of rotation is characterized among compact surfaces by $\lambda_1 = c\lambda_2^3$ where λ_i are the principal curvatures and c is a constant. In general, Voss was able to establish that a compact real analytic Weingarten surface of genus zero is a surface of revolution (see Hopf [Ho]). What are the compact, real analytic Weingarten surfaces of higher genus? Must they have genus = 1 and be either a tube surface or a surface of revolution?

Hopf proved that for a closed real analytic Weingarten surface of genus zero, the number $\frac{dk_2}{dk_1}$ (where k_1 and k_2 are the principal curvatures) must take the following discrete values at the umbilic point: $0, -1, (2k+1)^{\pm 1}$ for $k \geqslant 1$ and ∞. Is the same statement valid for compact smooth Weingarten surfaces of genus zero?

Another problem for surfaces in $\mathbb{R}^3$ is to give an intrinsic characterization of compact surfaces defined by a real algebraic polynomial. How does one express the degree of the polynomial in terms of the invariants of the metric?

Status. Partially resolved, but the full classification questions remain open. The rotational-ellipsoid subproblem has a sharp regularity split: it is affirmative for closed real analytic surfaces, by Voss together with Kühnel–Steller or Simon, but Fernández–Mira constructed compact C^2 counterexamples. Thus the unrestricted low-regularity statement is false, while the smooth nonanalytic and strictly convex regimes still require separate analysis. Guan–Lin–Ma developed curvature-equation methods for Weingarten-type Christoffel–Minkowski problems. The higher-genus analytic classification, the smooth replacement for Hopf's discrete umbilic statement, and the intrinsic characterization of real algebraic surfaces are still open in the stated generality.

The natural historical order is the following. Hopf's theorem for constant mean curvature spheres and its Weingarten variants show that genus-zero hypotheses are extremely restrictive when the Weingarten relation is analytic or elliptic. Voss's result, as recorded by Hopf, fits this circle: compact real analytic genus-zero Weingarten surfaces are forced toward rotational symmetry. Modern Codazzi-equation methods, including the work of Aledo–Espinar–Gálvez, give a flexible framework for uniqueness and sphere theorems for elliptic Weingarten surfaces in space forms [AEG]. This reinforces the distinction between genus-zero elliptic theory and the much less settled higher-genus or merely smooth cases.

The ellipsoid question is more specific. It asks whether the relation $\lambda_1 = c\lambda_2^3$ characterizes an ellipsoid of revolution among compact surfaces. In the real analytic category the answer is yes. In contrast, Fernández–Mira glue two rotational caps, each containing a flat disk and a nonflat annular part, to obtain a compact C^2 sphere which satisfies the relation in some ordering but is not an ellipsoid [FM, Theorem 8.1]. Their example also corrects an earlier rotational claim used in the analytic literature. Since this relation is non-elliptic, elliptic Weingarten uniqueness theorems do not apply to it directly. Guan–Lin–Ma's work instead supplies a continuity method and a priori estimates for genuinely elliptic prescribed Weingarten-curvature equations on star-shaped hypersurfaces.

The higher-genus question is substantially less settled. A torus tube or a surface of revolution gives natural model examples, but there is no general classification of compact real analytic Weingarten surfaces of higher genus. In the smooth genus-zero question, Hopf's discrete list for the derivative ratio at an umbilic is tied to analyticity and to the local complex-analytic structure of the curvature-line equation; whether the same discreteness survives in the smooth category is still a delicate regularity question.

Recent work also clarifies how much of the genus-zero theory is really elliptic rather than merely analytic. Fernández–Mira classified the possible phase portraits of rotational elliptic Weingarten surfaces in $\mathbb{R}^3$, proved a sharp halfspace theorem for finite-order elliptic Weingarten equations, and showed that outside the elliptic setting Yau's rotational-ellipsoid uniqueness question has a negative answer [FM]. Fernández then proved a Hopf-type uniqueness theorem for uniformly elliptic Weingarten spheres in $\mathbb{S}^2 \times \mathbb{R}$, using rotational examples and the Gálvez–Mira framework [Fe]. These results suggest that the right modern formulation should separate elliptic Weingarten equations, non-elliptic relations, and singular or peaked examples.

The final algebraic-surface question is also open in a different direction. It asks whether one can recognize compact real algebraic surfaces in $\mathbb{R}^3$ from the intrinsic metric and whether the degree of the defining polynomial has an intrinsic expression. This is not simply a Weingarten problem: it mixes rigidity, algebraic geometry, and inverse spectral or inverse curvature questions.

Related open problems.

1. *Original direction.* Complete the classification of compact real analytic Weingarten surfaces of genus at least one under ellipticity assumptions on the Weingarten relation.

2. *Original direction.* Decide whether the analytic rigidity theorem for $\lambda_1 = c\lambda_2^3$ extends to C^∞ nonanalytic compact surfaces; the C^2 statement is false [FM].

3. *Reference-driven formulation.* Determine whether strict convexity or strictly positive Gaussian curvature excludes the flat-core mechanism in the Fernández–Mira counterexample and restores ellipsoid rigidity.

4. *Original direction.* Determine whether Hopf's discrete umbilic derivative values for real analytic genus-zero Weingarten surfaces remain valid in the smooth category.

5. *Original direction.* Give intrinsic metric criteria for a compact surface in $\mathbb{R}^3$ to be congruent to a real algebraic surface, and relate those criteria to the algebraic degree.

6. *Reference-driven formulation.* Classify compact smooth elliptic Weingarten spheres with singularities or peaked points, extending the rotational phase-space classification of Fernández–Mira [FM].

7. *Reference-driven formulation.* Determine which Hopf-type rotational symmetry theorems for elliptic Weingarten spheres in homogeneous spaces descend to $\mathbb{R}^3$, $\mathbb{S}^2 \times \mathbb{R}$, and other space forms [GM, Fe].

8. *AI-generated formulation.* Find a curvature-line index formula for smooth elliptic Weingarten spheres which specializes to Hopf's analytic discreteness theorem when the relation is real analytic.

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Problem 59

Let h be a real-valued function on $\mathbb{R}^3$. Find (reasonable) conditions on h to insure that one can find a closed surface with prescribed genus in $\mathbb{R}^3$ whose mean curvature (or curvature) is given by h.

F. Almgren made the following comments:

For "suitable" h one can obtain a compact smooth submanifold ∂A in $\mathbb{R}^3$ having mean curvature h by maximizing over bounded open sets $A \subseteq \mathbb{R}^3$ the quantity

$$F(A) = \int_A h \, d\mathcal{L}^3 - \text{Area}(\partial A).$$

A function h would be suitable, for example, in case it were continuous, bounded, and $\mathcal{L}^3$ summable, and $\sup F > 0$. However, the relation between h and the genus of the resulting extreme ∂A is not clear.

In fact, the problem in this context is a special case of a variety of minimal partitioning problems. One can see [A] for this type of problem, and there is one of interest in the work of Sir W. Thomson (Lord Kelvin) [Th]. With a suitable restriction on h, Bakel'man [Ba] and Treibergs–Wei [TW] have found solutions of this problem for the closed surface of genus zero.

Status. Open in the stated genus-prescribed form. There are strong existence theorems for special topologies, for variational weak solutions, and for prescribed mean curvature hypersurfaces under generic or sign hypotheses, but there is no general intrinsic condition on a function h which determines the existence of a closed surface of each prescribed genus in $\mathbb{R}^3$ with mean curvature h.

The variational starting point is Almgren's observation. If h is continuous, bounded, integrable, and the functional

$$F(A) = \int_A h \, d\mathcal{L}^3 - \text{Area}(\partial A)$$

has positive supremum, then maximizing over bounded open sets suggests a surface whose prescribed mean curvature is h. This is powerful because it allows geometric-measure-theoretic compactness, but it does not by itself control the topology of the resulting boundary. Thus the prescribed-genus part of the problem is the genuinely difficult extra constraint.

For genus zero, Bakel'man and Treibergs–Wei obtained embedded sphere results under suitable assumptions. The work of Spruck, Nirenberg, F. Lin, Walter Wei, Treibergs and others developed the elliptic PDE technology around the prescribed mean curvature equation and its Dirichlet problems. Yau later gave a general condition for a function to be the mean curvature of a closed sphere, using ideas related to the Schoen–Yau general relativity method and the Jang equation.

In 2026 Yin obtained a new embedded-sphere theorem directly in the Euclidean setting. For a positive C^2 function $f : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}$, radial inner and outer barriers together with radial monotonicity on the intervening annulus produce a $C^{2,\alpha}$ embedded sphere which is a radial graph and has mean curvature f. The proof derives the quasilinear equation on $\mathbb{S}^n$, applies Leray–Schauder continuation, and obtains the needed global gradient estimate by a maximum-principle argument [Yi]. This materially strengthens the explicit sphere branch, but it does not control higher genus.

More recent work clarifies both the strength and the limitation of the classical problem. Anderson proved existence of branched conformal immersions of $\mathbb{S}^2$ into $\mathbb{R}^3$ with prescribed positive mean curvature, up to a finite-dimensional affine indeterminacy, and obtained partial higher-genus results. In closed ambient manifolds, Zhou–Zhu's min–max theory gives prescribed mean curvature hypersurfaces for generic prescription functions. These results show that prescribed mean curvature is now a robust variational subject, but they do not yet answer the original Euclidean question with prescribed embedded topology. Mazurowski–Zhou recently developed an Almgren–Pitts min–max theory for nonlocal functionals of the form

$$\Omega \mapsto \text{Area}(\partial\Omega) - \int_\Omega h + f(\text{Vol}(\Omega)),$$

and used it to construct infinitely many half-volume constant mean curvature hypersurfaces in closed manifolds under generic or positive Ricci assumptions [MZ]. Gaspar–Marx-Kuo proved infinitely many prescribed mean curvature hypersurfaces for broad classes of prescription functions when the ambient closed manifold contains a strictly stable minimal hypersurface or a contracting neighborhood [GMK]. These results are not Euclidean prescribed-genus theorems, but they show that Almgren's variational idea has a modern min–max continuation.

The related prescribed Gaussian curvature problem has a different PDE character, closer to Monge–Ampère type equations and convex geometry in the positive curvature case. For the present problem, the mean curvature case is especially subtle because topology can change under varifold or set-theoretic compactness, and because a closed surface imposes global balance conditions not visible in local elliptic theory.

Related open problems.

1. *Original direction.* Find necessary and sufficient conditions on $h : \mathbb{R}^3 \to \mathbb{R}$ for the existence of an embedded closed surface of a fixed genus with mean curvature h.

2. *Original direction.* Determine which additional assumptions on h force Almgren-type maximizers to have prescribed genus, or at least give effective genus bounds.

3. *Reference-driven formulation.* Extend the sphere results of Bakel'man, Treibergs–Wei, and Yau to higher genus while preserving embeddedness.

4. *Reference-driven formulation.* Test whether Yin's radial-barrier and monotonicity hypotheses can be replaced by degree-theoretic or variational conditions that survive for prescribed genus, and determine whether the radial-graph conclusion is unique [Yi].

5. *Reference-driven formulation.* Relate min–max prescribed mean curvature hypersurfaces to the Euclidean prescribed-genus problem, especially when h has compact support or prescribed sign.

6. *Reference-driven formulation.* Adapt the nonlocal min–max functionals used for half-volume CMC hypersurfaces to Euclidean prescription functions and test whether they can impose topological constraints [MZ].

7. *Reference-driven formulation.* Determine whether the infinitely many PMC constructions in closed manifolds have Euclidean analogues after compactification, barriers, or weighted volume constraints [GMK].

8. *AI-generated formulation.* Develop a genus-sensitive replacement for Almgren's functional by adding a controllable topological penalty whose critical points still solve the prescribed mean curvature equation.

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Problem 60

(Willmore [Wi]). Let M be a compact, two-dimensional torus embedded in $\mathbb{R}^3$. Let H be its mean curvature. Is it true that $\int_M H^2 \geqslant 2\pi^2$ with equality implying that M is obtained from the circular torus by a Möbius transformation? Recently, Li–Yau [LY] defined the concept of conformal area for a conformal structure on a surface. They prove that $\int_M H^2$ is not less than this area. Using this, they show that $\int_{RP^2} H^2 \geqslant 6\pi$ and $\int_M H^2 \geqslant 2\pi^2$ if M is conformally equivalent to the square torus.

Status. Solved. Marques and Neves proved the Willmore conjecture: every embedded torus in $\mathbb{R}^3$ satisfies

$$\int_M H^2 \, dA \geqslant 2\pi^2,$$

with equality precisely for the stereographic images of the Clifford torus, equivalently the circular torus up to Möbius transformation in the Euclidean formulation.

The historical development is especially important here. Willmore formulated the torus inequality in 1965 after computing the energy of standard tori. Li–Yau then introduced their conformal invariant and proved strong lower bounds for Willmore energy, including the non-embedded case and the square-torus conformal class mentioned in the problem. This reduced the hardest part of the conjecture to the embedded torus case.

The final proof by Marques–Neves used Almgren–Pitts min–max theory in $\mathbb{S}^3$. The essential bridge is conformal invariance of the Willmore energy: after stereographic projection, one studies a canonical family of cycles associated with an embedded surface in $\mathbb{S}^3$ and applies the min–max theory for minimal surfaces. The Clifford torus appears as the min–max surface with area $2\pi^2$, and the equality case follows from the rigidity of the argument.

Although the original problem is solved, it opened a larger Willmore program. For genus $g > 1$, the natural analogue asks for the minimizer of Willmore energy among surfaces of that genus in $\mathbb{S}^3$ or $\mathbb{R}^3$. Lawson's minimal surfaces are expected to play the role of the Clifford torus in higher genus, and recent work has proved such statements under additional symmetry assumptions, but the full higher-genus Willmore problem remains open.

Kusner–Lü–Wang have sharpened the symmetry-restricted picture by showing that Lawson surfaces $\xi_{m,k}$ minimize the Willmore energy among competitors sharing suitable ambient symmetry groups, while also noting genus-two phenomena where known methods do not guarantee a minimizer in the symmetric class [KLW]. Their revised paper supersedes a withdrawn 2021 preprint, so only the corrected theorem is used here. Quantitative stability has also begun to catch up with the qualitative theorem: Bi–Zhou proved that an integral two-varifold in $\mathbb{S}^3$ of genus at least one with Willmore energy at most $2\pi^2 + \delta^2$ is, after conformal normalization, $W^{2,2}$-close to the Clifford torus [BZ]. Thus the original Willmore conjecture is solved, but its stability and higher-genus descendants remain active.

Related open problems.

1. *Original direction.* Determine the Willmore minimizer among embedded surfaces of each genus $g > 1$ in $\mathbb{S}^3$ or $\mathbb{R}^3$.
2. *Reference-driven formulation.* Prove or disprove the expectation that Lawson's minimal surface $\xi_{g,1} \subset \mathbb{S}^3$ minimizes Willmore energy among genus g surfaces.
3. *Reference-driven formulation.* Develop a full Morse-theoretic picture of the Willmore functional near the Clifford torus and higher-genus candidate minimizers.
4. *Reference-driven formulation.* Extend the min–max and canonical-family methods to constrained Willmore problems, for example fixed conformal class or imposed symmetry.
5. *Reference-driven formulation.* Remove or weaken the symmetry assumptions in the Kusner–Lü–Wang higher-genus Willmore minimization theorem [KLW].
6. *Reference-driven formulation.* Strengthen quantitative Clifford-torus stability from $W^{2,2}$ conformal parametrization to geometric distances suited to embedded smooth tori, and identify sharp exponents [BZ].
7. *AI-generated formulation.* Construct a quantitative stability inequality for the Willmore conjecture measuring the distance of an embedded torus from the Möbius orbit of the Clifford torus by the energy gap $\int_M H^2 - 2\pi^2$.

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Problem 61

(Alexandrov [A]). Let S be the boundary surface of a convex body in $\mathbb{R}^3$. If the intrinsic radius of S is bounded by 1, what is the largest surface area of S possible?

Status. Solved sharply by Grove and Petersen [GP]. If S is a closed locally convex surface in $\mathbb{R}^3$ with intrinsic radius at most r, then

$$\mathrm{Area}(S) \leqslant \pi r^2.$$

Consequently the answer for $r = 1$ is the sharp supremum π. In the smooth nondegenerate class the inequality is strict, but there are smooth strictly convex surfaces with area arbitrarily close to πr^2. Their intrinsic metrics converge to the two-dimensional Euclidean *purse*: a singular Alexandrov sphere obtained by doubling a half-disc, equivalently by identifying boundary points of a disc by reflection. Thus "largest" is naturally interpreted as a supremum unless degenerate convex surfaces are admitted.

The proof is a direct instance of Grove–Petersen volume comparison à la Alexandrov. For an n-manifold with sectional curvature at least k and radius at most r, ordinary ball comparison gives $\mathrm{vol}\, M \leqslant v_k^n(r)$, the volume of the radius-r ball in the simply connected space form. Grove and Petersen prove that this bound is optimal and analyze almost-equality sequences by Gromov–Hausdorff convergence. In the range containing $k = 0$, the only maximal singular limit models are the crosscap and the purse; for a spherical convex hypersurface the topology selects the purse. Setting $n = 2$, $k = 0$, and $r = 1$ gives $v_0^2(1) = \pi$. Their Corollary D states explicitly that this supplies a solution to Problem 61 as worded in Yau's list.

The radius problem should not be confused with Alexandrov's older and still open *intrinsic-diameter* conjecture. That conjecture predicts that a convex surface of intrinsic diameter D satisfies $\mathrm{Area}(S) \leqslant \pi D^2/2$, with the doubled disc as the degenerate extremizer. Freitas–Krejčiřík prove the conjecture for general ellipsoids and obtain a deficit estimate involving the length of a cut locus [FK]. The contrast is instructive: a radius bound places the whole surface inside one comparison ball, whereas a diameter bound does not specify a center and requires additional global control of the cut locus.

Related open problems.

1. *Reference-driven formulation.* Prove a quantitative stability theorem: control the Gromov–Hausdorff distance to the Euclidean purse by the deficit $\pi r^2 - \mathrm{Area}(S)$, with a sharp rate.

2. *Reference-driven formulation.* Convert intrinsic near-maximality into an extrinsic statement for convex bodies, for example Hausdorff closeness after rigid motion and normalization to a thin neighborhood of a half-disc.

3. *Original direction.* Determine the optimal upper bound under simultaneous lower-curvature, radius, and noncollapsing constraints which exclude the singular purse limit.

4. *Reference-driven formulation.* Resolve Alexandrov's fixed intrinsic-diameter conjecture beyond the known special classes, and extend the Freitas–Krejčiřík cut-locus deficit method [FK].

5. *Reference-driven formulation.* Classify the possible near-maximal limits in higher dimensions and determine when topological or extrinsic convexity assumptions rule out the crosscap model [GP].

6. *AI-generated formulation.* Find a sharp bridge between radius and diameter deficits: does simultaneous near-equality in the radius bound and the conjectural diameter bound force a unique geometric degeneration?

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Problem 62

(Milnor [KO, M]). Let Σ be a complete noncompact surface immersed in $\mathbb{R}^3$, and let λ_1, λ_2 be its principal curvatures. Prove that either $|\lambda_1 \cdot \lambda_2|$ is not bounded away from zero on Σ, or K changes sign, or $K \equiv 0$.

Status. Solved in the smooth complete setting, as a direct consequence of Bonnet–Myers in the positive-curvature case and Efimov's theorem in the negative-curvature case. Since

$$K = \lambda_1 \lambda_2,$$

the failure of the first alternative means that $|K| \geqslant c > 0$ for some constant c. If K does not change sign and $K \not\equiv 0$, then either $K \geqslant c > 0$ or $K \leqslant -c < 0$.

In the first case, the intrinsic metric on Σ has Gaussian curvature bounded below by a positive constant. Bonnet–Myers then gives finite diameter, and Hopf–Rinow implies compactness, contradicting the assumption that Σ is complete and noncompact. In the second case, Efimov's theorem rules out a complete immersed surface in $\mathbb{R}^3$ whose Gaussian curvature is bounded above by a negative constant. Therefore the stated trichotomy follows.

The historical difficulty lies almost entirely in the negative-curvature case. Efimov's proof is deep and technically subtle, and this explains the comment that a transparent proof would be valuable. Klotz–Osserman and Milnor placed the result in the broader study of complete open surfaces in Euclidean three-space. More recent work of Gálvez–Martínez–Teruel gives extensions and partial Milnor-type results for complete surfaces whose ends have nonpositive curvature and whose principal curvatures satisfy lower-bound hypotheses. Mendonça sharpened the end-wise form of Efimov's theorem: under negative curvature bounded away from zero and infinite total curvature, complete immersed ends are excluded, and in particular complete Hadamard immersed surfaces with curvature bounded away from zero outside a compact set are ruled out [Me].

The complementary existence theory clarifies the sharp geometry behind the trichotomy. Cao–Han–Huang–Wang construct smooth isometric immersions of complete simply connected negatively curved surfaces of finite total curvature, allowing controlled angular oscillation and radial decay of the Gauss curvature [CHHW]. Their proof rewrites the Gauss–Codazzi equations in geodesic polar coordinates as a symmetric quasilinear hyperbolic system and obtains global estimates in two overlapping coordinate regions. Thus Efimov's uniform negative upper bound is not a technical artifact: once the curvature decays sufficiently toward zero, complete immersions can reappear.

Thus the original problem should be recorded as solved, but the surrounding program remains active: one wants simpler proofs, sharper end-wise hypotheses, and low-regularity or finite-total-curvature versions.

Related open problems.

1. *Reference-driven formulation.* Find a short and conceptually transparent proof of Efimov's nonimmersion theorem for complete surfaces with $K \leqslant -c < 0$ in $\mathbb{R}^3$.

2. *Reference-driven formulation.* Determine the weakest assumptions at infinity under which a complete immersed surface with eventually nonpositive curvature must satisfy Milnor-type alternatives.

3. *Original direction.* Classify complete noncompact immersed surfaces for which K has one sign outside a compact set and one principal curvature is bounded away from zero.

4. *Reference-driven formulation.* Extend the trichotomy to lower regularity immersions or weak Gauss–Codazzi solutions.

5. *Reference-driven formulation.* Relate finite-total-curvature smooth immersion theorems for negatively curved surfaces to Milnor's alternatives, especially when the principal curvatures have prescribed asymptotic behavior [CHHW].

6. *Reference-driven formulation.* Determine the borderline between the Efimov nonimmersion regime and the Cao–Han–Huang–Wang existence regime when $|K|$ has inverse-square decay with unrestricted angular oscillation [CHHW].

7. *AI-generated formulation.* Develop an Efimov-type "curvature escape" theorem measuring how fast $|K|$ must approach zero along every end of a complete noncompact surface with nonchanging curvature sign.

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Problem 63

(Hopf). Prove that a closed surface Σ immersed in $\mathbb{R}^3$ with constant mean curvature is a round sphere.

Status. False as stated. The positive genus immersed case was disproved by Wente's constant-mean-curvature torus [W]. The statement remains true under the classical additional hypotheses that Σ is a topological sphere, or that Σ is embedded.

Hopf proved that a constant-mean-curvature immersion of a closed surface homeomorphic to $\mathbb{S}^2$ is totally umbilic, hence a round sphere [H]. Alexandrov's reflection method later proved the stronger embedded "soap bubble" theorem: every compact embedded hypersurface in Euclidean space with constant mean curvature is a sphere [A2]. Reilly gave another proof of the embedded case by integral identities [R].

The immersed problem changed substantially after Wente constructed an immersed constant-mean-curvature torus in $\mathbb{R}^3$ [W]. Thus the original conjecture is not an open rigidity theorem for immersed surfaces, but it left a large classification problem. In the torus case, the subsequent integrable-systems theory gives a powerful algebro-geometric description of constant-mean-curvature tori, through finite type and spectral-curve methods [PS, Hi, Bo]. In higher genus, gluing methods and perturbative constructions produced many compact immersed examples and showed that the landscape is much richer than the sphere theorem suggests [K1, K2]. A complete geometric classification for arbitrary topology is still not known, especially if one asks for geometric invariants rather than an existence theorem alone.

For Alexandrov-embedded complete finite-topology CMC surfaces, the modern picture is also moduli-theoretic rather than rigid. Mazzeo–Pacard–Pollack prove gluing theorems attaching Delaunay ends to nondegenerate CMC surfaces and study the forgetful map from CMC moduli spaces to conformal structures; even in genus zero the resulting moduli spaces have nontrivial topology [MPP]. Breiner–Kapouleas extend graph-to-Delaunay gluing constructions to complete CMC hypersurfaces in higher-dimensional Euclidean spaces [BK], and the survey of Kapouleas–Breiner–Kleene emphasizes conservation laws as a central organizing tool for such gluing constructions [KBK]. These works do not classify compact immersed CMC surfaces, but they explain why the post-Wente problem should be formulated in terms of moduli spaces, degenerations, flux/conservation laws, and compactification.

Related open problems.

1. *Original direction.* Give a geometric classification of compact immersed constant-mean-curvature surfaces in $\mathbb{R}^3$ of fixed genus.

2. *Reference-driven formulation.* Describe the moduli space of compact immersed constant-mean-curvature surfaces and its compactification, including degenerations to bubble trees or Delaunay-type pieces.

3. *Reference-driven formulation.* Relate the spectral-curve classification of constant-mean-curvature tori to more directly geometric invariants, such as symmetry, area, enclosed algebraic volume, and umbilic data.

4. *Reference-driven formulation.* Determine which topological types occur under additional constraints such as Alexandrov embeddedness, finite symmetry group, or fixed Willmore energy.

5. *Reference-driven formulation.* Compare Alexandrov-embedded CMC moduli spaces and their conformal forgetful maps with the compact immersed CMC moduli problem [MPP].

6. *AI-generated formulation.* Develop a unified classification scheme which compares the integrable-systems moduli of tori with the gluing moduli of higher-genus compact constant-mean-curvature surfaces.

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Problem 64

Prove the Carathéodory conjecture that every closed convex surface in $\mathbb{R}^3$ has at least two umbilic points. In the real analytic case, this was asserted by Bol [B] and Hamburger [H], but doubts were later expressed about these papers. See Klotz [K] for corrections.

Status. Needs verification. Claimed solved for $C^{3,\alpha}$ surfaces by Guilfoyle and Klingenberg, but the acceptance status remains controversial and should be tracked explicitly rather than recorded as an unconditional settled theorem. Their original proof preprint [GK0] was followed by published components establishing Fredholm regularity, interior mean-curvature-flow estimates, and the boundary/existence step used also for the Toponogov conjecture [GK1, GK2, GK3]. The authors now state that these papers complete the proof [GK5]. On the other hand, Guckenheimer's independent 2024 analysis describes the status as controversial and refers to the existing proofs as purported [Gu]. In the absence of an independent verification located in this audit, the conservative label is therefore *claimed solution; needs verification*. The borderline C^2 formulation remains open in any case.

The classical route reduces the global conjecture to a local index bound for isolated umbilic points: on a topological sphere the sum of the indices of the principal curvature line field is 2, so if every isolated umbilic has index at most 1, at least two umbilics must occur. In the analytic case, Hamburger gave a long proof of the index bound, and Bol proposed a shorter proof [B, H]. Klotz later corrected important gaps in Bol's argument [K], while the analytic literature remained delicate because the central issue is the resolution of degenerate umbilics.

Modern approaches reformulate the problem in complex and symplectic geometry. The surface of oriented normal lines of a convex surface is a Lagrangian surface in the neutral Kähler geometry of the space of oriented lines. In this language, umbilics become complex points, and the global conjecture is attacked through a Riemann–Hilbert problem for holomorphic discs with Lagrangian boundary conditions. Guilfoyle and Klingenberg's proof combines Fredholm regularity of these boundary value problems with higher-codimensional mean curvature flow estimates [GK0, GK1, GK2, GK3]. Independently, Ghomi and Howard gave a useful reformulation of the C^2 global conjecture in terms of asymptotically constant graphs [GH].

The most recent literature also shows that the problem should still be tracked at two levels. Guilfoyle and Klingenberg present the global $C^{3,\alpha}$ claim through the incompatibility between Fredholm regularity and the existence of a holomorphic disc produced by spacelike mean curvature flow [GK0, GK5]. Guckenheimer instead studies analytic degenerate umbilics directly: merging and splitting of singularities give a geometric obstruction to integer indices larger than one and relate index bounds to the order of degeneracy [Gu]. His paper both supplies independent local progress and documents the continuing disagreement about the global status. Neither line settles the sharp C^2 problem.

Related open problems.

1. *Original direction.* Obtain an independent verification or a precise gap analysis of the Guilfoyle–Klingenberg $C^{3,\alpha}$ argument, especially the compactness, bubbling, and boundary-control steps for holomorphic discs [GK0, GK3].

2. *Original direction.* Decide the sharp regularity threshold for the global Carathéodory conjecture, especially in the C^2 category where umbilics are naturally defined.

3. *Original direction.* Prove or disprove the sharp smooth local index conjecture: whether every isolated umbilic on a smooth convex surface has index at most 1.

4. *Reference-driven formulation.* Classify possible degenerate smooth non-analytic umbilics, including whether index $3/2$ can occur.

5. *Reference-driven formulation.* Find a purely real-variable proof of the $C^{3,\alpha}$ theorem, or a proof closer to the classical principal-foliation picture.

6. *Reference-driven formulation.* Extend Guckenheimer's analytic degeneracy analysis to a systematic normal form for smooth finite-type umbilics [Gu].

7. *AI-generated formulation.* Build a quantitative Carathéodory theory which gives lower bounds for the separation, stability, or total multiplicity of umbilics under small perturbations of a convex surface.

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Problem 65

Can one define the rank of a compact C^ω-manifold M with nonpositive curvature so that if M is a locally symmetric space the definition agrees with the standard one? Suppose there is a totally geodesic, immersed, flat 2-plane in M. Can one find an immersed totally geodesic torus in M? (See Gromoll and Wolf [GW], Lawson and Yau [LY].) If the "rank" of M is greater than one, one expects that M is very rigid metrically. How do we describe this rigidity?

Status. Partially resolved. The standard higher-rank rigidity theorem for compact nonpositively curved manifolds is now a central rigidity theorem [Ba, BBE, BBS, BS, EH]. The flat-closing question has a positive answer in the real analytic setting of the original problem by Bangert–Schroeder [BSc]. In the smooth Riemannian setting, however, whether every flat plane in the universal cover must produce an immersed totally geodesic flat torus in the quotient remains open in this level of generality.

The rank used in modern rigidity theory is the geodesic rank: the dimension of the space of parallel Jacobi fields along a geodesic, with the velocity field included. Locally symmetric spaces of noncompact type have rank in this sense equal to their usual symmetric space rank. For compact manifolds of nonpositive curvature, the higher-rank rigidity theorem says, roughly, that an irreducible manifold whose geodesics all have rank at least two is locally symmetric; otherwise the universal cover splits as a nontrivial Riemannian product [Ba, BBE, BBS, BS, EH]. Later work shows that the same philosophy persists beyond nonpositive curvature: Watkins proved a higher-rank rigidity theorem for manifolds without focal points [Wa]. This gives the expected metric rigidity once "rank greater than one" is interpreted as a global geodesic-rank hypothesis.

The flat-plane-to-flat-torus part is different. The flat torus theorem proves one direction: if the fundamental group contains a free abelian subgroup of rank k, then a nonpositively curved compact manifold contains a corresponding flat k-torus [BH]. The converse, often called the flat closing problem, asks whether the existence of a geometric k-flat forces such a periodic flat. Bangert–Schroeder proved the exact k-dimensional statement for compact real analytic manifolds of nonpositive curvature: every k-flat is accompanied by a closed k-flat [BSc]. Their introduction identifies this theorem explicitly as an answer to Yau's Problem 65. The proof combines three ingredients which explain the analyticity hypothesis. First, maximal flats are organized by flags of singular subspaces. Second, real analyticity makes the resulting set of well-structured flats subanalytic and yields a compact invariant manifold in the frame bundle. Third, after eliminating normal affine Jacobi fields, the geodesic flow is normally hyperbolic on a saturated compact set, and a shadowing argument produces a periodic flat. Thus the analytic version of Yau's flat-plane question is settled. The smooth problem is subtler: in general CAT(0) complexes, flat planes need not behave periodically [Wi], although positive results exist for special classes such as Euclidean buildings [Ci]. Thus the original question has split into a solved higher-rank rigidity theorem, an analytic flat-closing theorem, and an open smooth flat-closing problem.

Recent literature does not appear to have closed the smooth Riemannian flat-closing problem. The most useful progress for this problem is therefore conceptual: the no-focal-points rank-rigidity theorem of Watkins shows that the rank-rigidity part survives beyond sectional curvature $\leqslant 0$ [Wa], while Ciobotaru's building result and Wise's CAT(0) example show that the flat-closing problem is sensitive to the category in which one works [Ci, Wi]. A modern report of this problem should therefore separate "rank rigidity" from "flat closing" and state clearly which category is being used: analytic Riemannian, smooth Riemannian, CAT(0) complex, Euclidean building, or group action.

Related open problems.

1. *Original direction.* Prove or disprove the Riemannian flat closing conjecture for closed nonpositively curved manifolds: does every k-flat in the universal cover force a virtually $\mathbb{Z}^k$ subgroup?

2. *Reference-driven formulation.* Decide the smooth flat closing problem under additional hypotheses, such as finite volume, rank-one/non-rank-one decomposition, or restrictions on the Tits boundary.

3. *Reference-driven formulation.* Describe quantitative versions of rank rigidity: if most geodesics carry almost-parallel Jacobi fields, must the metric be close to a locally symmetric or product metric?

4. *Reference-driven formulation.* Extend rank-rigidity phenomena to manifolds without focal points, or identify the exact hypotheses under which such extensions hold.

5. *Reference-driven formulation.* Compare the analytic Bangert–Schroeder proof with the building and CAT(0) literature to isolate which recurrence, regularity, or boundary-at-infinity input creates periodic flats [BSc, Ci, Wi].

6. *Reference-driven formulation.* Replace the subanalytic compactness step in Bangert–Schroeder by a C^∞ tame-geometry or dynamical substitute, while retaining enough normal hyperbolicity for the shadowing construction [BSc].

7. *AI-generated formulation.* Find a dynamical criterion on the geodesic flow which detects when an aperiodic flat plane can be approximated by periodic flat tori.

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Problem 66

(Kuiper). Let M be the surface obtained by attaching a handle to $\mathbb{RP}^2$. Can M be immersed into $\mathbb{R}^3$ with a "two-piece" property, i.e. every plane that cuts the surface divides the image into exactly two components?

See Kuiper's survey paper [K] in the Chern Symposium volume.

Status. Solved negatively in the smooth category. Haab proved that the real projective plane with one handle admits no smooth tight immersion into $\mathbb{R}^3$ [H]. The distinction between smooth and piecewise-linear geometry is essential: Cervone later constructed a tight polyhedral immersion of exactly this surface [C2].

The two-piece property is one of the geometric forms of tightness. For a smooth immersion it is equivalent to attaining the Chern–Lashof lower bound for total absolute curvature, and generic height functions are perfect Morse functions [CL, K, CR]. The surface in the question is the nonorientable surface of genus three, with Euler characteristic -1. Haab's theorem says more precisely that its normalized total absolute curvature has infimum 5, but no smooth immersion attains this infimum [H].

The mechanism of Haab's proof is global. A generic planar projection of a hypothetical tight immersion has fold curves with several types of saddle singularities. Cutting the surface into the regions determined by these curves and counting their incidences forces two regions to contain local maxima for one height direction. A plane just below the lower maximum then separates those two caps from the rest of the surface, producing at least three pieces and contradicting tightness [C1, C2]. Smooth fold and profile curves are therefore not a technical convenience but the point at which the obstruction enters.

Cervone's polyhedral construction shows exactly why the category matters. His model has 13 vertices, 42 edges, and 28 triangular faces, hence Euler characteristic -1. It is an immersion because every vertex star is embedded. Its tightness is checked combinatorially: each locally extreme vertex lies on the global convex hull, every other vertex lies inside the convex hull of its neighbors, and the induced map on fundamental groups has the required injectivity property [C2]. The positive curvature that Haab's smooth argument forces can be avoided at a non-extreme polyhedral vertex. Nevertheless, Cervone showed that this is not merely a local smoothing issue [C1], and subsequently proved that every tight polyhedral immersion of this surface is asymmetric [C3].

Related open problems.

1. *Original direction, resolved.* Haab's theorem gives a negative answer for smooth immersions; determine the sharp quantitative gap above total absolute curvature 5 under additional geometric bounds [H].

2. *Reference-driven formulation.* Explain intrinsically why Cervone's polyhedral immersion cannot be smoothed through tight immersions, despite the absence of a local smoothing obstruction [C1, C2].

3. *Reference-driven formulation.* Classify tight polyhedral immersions of this surface up to image homotopy and determine the minimum number of vertices and faces.

4. *Reference-driven formulation.* Quantify the instability of the two-piece property under smoothing: how many plane sections must acquire a third component, and at what scale?

5. *Reference-driven formulation.* Extend Cervone's asymmetry theorem to a classification of possible finite symmetry groups and regular homotopy classes [C3].

6. *AI-generated formulation.* Construct a computable certificate, from a triangulation and its oriented matroid, which verifies the two-piece property without checking all affine half-spaces.

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IV. The Spectrum

Problem 67

Let Ω_1 and Ω_2 be two bounded smooth domains in $\mathbb{R}^2$ so that the eigenvalues (counting multiplicity) of the Laplacian acting on functions defined on Ω_1 and Ω_2 having zero boundary conditions are the same. Is Ω_1 isometric to Ω_2?

This is an old problem. For closed manifolds one can formulate an analogous problem, however, the answer is negative. This is by virtue of examples of Milnor [M] and Vignéras [V], the latter providing a two-dimensional counterexample with negative curvature.

Status. Open for bounded smooth planar domains in the stated generality. The answer is negative for planar domains with corners, by Gordon–Webb–Wolpert [GWW], and positive results are known under strong analytic, symmetry, or special-shape hypotheses [Z2, HZ1, HZ2].

This is Kac's question, "Can one hear the shape of a drum?" [K]. For closed manifolds, Milnor's flat tori and Vignéras' arithmetic hyperbolic surfaces show that the spectrum does not determine the metric in general [M, V]. In the planar domain setting, the famous counterexamples of Gordon–Webb–Wolpert are isospectral but not isometric, but their boundaries have corners [GWW]. Thus they do not settle the smooth version stated here.

The strongest positive results come from wave trace invariants and billiard dynamics. Zelditch proved spectral determination for certain real analytic plane domains with a reflection symmetry and a nondegenerate bouncing-ball orbit [Z2]. Hezari–Zelditch extended this circle of ideas to analytic domains with higher symmetry and to ellipses of sufficiently small eccentricity, which are spectrally unique among all smooth domains [HZ1, HZ2]. Nevertheless, the general smooth problem, the general real analytic problem without symmetry, and broad convex smooth classes remain open.

Recent work continues to strengthen the billiard-dynamical side of the problem. De Simoi–Kaloshin–Wei proved dynamical spectral rigidity for sufficiently smooth $\mathbb{Z}_2$-symmetric strictly convex domains near a circle [DKW]. Kaloshin–Sorrentino proved a local Birkhoff conjecture near ellipses, with applications to spectral rigidity of elliptic domains [KS], and Koval obtained local spectral rigidity for almost every ellipse under a boundary-integrability hypothesis [Ko]. Fierobe–Kaloshin–Sorrentino recently proved deformational length spectral rigidity for dihedrally symmetric domains close to an ellipse [FKS]. These results do not produce smooth isospectral counterexamples or solve Kac's problem, but they indicate that wave-trace and billiard length-spectrum methods are now most effective near highly structured convex tables.

Related open problems.

1. *Original direction.* Construct, or rule out, nonisometric isospectral bounded smooth domains in $\mathbb{R}^2$.
2. *Original direction.* Decide whether every real analytic bounded planar domain is spectrally determined, perhaps under generic billiard nondegeneracy assumptions.
3. *Reference-driven formulation.* Prove spectral determination or spectral rigidity for broad classes of convex smooth domains.
4. *Reference-driven formulation.* Determine which billiard invariants of a smooth planar domain are encoded by the Dirichlet or Neumann spectrum.
5. *Reference-driven formulation.* Bridge dynamical length spectral rigidity for convex billiards with Laplace spectral rigidity, especially near ellipses and near the circle [DKW, KS, Ko, FKS].
6. *AI-generated formulation.* Find a smoothing principle which starts from a polygonal isospectral pair and decides whether the isospectrality can survive any smoothing of the corners.

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Problem 68

In Problem 67, suppose the spectra of Ω_1 and Ω_2 are equal except for a finite number of exceptions. Are the two spectra in fact identical? One can ask a similar question if the set of exceptions is infinite but has density zero.

Status. Open. No general theorem is known which forces equality of the full Dirichlet spectra from equality outside a finite set, or from equality outside a density-zero exceptional set, in the smooth planar domain setting.

The question is a refinement of the inverse spectral problem in Problem 67. Most standard spectral-geometric tools are asymptotic: Weyl's law, heat trace coefficients, and wave trace singularities measure high-energy distributional information rather than the exact placement of a finite list of eigenvalues [K, Z1, DH]. A finite spectral discrepancy changes either trace by a globally smooth term, so it is invisible to the singular-support information used in most inverse results.

The density-zero formulation is even more delicate. A sparse exceptional subsequence may still contribute to lower-order or oscillatory terms in trace formulae, depending on its growth and on where it lies in the spectrum. For this reason one must specify whether the spectra are compared as sets, ordered sequences with multiplicity, counting functions with an error term, or trace distributions. Under these different interpretations the problem may have different analytic strength.

A useful way to phrase the finite-exception case is through the heat trace. Let A and B be the two finite exceptional multisets. Then

$$D(t) = \operatorname{Tr}(e^{-t\Delta_1}) - \operatorname{Tr}(e^{-t\Delta_2}) = \sum_{a\in A} e^{-at} - \sum_{b\in B} e^{-bt}$$

is entire in t, and

$$D(t) = \sum_{m\geqslant 0} \frac{(-t)^m}{m!} \left(\sum_{a\in A} a^m - \sum_{b\in B} b^m \right).$$

For a smooth planar domain the Dirichlet heat trace has a complete expansion in powers $t^{-1+j/2}$ [Gr, Va]. Consequently, all half-integer heat coefficients of the two domains must agree, whereas the differences of the integer coefficients are precisely constrained by the power sums of A and B. This is substantially stronger than equality of area and perimeter, but it does not by itself make D vanish. If one could prove equality of all integer heat coefficients, then all power sums would agree and the two finite multisets would be identical.

The same obstruction appears in the other standard transforms. The difference of spectral zeta functions is a finite sum of entire functions $a^{-s} - b^{-s}$, so poles and residues do not see it. The difference of wave traces is a finite trigonometric sum and therefore smooth; hence the length-spectrum singularities also agree. Problem 68 therefore asks for a genuinely global uniqueness principle, not a formal consequence of the local trace expansions [DH, Gr, Va].

Related open problems.

1. *Original direction.* Decide whether two smooth planar domains whose Dirichlet spectra agree outside finitely many eigenvalues must be fully isospectral.

2. *Original direction.* Formulate sharp density-zero hypotheses under which missing or altered eigenvalues cannot change the geometric inverse problem.

3. *Reference-driven formulation.* Determine whether equality of all sufficiently large eigenvalues forces equality of heat invariants, wave invariants, and then the missing low spectrum.

4. *Reference-driven formulation.* Decide whether the integer heat coefficients of two smooth planar domains can differ by the moment sequence of two distinct finite positive multisets while all half-integer coefficients agree [Gr, Va].

5. *Reference-driven formulation.* Study the analogous question for closed negatively curved manifolds, where the wave trace is tied to the length spectrum.

6. *Reference-driven formulation.* Classify finite exponential errors in heat traces which are compatible with smooth-domain heat asymptotics and boundary wave invariants [BGM, Gi, Gr, Va].

7. *AI-generated formulation.* Develop a "spectral unique-continuation" principle: if two spectral measures agree at high energy up to a finite-rank error, determine when the error must vanish.

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Problem 69

Let $g(t)$ be a one-parameter family of metrics on a compact manifold with the same spectrum for the Laplacian. Prove that the metrics $g(t)$ are isometric to each other.

Guillemin and Kazhdan [GK] proved that this is the case if the manifold is a surface of negative curvature, or if the manifold is suitably negatively pinched when the dimension of the manifold is greater than two.

Status. False in this generality. There are nontrivial continuous isospectral deformations of compact Riemannian metrics [GW, Sc1, Sc2]. The statement is true in important rigid classes, especially under negative-curvature hypotheses [GK, CS].

The problem asks for spectral rigidity of a one-parameter isospectral family. Guillemin–Kazhdan proved such rigidity for negatively curved surfaces and for suitably pinched higher-dimensional negatively curved manifolds [GK]. Croke–Sharafutdinov later proved spectral rigidity for compact negatively curved manifolds in all dimensions [CS], and Sharafutdinov obtained related local audibility results for hyperbolic metrics [Sh]. These results use the relation between the Laplace spectrum, the wave trace, the marked length spectrum, and tensor tomography on negatively curved manifolds.

Without curvature or dynamical assumptions, the conclusion fails. Gordon–Wilson constructed continuous families of Dirichlet- and Neumann-isospectral metrics with boundary whose local geometry varies [GW]. Schueth then constructed continuous isospectral families on closed simply connected manifolds, including products of spheres, and on S^7 [Sc1, Sc2]. In her constructions changes in scalar-curvature critical values prove nonisometry, while a torus-action transplantation criterion preserves the function spectrum. Thus the modern problem is not to prove rigidity for all compact manifolds, but to identify the geometric classes in which isospectral deformations are forced to be trivial. In the planar-domain direction, the ellipse is C^∞-spectrally rigid in the appropriate setting [HZ], building on earlier work of Guillemin–Melrose [GM] and later wave-invariant methods [PT].

Recent rigidity work continues this deformation-theoretic viewpoint in more specialized settings. The billiard literature has obtained local and deformational length-spectral rigidity near circles and ellipses [DKW, KS, Ko, FKS]. On closed manifolds, Henheik–Kaloshin–Li–Vig proved rigidity for certain Laplace-isospectral deformations inside a generic Liouville conformal class on the two-torus, assuming linear dependence on the deformation parameter [HKLV]. Their proof first uses a wave-trace noncancellation theorem to recover rational invariant tori of the geodesic flow, and then applies the second variation of the closed-geodesic energy: preservation of one rational torus forces the linear conformal variation to vanish. These results confirm that the right modern replacement for the false global statement is a collection of rigidity theorems in dynamically controlled or integrable-nearby classes.

Related open problems.

1. *Original direction.* Characterize geometric or dynamical hypotheses under which every smooth isospectral deformation is trivial.

2. *Reference-driven formulation.* Extend negative-curvature spectral rigidity to broader classes, such as manifolds with no conjugate points or rank-one nonpositive curvature.

3. *Reference-driven formulation.* Decide which convex planar domains are C^∞-spectrally rigid under smooth domain deformations.

4. *Reference-driven formulation.* Understand whether known isospectral deformation mechanisms can occur under curvature sign restrictions, symmetry restrictions, or simple topology.

5. *Reference-driven formulation.* Remove linearity, genericity, or conformal-class restrictions from Liouville-torus spectral rigidity [HKLV].

6. *AI-generated formulation.* Develop a deformation theory whose infinitesimal kernel separates genuine geometric symmetries from Sunada-type or torus-action isospectral mechanisms.

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Problem 70

Let Ω be a bounded domain in $\mathbb{R}^2$. Let λ_i be the spectrum of the Laplacian acting on functions with zero boundary data (again, and henceforth, counting with multiplicity). Prove that

$$\lambda_i \geqslant \frac{4\pi i}{\text{area}(\Omega)} .$$

This was conjectured by Pólya [Pól] and was proved by him in the case that Ω can tile $\mathbb{R}^2$. One can formulate a similar question for eigenvalues of the Neumann problem (with the inequality in the opposite direction).

Status. Open in general. Pólya proved the conjecture for domains which tile the Euclidean plane [Pól], and Li–Yau proved the corresponding sharp lower bound for averages of Dirichlet eigenvalues [LY]. These average estimates do not imply the individual estimate above.

Historically, Pólya's theorem for tiling domains is the first main case where the Weyl constant gives a term-by-term bound. The Li–Yau inequality gives, in dimension two,

$$\sum_{j=1}^{k} \lambda_j \geqslant \frac{2\pi}{\text{area}(\Omega)} k^2,$$

which is the sharp Weyl-order lower bound for the sum of the first k eigenvalues. This is often described as the averaged form of Pólya's conjecture.

More recent work has proved the conjecture for additional special classes and has produced stronger universal inequalities short of the full termwise conjecture. Filonov–Levitin–Polterovich–Sher proved the conjecture for the disk, making it the first non-tiling planar domain known to satisfy it, as well as for arbitrary planar sectors and, in the Dirichlet case, Euclidean balls in every dimension [FLPS1]. Their proof converts Bessel-zero counts into uniform weighted lattice-point estimates; only a short Neumann-disk interval requires a rigorous computer-assisted verification. The same authors then proved the Dirichlet conjecture for annuli by combining variational comparison, Bessel phase functions, refined trapezoidal lattice sums, and another certified finite computation [FLPS2]. Steinerberger obtained new two-point lower bounds for Dirichlet eigenvalues; for instance, in dimension two one has

$$2\lambda_n + \lambda_{2n} \geqslant \frac{10\pi n}{|\Omega|},$$

which strengthens the information supplied by the Berezin–Li–Yau method but still does not settle the individual Pólya inequality [S].

The Neumann analogue, with the reverse inequality for the counting function or for the individual eigenvalues, is a parallel open problem in general. For planar convex domains, Filonov obtained the improved universal lower estimate

$$N_N(\Omega, \lambda) \geqslant \frac{|\Omega|\lambda}{2\sqrt{3}\, j_0^2},$$

where j_0 is the first positive zero of J_0 [Fi]. This improves Kröger's general constant but remains below the conjectural $|\Omega|\lambda/(4\pi)$. Guo–Miao–Wang–Zhan subsequently sharpened the ball estimates and verified the Neumann conjecture for circular cylinders in $\mathbb{R}^3$ [GMWZ].

Two recent approaches clarify how close present methods come to the termwise conjecture. Jiang–Lin proved an explicit ε-loss version for every bounded Lipschitz domain: above a computable geometric threshold,

$$k \leqslant (1+\varepsilon)\frac{|\Omega|\omega_n}{(2\pi)^n}\lambda_k^{n/2}.$$

Their proof combines a minimal containing rectangle, Whitney decomposition, domain monotonicity, and an explicit Weyl-remainder estimate [JL1]. Ji–Luo obtain improved Berezin–Li–Yau type lower bounds for averaged sums of Dirichlet eigenvalues of the Laplacian and poly-Laplacian on bounded Euclidean domains [JL2]. Such results sharpen the analytic information available below Pólya's conjecture, but they also reinforce the gap between sharp averaged bounds and the individual inequality $\lambda_i \geqslant 4\pi i/|\Omega|$.

Related open problems.

1. *Original direction.* Prove Pólya's Dirichlet conjecture for all bounded planar domains, preferably under minimal boundary regularity assumptions.

2. *Original direction.* Prove or disprove the Neumann analogue in the same generality.

3. *Reference-driven formulation.* Extend the known positive cases from tiling domains, balls, circular sectors, and annuli to broader families such as convex domains or simply connected smooth domains.

4. *Reference-driven formulation.* Find a mechanism which upgrades sharp averaged estimates, such as the Li–Yau inequality, to term-by-term estimates under additional geometric hypotheses.

5. *Reference-driven formulation.* Compare the Filonov–Levitin–Polterovich–Sher annulus method with refined lattice-counting and Bessel-phase estimates for other separable planar domains [FLPS1, FLPS2].

6. *Reference-driven formulation.* Determine whether the explicit ε-loss threshold of Jiang–Lin can be combined with a finite verified computation for broad geometric classes, as in the disk and annulus proofs [JL1, FLPS1, FLPS2].

7. *AI-generated formulation.* Identify a geometric correction term which interpolates between the Berezin–Li–Yau average inequality and the individual Pólya bound, and determine whether its nonnegativity is equivalent to Pólya's conjecture.

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Problem 71

Let M be a two-dimensional, closed, compact surface. Can one find a universal constant C so that

$$\frac{\lambda_i}{i} \leqslant \frac{C(g+1)}{\text{area}(M)} \ ?$$

Here, g is the genus of M. If M is diffeomorphic to $\mathbb{S}^2$, is $\lambda_i(M)$ no greater than $\lambda_i(\mathbb{S}^2)$ where $\mathbb{S}^2$ is equipped with a metric of constant curvature $4\pi/\text{area}(M)$?

In the case $i = 1$, this is known to be true. The case when M is diffeomorphic to $\mathbb{S}^2$ was proved by Hersch [He]. For M orientable and $g > 0$, this was proved by Yang and Yau [YY]. Recently, Li and Yau [LY] were able to find similar bounds for nonorientable surfaces.

Status. Partially resolved. The existence of a universal upper bound of the stated form was solved by Korevaar [K]. The optimal constants, and especially the sharp dependence on the genus and on the index, remain active questions. The proposed higher-eigenvalue comparison with the round metric on $\mathbb{S}^2$ is true for the first positive eigenvalue by Hersch [He], but is false as a literal statement for higher eigenvalues; the sharp maximizers on the two-sphere are obtained only in a limiting sense by degeneration to several touching round spheres [KNPP].

The historical line begins with Hersch's sharp bound $\lambda_1 \,\text{area} \leqslant 8\pi$ on the two-sphere [He]. Yang–Yau then proved the orientable genus-g first-eigenvalue bound using meromorphic maps to the sphere [YY], and Li–Yau introduced conformal volume, giving a flexible framework which also applies to nonorientable surfaces [LY]. Korevaar's theorem extended the picture from the first eigenvalue to all eigenvalues, proving a bound of the form

$$\lambda_i(M)\ \text{area}(M) \leqslant C(g+1)i$$

for a universal constant C [K]. Grigoryan–Yau later gave an efficient capacitary decomposition method for proving such inequalities [GY].

The sharp theory is more subtle. On $\mathbb{S}^2$, the round metric maximizes only the first positive eigenvalue. For the k-th nonzero eigenvalue, Karpukhin–Nadirashvili–Penskoi–Polterovich proved the sharp isoperimetric inequality; the supremum is approached by metrics degenerating to a union of k identical round spheres, and for $k \geqslant 2$ it is not attained by a smooth round metric [KNPP]. Thus the original sphere-comparison question should be recorded as solved positively only for $i = 1$ and negatively for higher eigenvalues in the literal form stated.

Recent work has also clarified the variational side of the problem. For a fixed conformal class, Karpukhin–Nadirashvili–Penskoi–Polterovich show that conformal maximizers are either conical extremal metrics or limits with bubble trees [KNPP2]. Karpukhin–Kusner–McGrath–Stern developed equivariant eigenvalue-optimization methods which produce minimal surfaces in balls and spheres and give new existence tools for maximizing metrics under symmetry [KKMS]. Petrides recently proved existence of optimal metrics for a wide class of combinations of Laplace eigenvalues on closed orientable surfaces, relating them to minimal eigenmaps into ellipsoids [Pe]. Karpukhin–Petrides–Stern then completed the first-eigenvalue existence theory: every closed surface, orientable or nonorientable, admits a metric maximizing λ_1 area, smooth except at finitely many conical points [KPS]. Their key step is the strict topological monotonicity

$$\Lambda_1(M \# \mathbb{RP}^2) > \Lambda_1(M), \qquad \Lambda_1(M \# T^2) > \Lambda_1(M).$$

Refined extension operators on shrinking cross-caps and handles force a limiting first-eigenfunction map with vanishing differential at the attachment point, contradicting the constant-energy identity for such an eigenmap. This theorem settles existence for $i = 1$, but it does not give the sharp value, uniqueness, or a corresponding result for all higher eigenvalues. These results do not replace Korevaar's universal bound, but they make the sharp version of the problem a question about compactness, bubbling, conical singularities, and eigenmap geometry.

Related open problems.

1. *Original direction.* Determine the optimal universal constant in Korevaar's inequality, or the optimal asymptotic constant, for each fixed genus.

2. *Reference-driven formulation.* For each closed surface and each k, describe whether the sharp supremum of λ_k area is attained by a smooth metric, by a metric with conical singularities, or only by a degenerating sequence. For $k = 1$, existence of a conical maximizer is now known for every closed surface [KPS].

3. *Reference-driven formulation.* Extend the sharp higher-eigenvalue theory from $\mathbb{S}^2$ to surfaces of positive genus and to nonorientable surfaces.

4. *Reference-driven formulation.* Clarify the best possible linear dependence on g and k in upper bounds for λ_k area.

5. *Reference-driven formulation.* Relate optimal metrics for combinations of eigenvalues to minimal eigenmaps into spheres or ellipsoids, and determine which topological types admit smooth maximizers [Pe, KKMS].

6. *AI-generated formulation.* Develop a compactification of the space of extremal metrics for surface eigenvalues in which bubbling limits and smooth extremal metrics appear as different boundary strata.

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Problem 72

Study the discrete spectrum of a complete manifold whose curvature is bounded and negative and whose volume is finite. When is it nonempty? What is its asymptotic behavior and relation to the closed geodesics?

Let $M = \{(x, y) \in \mathbb{R}^2 \mid y > 0\}/\Gamma$, where Γ is a congruence subgroup of $\mathrm{SL}(2, \mathbb{Z})$. It is an old conjecture that λ_1 for M is at least $\frac{1}{4}$. Selberg [Se] proved that $\lambda_1 \geqslant \frac{3}{16}$.

It will also be important to study the continuous spectrum of a general complete manifold with finite volume. Hopefully, one can obtain some kind of L^2 index theorem for elliptic operators for these manifolds.

Status. Open in this generality, with substantial partial results in locally symmetric and arithmetic cases. For congruence quotients of the upper half-plane, Selberg's $\lambda_1 \geqslant 1/4$ conjecture remains open. Selberg proved the classical lower bound $\lambda_1 \geqslant 3/16$ [Se], and the best known general bound toward the conjecture is the Kim–Sarnak bound $\lambda_1 \geqslant 975/4096$ [KS], following earlier progress of Luo–Rudnick–Sarnak [LRS].

There are two intertwined issues. First, in finite-volume hyperbolic surfaces and more general cusp manifolds, the Laplacian has continuous spectrum as well as possible discrete spectrum. Thus one cannot usually study the asymptotics of the discrete spectrum in isolation from the continuous spectrum and the scattering matrix. Second, embedded eigenvalues can be unstable under perturbation and may deform into resonances; this phenomenon is already visible for cofinite hyperbolic surfaces in the work of Phillips–Sarnak [PhS].

For this reason, resonances are often the more stable replacement for the discrete spectrum in noncompact finite-volume spectral geometry. They encode scattering poles and are tied to trace formulae, closed geodesics, and the analytic continuation of the resolvent. General finite-volume negatively curved manifolds with bounded curvature do not yet have a single clean theory parallel to the compact Weyl law; the correct formulation depends on cusp geometry, scattering theory, and the choice

of operator. The hoped-for L^2 index theorem is also part of a broader program: strong results are known in special settings, but the fully geometric variable-curvature finite-volume formulation remains a natural open direction.

For general finite-volume negative curvature, Lott's analysis remains one of the important geometric references: for pinched finite-volume manifolds he proves that every space of L^2 harmonic p-forms is finite dimensional, and the essential spectrum of the p-form Laplacian can be described through ordinary differential operators associated to the ends, and examples show that gaps can occur even for metrics with curvature pinched close to -1 [Lo]. On the geometric side, Nguyen Phan constructed finite-volume negatively curved manifolds with unexpected cusp and fundamental-group behavior, illustrating why a single locally symmetric model is too restrictive for the general problem [NP]. On the index-theory side, Albin–Rochon proved families index formulae for manifolds with hyperbolic cusp singularities, extending earlier fibered cusp methods and giving a natural framework for the L^2 index direction in cusp geometries [AR].

Related open problems.

1. *Original direction.* Prove Selberg's eigenvalue conjecture $\lambda_1 \geqslant 1/4$ for congruence subgroups of $\mathrm{SL}(2,\mathbb{Z})$.
2. *Reference-driven formulation.* Develop a Weyl-type asymptotic or trace formula for resonances on finite-volume negatively curved manifolds with variable curvature and controlled cusp geometry.
3. *Reference-driven formulation.* Determine which embedded eigenvalues of finite-volume negatively curved manifolds are stable under geometric perturbation, and describe precisely how unstable eigenvalues become resonances.
4. *Reference-driven formulation.* Formulate and prove an L^2 index theorem for broad classes of finite-volume complete manifolds with bounded negative curvature.
5. *Reference-driven formulation.* Classify how cusp geometry controls essential spectrum and spectral gaps for the p-form Laplacian in variable negative curvature [Lo, NP].
6. *AI-generated formulation.* Find a geometric condition on the cusps under which the resonance counting function, the closed geodesic data, and the L^2 index contribution fit into one uniform trace formula.

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Problem 73

The behavior of the spectrum of a compact manifold of negative curvature is quite different for dimension two and dimension three. For example, R. Schoen [S] proved that for a three-dimensional hyperbolic space form (with curvature -1), $\lambda_1 \geqslant \frac{c}{\mathrm{vol}(M)^2}$ where c is a universal constant. This is certainly false for surfaces (See Schoen–Wolpert–Yau [SWY]).

Is it true that $\lambda_1 \mathrm{vol}(M)^2$ has an upper bound if M is a three-dimensional hyperbolic space?

Status. False in general for compact hyperbolic three-manifolds. Schoen's estimate gives a universal lower bound of order $\mathrm{vol}(M)^{-2}$ [S], but there are towers of compact arithmetic hyperbolic three-manifolds with volume tending to infinity and a uniform positive spectral gap. Along such a tower, $\lambda_1 \mathrm{vol}(M)^2$ tends to infinity, so no universal upper bound of the proposed form can hold.

The contrast with surfaces is important. For compact hyperbolic surfaces, degeneration in moduli space can force many small eigenvalues, and Schoen–Wolpert–Yau gave geometric bounds for low eigenvalues in terms of the topology and short geodesics [SWY]. In dimension three, Mostow rigidity removes Teichmüller-type deformations of a fixed finite-volume hyperbolic structure, and arithmetic congruence towers give a different phenomenon: the volume can grow while the first positive eigenvalue stays uniformly bounded away from zero. Uniform spectral gaps for congruence covers follow from the representation-theoretic theory of automorphic forms and property τ; see Burger–Sarnak [BS], Clozel [Cl], and the effective treatment of Kelmer–Silberman [KS].

The nonarithmetic case remains much less rigid from the viewpoint of available estimates. It is still natural to ask for geometric upper and lower bounds for λ_1 in terms of volume, diameter, Cheeger constants, Heegaard or congruence data, or the topology of finite covers. The Schoen–Yau scalar-curvature estimate [SY] belongs to a related but different circle of questions, where lower scalar curvature and minimal-surface methods control geometric concentration.

Recent work suggests two useful refinements of this discussion. First, Benjamini–Schramm convergence has become a natural language for large volume locally symmetric spaces: Abert–Bergeron–Biringer–Gelander– Nikolov–Raimbault–Samet developed the framework for asymptotic L^2-invariants [ABBGNRS], and Fraczyk–Hurtado–Raimbault recently proved Benjamini–Schramm convergence for arithmetic locally symmetric spaces in broad generality [FHR]. Second, spectral gaps in dimension three also interact with topology beyond functions: Abdurrahman–Adve– Giri–Lowe–Zung constructed closed hyperbolic integer homology spheres with volume tending to infinity and a uniform spectral gap for coclosed one-forms [AAGLZ]. These results do not restore the false upper bound for $\lambda_1 \mathrm{vol}(M)^2$, but they point to refined questions involving towers, local convergence, homology, and spectra of differential forms.

A complementary result gives an upper bound of a quite different kind. Bonifacio–Mazáč–Pal combined associativity identities for products of Laplace eigensections with the Selberg trace formula and rigorous linear programming to prove

$$\lambda_1(M) < 47.32$$

for every closed hyperbolic three-manifold [BMP]. This universal bound on λ_1 is compatible with the negative answer: after multiplication by $\mathrm{vol}(M)^2$, congruence towers with a uniform positive gap still make the original quantity diverge.

Related open problems.

1. *Original direction.* Characterize which towers of closed hyperbolic three-manifolds have a uniform positive lower bound for λ_1.

2. *Reference-driven formulation.* Determine sharp upper and lower bounds for λ_1 in nonarithmetic towers, especially under geometric convergence or Benjamini–Schramm convergence assumptions.

3. *Reference-driven formulation.* Determine the optimal universal upper bound for λ_1 on closed hyperbolic three-manifolds and sharpen the trace-formula bound 47.32 [BMP].

4. *Reference-driven formulation.* Relate spectral gap quantitatively to Heegaard genus, Cheeger constant, injectivity radius distribution, and the topology of covers.

5. *Reference-driven formulation.* Find analogues of the Schoen–Wolpert–Yau low-eigenvalue description for degenerating sequences of hyperbolic three-manifolds.

6. *Reference-driven formulation.* Compare function spectral gaps with coclosed one-form spectral gaps and torsion-homology growth in large-volume hyperbolic three-manifold towers [AAGLZ].

7. *AI-generated formulation.* Separate the possible growth rates of $\lambda_1 \mathrm{vol}(M)^2$ into arithmetic, congruence, and nonarithmetic classes, and decide which rates are realizable.

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Problem 74

Let M be a compact surface. Let $\lambda_1 \leqslant \lambda_2 \leqslant \cdots$ be the spectrum of M and $\{\phi_i\}$ be the corresponding eigenfunctions. For each i, the nodal set $\{x \mid \phi_i(x) = 0\}$ is a one-dimensional rectifiable simplicial complex. Let L_i be the length of such a set. It is not difficult to prove that $\liminf_{i\to\infty} L_i/\sqrt{\lambda_i}$ has a positive lower bound depending only on the area of M. (This was independently observed by Brüning [B].)

It seems more difficult to find an upper bound of $\limsup_{i\to\infty} L_i/\sqrt{\lambda_i}$.

Status. Partially solved. The lower bound $L_i \geqslant c\sqrt{\lambda_i}$ is now known for smooth compact manifolds in all dimensions by Logunov [Lo2]. The corresponding sharp upper bound $L_i \leqslant C\sqrt{\lambda_i}$ is solved for real analytic metrics by Donnelly–Fefferman [DF1], but remains open for general smooth surfaces. Thus the stated upper-bound problem for smooth compact surfaces is still open.

The early surface theory begins with Brüning's lower bound [B] and Dong's analysis of nodal sets on Riemann surfaces [D]. Donnelly and Fefferman proved Yau's conjectural upper and lower bounds in the real analytic case [DF1], and obtained general smooth-surface upper bounds which are weaker than the conjectural order [DF2]. In the convention where the eigenvalue is written as λ^2, the conjectural length bound is $O(\lambda)$, while the classical smooth bound was of larger order.

The modern breakthrough concerns lower bounds and quantitative unique continuation. Colding–Minicozzi and Sogge–Zelditch obtained important lower bounds in higher dimensions [CM, SoZ]. Logunov then proved Nadirashvili's conjecture and the lower bound in Yau's conjecture for smooth compact manifolds [Lo2], and also established polynomial upper bounds for smooth nodal sets [Lo1]. In dimensions two and three, Logunov–Malinnikova obtained refined estimates; in dimension two they improved the known smooth upper exponent to $H^1(\{u = 0\}) \leqslant C\lambda^{3/4-\beta}$ when $\Delta u + \lambda u = 0$ [LM]. This is still not the conjectural $C\lambda^{1/2}$ upper bound.

Two related literatures help clarify what the sharp bound should mean. Logunov–Malinnikova's review emphasizes that the smooth upper bound is still the missing half of Yau's conjecture and that two-dimensional methods are tied to doubling indices and propagation of smallness [LM2]. Random-wave models and arithmetic random waves typically have nodal length of order $\sqrt{\lambda}$, with refined fluctuation theory surveyed by Rossi [Ro]; this supports the expected order but does not provide a deterministic upper bound. Results on quantum ergodic sequences, such as Hezari's improvements for inner radii of nodal domains under small-scale equidistribution [He], show that dynamical assumptions can improve some nodal-domain geometry along density-one subsequences, but they do not yet give the full sharp nodal-length upper bound.

Related open problems.

1. *Original direction.* Prove the sharp upper bound $H^1(\{\phi_i = 0\}) \leqslant C\sqrt{\lambda_i}$ for smooth compact surfaces.
2. *Reference-driven formulation.* Determine whether additional geometric assumptions, such as negative curvature or quantum ergodicity, imply sharp upper bounds for a full-density subsequence of eigenfunctions.
3. *Reference-driven formulation.* Find the optimal smooth upper exponent for nodal length on a compact surface if the full Yau upper bound remains out of reach.
4. *Reference-driven formulation.* Understand the distributional limits of nodal length measures for special eigenfunction sequences on smooth surfaces.
5. *Reference-driven formulation.* Compare deterministic doubling-index methods with random-wave predictions and quantum ergodic subsequence estimates [LM2, Ro, He].
6. *AI-generated formulation.* Develop a scale-by-scale doubling-index criterion which is both necessary and sufficient for the conjectural upper nodal-length bound on smooth surfaces.

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Problem 75

S.-Y. Cheng [C] proved that for a compact surface, the multiplicity of λ_i has an upper bound depending only on the genus of the surface. Can one generalize this to higher dimensions? Most likely this is not true without modification. What is the correct statement? For a compact surface with fixed genus g, can one exhibit explicitly a metric for which λ_i has maximal multiplicity?

Status. Partially resolved and partly false in the naive higher-dimensional form. On compact surfaces, Cheng's nodal set argument gives multiplicity bounds in terms of the genus and the eigenvalue index [C], with later refinements and extensions to Schrödinger operators and singular or bordered surfaces [Ko, Be]. In dimensions at least three, Colin de Verdière's prescribed finite spectrum theorem shows that no bound for the multiplicity of the j-th eigenvalue can depend only on topology and j [CdV].

The two-dimensional result is special because the nodal set of an eigenfunction is a graph on the surface. Euler characteristic and Courant's nodal domain theorem then constrain the possible order of vanishing and hence the dimension of an eigenspace. This is the mechanism behind Cheng's theorem and its later improvements. For example, Berdnikov–Nadirashvili–Penskoi proved the refined estimate

$$m(\mathbb{RP}^2, g, \lambda_{2\ell}) \leqslant 4\ell + 1$$

and analogous bounds for bordered and Steklov problems [BNP].

In higher dimensions the topology of nodal sets is too flexible for this argument to survive. Colin de Verdière proved that on any closed manifold of dimension at least three one can prescribe an arbitrary finite part of the Laplace spectrum, with multiplicities, by choosing a suitable smooth metric [CdV]. Thus the correct higher-dimensional replacement must impose extra geometric conditions such as curvature bounds, diameter or volume normalization, analytic metrics, locally symmetric structure, or restrictions to a fixed geometric class.

For fixed surfaces, the explicit extremal-metric problem remains interesting. It is related to the theory of extremal metrics for Laplace eigenvalues, minimal immersions by eigenfunctions, and degeneration by bubbling. The modern variational theory for maximizing eigenvalues on surfaces, including work of Nadirashvili, Sire, Petrides, and Karpukhin–Nadirashvili–Penskoi–Polterovich, gives a natural setting in which high multiplicities can occur [KNPP]. There is a useful logical distinction here. For a fixed surface and index, Cheng-type bounds imply that the possible multiplicities form a nonempty bounded set of integers, so some smooth metric realizes the largest integer. What remains difficult in Yau's question is to find such a metric explicitly and to determine the sharp integer.

Recent spectral-optimization work adds two useful refinements. Stability results of Karpukhin–Nahon–Polterovich–Stern show that metrics nearly maximizing the first or second normalized eigenvalue on several surfaces must be close, in a weak Sobolev sense, to the corresponding maximizing measures [KNPS]. The existence theory of Petrides for broad combinations of eigenvalues on orientable surfaces relates optimal metrics to minimal eigenmaps into ellipsoids [Pe]. These results suggest that maximal multiplicity should be studied together with extremal metrics, stability, and the representation theory of the eigenmap target, not only through nodal graphs. Karpukhin–Petrides–Stern have now proved that every closed surface has a metric maximizing the normalized first eigenvalue, smooth up to finitely many conical points [KPS]. This supplies a canonical extremal eigenmap on every topology, but it does not assert that the first-eigenvalue multiplicity is maximal, nor does it treat arbitrary index.

Related open problems.

1. *Original direction.* Determine sharp multiplicity bounds for the k-th Laplace eigenvalue on a closed surface of fixed genus.

2. *Original direction.* For each genus and index, classify the smooth metrics which achieve maximal multiplicity, and determine whether extremal-eigenvalue or eigenmap methods select any of them.

3. *Reference-driven formulation.* Find explicit metrics realizing maximal multiplicity when such maximizers exist.

4. *Reference-driven formulation.* Formulate higher-dimensional multiplicity bounds under natural geometric hypotheses, for example bounded curvature, bounded diameter, or locally symmetric structure.

5. *Reference-driven formulation.* Relate maximal multiplicity to extremal eigenvalue metrics, minimal eigenmaps, and stability of maximizers [KNPP, KNPS, Pe].

6. *AI-generated formulation.* Classify the geometric mechanisms which force high multiplicity on surfaces, separating symmetry, minimal-immersion, and degeneration mechanisms.

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Problem 76

Let M be a compact manifold and denote by f_i, $i = 1, 2, \ldots$, the eigenfunctions for the Laplacian on M. Show that the number of critical points of f_i is increasing with i.

Status. False in this generality. The literal monotonicity statement is not a valid general theorem. The number of critical points depends on the choice of eigenbasis inside eigenspaces, on degeneracy of critical points, and on the regularity of the metric. Moreover, Jakobson–Nadirashvili constructed metrics on $\mathbb{T}^2$ with a sequence of eigenfunctions whose eigenvalues tend to infinity but which have only 16 critical points [JN]. Thus no general theorem can say that the number of critical points must increase to infinity along every high-energy sequence.

The problem should therefore be reformulated. A first issue is regularity: for smooth metrics, critical sets can behave pathologically unless one imposes nondegeneracy or analyticity. Buhovsky–Logunov– Sodin constructed a metric on the two-torus for which infinitely many eigenfunctions have infinitely many isolated critical points [BLS]. For real analytic metrics, critical sets are more rigid, and Bakri proved an upper bound for the Hausdorff measure of critical sets of eigenfunctions on compact analytic manifolds [Ba]. This is closer in spirit to Donnelly–Fefferman type estimates for nodal sets.

Another related circle concerns eigenfunction growth. Sogge's L^p and sup-norm estimates give general upper bounds for eigenfunctions [So], and the sharpness of maximal growth is tied to geodesic recurrence and focal behavior. Sogge–Zelditch and later work show that saturation of maximal sup-norm growth is geometrically restricted [SoZ]. Toth–Zelditch studied uniformly bounded eigenfunctions in completely integrable settings and showed strong rigidity for flat tori under additional hypotheses [TZ]. These results do not count critical points directly, but they help identify geometric situations where eigenfunctions are unusually concentrated or unusually regular.

Two more recent viewpoints sharpen the reformulated problem. Hofmann and Täufer use the graph structure of nodal sets on compact orientable surfaces to bound critical points lying on the nodal set and sums of vanishing orders in terms of the eigenvalue index and the topology [HT]. This does not count all critical points of an eigenfunction, but it gives a concrete topological replacement for the false monotonicity statement. A complementary probabilistic model is provided by random plane waves: Beliaev–Cammarota–Wigman computed one- and two-point statistics for critical points of the random plane wave [BCW]. These results suggest that for generic or chaotic systems one should ask for asymptotic laws, local statistics, and fluctuation estimates rather than deterministic monotonicity.

Related open problems.

1. *Original direction.* For real analytic metrics, obtain sharp upper bounds for the number of isolated critical points, or for the complexity of the critical set, in terms of the eigenvalue.

2. *Reference-driven formulation.* Under generic metric assumptions, determine whether eigenfunctions have nondegenerate critical points and whether their number satisfies almost-sure or density-one asymptotic laws.

3. *Reference-driven formulation.* Classify compact manifolds or metrics which admit infinitely many eigenfunctions with uniformly bounded numbers of critical points.

4. *Reference-driven formulation.* Relate critical-point counts to nodal geometry, doubling indices, and eigenfunction concentration.

5. *Reference-driven formulation.* Bound critical points on the nodal set, and the total vanishing order there, using nodal-graph methods on surfaces [HT].

6. *AI-generated formulation.* Replace the false monotonicity question by a stable invariant: define a regularized critical complexity of an eigenspace and ask whether its average has a Weyl-law asymptotic.

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Problem 77

Let Ω be a bounded domain in $\mathbb{R}^2$. Denote by $\lambda_1(\Omega)$ and $\lambda_2(\Omega)$ the first and second (nonzero) eigenvalues of the Laplacian for functions with zero boundary values. Show

$$\frac{\lambda_2(\Omega)}{\lambda_1(\Omega)} \leqslant \frac{\lambda_2(D)}{\lambda_1(D)}$$

where D is the disk in $\mathbb{R}^2$, and that equality implies Ω is a disk.

This will mean that one can determine whether the drum is circular or not by knowing the first two tones of the drum. For more details, see [PPW].

Status. Solved. This is the Payne–Pólya–Weinberger conjecture for the first two Dirichlet eigenvalues, and it was proved sharply by Ashbaugh–Benguria [AB]. The equality case is rigid: equality holds only for the disk in the planar case, and more generally for the ball in Euclidean space.

The historical development begins with the universal eigenvalue inequalities of Payne–Pólya–Weinberger [PPW], who conjectured that the ratio λ_2/λ_1 for the Dirichlet Laplacian is maximized by the disk. Ashbaugh–Benguria proved the conjecture by a symmetrization argument combined with delicate Bessel-function monotonicity estimates [AB]. Their result is sharp and gives a spectral way to recognize circularity from the first two Dirichlet tones.

The theorem belongs to the broader family of spectral isoperimetric inequalities, alongside the Faber–Krahn theorem for λ_1 and the Szegő–Weinberger theorem for the first nonzero Neumann eigenvalue. Ashbaugh–Benguria also proved a spherical analogue for domains contained in a hemisphere [AB2]. Variants for other boundary conditions, other ambient spaces, and higher eigenvalue ratios remain active sources of problems.

Recent work illustrates how the original solved theorem continues to generate new problems. In the Robin setting, Dai–Sun recently proved a sharp comparison for the ratio of the first two Robin eigenvalues under an interior-sphere hypothesis and an additional distribution-radius condition on the first eigenfunction [DS]. Equality is rigid and their theorem applies for every positive Robin parameter, but these extra hypotheses mean that the unrestricted Payne–Schaefer analogue is not settled by that result. In a different direction, Eddaoudi studied PPW-type inequalities on closed Riemannian manifolds of dimension at least three, with bounds involving scalar curvature, the Yamabe constant, isoperimetric constants, and conformal volume [E]. These developments suggest that the original disk theorem is best viewed as the model case for a family of ratio inequalities whose sharp form depends strongly on boundary condition and ambient geometry.

Related open problems.

1. *Original direction.* Find sharp analogues of the PPW ratio inequality for other boundary conditions, such as Robin or mixed boundary conditions, with full equality and stability statements.

2. *Reference-driven formulation.* Determine optimal spectral-isoperimetric inequalities for higher ratios λ_{k+1}/λ_k or for finite collections of low Dirichlet eigenvalues.

3. *Reference-driven formulation.* Prove quantitative stability: if $\lambda_2(\Omega)/\lambda_1(\Omega)$ is close to the disk value, how close is Ω to a disk in a geometric metric?

4. *Reference-driven formulation.* Extend sharp ratio inequalities to domains in curved ambient surfaces or manifolds under natural curvature hypotheses.

5. *Reference-driven formulation.* Determine the sharp Robin, mixed, and curved-space analogues of the Ashbaugh–Benguria comparison, including removal of the extra geometric and first-eigenfunction hypotheses in the current Robin theorem [DS].

6. *AI-generated formulation.* Develop a unified "two-tone rigidity" theory comparing how much of the shape of a domain is detected by the first two eigenvalues under different boundary conditions.

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Problem 78

Let Ω be a bounded convex domain in $\mathbb{R}^2$. Let f_2 be the second eigenfunction for the Laplacian with zero boundary conditions. Show that the nodal line of f_2 cannot enclose a compact sub-region of Ω.

In general, one likes to know the qualitative behavior of the nodal line. This conjecture has been around for a long time.

Status. Solved for bounded convex planar domains. This is the convex-domain case of Payne's nodal line conjecture for the second Dirichlet eigenfunction. Lin proved important special cases and structural results [L], and Melas proved that for a bounded convex domain in $\mathbb{R}^2$ the nodal line of a second Dirichlet eigenfunction meets the boundary and hence cannot enclose a compact subdomain [M].

The convexity assumption is essential for the known positive theorem. The broader form of Payne's conjecture, asserting that the nodal line of the second Dirichlet eigenfunction must meet the boundary for every bounded planar domain, is false. Hoffmann-Ostenhof–Hoffmann-Ostenhof– Nadirashvili constructed planar domains with holes for which the second nodal line is closed [HHN]. More recent work has reduced the number of holes needed for such counterexamples [DGH].

The counterexample side has recently become sharper. Freitas and Leylekian constructed bounded planar domains with a single hole for which the nodal line of a second Dirichlet eigenfunction is closed and does not touch the boundary [FL]. This shows that if Payne's conjecture has a surviving general form, it must live at the level of simply connected or quantitatively near-convex domains. On the positive and quantitative side, Beck–Canzani–Marzuola studied nodal curves of low Dirichlet eigenfunctions in generalized curvilinear quadrilaterals, proving curvature and boundary-intersection estimates for the second nodal curve in model geometries [BCM]. These results move the problem from mere boundary-touching toward quantitative control of the nodal line.

Bobkov extended a complementary positive direction to the nonlinear p-Laplacian: on circularly symmetric domains in arbitrary dimension, moving polarization forces the nodal set of a second eigenfunction to meet the boundary under explicit symmetry and regularity assumptions [Bo]. This does not settle the remaining simply connected planar case, but it identifies a mechanism that does not depend on linearity.

Thus the qualitative picture is now split into three directions: convex domains, where the theorem is true; general multiply connected domains, where it is false; and intermediate geometric classes, where one asks how much convexity, symmetry, or simple connectivity is needed to force the nodal line to hit the boundary.

Related open problems.

1. *Original direction.* Determine the weakest geometric assumptions on a simply connected planar domain which force the nodal line of the second Dirichlet eigenfunction to meet the boundary.

2. *Reference-driven formulation.* Classify multiply connected planar domains which admit a closed second nodal line.

3. *Reference-driven formulation.* Prove quantitative versions for convex domains: estimate where the second nodal line hits the boundary and how its geometry depends on the shape of the domain.

4. *Reference-driven formulation.* Extend the convex-domain theorem, or identify counterexamples, for higher-dimensional convex domains and for other boundary conditions.

5. *Reference-driven formulation.* Decide whether a simply connected counterexample exists, or prove that simple connectivity plus a quantitative geometric hypothesis rules out closed second nodal lines [FL].

6. *AI-generated formulation.* Define a convexity defect which controls whether Payne's nodal-line conclusion remains true, and seek a sharp threshold for this defect.

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Problem 79

Let M be a compact manifold without boundary. Then we can define the eigenvalues for the Laplacian acting on the differential forms. How can we estimate the first nonzero eigenvalue in terms of computable geometric quantities? See Li [L], Li–Yau [LY] and the recent work of Gromov [G] where the estimates of λ_n on functions depend on the diameter of M and a lower bound for the Ricci curvature. See also the paper of Uhlenbrock [U].

Status. Partially solved. For functions, the problem is now well understood under the standard assumptions of a lower Ricci-curvature bound and a diameter bound. For the Hodge Laplacian on differential forms, the corresponding sharp theory is not complete in this generality; the estimates depend on the degree of the form, on the harmonic part determined by cohomology, and on curvature terms which are stronger than scalar function estimates.

For functions, Cheng's eigenvalue comparison gave the fundamental upper comparison estimates, while Li [L] and Li–Yau [LY] developed lower estimates for the first nonzero eigenvalue in terms of lower Ricci curvature and diameter. Gromov's work on Lévy–Gromov isoperimetry [G] placed these estimates in the broader comparison geometry framework. In the case $\mathrm{Ric} \geqslant 0$, Zhong and Yang [ZY] proved the sharp lower bound

$$\lambda_1(M) \geqslant \frac{\pi^2}{D^2},$$

where D is the diameter. Later work on sharp moduli of continuity and the fundamental gap, especially Andrews–Clutterbuck [AC, AC2], clarified the sharp one-dimensional model mechanism behind these estimates.

For differential forms, the first point is that the zero eigenspace of the Hodge Laplacian consists of harmonic forms, hence is governed by de Rham cohomology. Thus one must estimate the first positive eigenvalue after removing the harmonic forms. The Bochner–Weitzenbock formula then involves curvature operators acting on $\Lambda^p T^*M$, not only the Ricci tensor appearing in the function case. Classical work of Gallot–Meyer [GM] gives estimates under curvature-operator positivity assumptions, and Uhlenbrock's work [U] belongs to the same analytic circle through Kato-type inequalities and spectral distribution estimates. On the other hand, spectral prescription results for Hodge Laplacians, such as those of Jammes [J], show that without appropriate geometric hypotheses one should not expect a simple topological or dimensional bound.

Recent work makes this boundary more precise. Bhattacharya–Maity– Tiwari announced uniform Hodge-eigenvalue bounds under lower Ricci curvature, positive injectivity radius, and bounded diameter, but that preprint was withdrawn after an error was found in Proposition 3.1 and is therefore not used as evidence here. Huang–Wang treat the rough Laplacian on 1-forms and show both why a direct Li–Yau type estimate can fail and how lower bounds can be recovered after adding curvature-integrability and topological assumptions such as nonzero Euler characteristic [HW]. Huang–Zhang then obtained explicit rough and Hodge 1-form estimates under almost-nonnegative curvature-operator hypotheses and nonvanishing Euler characteristic [HZ]. Their argument combines Schrödinger operator estimates with a twisted Hodge–Dirac operator whose index is $\chi(M)$. For coexact 1-forms, Rudd proved Cheeger-type lower bounds in terms of stable filling or stable commutator-length constants by studying the magnetic geodesic flow of an eigenform [Ru]. These results support the view that the form-valued problem should be split into Hodge, rough, exact, coexact, and connection-Laplacian versions, each with its own natural geometric hypotheses.

Related open problems.

1. *Original direction.* Find sharp lower bounds for the first positive Hodge Laplacian eigenvalue on p-forms under natural lower curvature and diameter assumptions.

2. *Reference-driven formulation.* Determine precisely which estimates for forms require curvature-operator bounds and which can be obtained from lower Ricci curvature plus non-collapsing hypotheses.

3. *Reference-driven formulation.* Separate the estimates for exact and coexact forms, especially in degrees 1 and $n-1$, where comparison with the scalar Laplacian and the Hodge star gives extra structure.

4. *Reference-driven formulation.* Study rigidity and stability: if equality, or near equality, holds in a sharp lower bound for forms, must the manifold be close to a one-dimensional or symmetric model?

5. *Reference-driven formulation.* Find the weakest noncollapsing and curvature-integrability assumptions under which uniform estimates for Hodge and rough Laplacians on 1-forms hold [HW, HZ], and relate the coexact gap to computable filling invariants [Ru].

6. *AI-generated formulation.* Is there a Zhong–Yang type sharp diameter lower bound for the first positive exact 1-form eigenvalue under $\mathrm{Ric} \geqslant 0$, or can one construct a counterexample showing that such a bound is not the right formulation?

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Problem 80

(The Schiffer conjecture, or Pompeiu problem). Let Ω be a smooth bounded domain in $\mathbb{R}^2$ with compact closure. Suppose there exists an f which is an eigenfunction for the Laplacian with Neumann boundary conditions. If also f is constant on the boundary of Ω, prove that Ω is a disk.

This problem is relevant to the following classical problem:

Given a function f defined on $\mathbb{R}^2$, and a bounded domain Ω, if one knows the value of the integral of f over all images of Ω under Euclidean motions of the plane, can the function f be recovered?

Status. Open for smooth bounded simply connected planar domains in the stated Schiffer form. The disk is the model case. The full Pompeiu–Schiffer conjecture is still unresolved, although many special cases and related variants are known.

Pompeiu's original problem is an integral geometry problem: if a nonzero function has zero integral over every rigid motion of a fixed domain, must the domain be a ball? The Schiffer formulation is the corresponding overdetermined Neumann eigenvalue problem: a nonconstant Neumann eigenfunction is also constant on the boundary. Fourier transform methods relate the failure of the Pompeiu property to the zero set of the Fourier transform of the indicator function of the domain; in the smooth contractible setting this is equivalent to the Schiffer conjecture. The work of Brown–Schreiber–Taylor [BST] and Williams [W] established important early structural results and proved the Pompeiu property for large non-spherical classes such as polytopal regions.

In the planar smooth case, the problem remains difficult because the boundary condition is overdetermined but the spectral parameter is not fixed in advance. Deng [De] proved disk rigidity under low-eigenvalue or symmetry assumptions. Ramm [Ra] gives a useful modern account of the equivalent formulations and known reductions. Liu [Li] connects Schiffer's conjecture with interior transmission eigenvalues and invisibility cloaking. Recent work continues to clarify the boundary between the original conjecture and nearby problems: for example, polytopal regions have the Pompeiu property by Fourier-zero arguments [MR, KS], while the classical simply connected hypothesis cannot be discarded: Enciso–Fernández–Ruiz–Sicbaldi constructed analytic noncircular doubly connected domains with a nonradial Neumann eigenfunction locally constant on the boundary [EFRS]. Their local bifurcation starts from radial eigenfunctions on annuli and separates this Schiffer-type counterexample from the still-open simply connected conjecture.

One should also distinguish this classical overdetermined-eigenvalue problem from the Schiffer problem in inverse scattering. In that nearby setting, Hongyu Liu recently proved an almost-sure uniqueness result for determining an obstacle from a single far-field measurement [HL]. This does not solve the deterministic Pompeiu–Schiffer conjecture stated here, but it shows that probabilistic or generic versions of Schiffer-type inverse problems may be more accessible than the full rigidity problem.

Related open problems.

1. *Original direction.* Prove Schiffer's conjecture for smooth simply connected planar domains without extra symmetry or eigenvalue-position assumptions.

2. *Reference-driven formulation.* Determine whether analyticity, convexity, or finite-type assumptions on the boundary are sufficient to force disk rigidity.

3. *Reference-driven formulation.* Classify smooth bounded domains whose Fourier transform of the indicator function vanishes on a complex sphere centered at the origin.

4. *Reference-driven formulation.* Understand which multiply connected Schiffer-type formulations should have disks and annuli as the only solutions, and which admit genuine counterexamples.

5. *Reference-driven formulation.* Compare the deterministic Pompeiu–Schiffer conjecture with almost-sure or generic uniqueness versions in inverse scattering [HL].

6. *AI-generated formulation.* Develop a quantitative stability theorem: if a Neumann eigenfunction is nearly constant on the boundary, is the domain close to a disk in a geometric norm?

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V. Problems Related to Geodesics

Problem 81

Prove that every compact manifold M has an infinite number of closed geodesics.

This is an old problem. Klingenberg has studied this extensively, and has obtained many deep results. See his book [Kl] for the case where $\pi_1(M)$ is finite.

Status. Open in full generality, if "infinite" means infinitely many geometrically distinct closed geodesics. The existence of at least one closed geodesic on every compact Riemannian manifold was proved by Lyusternik–Fet [LF].

The variational formulation regards closed geodesics as critical points of the energy functional on the free loop space. Gromoll and Meyer [GM] proved that a closed Riemannian manifold has infinitely many geometrically distinct closed geodesics if the Betti numbers of its free loop space are unbounded. Vigué-Poirrier and Sullivan [VPS], using Sullivan's minimal model theory [S], gave a powerful topological criterion: for simply connected closed manifolds whose rational cohomology algebra is not generated by one element, the free-loop-space homology is large enough to force infinitely many closed geodesics.

Other major cases were settled by dynamical and low-dimensional methods. Bangert–Hingston [BH] proved strong results for manifolds with infinite abelian fundamental group. For $\mathbb{S}^2$, the classical three-geodesics theorem gives at least three simple closed geodesics, and the combined work of Franks, Bangert, and Hingston [B, F, H] implies infinitely many closed geodesics for every Riemannian 2-sphere. Nevertheless, the general case of compact manifolds, especially rationally elliptic spaces with cohomology generated by one element such as spheres and projective spaces in higher dimensions, remains a central open problem.

Recent progress continues to enlarge the solved classes rather than closing the general problem. Rademacher–Taimanov proved growth results for closed geodesics on connected sums and applied them to compact 3-manifolds with infinite fundamental group; in particular, for generic Riemannian metrics on compact 3-manifolds in that setting one obtains infinitely many geometrically distinct closed geodesics [RT]. De Philippis–Marini–Mazzucchelli–Suhr extended both the three-geodesics theorem and the Bangert–Franks–Hingston infinite-geodesics theorem from Riemannian 2-spheres to reversible Finsler 2-spheres [DMMS]. These results emphasize that the remaining hard cases are concentrated in simply connected or rationally elliptic higher-dimensional manifolds and in nonreversible Finsler settings.

Contreras–Mazzucchelli obtained a further generic result in 2025 [CM]. If a closed manifold of dimension at least two has nonzero first Betti number, then an open dense set of metrics in every C^k topology, $2 \leqslant k \leqslant \infty$, has infinitely many closed geodesics of arbitrarily large length. Their mechanism is dynamical: an Aubry–Mather minimizing closed geodesic can be made hyperbolic with a transverse homoclinic by an arbitrarily C^∞-small perturbation, producing a horseshoe; if no such minimizer exists, a theorem of Mañé already supplies infinitely many closed geodesics. This is strong generic progress, but it does not prove the conjecture for every metric.

Related open problems.

1. *Original direction.* Prove infinitely many geometrically distinct closed geodesics on every closed Riemannian sphere $\mathbb{S}^n$, $n \geqslant 3$.

2. *Reference-driven formulation.* Prove the conjecture for compact rank-one symmetric spaces, especially complex, quaternionic, and Cayley projective spaces.

3. *Reference-driven formulation.* Find sharp growth lower bounds for the number of closed geodesics of length at most L under generic and nongeneric metrics.

4. *Reference-driven formulation.* Clarify which Finsler analogues remain true without reversibility assumptions.

5. *Reference-driven formulation.* Extend the Rademacher–Taimanov growth results from connected sums and 3-manifold prime decomposition to broader classes of manifolds with large fundamental group [RT].

6. *Reference-driven formulation.* Determine whether the Aubry–Mather/homoclinic mechanism of Contreras–Mazzucchelli can be made nonperturbative, or extended to manifolds with $b_1 = 0$ through finite covers or other minimizing classes [CM].

7. *AI-generated formulation.* Develop a quantitative Morse-theoretic criterion which detects when all high iterates of finitely many prime closed geodesics cannot account for the loop-space homology.

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Problem 82

Let M be a compact manifold without conjugate point. If M is homotopically equivalent to the torus, prove that M is flat.

This was conjectured by E. Hopf and proved by him for two-dimensional M. L. Green [G] has proved that the total scalar curvature of M must be nonpositive, and is zero only if M is flat. It is believable that the fundamental group of a compact manifold without conjugate point has exponential growth unless the manifold is flat.

Status. Solved. Burago and Ivanov proved in 1994 that every Riemannian metric without conjugate points on a torus is flat [BI].

The two-dimensional case was proved earlier by E. Hopf [H], following the Morse–Hedlund study of manifolds without conjugate points [MH]. Green [G] proved that in the compact case without conjugate points the total scalar curvature is nonpositive, with equality only in the flat case. Burago–Ivanov's proof introduced a higher-dimensional rigidity mechanism using minimal geodesics and stable norms for periodic metrics, and it settled the torus version of the Hopf problem completely.

The broader expectation, mentioned in the original problem, remains a major theme: compact manifolds without conjugate points should behave like nonpositively curved manifolds in many algebraic and dynamical ways, even though no curvature sign is assumed. Ivanov–Kapovitch [IK] proved structural restrictions on fundamental groups of closed manifolds without conjugate points, for instance virtual splitting properties of centralizers. Related rigidity questions also appear for cylinders, noncompact quotients, and Lorentzian analogues, where the Riemannian rigidity can fail [BE, Me].

Recent work on manifolds without conjugate points is less about the already solved torus theorem and more about the dynamical and group theoretic environment around it. For uniform visibility manifolds without conjugate points, Wu studied Patterson–Sullivan measures, measures of maximal entropy, equilibrium states, counting of free-homotopy classes containing closed geodesics, and rigidity features of the Margulis function [Wu]. Liu studied Myrberg limit sets and Bowen–Margulis–Sullivan measures in the same visibility setting [Li]. These results reinforce the idea that the correct extension of the Hopf–Burago–Ivanov theorem is not merely "flatness versus nonflatness," but a package of entropy, counting, boundary dynamics, and group-growth rigidity for manifolds without conjugate points.

Related open problems.

1. *Original direction.* Classify closed manifolds without conjugate points whose fundamental group has polynomial growth.
2. *Reference-driven formulation.* Determine whether a closed manifold without conjugate points and virtually abelian fundamental group must be finitely covered by a flat torus.
3. *Reference-driven formulation.* Prove entropy or marked-length-spectrum rigidity results for metrics without conjugate points under hypotheses weaker than nonpositive curvature.
4. *Reference-driven formulation.* Understand which group-theoretic properties of nonpositively curved manifolds extend to the no-conjugate-points setting.
5. *Reference-driven formulation.* Relate polynomial or subexponential group growth to entropy, Patterson–Sullivan theory, and closed-geodesic counting in visibility manifolds without conjugate points [Wu, Li].
6. *AI-generated formulation.* Formulate a quantitative almost-flat theorem: if a torus has no conjugate points after a small controlled perturbation of the Jacobi equation, must the metric be close to flat?

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Problem 83

Prove the Blaschke conjecture for other symmetric spaces of rank one besides $\mathbb{S}^N$. For the sphere, this was established through the efforts of Green, Weinstein, Berger, Kazdan, and Yang (see [B] for the precise history of the problem).

Status. Open for the complex, quaternionic, and Cayley projective-space models. The metric Blaschke conjecture is solved for homology spheres and homology real projective spaces, but not for the other compact rank-one symmetric spaces [M].

The Blaschke problem asks for rigidity of compact manifolds whose injectivity radius equals their diameter, or equivalently in the classical setting, manifolds whose geodesics have the same strong cut-locus behavior as compact rank-one symmetric spaces. Besse's monograph [B] remains the standard source for the history and for the connection with manifolds all of whose geodesics are closed. The sphere and real-projective-space branches were proved through the sequence of works of Berger, Kazdan, Weinstein, and Yang. Yang's paper [Y] belongs to this sphere/Wiedersehen branch; it does not prove the complex, quaternionic, or Cayley projective cases asked for in the present problem.

In modern terminology, the broader Blaschke-manifold program is often stated as the classification of compact Riemannian manifolds with injectivity radius equal to diameter as compact rank-one symmetric spaces, up to scale. McKay's corrected survey [M] gives the essential status audit. A Blaschke manifold modelled on $\mathbb{C}P^2$, $\mathbb{H}P^2$, or $\mathbb{O}P^2$ is known to be diffeomorphic to its model, and in the $\mathbb{H}P^n$ case it is known to be homotopy equivalent to the model, but these topological conclusions do not identify the metric. The survey also records flaws in earlier claimed proofs for great sphere fibrations, so those claims cannot be used to mark the projective cases solved. Recent work studies nearby Besse and diameter-rigidity hypotheses, for example Besse projective spaces with many diameter-realizing directions [AV].

The recent Besse-manifold perspective gives a useful way to report progress on the problem. Blaschke manifolds are Besse, but the converse is false without additional diameter information. Adelstein and Vargas Pallete prove that Besse projective spaces with sufficiently many diameter-realizing directions are Blaschke [AV]. Thus one current direction is to replace the exact injectivity-radius-equals- diameter hypothesis by measurable or incidence-type assumptions on diameter-realizing geodesics, and then recover the Blaschke conclusion.

Related open problems.

1. *Original direction.* Prove that a Blaschke manifold modelled on $\mathbb{C}P^n$, $\mathbb{H}P^n$, or $\mathbb{O}P^2$ is isometric, after rescaling, to its compact rank-one symmetric model.
2. *Reference-driven formulation.* Upgrade the known diffeomorphism or homotopy classifications in the projective cases to metric rigidity without relying on the flawed great-sphere-fibration arguments identified by McKay [M].
3. *Reference-driven formulation.* Develop quantitative stability: if $\mathrm{inj}(M, g)$ is close to $\mathrm{diam}(M, g)$, is (M, g) close to a compact rank-one symmetric space?
4. *Reference-driven formulation.* Study Finsler and low-regularity analogues of the Blaschke conjecture.
5. *Reference-driven formulation.* Determine the minimal "many diameters" condition under which a Besse projective space must be Blaschke [AV].
6. *AI-generated formulation.* Classify Besse manifolds which satisfy a weakened "many diameters" condition, interpolating between Zoll metrics and Blaschke metrics.

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Problem 84

Prove that a compact harmonic manifold is symmetric.

A manifold is defined to be harmonic if geodesic spheres of small radius have constant mean curvature (see [B]).

Status. Partially resolved in the formulation stated. Szabó proved the Lichnerowicz conjecture when the universal cover is compact, in particular for compact harmonic manifolds with finite fundamental group [Sz]. For a compact harmonic manifold whose universal cover is noncompact, symmetry is known under additional hypotheses such as no focal points or Gromov-hyperbolicity of the fundamental group [K], but the unrestricted compact-quotient case remains open [KP].

Harmonicity means that the volume density in geodesic polar coordinates, equivalently the mean curvature of sufficiently small geodesic spheres, depends only on the radius. Flat spaces and rank-one symmetric spaces are the basic examples. Lichnerowicz conjectured that harmonic manifolds should be flat or rank-one locally symmetric. Szabó's spectral and "nice embedding" argument establishes this for compact simply connected harmonic spaces and hence for compact universal covers. Compactness of the base manifold must not be confused with compactness of its universal cover.

The noncompact theory is subtler. Damek–Ricci spaces give nonsymmetric homogeneous harmonic manifolds, showing that the unrestricted noncompact Lichnerowicz conjecture is false [DR]. However, the classification of noncompact simply connected harmonic manifolds remains a major topic. Knieper proves that rank one, Anosov geodesic flow, purely exponential volume growth, and Gromov hyperbolicity are equivalent in the relevant noncompact harmonic setting; when such a space admits a compact quotient, dynamical and entropy rigidity identify it with a rank-one symmetric space [K]. He also proves rank one under the no-focal-points assumption. The missing bridge in the general compact-quotient case is to derive one of these equivalent rank-one conditions from harmonicity alone.

Two nearby developments help clarify the frontier. Zimmer proved rigidity results for compact asymptotically harmonic manifolds under additional assumptions, including nonpositive curvature or Gromov hyperbolic fundamental group [Z]. Kim–Park–Shah revisited the three-dimensional asymptotically harmonic case with minimal horospheres and supplied a corrected proof completing the classification in dimension three: the model is flat or real hyperbolic [KPS]. These asymptotic variants are not identical to the compact-universal-cover problem solved by Szabó, but they identify which horosphere and boundary-at-infinity hypotheses can force symmetric geometry.

Related open problems.

1. *Original direction.* Prove that a noncompact simply connected harmonic manifold admitting a compact quotient is flat or a rank-one symmetric space.
2. *Reference-driven formulation.* Prove that every nonflat noncompact harmonic universal cover of a compact manifold has purely exponential volume growth, equivalently rank one, an Anosov geodesic flow, or Gromov hyperbolicity [K, KP].
3. *Reference-driven formulation.* Understand the boundary between rank-one symmetric spaces and nonsymmetric Damek–Ricci examples through asymptotic geometry.
4. *Reference-driven formulation.* Develop stability versions of Szabó's compact theorem.
5. *Reference-driven formulation.* Extend the three-dimensional classification of asymptotically harmonic manifolds to higher dimensions under minimal horosphere, visibility, or entropy hypotheses [KPS, Z].
6. *AI-generated formulation.* Formulate a compactness theorem for harmonic metrics near a fixed locally symmetric metric in terms of the radial volume-density function.

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Problem 85

Let M be a compact manifold with finite fundamental group. Can one find a non-hyperbolic closed geodesic? For the details of this problem see [Kl], and the paper of Ballmann–Thorbergsson–Ziller [BTZ].

Status. Open in general. Important partial results are known under additional curvature, homotopy, or low-dimensional hypotheses, but there is no general theorem proving that every compact Riemannian manifold with finite fundamental group has a non-hyperbolic closed geodesic.

The problem belongs to the stability theory of closed geodesics. A closed geodesic is hyperbolic when the linearized Poincaré return map has no eigenvalue on the unit circle. Thus a non-hyperbolic closed geodesic is a closed geodesic with elliptic or degenerate linearized behavior. Klingenberg's book [Kl] gives the classical background. Ballmann–Thorbergsson–Ziller [BTZ, BTZ2, BTZ3] proved strong existence and stability results for closed geodesics, in particular under positive curvature or homotopy-theoretic assumptions. These results show that topology and index iteration can force special closed geodesics, but they do not settle the full finite-fundamental-group problem.

For surfaces, there is stronger evidence. Contreras–Oliveira [CO] proved that every Riemannian metric on $\mathbb{S}^2$ or $\mathbb{R}P^2$ can be C^2-approximated by a smooth metric with an elliptic closed geodesic. Later, Contreras–Mazzucchelli [CM] proved a C^2-generic dichotomy for closed surfaces: a generic geodesic flow has either an elliptic closed geodesic or is Anosov. Since Anosov geodesic flows cannot occur on a closed surface of genus at most one, this gives a C^2-open dense elliptic-geodesic result on $\mathbb{S}^2$ and $\mathbb{R}P^2$. It remains a generic or approximation theorem, not an existence theorem for every metric, so it does not settle the exact finite-fundamental-group question.

The generic and Finsler viewpoints help sharpen what is missing. Rademacher–Taimanov explain how generic Riemannian statements about closed geodesics can be transferred to Finsler metrics and prove second-closed-geodesic results under topological or hyperbolicity assumptions [RT]. Their discussion also recalls the classical generic alternative: for sufficiently regular generic metrics, either all closed geodesics are hyperbolic or a non-hyperbolic geodesic of twist type appears, in which case Birkhoff–Lewis type arguments produce many closed geodesics. Thus the hard case of the present problem is the possibility of a metric on a finite-fundamental-group manifold whose entire closed-geodesic dynamics is hyperbolic in a way compatible with the topology.

Related open problems.

1. *Original direction.* Prove the existence of a non-hyperbolic closed geodesic on every Riemannian sphere $\mathbb{S}^n$, $n \geqslant 2$.
2. *Reference-driven formulation.* Decide whether a compact Riemannian manifold with finite fundamental group can have all closed geodesics hyperbolic.
3. *Reference-driven formulation.* Understand whether bumpy metrics on simply connected compact manifolds can force elliptic closed geodesics through index-growth or resonance identities.
4. *Reference-driven formulation.* Extend the surface dichotomy between elliptic closed geodesics and Anosov dynamics to higher-dimensional geodesic flows under topological restrictions.
5. *Reference-driven formulation.* Compare the Riemannian finite-fundamental-group question with generic Finsler alternatives and second-closed-geodesic criteria [RT].
6. *AI-generated formulation.* Develop a quantitative criterion which detects non-hyperbolicity from the failure of the closed-geodesic index spectrum to match an Anosov-type model.

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Problem 86

If M is diffeomorphic to the N-sphere, give a lower estimate on the number of embedded closed geodesics.

It is well known that Lusternik–Schnirelmann proved the existence of three distinct embedded closed geodesics if $N = 2$ (See Lusternik–Schnirelmann [LS].). For contributions to this type of problem, see [Kl].

Status. Partially resolved. For $\mathbb{S}^2$, the Lusternik–Schnirelmann theorem gives at least three embedded closed geodesics for every metric [LS]. For $\mathbb{S}^N$, $N \geqslant 3$, generic metrics have infinitely many embedded closed geodesics, but no positive every-metric lower bound for embedded closed geodesics is known in comparable generality [R].

The two-dimensional case is special because closed geodesics can be studied through curve-shortening and planar topology. Bangert [B], Franks [F], and Hingston [H] imply that every Riemannian 2-sphere has infinitely many geometrically distinct closed geodesics, but this does not mean infinitely many embedded ones. The classical Lusternik–Schnirelmann lower bound of three simple closed geodesics is the robust embedded statement on $\mathbb{S}^2$.

In higher dimensions the variational theory on the free loop space gives closed geodesics, but controlling embeddedness is much harder. Hingston–Rademacher [HR] obtained resonance results for loop homology of spheres. More recently, Rademacher proved a sharp generic embeddedness statement in dimensions at least three: for a C^r-generic Riemannian metric, $r \geqslant 2$, every prime closed geodesic is simple and distinct prime geodesics are disjoint [R]. Combining this with generic existence and hyperbolic-dynamics results, a C^2-generic metric on any compact simply connected manifold of dimension at least three has exponentially many simple closed geodesics. The proof perturbs metrics locally near intersection segments and then uses the bumpy-metric theorem and a Baire argument. This is a strong generic solution, but it does not provide an every-metric lower estimate.

Recent work mainly strengthens the two-dimensional and Finsler versions rather than solving the higher-dimensional embedded problem. Liokumovich–Nabutovsky–Rotman proved that every Riemannian 2-sphere has three simple periodic geodesics whose lengths are bounded by a universal multiple of the diameter, and obtained sharper area-diameter estimates in thin cases [LNR]. De Philippis–Marini– Mazzucchelli–Suhr extended both the Lusternik–Schnirelmann three-simple-geodesics theorem and the Bangert–Franks–Hingston infinite-geodesics theorem to reversible Finsler 2-spheres [DMMS]. These results show that the 2-dimensional theory now has both quantitative and Finsler refinements, while the higher-dimensional embeddedness issue remains the main gap. For bumpy Riemannian 2-spheres, Ko further shows that there are simple closed geodesics of Morse indices 1, 2, and 3, and develops strong Morse inequalities directly on the space of simple closed curves [Ko].

Related open problems.

1. *Original direction.* Prove that every Riemannian $\mathbb{S}^N$, $N \geqslant 3$, has at least one embedded closed geodesic, and then seek a nontrivial dimension-dependent lower bound.

2. *Reference-driven formulation.* Upgrade Rademacher's generic infinite and exponential-growth theorem to arbitrary metrics, or identify the precise nongeneric obstruction [R].

3. *Reference-driven formulation.* Separate the problem of simple embedded closed geodesics from the general closed-geodesic problem in loop-space Morse theory.

4. *Reference-driven formulation.* Extend the three-geodesics theorem to natural higher-dimensional notions such as embedded closed geodesics with prescribed index or minimax origin.

5. *Reference-driven formulation.* Seek diameter, volume, or curvature-controlled length bounds for embedded closed geodesics in higher-dimensional spheres, modeled on the 2-sphere estimates [LNR].

6. *AI-generated formulation.* Develop a minimax procedure for sweepouts by embedded loops in $\mathbb{S}^N$ which prevents self-intersection in the limiting geodesic.

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Problem 87

Generalize Loewner and Pu's inequality [P] to higher dimensions. The Loewner inequality says that for the two-torus,

$$\frac{A}{l^2} \geqslant C$$

where l is the length of the shortest closed homotopically nontrivial loop and C is a universal constant. In this regard, consult the work of Berger [B] and Gromov [G1].

Status. Partially resolved. Gromov's systolic inequality for essential manifolds gives a fundamental higher-dimensional generalization of Loewner's and Pu's inequalities, but sharp constants, equality cases, and the relations among distinct higher systoles remain open in many natural classes [G2].

Loewner's inequality for $\mathbb{T}^2$ and Pu's inequality for $\mathbb{R}P^2$ are optimal two-dimensional estimates relating area to the square of the shortest noncontractible loop. Gromov introduced the filling radius and proved that if M^n is essential, then

$$\operatorname{sys}_{\pi_1}(M, g)^n \leqslant C_n \operatorname{Vol}(M, g),$$

where C_n depends only on the dimension [G2]. This places the Loewner–Pu phenomenon in a topological framework: the fundamental class must remain nontrivial in the homology of the classifying space of the fundamental group. The proof and its later expositions are a central part of modern systolic geometry [G1, K].

The general theorem is not usually sharp. Even for natural families such as projective spaces, lens spaces, complex projective spaces, and manifolds with nontrivial cup products, the search for optimal constants and equality cases is delicate. Stable systolic inequalities, filling-radius inequalities, and higher-dimensional systoles give several different extensions of the Loewner–Pu picture. Recent work continues this direction by extending sharp Gromov-type inequalities from $\mathbb{C}P^n$ to broader cohomological settings [GHK].

The point of the Goodwillie–Hebda–Katz extension is that some cohomological hypotheses do more than produce a nonsharp systolic inequality: when the fundamental class is a cup product of two-dimensional classes, one obtains a natural extension of Gromov's optimal inequality for complex projective space [GHK]. This suggests that one should organize higher-dimensional Loewner–Pu problems by the structure of the cohomology ring and by whether the resulting systolic inequality is merely qualitative, quantitatively sharp, or sharp with a rigid equality case.

Tsiamis obtained a complementary sharp development in 2026 [T]. Under positive scalar-curvature hypotheses he proves sharp bounds for the 2-systole, spherical 2-systole, and stable 2-systole in Kähler and spin^c settings. Equality identifies Fubini–Study $\mathbb{C}P^n$ in the principal Kähler theorem, with further sharp refinements distinguishing complex quadrics and other Fano models. These results concern higher-dimensional two-systoles rather than the homotopy one-systole in Loewner's inequality, and so they enlarge rather than close the original program.

Related open problems.

1. *Original direction.* Determine the optimal constants in Gromov's systolic inequality for important essential manifolds beyond $\mathbb{T}^2$ and $\mathbb{R}P^2$.

2. *Reference-driven formulation.* Classify equality cases for sharp systolic inequalities in higher dimensions.

3. *Reference-driven formulation.* Understand when cup products, Massey products, or other cohomological structures force optimal systolic inequalities.

4. *Reference-driven formulation.* Relate homotopy systoles, stable systoles, and filling radius in a way that preserves sharp constants.

5. *Reference-driven formulation.* Extend the Goodwillie–Hebda–Katz optimal inequality to cohomology rings involving mixed degrees, torsion, or Massey products [GHK].

6. *Reference-driven formulation.* Determine which of the sharp scalar-curvature two-systolic inequalities extend beyond the Kähler or spin^c hypotheses, and whether their equality cases remain rigid [T].

7. *AI-generated formulation.* Develop a quantitative stability theorem for Gromov's essential-manifold inequality: if the ratio $\operatorname{Vol}/\operatorname{sys}^n_{\pi_1}$ is nearly minimal, must the metric be close to a canonical extremal metric?

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VI. Minimal Submanifolds

Problem 88

Prove that any three-dimensional manifold must contain an infinite number of immersed minimal surfaces. Sacks and Uhlenbeck [SU] proved the existence of a minimal sphere in any compact manifold which is not covered by a contractible space. Sacks–Uhlenbeck and Schoen–Yau [SY] independently proved that any incompressible surface can be deformed into a minimal surface. When the ambient manifold is three-dimensional, an argument of Osserman [O] shows that they are immersed. In most cases, they are in fact embedded by the results of Meeks and Yau [MY] and more recent work of Freedman, Hass and Scott [FHS]. The work of Meeks–Simon–Yau [MSY] also showed that, starting from any compact surface in a compact three-dimensional manifold, one can minimize its area "within its isotopy class."

For a general three-dimensional manifold, Pitts [P] proved the existence of one such minimal surface. However, one does not know the genus of the surface from his method.

Status. Solved for closed three-dimensional Riemannian manifolds. Song proved Yau's conjecture in full generality: every closed Riemannian manifold of dimension between 3 and 7 contains infinitely many smooth closed embedded minimal hypersurfaces [So]. In dimension three this gives infinitely many embedded minimal surfaces, hence also immersed minimal surfaces.

The early theory provided several different sources of minimal surfaces. Sacks–Uhlenbeck [SU] proved existence of minimal 2-spheres in many compact manifolds. Schoen–Yau [SY] and Sacks–Uhlenbeck proved that incompressible surfaces can be represented by minimal surfaces. In dimension three, regularity and embeddedness were developed through the work of Osserman [O], Meeks–Yau [MY], Freedman–Hass–Scott [FHS], and Meeks–Simon–Yau [MSY]. Pitts [P] used Almgren–Pitts min-max theory to prove the existence of at least one minimal surface in a closed three-manifold.

The modern solution proceeds through min-max theory for the area functional. Marques–Neves proved the conjecture under positive Ricci curvature, in dimensions at most seven [MN]. Irie–Marques–Neves proved the conjecture for C^∞-generic metrics by showing that minimal hypersurfaces are dense [IMN]. Song then removed the genericity and curvature hypotheses [So]. Thus the original existence question is now settled, while quantitative questions about genus, index, area growth, multiplicity, and distribution remain very active.

Song's proof also supplies a reusable mechanism [So]. Assuming finitely many minimal hypersurfaces, one cuts maximally along stable ones to obtain a compact core with stable boundary and attaches cylindrical ends. Min-max widths on this completion localize closed minimal hypersurfaces inside the core and satisfy a linear "cylindrical Weyl law." The resulting arithmetic behavior of the widths contradicts the assumed finite list. This isolates the main ideas that remove both the Frankel and genericity hypotheses.

The post-solution literature has turned the conjecture into a quantitative theory of the area spectrum. Liokumovich–Marques–Neves proved the Weyl law for the volume spectrum [LMN], giving the correct asymptotic scale for the Almgren–Pitts widths. Marques–Neves–Song proved equidistribution of minimal hypersurfaces for generic metrics [MNS], and Chodosh–Mantoulidis and Zhou supplied multiplicity-one and Allen–Cahn/min-max regularity inputs which clarify why the generic minimal hypersurfaces counted by the theory should behave like genuine Morse critical points of area [CM], [Z]. For Yau's original three-dimensional problem, the main frontier is therefore no longer existence but the structure of the infinite family: how the areas obey the Weyl law, whether genus and index have predictable asymptotics, and how much of the generic picture survives for degenerate metrics.

Related open problems.

1. *Original direction.* Give sharp genus and index bounds, or asymptotic laws, for the minimal surfaces produced by the min-max widths in a closed three-manifold.

2. *Original direction.* Extend abundance results to noncompact three-manifolds under natural bounded-geometry or finite-volume hypotheses.

3. *Reference-driven formulation.* Prove non-generic versions of the Weyl-law and equidistribution picture: determine when the min-max minimal surfaces are equidistributed and how their areas grow with the width parameter [LMN], [MNS].

4. *Reference-driven formulation.* Classify possible multiplicities and bubbling behavior in degenerate metrics, comparing Almgren–Pitts, Simon–Smith, and Allen–Cahn constructions [CM], [Z].

5. *Reference-driven formulation.* Determine whether the generic density theorem can be strengthened to prescribed topology or bounded index in dimension three, rather than only producing an uncontrolled infinite set [IMN], [MNS].

6. *AI-generated formulation.* Develop a canonical "minimal-surface spectrum" for closed three-manifolds which records area, genus, index, and multiplicity together, analogous to the Laplace spectrum.

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Problem 89

Prove that there are four distinct embedded minimal spheres in any manifold diffeomorphic to $\mathbb{S}^3$. One should study the work of Sacks and Uhlenbeck [SU] in this regard.

Status. Open in full generality, but solved in two major cases: for bumpy metrics and for metrics with positive Ricci curvature.

The harmonic-map min-max theory of Sacks–Uhlenbeck produced minimal immersions of two-spheres. In the embedded category, the first basic existence theorem for an arbitrary Riemannian three-sphere is the Simon–Smith theorem, which gives at least one embedded minimal two-sphere by a one-parameter sweepout. Jost developed related embedded minimal-sphere existence results for manifolds diffeomorphic to a ball or a sphere.

The number four is topologically natural: the space of embedded two-spheres in $\mathbb{S}^3$ is related, through the Smale conjecture and Hatcher's work, to the space of great spheres, which is homotopy equivalent to $\mathbb{RP}^3$. Thus one expects four critical points of the area functional on this space. The main analytic obstruction is that different min-max families may produce the same minimal sphere, possibly with multiplicity.

Important partial results appeared before the recent solution in generic and positively curved cases. White proved multiplicity results under positive Ricci curvature and near the round metric. Haslhofer and Ketover proved that a three-sphere with a generic metric contains at least two embedded minimal two-spheres. Wang and Zhou then proved that a bumpy metric on $\mathbb{S}^3$, or a metric with positive Ricci curvature, contains at least four distinct embedded minimal two-spheres. Their proof uses a multiplicity-one theorem for the Simon–Smith min-max theory: unstable two-sided components occur with multiplicity one, preventing the four natural sweepout classes from recycling a single unstable sphere [WZ]. Talipov obtained a quantitative refinement in 2026 [T]. If $\mathrm{Ric} > 0$ and $R_g \geqslant \Lambda_0 > 0$, the four spheres may be chosen with

$$\mathrm{area}(\Sigma_i) \leqslant \frac{12\pi(i+1)}{\Lambda_0}, \qquad 1 \leqslant i \leqslant 4.$$

His proof combines Simon–Smith widths, Ricci flow, and a sharp eigenvalue inequality on $\mathbb{S}^2$; it also yields at least three nonplanar minimal spheres in sufficiently elongated ellipsoids. The remaining case is the arbitrary non-bumpy metric without a positivity assumption.

Recent work also shows that the four-sphere theorem is part of a broader Morse-theoretic program for spaces of embedded surfaces in three-manifolds. Li–Wang proved strong Morse inequalities for the area functional on embedded tori and spheres in a three-sphere with positive Ricci curvature, giving at least four embedded minimal tori and, for bumpy metrics, at least nine [LW]. Li–Wang–Yao developed analogous equivariant Simon–Smith arguments in lens spaces and $\mathbb{RP}^3$, where projective planes, Klein bottles, spheres, and tori interact [LWY]. These results do not close the arbitrary metric case of Yau's Problem 89, but they strongly support the view that the correct proof should be a robust Morse theory for the area functional on embedded surface spaces, with multiplicity-one and degenerate-critical-set issues under control. In the same positive-Ricci setting, Chu proved in 2025 that for every $g \geqslant 0$ there is an embedded minimal surface of genus g, with area at most twice the first Simon–Smith width [C]. This topological abundance does not by itself produce four genus-zero surfaces for a completely arbitrary metric, but it strengthens the Morse-theoretic picture behind the conjecture.

Related open problems.

1. *Original direction.* Prove the four-sphere conclusion for every smooth Riemannian metric on $\mathbb{S}^3$, without assuming bumpiness or positive Ricci curvature.

2. *Original direction.* Develop a non-generic Morse theory for the area functional on the space of embedded two-spheres in $\mathbb{S}^3$ which can count degenerate critical points with the correct multiplicity.

3. *Reference-driven formulation.* Determine whether every Riemannian three-sphere contains quantitatively controlled embedded minimal two-spheres, with area or index bounds depending only on natural curvature, volume, or diameter data; compare Talipov's positive-Ricci scalar-curvature bounds [T].

4. *Reference-driven formulation.* Extend the Wang–Zhou multiplicity-one Simon–Smith method from bumpy and positive-Ricci metrics to arbitrary metrics by a controlled degeneration or perturbation argument [WZ].

5. *Reference-driven formulation.* Build the analogous counting theory for embedded minimal tori, projective planes, and Klein bottles in spherical space forms, continuing the programs of Li–Wang and Li–Wang–Yao [LW], [LWY].

6. *AI-generated formulation.* Construct a "sphere-width spectrum" for embedded two-spheres in $\mathbb{S}^3$ whose first four critical values always correspond to four geometrically distinct embedded minimal spheres.

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Problem 90

Is it true that every compact differentiable manifold can be minimally embedded into $\mathbb{S}^N$ for some N?

Recently, W.-T. Hsiang and W.-Y. Hsiang studied the problem of minimally embedding some exotic spheres in $\mathbb{S}^M$.

Status. Partially resolved. Lawson proved that every closed orientable surface admits a minimal embedding into the round $\mathbb{S}^3$ [L]. Important further families are known, but no general theorem covers all compact differentiable manifolds, and even the analogous universal statement for closed nonorientable surfaces remains open [KKMS].

The problem should be separated from the Whitney and Nash embedding theorems. Those theorems give smooth embeddings, or isometric embeddings after a Riemannian metric has been chosen, but they do not force the image to be minimal in a round sphere. Takahashi's theorem gives the basic analytic characterization: an isometric immersion into a sphere is minimal exactly when its coordinate functions are eigenfunctions of the Laplacian with the same eigenvalue. Thus the problem asks whether every compact smooth manifold admits some Riemannian metric and some finite eigenspace which gives an embedding into a round sphere with constant length.

Classical examples come from homogeneous spaces and from eigenmap constructions. Lawson's conjugate-surface and reflection construction settled the topological realization problem for closed orientable surfaces in $\mathbb{S}^3$. The work of W.-T. Hsiang, W.-Y. Hsiang and Sterling shows that even exotic spheres can sometimes be realized by minimal immersions or by related constant-mean-curvature embeddings in spheres. These results show that the smooth structure itself is not an obvious obstruction. However, the known constructions still depend on special symmetry, special eigenfunctions, or explicit topology.

The present form of the problem remains a broad existence problem at the interface of differential topology, spectral geometry, and the theory of minimal submanifolds. One also has to distinguish immersion from embedding: Takahashi-type spectral constructions often produce minimal immersions first, while embeddedness may require additional separation properties of the chosen eigenfunctions.

Recent progress has mostly clarified adjacent spectral problems rather than solved Yau's universal embedding question. El Soufi–Ilias connected minimal immersions by eigenfunctions with extremal metrics for Laplace eigenvalues [ESI], and this point of view has become a productive source of examples. On surfaces, Petrides proved broad existence theorems for optimal metrics and Laplace minimal eigenmaps, giving branched minimal immersions into spheres or ellipsoids in many cases [Pe]. Karpukhin–Metras formulated higher-dimensional extremal Laplace and Steklov problems using n-harmonic maps [KM], showing that the natural variational framework in dimensions greater than two is subtler than the classical surface theory. Kislitsyn's recent work on complex-valued eigenfunctions gives another construction mechanism for minimal submanifolds in spheres [Ki]. These results suggest that the original problem may be attacked by first producing a metric with a sufficiently rich eigenspace and then proving a separation theorem turning the corresponding eigenmap into an embedding.

The surface theory has recently become much more quantitative. Karpukhin–Kusner–McGrath–Stern use equivariant Laplace and Steklov eigenvalue optimization to construct many prescribed-topology minimal embeddings in $\mathbb{S}^3$ with area below 8π [KKMS]. Their paper also formulates the still-open nonorientable-surface realization problem. In a higher-dimensional direction, Wang constructed in 2026 infinitely many embedded minimal hypersurfaces of pairwise distinct 3-manifold types in $\mathbb{S}^4$, including principal S^1-bundles and Seifert fibered manifolds; in particular, $S^1 \times \Sigma_g$ is obtained for every odd genus g [W]. The method uses equivariant min-max theory for the suspended Hopf action. This is substantial new evidence in dimension three, but far from a universal construction for arbitrary compact manifolds.

Related open problems.

1. *Original direction.* Determine whether every compact smooth manifold admits a Riemannian metric whose eigenfunctions for a single eigenvalue give a minimal embedding into some round sphere.

2. *Original direction.* Decide whether the weaker statement with "minimal immersion" in place of "minimal embedding" is true for every compact smooth manifold.

3. *Reference-driven formulation.* Prove that every closed nonorientable surface admits a minimal embedding into some round sphere, as conjectured in [KKMS].

4. *Reference-driven formulation.* Find purely topological or smooth-structure obstructions, if any, to minimal embedding into a sphere.

5. *Reference-driven formulation.* Classify which exotic spheres admit minimal embeddings, rather than merely minimal immersions, into round spheres.

6. *Reference-driven formulation.* Determine whether extremal-eigenvalue metrics on surfaces, and their higher-dimensional n-harmonic analogues, can be desingularized or perturbed from branched eigenmaps to genuine minimal embeddings [Pe], [KM].

7. *Reference-driven formulation.* Characterize when a finite eigenspace of a chosen metric separates points and tangent directions strongly enough to convert Takahashi's minimal immersion criterion into an embedding criterion.

8. *AI-generated formulation.* Develop a flexible Takahashi–Nash principle: characterize when a smooth embedding can be deformed, after changing the intrinsic metric, into an eigenmap embedding with minimal image in a sphere.

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Problem 91

Is there any complete minimal surface of $\mathbb{R}^3$ which is a subset of the unit ball?

This was asked by Calabi [C]. There is an example of a complete minimally immersed surface between two planes due to Jorge and Xavier [JX]. Calabi has also shown that such an example exists in $\mathbb{R}^4$. (One takes an algebraic curve in a compact complex surface covered by the ball and lifts it up.)

Status. Solved affirmatively for immersed minimal surfaces. Nadirashvili constructed a complete bounded immersed minimal surface in $\mathbb{R}^3$ in 1996. The embedded versions are much more rigid and have different answers under standard hypotheses.

The first step toward the bounded problem was the Jorge–Xavier construction of a complete minimally immersed surface between two parallel planes. This showed that completeness did not force extrinsic properness even in a slab. Nadirashvili then used Runge approximation methods to construct a complete bounded minimal disk in $\mathbb{R}^3$, thereby answering Calabi's question in the immersed category.

After Nadirashvili's construction, the problem split into several more precise versions. Martin and Morales constructed complete properly immersed minimal surfaces in bounded convex bodies, showing that properness in a bounded domain is possible when the boundary of the domain is included as the limiting boundary. In contrast, Colding and Minicozzi proved the embedded Calabi–Yau conjectures for embedded surfaces, showing that embeddedness imposes strong properness and unboundedness phenomena. The later work of Meeks, Pérez and Ros refined the finite-genus and finite-topology embedded theory. Thus the original immersed question is solved, while the modern subject concerns topology, embeddedness, properness, and quantitative control of complete minimal surfaces in bounded or convex regions.

The conformal version has also developed strongly. Alarcón and collaborators showed that many prescribed open Riemann surfaces occur as complete bounded minimal surfaces, and recent work of Alarcón extends this flexibility to complete proper minimal surfaces with Cantor ends in minimally convex domains [A]. More precisely, for every compact Riemann surface R and every minimally convex domain $\Omega \subset \mathbb{R}^3$, he constructs a Cantor set $C \subset R$ such that $R \setminus C$ is the conformal type of a complete proper minimal surface in Ω. Thus, in the immersed category, boundedness is compatible not only with completeness but also with very complicated conformal type and end structure.

At the same time, recent higher-dimensional work shows that the embedded Calabi–Yau picture is special to surfaces. Aryan–McWeeney construct complete improperly embedded minimal hypersurfaces in $\mathbb{R}^{n+1}$ for every $n \geqslant 3$ by iteratively gluing catenoidal necks [AM]. Their examples have unbounded curvature, infinite genus, infinitely many ends, and are unbounded; hence they disprove a general properness principle but not the bounded embedded version of Calabi's question. They also prove that every complete embedded minimal hypersurface with bounded second fundamental form is unbounded, and obtain polynomial chord–arc control for bounded-curvature embedded minimal disks when $3 \leqslant n \leqslant 5$. This sharpens the modern formulation of Problem 91: the immersed surface question is solved, the finite-topology embedded surface question is rigid, and the right general problem is to identify exactly which extra hypotheses force properness or unboundedness.

Related open problems.

1. *Original direction.* Classify the conformal types and topological types which can occur for complete bounded immersed minimal surfaces in $\mathbb{R}^3$.

2. *Original direction.* Determine sharp conditions under which a complete embedded minimal surface in $\mathbb{R}^3$ must be proper.

3. *Reference-driven formulation.* Understand which bounded domains in $\mathbb{R}^3$ admit complete properly immersed minimal surfaces with prescribed topology.

4. *Reference-driven formulation.* Develop quantitative versions of the embedded Calabi–Yau phenomenon: for example, relate intrinsic completeness, curvature blow-up, and extrinsic escape in embedded minimal surfaces.

5. *Reference-driven formulation.* Determine the exact complex structures and end spaces, beyond Cantor complements of compact Riemann surfaces, that can occur for complete proper minimal surfaces in minimally convex bounded domains [A].

6. *Reference-driven formulation.* Compare the surface Calabi–Yau theorem of Colding–Minicozzi with higher-dimensional counterexamples to properness; identify the dimension-dependent hypotheses, such as bounded curvature or disk topology, under which a properness theorem survives [AM], [CM].

7. *Reference-driven formulation.* Decide whether every complete embedded finite-genus minimal surface in $\mathbb{R}^3$ is proper when its end space is uncountable; countable end space is covered by Meeks–Pérez–Ros [MPR].

8. *Reference-driven formulation.* Determine whether a complete bounded embedded minimal hypersurface can exist in $\mathbb{R}^{n+1}$, $n \geqslant 3$, when curvature is allowed to be unbounded; the examples in [AM] are nonproper but unbounded.

9. *AI-generated formulation.* Find a moduli-theoretic compactness theory for complete bounded immersed minimal surfaces which records the Runge-type approximation data responsible for nonproperness.

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Problem 92

What are the complete, embedded, minimal surfaces (with finite genus) in $\mathbb{R}^3$?

The only known examples are the catenoid and the helicoid. It is possible to prove that any such surface is standardly embedded, in the topological sense.

Status. Open as a full finite-genus classification problem, but many fundamental subcases have been solved. The simply-connected case was solved by Meeks and Rosenberg: the only complete embedded simply-connected minimal surfaces in $\mathbb{R}^3$ are the plane and the helicoid.

The historical picture changed after the discovery of the Costa surface. The older expectation that plane, catenoid and helicoid exhaust the embedded finite-topology examples became false: Costa's genus-one finite-total-curvature surface, and the subsequent Hoffman–Meeks families, showed that there are many embedded examples with finite topology. Thus the problem is not merely to prove uniqueness of the classical examples, but to organize the full moduli space.

Several decisive classification theorems now structure the subject. Collin proved that a properly embedded minimal surface in $\mathbb{R}^3$ with finite topology and more than one end has finite total curvature. Meeks and Rosenberg proved the uniqueness of the helicoid in the simply-connected case. Meeks, Pérez and Ros completed the classification of properly embedded minimal planar domains, showing that the infinite-topology genus-zero examples are the Riemann minimal examples. Later finite-topology and finite-genus results describe conformal type, asymptotics, ends, flux, and properness under broad hypotheses.

The remaining finite-genus classification is best viewed as a moduli problem. One wants to understand all properly embedded minimal surfaces of fixed finite genus, with arbitrary end structure, and to describe their possible limits, degenerations, and Weierstrass data.

The current organizing framework is the Meeks–Pérez–Ros program, which combines lamination theory, limit-leaf stability, local removable-singularity theorems, dynamics of moduli spaces, and flux control. Pérez's recent survey emphasizes that the genus-zero infinite-topology classification is now a four-step theorem ending with the Riemann minimal examples, while finite positive genus remains the natural next frontier [P]. The same program relates properness to the number of limit ends and proves that finite genus with countably many ends falls under strong properness and structural constraints [MPR2], [P]. Thus the old phrase "finite genus" now hides several separate problems: finite total curvature, finitely many ends, one or two limit ends, uncountably many ends, and the compactification of moduli spaces under curvature blow-up.

Two precise frontiers emerge from this program [P]. For finite topology, the Hoffman–Meeks conjecture predicts that a connected embedded minimal surface of genus g with $r > 2$ ends exists only if $r \leqslant g + 2$. The best general theorem currently gives some finite bound $r \leqslant r(g)$, without a sharp formula; the conjecture remains open even for $g = 1$. For finite genus with possibly infinitely many ends, Meeks–Pérez–Ros prove properness when the end space is countable, equivalently when there are at most two limit ends [P]. Whether a complete embedded finite-genus surface with an uncountable end space can be nonproper remains open. These statements locate the surviving classification problem more sharply than the older phrase "finite genus."

Related open problems.

1. *Original direction.* Classify properly embedded minimal surfaces in $\mathbb{R}^3$ of a fixed finite genus and arbitrary end space.

2. *Original direction.* Describe the moduli spaces of finite-total-curvature embedded minimal surfaces with prescribed genus and number of ends.

3. *Reference-driven formulation.* Determine sharp compactness and degeneration theorems for finite-genus embedded minimal surfaces with curvature concentration.

4. *Reference-driven formulation.* Classify possible flux data and asymptotic configurations for finite-genus properly embedded minimal surfaces.

5. *Reference-driven formulation.* Complete the Meeks–Pérez–Ros program for positive finite genus with one or two limit ends, including properness, dynamics, and lamination limits [P].

6. *Reference-driven formulation.* Decide whether every finite-genus properly embedded minimal surface with countably many ends is controlled by a finite collection of model ends and flux parameters [MPR2].

7. *Reference-driven formulation.* Prove the Hoffman–Meeks bound $r \leqslant g+2$ for finite-topology embedded minimal surfaces, beginning with the still-open genus-one case [P].

8. *Reference-driven formulation.* Classify one-ended genus-g helicoids and prove or disprove the expected uniqueness for each $g \geqslant 1$ [P].

9. *AI-generated formulation.* Develop a dictionary between the topology of the end space and the dynamical/lamination limits of finite-genus embedded minimal surfaces in $\mathbb{R}^3$.

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Problem 93

Prove that every smooth, regular Jordan curve in $\mathbb{R}^3$ can bound only a finite number of stable minimal surfaces.

If the Jordan curve is real analytic, Tomi [T] proved that it can bound only a finite collection of locally minimal disks. Tomi's argument is quite general, and the basic point that he requires to generalize the theorem to the smooth case is the proof of the absence of boundary branch points for stable minimal surfaces whose boundary is a smooth regular curve. To date, this last assertion is unestablished.

If we assume that the minimal surface has least area in the strong sense, Hardt–Simon [HS] established the absence of boundary branch points, thus proving the finiteness in this case.

There are various uniqueness theorems after suitable perturbation of the boundary. These were due to Böhme, Morgan, Tomi, Tromba and others.

Status. Open in the stated smooth-boundary form. Important special cases and generic variants are known, but there is no general finiteness theorem for all stable minimal surfaces bounded by an arbitrary smooth regular Jordan curve.

The classical finiteness statements require careful separation [J]. Tomi's 1973 theorem says that the global area minimizers are finite provided every such minimizer is immersed; real analyticity of the boundary, or a total-curvature bound below 4π, supplies this premise through the classical branch-point theory [T]. His later stable-disk theorem assumes the absence of boundary branch points and allows only controlled interior branching [T2]. Hardt–Simon proved the boundary regularity needed for the strongly area-minimizing oriented Plateau problem [HS]. None of these theorems removes the branch-point hypothesis for every stable disk spanning an arbitrary smooth Jordan curve.

There is also a useful nonsmooth positive result. Jakob proved that a simple closed extreme polygon bounds only finitely many immersed stable minimal disks [J]. Each such disk is isolated in the C^0 Plateau space, while the collection of stable immersed disks is closed in the compact space of all Plateau solutions. The decisive boundary input is that an extreme polygon prevents boundary branch points. This theorem does not follow by smoothing the polygon: its corner expansions are part of the isolation argument, so it highlights rather than removes the missing smooth-boundary step.

The problem is subtle because existence and uniqueness behave non-generically. Morgan constructed a smooth curve in $\mathbb{R}^3$ bounding a continuum of minimal manifolds, showing that some finiteness statements fail if the stability or minimizing hypotheses are weakened. Morgan's separate fixed-boundary theory proves finiteness for stable or area-minimizing hypersurfaces under several extremality and analyticity hypotheses [M2]. On the other hand, later work of White and Hoffman–White gives strong generic finiteness and parity theorems for embedded minimal surfaces with prescribed boundary in mean-convex three-manifolds. In the Hoffman–White framework a curve is bumpy when no spanning minimal surface carries a nonzero normal Jacobi field vanishing on the boundary; compactness plus bumpiness gives finiteness, and degree theory gives the mod-two count [HW]. These results support the expectation that the finite-stable-surface statement should be true under suitable nondegeneracy hypotheses, while the fixed smooth-boundary problem remains unresolved.

Two nearby developments clarify what is missing. White's generic transversality theorem shows that, after generic perturbation of the ambient metric, simple minimal immersions are transverse and self-transverse, and it gives generic regularity for two-dimensional area-minimizing currents [W2]. This is the kind of Fredholm nondegeneracy one would want for a boundary "bumpy curve" theory. White's later work on boundary singularities in mean curvature flow also gives new information about total-curvature thresholds for minimal surfaces with a smooth simple closed boundary [W3]. These results do not remove boundary branch points for arbitrary stable Plateau solutions, but they show that the obstruction is now quite focused: one needs a fixed-boundary regularity and compactness theorem strong enough to rule out hidden branching or accumulation of stable solutions.

Related open problems.

1. *Original direction.* Prove absence of boundary branch points for stable minimal surfaces with arbitrary smooth regular Jordan boundary.

2. *Original direction.* Determine whether every smooth regular Jordan curve bounds only finitely many stable minimal disks.

3. *Reference-driven formulation.* Classify the degenerations by which a sequence of stable minimal surfaces with the same smooth boundary could fail to have a finite subsequence.

4. *Reference-driven formulation.* Extend generic parity and finiteness theorems to less generic fixed curves under local analytic or geometric conditions on the boundary.

5. *Reference-driven formulation.* Formulate a fixed-boundary analogue of White's generic transversality theorem: for a generic smooth Jordan curve, prove that every stable Plateau solution is nondegenerate and has no boundary branch points [W2].

6. *Reference-driven formulation.* Relate total curvature of the boundary curve to topological type and stability of spanning minimal surfaces, extending White's mean-curvature-flow estimates [W3].

7. *Reference-driven formulation.* Extend Jakob's compact-plus-isolation proof from extreme polygons to smooth curves, or identify exactly which corner expansion must be replaced by a smooth-boundary estimate [J].

8. *Reference-driven formulation.* Determine whether a bumpy smooth Jordan curve in $\mathbb{R}^3$, without first placing it on the boundary of a mean-convex three-ball, bounds only finitely many stable immersed disks [HW].

9. *AI-generated formulation.* Define a boundary bumpy curve condition for Plateau critical points and prove that finiteness of stable solutions holds exactly under this nondegeneracy condition.

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Problem 94

Given a single smooth, regular, Jordan curve, can one find a nontrivial, continuous family of minimal disks bounded by this curve?

There is a classical example due to Lévy [L] and Courant [C] of a rectifiable Jordan curve which is smooth except at one point and which bounds an uncountable number of minimal disks. (A proof of the validity of this example depends on the "bridge principle" which was first established by Kruskal [K]. A more rigorous proof of the bridge principle was independently established by Almgren–Solomon [AS], and Meeks–Yau [MY].) Morgan [M] has found an example of continuous family of minimal surfaces whose boundary consists of four disjoint circles.

Status. Open for a single smooth regular Jordan curve bounding a nontrivial continuous family of minimal disks. Known examples either allow a nonsmooth boundary point, use more than one boundary component, or produce continua of minimal surfaces rather than the precise disk-with-one-smooth-Jordan-boundary configuration.

The classical Lévy–Courant example shows that a rectifiable Jordan curve, smooth except at one point, can bound uncountably many minimal disks. The singular boundary point is essential in the known construction and prevents it from answering the smooth regular problem. The bridge principle, developed by Kruskal and later by Almgren–Solomon and Meeks–Yau, explains how new minimal surfaces can be created by joining boundary components with thin bridges. Morgan's example gives a continuous family of minimal surfaces whose boundary consists of four disjoint circles, showing that continuum phenomena are real in Plateau theory.

For a single smooth Jordan curve, however, the disk case remains much more rigid. Generic finiteness and parity theorems for minimal surfaces with prescribed boundary suggest that continuous families should be nongeneric. The unresolved issue is whether one can create such a family while keeping the boundary a single smooth embedded curve and keeping every spanning surface a disk.

Recent work around Plateau theory gives useful comparison cases but does not settle the Euclidean smooth-Jordan-curve question. In the asymptotic Plateau problem in hyperbolic three-space, Huang–Lowe–Seppi proved both uniqueness criteria and nonuniqueness phenomena; in particular they construct quasicircles at infinity which bound uncountably many stable minimal disks [HLS]. This is not an example in $\mathbb{R}^3$ with a compact smooth boundary, but it shows that continuum families of stable disks can persist in a geometrically natural Plateau problem with a single boundary curve. Their construction uses a self-similar quasicircle surrounding nested pairs of round circles; asymptotic Plateau compactness produces stable disks on alternating sides, and an infinite sequence of binary choices yields the uncountable family. The ideal boundary and this nested separation mechanism are precisely the features that would have to be replaced in a compact Euclidean construction. On the other side, Hoffman–White type parity theorems and White's generic regularity results suggest that any Euclidean example should come from a genuinely degenerate fixed-boundary Fredholm problem, not from a generic boundary curve [W], [Wh]. Thus the modern version of Problem 94 is best phrased as a bifurcation problem for Plateau disks at a smooth curve where the Jacobi operator has kernel.

Related open problems.

1. *Original direction.* Construct, or rule out, a smooth regular Jordan curve in $\mathbb{R}^3$ which bounds a nontrivial continuous family of minimal disks.

2. *Original direction.* Determine whether any such family must contain degenerate minimal disks with nontrivial Jacobi fields satisfying the fixed boundary condition.

3. *Reference-driven formulation.* Decide whether the Lévy–Courant nonsmooth example can be smoothed without destroying the continuum of disk solutions.

4. *Reference-driven formulation.* Understand whether bridge-principle constructions can be adapted from multiple boundary components to one smooth Jordan boundary component.

5. *Reference-driven formulation.* Compare compact Plateau families in $\mathbb{R}^3$ with asymptotic Plateau families in $\mathbb{H}^3$: which features of the Huang–Lowe–Seppi quasicircle examples can survive when the boundary is a compact smooth Jordan curve [HLS]?

6. *Reference-driven formulation.* Develop a boundary Fredholm theory in which Hoffman–White parity can be continued through degenerate curves and used to detect possible continua [W].

7. *Reference-driven formulation.* Determine whether a nonzero fixed-boundary Jacobi field on a Plateau disk can always be integrated to a local family, or identify the higher-order obstruction to such a bifurcation.

8. *Reference-driven formulation.* Decide whether a continuum is possible when all spanning disks are embedded, stable, and satisfy a uniform curvature bound.

9. *AI-generated formulation.* Define a local bifurcation model for Plateau disks at a degenerate boundary curve and classify which bifurcations can persist under smoothing of the boundary.

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Problem 95

Let σ be a smooth Jordan curve in $\mathbb{S}^3$ which bounds an embedded disk in the unit ball of $\mathbb{R}^4$. Prove that there is a curve σ' isotopic to σ in $\mathbb{S}^3$ which bounds an embedded minimal disk in the unit ball of $\mathbb{R}^4$. An application of this would be the proof that the slice knot is a ribbon knot.

Status. Open in the stated geometric form. The problem should be separated from the slice-ribbon conjecture itself: slice-ribbon asks whether every smoothly slice knot is ribbon, and that problem remains open. Yau's formulation proposes a stronger minimal-surface route toward such a conclusion, but no general theorem is known which replaces an arbitrary slice disk, after isotopy of the boundary knot in $\mathbb{S}^3$, by an embedded minimal disk in the Euclidean four-ball.

The geometric difficulty is twofold. First, Plateau theory in higher codimension naturally produces area-minimizing currents or branched minimal disks, while the problem asks for a smooth embedded disk with controlled boundary. Boundary regularity, branch points, and embeddedness are therefore central issues. Second, the disk must lie in the correct boundary-knot isotopy class. Once such a minimal disk $X\colon D \to B^4$ exists, however, the radial part of Yau's strategy is automatic. Since the coordinate functions of a minimal immersion are harmonic,

$$\Delta_D |X|^2 = 2|dX|^2 = 4.$$

Thus $|X|^2$ is strictly subharmonic and has no interior local maximum. After the standard generic Morse adjustment, this is precisely the no-maxima radial movie associated with a ribbon disk. The central unsolved step is therefore existence of a smooth embedded minimal representative, not exclusion of radial maxima after it has been found.

Gauge-theoretic and Floer-theoretic invariants show why any positive answer must be very delicate. Donaldson theory and the work of Kronheimer–Mrowka on embedded surfaces in four-manifolds give sharp constraints on smoothly embedded surfaces and on slice genus in many settings. These constraints do not by themselves settle the minimal-disk question above, but they show that a successful theorem would have to interact with modern smooth concordance invariants, not only with classical Plateau theory.

Recent knot-concordance work has sharpened the surrounding slice-ribbon landscape. Dai–Kang–Mallick–Park–Stoffregen proved that the $(2, 1)$-cable of the figure-eight knot is not smoothly slice [DKMPS], removing a long-standing potential counterexample to the slice-ribbon conjecture. Kang–Park–Taniguchi extended this direction to further figure-eight cables using real Seiberg–Witten Frøyshov invariants [KPT]. Hom–Park used bordered Heegaard Floer theory and the γ_0 invariant to show strong restrictions on when connected sums of certain fibered knots can be ribbon [HP]. Computationally, Dunfield–Gong studied knots through 19 crossings and found large-scale evidence consistent with slice-ribbon phenomena, while isolating a comparatively small list of unresolved sliceness cases [DG]. One of their most interesting knots has since become a sharper test. Oliveira–Smith proved in 2026 that the Dunfield–Gong homotopy four-sphere is standard and hence that the knot $18_{\mathrm{nh}00000601}$ is smoothly slice in the standard B^4 [OSm]. The knot is not known to be ribbon, although the constructed disk is fibered handle-ribbon, so it is now an explicit potential counterexample to slice–ribbon and a natural test for Yau's minimal disk proposal. In another direction, Lobb recasts ribbon concordance as a partial-order problem and proves that the $(4, 5)$ torus knot is a global minimum in its concordance class for reduced rational Khovanov homology [Lob]. None of these results produces Yau's desired minimal disk in B^4, but they show that any geometric proof of the slice-ribbon implication must be compatible with a very rich and computable concordance theory.

Related open problems.

1. *Original direction.* Determine which slice knots can be isotoped to bound embedded minimal disks in the Euclidean four-ball.
2. *Original direction.* Identify geometric conditions on a slice disk which force the boundary knot to be ribbon.
3. *Reference-driven formulation.* Relate the radial Morse theory of embedded disks in B^4 to gauge-theoretic and Floer-theoretic concordance invariants.
4. *Reference-driven formulation.* Decide whether special classes of slice knots admit minimal-disk representatives under additional geometric hypotheses.
5. *Reference-driven formulation.* Test Yau's minimal-disk strategy on families where slice-ribbon is known or strongly constrained, such as 3-stranded pretzel knots, figure-eight cables, and γ_0-sharp fibered knots [GJ], [DKMPS], [KPT], [HP].

6. *Reference-driven formulation.* Develop obstructions to minimal-disk representatives from modern concordance invariants: determine whether knot Floer, Khovanov, instanton, or equivariant Seiberg–Witten data can detect unavoidable branching, self-intersection, or failure of compactness in a codimension-two Plateau construction.

7. *Reference-driven formulation.* Decide whether the slice knot $18_{\mathrm{nh}00000601}$ is ribbon and whether its fibered handle-ribbon disk can be deformed to an embedded minimal disk [OSm].

8. *Reference-driven formulation.* Formulate the no-interior-maximum identity $\Delta_D|X|^2 = 4$ as a quantitative Morse theory for almost-minimal disks, stable under the limiting process used to solve Plateau's problem.

9. *AI-generated formulation.* Define a "minimal ribbon complexity" of a slice knot by minimizing the number of radial critical points among geometrically controlled embedded disks in B^4.

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Problem 96

What is the structure of the space of minimal surfaces of a fixed genus in $\mathbb{S}^3$? Lawson [L] has proved that, besides $\mathbb{RP}^2$, any closed surface can be minimally embedded in $\mathbb{S}^3$. Which conformal structures can be realized in such a way? What happens if we replace $\mathbb{S}^3$ by $\mathbb{S}^N$ with $N > 3$?

Status. Open in general. There is a complete answer in genus zero and a striking classification in genus one, but the fixed-genus moduli problem in higher genus, and the question of which conformal structures occur, remain largely open.

Lawson's 1970 construction produced embedded minimal surfaces in $\mathbb{S}^3$ of many topological types and showed that minimal surface theory in the round three-sphere is far richer than the Euclidean compact case. In genus zero, the only embedded minimal two-sphere in $\mathbb{S}^3$ is the totally geodesic equator. In genus one, Lawson conjectured that the Clifford torus is the only embedded minimal torus in $\mathbb{S}^3$; this was proved by Brendle. Thus the conformal structure in the embedded genus-one case is rigid: it is the conformal structure of the Clifford torus.

For higher genus there are many constructions but no classification. The Lawson surfaces give an infinite sequence of embedded examples, and later constructions of Karcher–Pinkall–Sterling, Kapouleas–Yang, Choe–Soret and others produced further families. If $\mathcal{M}_g$ denotes the space, modulo ambient isometries, of minimal embeddings of a genus-g surface into $\mathbb{S}^3$, then Choi–Schoen proved that $\mathcal{M}_g$ is compact in the smooth topology [ChS]. This is an important positive result, but it neither classifies $\mathcal{M}_g$ nor determines whether it is finite. The broader local theory of Colding–Minicozzi describes the two basic curvature-concentration mechanisms for fixed-genus sequences: double spiral staircases in the uniformly locally simply connected regime, and short-neck or pair-of-pants decompositions in the complementary regime [CM1, CM2].

The replacement of $\mathbb{S}^3$ by $\mathbb{S}^N$, $N > 3$, changes the problem substantially. Higher codimension allows many more harmonic-map and algebraic-geometric constructions, and the conformal type is no longer controlled by the same embedded hypersurface restrictions. Thus the higher-codimension version should be treated as a problem about minimal immersions, eigenmaps, and special projective or algebraic models, rather than only as a hypersurface moduli problem.

There are now several complementary modern approaches to this moduli problem. Integrable-systems methods, especially DPW and spectral data, give concrete access to conformal structures in low genus; Heller's work on genus two surfaces and Lawson's genus-two example shows how stable holomorphic bundles and meromorphic connections enter the description [He1], [He2]. From a different direction, Karpukhin–Kusner–McGrath–Stern used equivariant eigenvalue optimization to produce embedded minimal surfaces in $\mathbb{S}^3$ with prescribed topology and area below 8π, proving $|\mathcal{M}_{g,8\pi}| \geqslant \lfloor (g-1)/4 \rfloor + 1$ [KKMS]. Kapouleas–McGrath then constructed Clifford-torus doublings with catenoidal bridges along parallel torus knots and obtained the stronger quadratic count $|\mathcal{M}_g| \geqslant Cg^2$ for large g [KMg]. This is particularly relevant to Yau's question because it links the topological realization problem, the Laplace spectrum, and the image of the moduli space in Teichmüller space. The current picture is therefore not just construction versus classification: one wants to organize the fixed-genus space simultaneously by topology, conformal structure, spectrum, area, index, symmetry, and degeneration.

Related open problems.

1. *Original direction.* Classify embedded minimal surfaces of each fixed genus $g \geqslant 2$ in the round three-sphere.

2. *Original direction.* Determine the subset of the moduli space of Riemann surfaces of genus g which can be realized by embedded minimal surfaces in $\mathbb{S}^3$.

3. *Reference-driven formulation.* Describe the compactification of the fixed-genus moduli space: which degenerations, bubbling phenomena, or limiting laminations can occur when genus, ambient metric, or embeddedness is allowed to vary, and how do the Colding–Minicozzi neck and double-spiral models appear at the boundary [CM1, CM2]?

4. *Reference-driven formulation.* Determine whether the Lawson surfaces are unique within natural symmetry, index, or area classes.

5. *Reference-driven formulation.* Use integrable-systems data to characterize which genus-two and higher-genus conformal structures can occur for embedded minimal surfaces in $\mathbb{S}^3$ [He1], [He2].

6. *Reference-driven formulation.* Relate equivariant eigenvalue-optimization constructions to the classical Lawson and Karcher–Pinkall–Sterling families: determine whether the new low-area examples exhaust any natural part of the fixed-genus moduli space [KKMS].

7. *Reference-driven formulation.* Determine the asymptotic growth of $|\mathcal{M}_g|$: is the quadratic lower bound of Kapouleas–McGrath sharp, and is $\mathcal{M}_g$ finite for every fixed $g \geqslant 2$ [KMg]?

8. *AI-generated formulation.* Construct a period-map-like invariant from embedded minimal surfaces in $\mathbb{S}^3$ to Teichmüller space and describe its image, fibers, and boundary behavior.

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Problem 97

(Lawson). Is the only embedded minimal torus in $\mathbb{S}^3$ the Clifford torus?

There are many minimal tori in $\mathbb{S}^3$ which are not Clifford tori, but they are not embedded.

Status. Solved. Brendle proved in 2013 that the Clifford torus is the only embedded minimal torus in the round three-sphere.

Lawson's construction produced many compact minimal surfaces in $\mathbb{S}^3$ and led naturally to the conjecture that the embedded torus case is rigid. There are many immersed minimal tori in $\mathbb{S}^3$, and the spectral-curve theory of Pinkall–Sterling shows that the immersed theory has rich finite-type structure. The difficulty of Lawson's conjecture was therefore not minimality alone, but the combination of minimality and embeddedness.

Brendle's proof introduced a two-point function and an Alexandrov reflection argument adapted to the nonlocal geometry of an embedded minimal torus. The method shows that the surface must have the symmetries of the Clifford torus and hence must be congruent to it. The result gives the complete embedded genus-one classification in $\mathbb{S}^3$ and forms one of the central solved cases in the broader fixed-genus moduli problem.

After Brendle's theorem, the Clifford torus became not only the unique embedded torus but also a central local model for higher-genus constructions. Choe–Soret constructed new minimal surfaces desingularizing Clifford tori [CS]. Kapouleas–Wiygul constructed low-genus desingularizations of three Clifford tori and proved related characterization results for the generating disks [KW]. Kapouleas–McGrath later produced new Clifford-torus doublings and used them to obtain a quadratic lower bound for the number of embedded minimal surfaces in $\mathbb{S}^3$ of prescribed genus, as well as new examples relevant to Yau's first-eigenvalue question [KMg]. These developments do not reopen the genus-one problem, but they show that the Clifford torus remains the organizing singular model for many higher-genus moduli questions.

There is also a quantitative counterpart to the rigidity picture. Bi–Zhou proved that a unit-density integral two-varifold in $\mathbb{S}^3$, with support of genus at least one and Willmore energy at most $2\pi^2 + \delta^2$, is, after a conformal transformation, $W^{2,2}$-close to the Clifford torus with error $O(\delta)$ [BZ]. This is a stability theorem for the Willmore minimum rather than a new proof of the Lawson conjecture: conformal transformations need not preserve minimality, and the theorem applies to a larger varifold class. Its importance here is that it supplies a precise topology and rate in which near-equality forces geometric closeness.

Related open problems.

1. *Original direction.* Find analogous uniqueness theorems for embedded minimal surfaces of higher genus under natural symmetry, index, or area assumptions.
2. *Reference-driven formulation.* Classify immersed minimal tori in $\mathbb{S}^3$ with additional constraints such as embedded fundamental domain, low spectral genus, or low Morse index.
3. *Reference-driven formulation.* Determine whether Brendle's two-point-function method extends to other ambient three-manifolds or to free-boundary minimal tori.
4. *Reference-driven formulation.* Relate the Lawson conjecture to quantitative stability: compare the $W^{2,2}$ Willmore stability of Bi–Zhou with stronger smooth, spectral, or Morse-index stability statements inside the minimal-surface class [BZ].
5. *Reference-driven formulation.* Classify Clifford-torus doubling and desingularization families, and decide when gluing, min-max, and eigenvalue-optimization constructions produce the same embedded minimal surfaces [CS, KW, KMg].
6. *AI-generated formulation.* Develop an Alexandrov-type nonlocal maximum principle for higher-genus embedded minimal surfaces in $\mathbb{S}^3$ which detects hidden rotational symmetry.

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Problem 98

(Lawson). Let M be an embedded minimal surface in $\mathbb{S}^3$. Prove that the two domains in $\mathbb{S}^3$ divided by M have equal volume.

This is a delicate form of the Gauss–Bonnet theorem. Indeed, if $M^{2N-1} \subseteq \mathbb{S}^{2N}$ is a compact connected hypersurface such that all the odd elementary symmetric functions of the second fundamental form are zero, then the general Gauss–Bonnet theorem proves that the two components of $\mathbb{S}^{2N} \setminus M^{2N-1}$ have equal volume provided they have the same Euler characteristic. For a minimal surface in $\mathbb{S}^3$, these two components are always diffeomorphic (cf. Lawson [L]).

In the general case of $\mathbb{S}^N$ for $N > 3$, the conjecture fails. C.-L. Terng, for example, shows that $\mathbb{S}^P(\sqrt{P/N}) \times \mathbb{S}^{N-P}(\sqrt{(N-P)/N})$ does not divide $\mathbb{S}^{N+1}$ into two equal-volume pieces unless $P = N - P$.

Status. False. Karcher–Pinkall–Sterling constructed embedded minimal surfaces in $\mathbb{S}^3$ for which the two complementary domains do not have equal volume.

The conjecture was natural because of Lawson's unknottedness theorem and the Gauss–Bonnet-type calculation recalled above. In odd hypersurface dimension inside an even-dimensional sphere, the vanishing of the odd elementary symmetric functions can force equal volumes when the two complementary domains have the same Euler characteristic. For embedded minimal surfaces in $\mathbb{S}^3$, the two regions are diffeomorphic, so it was reasonable to expect a similar volume balance. The Karcher–Pinkall–Sterling paper proves more than the bare existence of a counterexample. It constructs a minimal patch inside a Coxeter tetrahedron, extends it by reflections, proves embeddedness by controlling the polar surface, and then estimates the patch inside a tube about an equatorial reflection sphere. For the surface generated by their $(\pi/3, \pi/2, \pi/5)$ patch, the tube contains less than half of $\mathbb{S}^3$, which forces unequal complementary volumes [KPS, §5]. By contrast, their self-dual examples contain great circles whose symmetries exchange the two sides, so those examples are balanced. Thus minimality and embeddedness alone do not impose volume balance; side-exchanging symmetry is the decisive extra feature in the balanced examples.

The higher-dimensional product examples also show that the failure is not accidental. Clifford-type products $\mathbb{S}^P(\sqrt{P/N}) \times \mathbb{S}^{N-P}(\sqrt{(N-P)/N})$ are minimal hypersurfaces in spheres, but they divide the ambient sphere into equal-volume pieces only in the symmetric case $P = N - P$. Thus any true equal-volume theorem must use stronger symmetry, parity, or curvature hypotheses than minimality alone.

Recent existence results make the false problem useful as a moduli question. New gluing, desingularization, and equivariant eigenvalue-optimization constructions produce large families of embedded minimal surfaces in $\mathbb{S}^3$ [CS, KMg, KKMS]. On such families one can ask whether the volume imbalance varies nontrivially, whether symmetry forces zeros of the imbalance, and what happens near singular limits built from Clifford tori or other highly symmetric configurations. Thus the modern replacement is not an equal-volume theorem for all embedded minimal surfaces, but a study of enclosed volume as an invariant on concrete moduli spaces.

Related open problems.

1. *Original direction.* Characterize embedded minimal surfaces in $\mathbb{S}^3$ which do divide the sphere into two equal-volume domains.

2. *Original direction.* Determine whether equality of the two volumes follows from additional assumptions such as antipodal symmetry, special isoparametric structure, or prescribed symmetry group.

3. *Reference-driven formulation.* For a symmetry group preserving M, determine from its action on the two components of $\mathbb{S}^3 \setminus M$ whether volume balance is forced; in particular, separate symmetries preserving each side from those exchanging the sides.

4. *Reference-driven formulation.* Study the volume imbalance as a functional on moduli spaces of embedded minimal surfaces in $\mathbb{S}^3$.

5. *Reference-driven formulation.* Find sharp geometric bounds for the possible volume imbalance in terms of genus, area, index, or symmetry.

6. *Reference-driven formulation.* Compute the imbalance on the Karcher–Pinkall–Sterling, Choe–Soret, Kapouleas–McGrath, and equivariant eigenvalue-optimization families, and decide whether balanced members occur by symmetry or bifurcation [KPS, CS, KMg, KKMS].

7. *AI-generated formulation.* Define an "enclosed-volume map" on the fixed-genus minimal-surface moduli space in $\mathbb{S}^3$ and describe its image and critical points.

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Problem 99

(Chern). Prove that the only embedded minimal hypersurface in $\mathbb{S}^{N+1}$ which is diffeomorphic to $\mathbb{S}^N$ is the totally geodesic sphere.

An affirmative answer will be interesting even for a special case where we assume the cone over the hypersurface is stable in $\mathbb{R}^{N+2}$. Hopefully this would mean that an area-minimizing hypersurface which is a topological manifold is smooth. Under the assumption of stability of the cone the conjecture is true for $N = 2, 3, 4, 5$, see [S]. For higher codimension the conjecture is false; see [LO].

Status. False in the stated generality, already for $N = 3$. Hsiang constructed infinitely many noncongruent, non-equatorial embedded minimal three-spheres in $\mathbb{S}^4$ [H].

The problem is a natural strengthening of the classical rigidity of the equatorial sphere. In low dimensions, and especially for embedded minimal two-spheres in $\mathbb{S}^3$, the totally geodesic sphere is indeed rigid. Simons' work on minimal varieties and stable cones also gives strong positive results for the stable-cone version in low dimensions. This explains why the conjecture was plausible from the viewpoint of regularity theory for area-minimizing hypersurfaces.

The higher-dimensional situation is different. In the notation of the present problem, Hsiang's embedded examples occur for $N \in \{3, 4, 5, 6, 7, 9, 11, 13\}$, and Tomter obtained a non-equatorial embedded minimal hypersphere for every odd $N \geqslant 3$. By contrast, Hsiang–Tomter's main theorem for every $N \geqslant 3$ is an *immersed* existence theorem; it should not be quoted as embedded in all dimensions [HT]. The even-dimensional embedded case $N \geqslant 8$ remains the unresolved parity range of the spherical Bernstein problem. In higher codimension, Lawson–Osserman showed that analogous minimal-surface-system uniqueness and regularity statements fail even more dramatically.

Recent work supplies new constructions rather than restoring rigidity. Carlotto–Schulz gave a self-contained dynamical-systems construction of infinitely many pairwise non-isometric embedded minimal hyperspheres in $\mathbb{S}^4$ and $\mathbb{S}^6$ [CaS]. In 2026, T. Wang–Z. Wang–Zhou developed an equivariant min–max theory with topological control and constructed another embedded non-equatorial minimal $\mathbb{S}^3 \subset \mathbb{S}^4$ [WwZ]. Unlike the classical ODE shooting construction, their variational method persists for a class of symmetric nonround metrics. This does not enlarge the known dimension range, but it changes the available mechanism and opens a route toward robust or multiplicity results.

Modern rigidity progress is therefore best read as a collection of replacement theorems. Lei–Xu–Xu obtain small-gap results for minimal hypersurfaces with constant scalar curvature [LXX]. Cheng–Wei–Yamashiro prove, under the additional constancy of the third power sum f_3, that $S > n$ forces $S > 1.8252n - 0.712898$ [CWY]. Magliaro–Mari–Roing–Savas-Halilaj prove the sharp complete pinching alternative $|A|^2 \leqslant n$: the hypersurface is totally geodesic or covers a Clifford torus [MMRS]. C. Wang–P. Wang show that, under reflection symmetry in the $n + 2$ coordinate hyperspheres, the average inequality $\overline{S} \geqslant n$ is rigid at Clifford tori [WW]. None of these theorems uses the topology $M \cong \mathbb{S}^N$ alone, so none revives the false original conjecture.

Related open problems.

1. *Original direction.* Classify embedded minimal hyperspheres in $\mathbb{S}^{N+1}$ under additional assumptions such as low index, curvature pinching, or prescribed symmetry.
2. *Original direction.* Construct, or rule out, a non-equatorial embedded minimal $\mathbb{S}^N \subset \mathbb{S}^{N+1}$ for every even $N \geqslant 8$.
3. *Original direction.* Determine sharp dimension bounds for stable-cone rigidity when the link is a smooth topological sphere.
4. *Reference-driven formulation.* Understand whether area-minimizing hypersurfaces which are topological manifolds must be smooth under natural extra hypotheses.
5. *Reference-driven formulation.* Classify Hsiang–Tomter type minimal hyperspheres and their deformations in higher-dimensional spheres.
6. *Reference-driven formulation.* Extend the equivariant min–max construction of [WwZ] to new group actions and dimensions, especially the unresolved even-N range, while retaining the hypersphere topology.
7. *Reference-driven formulation.* Prove sharp pinching and isoparametric rigidity theorems for minimal hyperspheres under constant scalar curvature or small $|A|^2$-gap assumptions, sharpening [LXX] and [CWY].
8. *Reference-driven formulation.* Extend symmetry-based Clifford-torus and totally-geodesic rigidity theorems to weaker symmetry, average-curvature, or low-index hypotheses [WW].

9. *AI-generated formulation.* Define a "Chern-defect" measuring the failure of a minimal hypersphere in $\mathbb{S}^{N+1}$ to be totally geodesic, and seek pinching theorems forcing this defect to vanish.

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Problem 100

Is it true that the first eigenvalue for the Laplace–Beltrami operator on an embedded minimal hypersurface of $\mathbb{S}^{N+1}$ is N?

This is not known even for $N = 2$. An affirmative answer will imply that the area of embedded minimal surfaces in $\mathbb{S}^3$ will have an upper bound depending only on the genus. This is a consequence of the theorem of Yang–Yau [YY].

Status. Open in the stated generality as of July 2026. There are strong lower bounds and many special cases. A 2025 preprint claims a proof, but the argument as posted has decisive gaps and does not justify a status change; later 2026 work continues to treat Yau's statement as a conjecture [Zh2, Yu].

The coordinate functions of an embedded minimal hypersurface $M^N \subset \mathbb{S}^{N+1}$ are eigenfunctions with eigenvalue N, so the problem is to prove that no smaller positive eigenvalue exists. Choi and Wang proved the fundamental lower bound $\lambda_1(M) \geqslant N/2$. In dimension two, the conjecture would combine with the Yang–Yau eigenvalue estimate to give area bounds for embedded minimal surfaces in $\mathbb{S}^3$ depending only on genus.

Many structured examples satisfy the conjecture. Choe and Soret proved $\lambda_1 = 2$ for reflection-generated surfaces whose simply connected fundamental patch has fewer than six edges; this covers the Lawson and Karcher–Pinkall–Sterling examples [CS]. Tang and Yan proved the conjecture for isoparametric minimal hypersurfaces, connecting it to Chern-type problems for minimal hypersurfaces with constant scalar curvature [TY]. Kapouleas–McGrath proved the stronger first-eigenspace statement for their Clifford-torus doublings and for a broad class of side-symmetric equatorial-sphere doublings [KMg].

The quantitative lower-bound program has also advanced. If $\Lambda = \max_M |A|$, Duncan–Sire–Spruck prove

$$\lambda_1(M) \geqslant \frac{N}{2} + \frac{a_N}{\Lambda^6 + b_N}$$

with explicit dimensional constants [DSS]. Yu improves this to

$$\lambda_1(M) \geqslant \frac{N}{2} + \frac{N^2}{27\left(\sqrt{\Lambda^2 + 2N} + \Lambda\right)^2}$$

[Yu]. In dimension two, Zhao proves $\lambda_1 \geqslant 1 + \varepsilon_g$ for a genus-g embedded minimal surface [Zh1]; his higher-dimensional work gives a positive correction $G(N, |A|_{\max}, |A|_{\min})$ to $N/2$ and a dimension-only gap when $|A|$ is constant [Zh2]. These results do not reach N, but they identify the loss in the harmonic-extension and Reilly-formula method.

The claimed proof in [Ze] was audited directly. Its trial functions u_β converge in H^1 to a coordinate function x_i, and the calculation $R[u_\beta] \to N$ therefore recovers only the already known upper bound $\lambda_1 \leqslant N$; the paper never proves that $\{u_\beta\}$ is a minimizing sequence for λ_1. Moreover, the asserted exact orthogonality $\int_M x_i e^{-\beta d(p,p_0)^2} = 0$ is deduced from a local parity argument that is not valid on a general embedded hypersurface. An independent check is provided by its stability corollary, which treats the variable potential $|A|^2$ as a constant spectral shift and concludes that every closed embedded minimal hypersurface in a sphere is stable. Thus the posted proof does not establish the main theorem.

Related open problems.

1. *Original direction.* Prove $\lambda_1 = N$ for every closed embedded minimal hypersurface in $\mathbb{S}^{N+1}$.
2. *Original direction.* Prove the conjecture for embedded minimal surfaces in $\mathbb{S}^3$ under weaker assumptions than symmetry or membership in known families.
3. *Reference-driven formulation.* Determine whether Chern-type constant-scalar-curvature conditions imply the first-eigenvalue conclusion.
4. *Reference-driven formulation.* Find sharp improvements of the Choi–Wang bound $\lambda_1 \geqslant N/2$ in terms of genus, area, index, or bounds for the second fundamental form.
5. *Reference-driven formulation.* Compare the claimed full solution of Zeng with the quantitative approaches of Duncan–Sire–Spruck, Zhao, and Yu, and isolate which analytic step would upgrade a gap estimate to $\lambda_1 = N$ [DSS, Ze, Zh2, Yu].
6. *Reference-driven formulation.* Decide whether a genuine mean-zero correction of the coordinate functions can be combined with harmonic extension and the Reilly formula to produce a lower, rather than merely an upper, estimate for λ_1.

7. *AI-generated formulation.* Develop a spectral compactness theory for sequences of embedded minimal hypersurfaces in spheres which tracks possible collapse of λ_1 below the coordinate eigenvalue.

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Problem 101

Is there a closed minimal surface in $\mathbb{S}^N$ with negative curvature?

Status. Solved affirmatively in high codimension. Ancona, Labourie, Roig-Sanchis and Toulisse proved that for every sufficiently large integer N there is a closed minimal surface in $\mathbb{S}^N$ whose induced Gaussian curvature is negative everywhere [ALRT]. In contrast, Lawson's Gauss–Codazzi argument rules out such a surface in $\mathbb{S}^3$ [L]. The smallest possible ambient dimension remains unknown.

The question asks for a compact minimal surface whose intrinsic Gaussian curvature is negative everywhere. For a minimal surface in a sphere this is a strong pointwise condition, not merely a topological condition. By Gauss–Bonnet, such a surface must have genus at least two, but the main difficulty is to realize a negatively curved metric by a minimal immersion into a round sphere.

The recent construction uses closed Riemann surfaces with large automorphism groups and a strategy inspired by Song's high-dimensional random-representation method. More precisely, the surfaces Σ_N satisfy

$$\|K_{\Sigma_N} + 8\|_{C^k} \longrightarrow 0 \qquad \text{for every fixed } k,$$

and their symmetry quotients can be chosen to be one fixed Riemann surface. Pointwise C^k convergence of curvature, rather than an average statement, is what eventually forces $K_{\Sigma_N} < 0$.

This should be separated from Song's precursor. Song constructed closed branched minimal surfaces in high-dimensional spheres which Benjamini–Schramm converge to a rescaled hyperbolic plane and for which $K+8$ tends to zero only in normalized L^1 [So]. That average control allows exceptional regions and does not answer the pointwise-sign question. Ancona–Labourie–Roig-Sanchis–Toulisse replace the cusped three-punctured sphere in Song's construction by a rigid compact orbifold, pass to suitable finite covers, and upgrade the convergence to C^k control [ALRT]. The method is genuinely high-dimensional. After Lawson's obstruction in $\mathbb{S}^3$, the first unresolved low-codimension cases are $\mathbb{S}^4$ and $\mathbb{S}^5$.

Related open problems.

1. *Original direction.* Determine the smallest dimension $N \geqslant 4$ for which there exists a closed minimal surface in $\mathbb{S}^N$ with negative curvature everywhere.
2. *Original direction.* Decide whether such examples exist in $\mathbb{S}^4$ or $\mathbb{S}^5$.
3. *Reference-driven formulation.* Construct embedded, rather than merely immersed, negatively curved closed minimal surfaces in spheres when possible.
4. *Reference-driven formulation.* Understand which negatively curved metrics on closed surfaces can arise as induced metrics of minimal immersions into round spheres.
5. *Reference-driven formulation.* Make the threshold "N sufficiently large" effective and quantify genus, symmetry-group order, branching, and the rate in the C^k curvature bounds [So, ALRT].
6. *AI-generated formulation.* Develop a compactness theory for almost-hyperbolic minimal surfaces in high-dimensional spheres and describe their limiting hyperbolic geometry.

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Problem 102

As a generalization of Bernstein's theorem, Fischer-Colbrie–Schoen [FCS] and do Carmo–Peng [DP] proved that any complete stable minimal surface in $\mathbb{R}^3$ is a plane. Can one generalize this statement to the case of a complete stable hypersurface in $\mathbb{R}^N$ for $N \leqslant 8$?

Status. False as stated, with exactly one intermediate dimension still open. For a complete, connected, two-sided stable minimal immersion $M^n \to \mathbb{R}^{n+1}$, flatness is known for $n \leqslant 5$, is open for $n = 6$, and is false for every $n \geqslant 7$. Equivalently, among the ambient dimensions asked for here, the theorem holds in $\mathbb{R}^3, \ldots, \mathbb{R}^6$, remains open in $\mathbb{R}^7$, and fails in $\mathbb{R}^8$.

The surface case in $\mathbb{R}^3$ was proved independently by do Carmo–Peng and by Fischer-Colbrie–Schoen. Their result says that a complete stable minimal surface in Euclidean three-space must be a plane. Schoen–Simon–Yau developed curvature estimates for stable minimal hypersurfaces, giving the basic analytic framework for higher-dimensional Bernstein-type rigidity. These estimates explain why the natural critical range is tied to the regularity threshold for minimal hypersurfaces and to the classical Bernstein theorem.

The modern positive progress is recent. Chodosh and Li proved that a complete two-sided stable minimal immersed hypersurface in $\mathbb{R}^4$ is flat. Chodosh, Li, Minter and Stryker then proved the analogous result in $\mathbb{R}^5$, and Mazet proved the $\mathbb{R}^6$ case using the strategy of Chodosh–Li–Minter–Stryker together with volume estimates [M]. Thus the only unresolved positive case is a six-dimensional hypersurface in $\mathbb{R}^7$.

The obstruction at the next dimension is concrete. The Simons cone in $\mathbb{R}^8$ is area-minimizing [BDG], and the Hardt–Simon foliation produces smooth complete nonflat area-minimizing hypersurfaces asymptotic to it [HS]. Area minimization implies two-sided stability, so these hypersurfaces disprove the requested statement in ambient dimension 8. This parametric stable counterexample occurs one dimension earlier than the non-affine entire-graph counterexamples, which begin with graphs over $\mathbb{R}^8$ in ambient $\mathbb{R}^9$.

One important feature of the recent work is methodological. The Chodosh–Li proof in $\mathbb{R}^4$ has a positive-scalar-curvature flavor, and their anisotropic version gives an alternative proof which is closer to scalar-curvature techniques [CL2]. This suggests that the remaining $\mathbb{R}^7$ case should be studied not only through classical curvature estimates, but also through stability inequalities, volume growth, blow-down cones, and scalar-curvature comparison ideas. Bellettini proves the required curvature estimate for six-dimensional stable immersions under extrinsic Euclidean volume growth [B]. Antonelli–Xu isolate an algebraic obstruction to extending the current μ-bubble parameter choices to $n = 6$ [AX], while a 2026 synthesis confirms that this is still the sole missing dimension [CCMMR].

Related open problems.

1. *Original direction.* Decide whether every complete two-sided stable minimal hypersurface in $\mathbb{R}^7$ is a hyperplane.
2. *Original direction.* Remove the extrinsic Euclidean-volume-growth hypothesis from the known $\mathbb{R}^7$ curvature estimate, or construct a counterexample.
3. *Reference-driven formulation.* Classify the blow-downs of complete stable $M^6 \to \mathbb{R}^7$ and determine whether stability alone forces Euclidean volume growth.
4. *Reference-driven formulation.* Extend the recent scalar-curvature and positive-mass methods used in low dimensions to the full stable Bernstein range.
5. *Reference-driven formulation.* Find a replacement for the matrix inequalities in the present μ-bubble arguments, which Antonelli–Xu show cannot close in hypersurface dimension 6 with the existing parameter family [AX].
6. *AI-generated formulation.* Develop a quantitative stable Bernstein theorem: if the stability inequality is almost sharp at all scales, then the hypersurface is quantitatively close to a hyperplane.

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Problem 103

If u is an entire solution of the minimal surface equation on $\mathbb{R}^N$, does u have polynomial growth?

Status. Open in this form. For $N \leqslant 7$ the Bernstein theorem implies that every entire solution is affine, hence has linear growth. For $N \geqslant 8$ non-affine entire solutions exist, but the general growth question remains a separate asymptotic problem.

Bombieri and Giusti [BG] gave a central analytic tool for this circle of questions. Bombieri also suggested a connection with the first eigenvalue of minimal hypersurfaces in $\mathbb{S}^N$; the blow-down analysis of Allard–Almgren [AA] is another important point of contact.

The background is the classical Bernstein problem. Bernstein proved the two-dimensional case; De Giorgi, Almgren, and Simons extended the flatness theorem up to graphs over $\mathbb{R}^7$. Bombieri–De Giorgi–Giusti then constructed non-affine entire solutions in dimension $N \geqslant 8$ using the minimizing property of the Simons cone. Thus polynomial growth is automatic in the affine range, while in the non-affine range the question asks for a much finer description of the asymptotic behavior of entire minimal graphs.

Moser's theorem shows that bounded gradient forces an entire solution to be affine. Bombieri–Giusti proved Harnack inequalities on minimal surfaces and generalized this rigidity under boundedness assumptions on many derivatives. More recently, Farina sharpened this direction: for $N \geqslant 8$, if $N - 7$ partial derivatives are bounded on one side, then the solution is affine; this is sharp with respect to the Bombieri–De Giorgi–Giusti examples. These results show that asymptotic control of slopes is powerful, but they do not by themselves settle the unrestricted polynomial-growth question.

The connection with Allard–Almgren is through blow-downs and tangent cones at infinity. If one could prove sufficient uniqueness or quantitative convergence of blow-downs for entire minimal graphs, the growth problem would become closer to the spectral and stability questions for the corresponding minimal cones in spheres. This is the reason the problem is naturally related to first-eigenvalue questions for minimal hypersurfaces in $\mathbb{S}^N$.

Recent work around the question suggests two nearby but distinct directions. Guo studies the much stronger ansatz that u itself is a polynomial [G]. No nonlinear real polynomial solution is currently known. Guo proves that there are no quadratic or cubic polynomial solutions and obtains a factorization theorem for the leading homogeneous term $P_m = p^k Q$, where k is odd, Q is nonnegative, and the zero set of the irreducible factor p is an area-minimizing cone. The tangent cone cannot be isoparametric. In domain dimension 8, existence of a nonlinear polynomial solution would force a non-strictly-minimizing cubic cone. These are strong algebraic restrictions, but polynomiality of u is far stronger than the polynomial-growth assertion in Yau's question.

Nonlocal and higher-codimension minimal graph problems have Bernstein-type analogues, but they do not settle the classical codimension-one problem. Weitsman proved an exponential upper growth estimate for positive minimal graphs over planar simply connected domains with zero boundary values [W]. This is a two-dimensional boundary-value theorem, not an estimate for arbitrary entire graphs on $\mathbb{R}^N$; it therefore supplies a comparison mechanism rather than direct progress on the conjecture.

Related open problems.

1. *Original direction.* Prove or disprove that every entire solution of the minimal surface equation on $\mathbb{R}^N$ has polynomial growth.

2. *Original direction.* Classify the possible growth rates of non-affine entire minimal graphs in the Bombieri–De Giorgi–Giusti range $N \geqslant 8$.

3. *Reference-driven formulation.* Determine whether uniqueness of blow-downs for entire minimal graphs implies polynomial asymptotics.

4. *Reference-driven formulation.* Relate polynomial growth of an entire minimal graph to the first eigenvalue, stability, and Jacobi fields of its tangent cone at infinity.

5. *Reference-driven formulation.* Compare polynomial growth with weaker a priori bounds, such as exponential growth estimates for minimal graphs under boundary hypotheses, and identify which extra entire-graph assumptions force an algebraic rate [W].

6. *Reference-driven formulation.* Decide whether any non-affine real polynomial can solve the codimension-one minimal surface equation; if so, classify the possible tangent-cone factors allowed by Guo's structure theorem [G].

7. *AI-generated formulation.* Develop a frequency-function or monotonicity theory for the minimal surface equation which detects polynomial growth directly from blow-down data.

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Problem 104

Classify the topological type of the seven-dimensional area minimizing cones in $\mathbb{R}^8$.

Status. Open. The regularity theory reduces the problem to the study of smooth stable minimal hypersurfaces $M^6 \subset \mathbb{S}^7$ which arise as links of area-minimizing hypercones in $\mathbb{R}^8$, but no complete topological classification of such links is known. As of 2026, the only known area-minimizing hypercones in $\mathbb{R}^8$ are the quadratic cones over $\mathbb{S}^2 \times \mathbb{S}^4$ and $\mathbb{S}^3 \times \mathbb{S}^3$; it is not known whether this list is complete [E, C].

Lawson observed that one expects only finitely many diffeomorphism classes and that effective bounds should be obtainable. Merely from stability, Simons [Ss] obtains an explicit L^2 estimate on the second fundamental form of the link $M^6 \subsetneq \mathbb{S}^7$. Comparable L^p estimates through the top dimension would give topological control, but the required effective classification of links has not been carried out.

The two known models are the quadratic cones over the minimal products $\mathbb{S}^2(\sqrt{2/6}) \times \mathbb{S}^4(\sqrt{4/6})$ and $\mathbb{S}^3(1/\sqrt{2}) \times \mathbb{S}^3(1/\sqrt{2})$ in $\mathbb{S}^7$. The symmetric second example is the usual Simons cone, whose area-minimizing property was proved by Bombieri–De Giorgi–Giusti. The third stable quadratic cone over $\mathbb{S}^1 \times \mathbb{S}^5$ is minimizing only on one side and is not area-minimizing [E]. More generally, an area-minimizing hypersurface cone in $\mathbb{R}^8$ has an isolated singularity at the origin; away from the origin its link is a smooth minimal hypersurface in the unit sphere. Thus the classification problem is equivalent to a classification of those smooth minimal hypersurfaces in $\mathbb{S}^7$ whose cones are area-minimizing.

Simons' work gives the fundamental stability identity and L^2 control for stable minimal hypersurfaces. Schoen–Simon–Yau and Schoen–Simon developed curvature estimates and regularity theory for stable minimal hypersurfaces, while Leon Simon obtained further curvature estimates and compactness tools relevant to possible links. These results explain why one expects finiteness of topological types, but they do not identify all possible types.

There are many explicit and semi-explicit constructions of minimizing cones in other dimensions and codimensions. Calibrated geometry gives important classes, and Lawlor's curvature criterion provides a powerful sufficient condition. Cui's 2026 survey and construction paper emphasizes that all currently known regular area-minimizing hypercones are isoparametric, while also recording Yau's $\mathbb{R}^8$ classification question as open [C]. This empirical isoparametric pattern is not a classification theorem.

Two more recent directions sharpen the surrounding analytic picture. Zhang revisited Lawson's homogeneous area-minimizing hypercones and showed that the full homogeneous list classified by Lawlor is area-minimizing, with coflat calibrations singular only at the origin [Z]. Székelyhidi constructed minimal hypersurfaces with cylindrical tangent cones and proved a strong unique-continuation theorem for such tangent cones; many of the examples are area-minimizing [Sz]. These works do not classify seven-dimensional minimizing hypercones in $\mathbb{R}^8$, but they clarify which analytic and calibration mechanisms can produce or rigidify minimizing cones. Edelen proves a cone-decomposition and bi-Lipschitz compactness theory for stable or bounded-index seven-dimensional minimal hypersurfaces and obtains finiteness of diffeomorphism types under global area and index bounds in a closed eight-manifold [E]. He explicitly does not classify the tangent cones; rather, his theorem shows what controlled degeneration can be proved without such a classification.

Related open problems.

1. *Original direction.* Classify all smooth stable minimal hypersurfaces $M^6 \subset \mathbb{S}^7$ whose cones are area-minimizing.

2. *Original direction.* Decide whether the two known quadratic minimizing cones, with links $\mathbb{S}^2 \times \mathbb{S}^4$ and $\mathbb{S}^3 \times \mathbb{S}^3$, exhaust all area-minimizing hypercones in $\mathbb{R}^8$.

3. *Original direction.* Obtain effective bounds on the Betti numbers and diffeomorphism types of the links of seven-dimensional area-minimizing cones in $\mathbb{R}^8$.

4. *Reference-driven formulation.* Decide whether every such cone is forced by a known symmetry, calibration, or Lawlor-type minimizing criterion.

5. *Reference-driven formulation.* Classify the possible singular tangent cones of area-minimizing hypersurfaces in dimension 8.

6. *Reference-driven formulation.* Relate Lawson–Lawlor homogeneous minimizing cones and cylindrical tangent-cone constructions to the topology of links in $\mathbb{S}^7$ [L, Sz, Z], and determine whether Edelen's cone-decomposition invariants can be made effective for the two known cone models [E].

7. *AI-generated formulation.* Construct computable invariants of stable minimal links in $\mathbb{S}^7$ which distinguish the finite list of possible topological types, if Lawson's finiteness expectation is made effective.

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Problem 105

(Chern). Consider the set of all compact minimal hypersurfaces in $\mathbb{S}^N$ with constant scalar curvature. Think of the scalar curvature as a function on this set. Is the image of this function a discrete set of positive numbers?

Status. Open in general. For a minimal hypersurface $M^n \subset \mathbb{S}^{n+1}$, constant scalar curvature is equivalent to constant $S = |A|^2$, since

$$R = n(n-1) - S.$$

Thus Chern's question is usually formulated as the discreteness of the possible constant values of $|A|^2$.

There have been the works of Simons [S], Chern–do Carmo–Kobayashi [CDK], Lawson [L] and Yau [Y]. More recently, Terng and Peng [TP] made a breakthrough on this problem.

The first decisive results are gap theorems. Simons' identity gives a strong integral inequality for minimal submanifolds in spheres. Chern–do Carmo–Kobayashi and Lawson used this circle of ideas to show rigidity in the first pinching range: in the hypersurface case, if $S \leqslant n$, then the constant values force either the totally geodesic sphere or the Clifford examples. These results identify the first possible values but do not prove discreteness of the whole image.

Peng–Terng introduced higher-order integral identities and invariants which gave important progress toward the next gaps. Their work became the standard route to the stronger form of the conjecture: a compact minimal hypersurface in $\mathbb{S}^{n+1}$ with constant scalar curvature should be isoparametric. If this stronger statement were true, discreteness would follow from the classification and structure theory of isoparametric hypersurfaces.

Modern work has obtained sharper partial gap theorems. For example, Lei–Xu–Xu proved a rigidity theorem near the first Clifford value: under $0 \leqslant S - n \leqslant n/18$, one has $S \equiv n$ and the hypersurface is a Clifford torus. Cheng–Wei–Yamashiro proved further rigidity under additional constancy assumptions, such as constancy of the third mean curvature invariant f_3, in an improved numerical range. These results support Chern's prediction but still leave the full discreteness problem open.

Very recent papers continue to attack special curvature-invariant regimes without settling the unrestricted conjecture. Ge–Tan–Yan–Zhang proved that a closed minimal Willmore hypersurface of dimension $n \geqslant 5$ with constant scalar curvature and $S > n$ satisfies an explicit lower bound tending to $2n - 3$ as $n \to \infty$; under either of two pointwise lower bounds on f_4 they obtain the expected conclusion $S \geqslant 2n$ [GTYZ]. Their proof refines Yang–Cheng's Peng–Terng integral estimates and closes them with a pointwise upper estimate for f_4.

He–Xu–Zhao proved a classification theorem for closed minimal hypersurfaces $M^4 \subset \mathbb{S}^5$ with constant scalar curvature and constant third mean curvature: the examples are the equatorial sphere, Clifford products, or Cartan's minimal hypersurface, with $S = 0, 4, 12$ [HXZ]. Their argument studies the characteristic polynomial of the shape operator and the image of the Gauss–Kronecker curvature, then invokes the low-dimensional isoparametric classification.

Ge–Liu–Luo–Yan proved a 2026 rigidity theorem in $\mathbb{S}^5$ under constant Gauss–Kronecker curvature, replacing the usual consecutive trace hypotheses by the invariant set $\{H, S, K\}$ [GLLY]. These authors construct weighted 3-forms adapted to S and K and prove the Euler characteristic vanishes in order to close the global curvature estimates across the repeated-principal-curvature locus. These preprints reinforce the trend that Chern's discreteness problem is being approached through increasingly precise finite-dimensional curvature invariants, but each still assumes an invariant beyond constant scalar curvature.

Related open problems.

1. *Original direction.* Prove that the set of constant values of $S = |A|^2$ for compact minimal hypersurfaces in $\mathbb{S}^{n+1}$ is discrete.
2. *Original direction.* Prove the stronger Chern conjecture: compact minimal hypersurfaces in $\mathbb{S}^{n+1}$ with constant scalar curvature are isoparametric.
3. *Reference-driven formulation.* Establish the second and higher gap theorems predicted by the isoparametric examples.
4. *Reference-driven formulation.* Relate Chern's conjecture to Yau's first eigenvalue conjecture for embedded minimal hypersurfaces in spheres.
5. *Reference-driven formulation.* Determine which additional constant curvature invariants, such as f_3, fourth mean curvature, or Gauss–Kronecker curvature, force isoparametricity in low dimensions and then in general [CWY, GTYZ, HXZ, GLLY].
6. *Reference-driven formulation.* In $M^4 \subset \mathbb{S}^5$, remove first the constant-f_3 or constant-K hypothesis from the 2026 classifications, and identify which part of the characteristic-polynomial or weighted-form arguments genuinely requires it [HXZ, GLLY].

7. *AI-generated formulation.* Develop an effective "Chern spectrum" for constant-scalar-curvature minimal hypersurfaces, including numerical invariants which would isolate each possible value of S.

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Problem 106

Let M be a compact, three-dimensional manifold with curvature -1. Let Σ be a surface of genus g such that there is some continuous $f : \Sigma \to M$ with $f_* : \pi_1(\Sigma) \to \pi_1(M)$ an injection. It was known ([SY], [SU]) that such a map can be deformed to be a minimal immersion. Is it true that for most M, the resulting immersion would be unique?

Status. Open as stated, and not yet a precise genericity question. Almost-Fuchsian surface-group covers have a unique closed incompressible minimal surface, whereas quasi-Fuchsian manifolds can have arbitrarily many. The latter manifolds are noncompact, however, so their multiplicity does not by itself disprove the original question for compact hyperbolic 3-manifolds.

The existence part belongs to the classical interaction between minimal surface theory and 3-manifold topology. Schoen–Yau and Sacks–Uhlenbeck gave analytic existence theorems for least-area or harmonic representatives of incompressible surface classes. The topological theory of Freedman–Hass–Scott further shows that least-area incompressible surfaces behave optimally when their homotopy classes contain embedded representatives [FHS].

For a surface subgroup $\Gamma < \pi_1(M)$, a minimal immersion lifts to the associated cover $\mathbb{H}^3/\Gamma$. When this cover is almost-Fuchsian—that is, it contains a closed incompressible minimal surface with principal curvatures in $(-1, 1)$—the surface is unique [U, HW1]. The proof uses the normal exponential flow and the strict principal-curvature bound to obtain barriers and rule out a second closed minimal surface.

Outside the almost-Fuchsian region, Biao Wang constructed quasi-Fuchsian manifolds with at least two closed incompressible minimal surfaces [W]. Huang–Wang then produced, for every $N > 0$, a quasi-Fuchsian manifold containing at least 2^N such surfaces [HW2]. Their min–max and barrier construction proves that the curvature condition is genuinely restrictive, but these examples have topology $\Sigma \times \mathbb{R}$ and infinite volume. No source checked in this audit supplies a compact hyperbolic 3-manifold with two minimal immersions in the same prescribed surface-subgroup homotopy class.

Agol's virtual Haken theorem [A] greatly enlarges the supply of embedded incompressible surfaces in finite covers of closed hyperbolic 3-manifolds, but it does not compare minimal representatives in one fixed homotopy class. Recent work of Coskunuzer–Huang–Lowe–Vargas Pallete gives Schottky manifolds with infinitely many closed minimal surfaces [CHLV], again in a noncompact setting. These results clarify that abundance across homotopy classes and uniqueness within one class are different questions.

Finally, Mostow rigidity prevents one from varying the hyperbolic metric on a fixed closed 3-manifold. Thus "most M" must mean a probability model or density on closed hyperbolic manifolds, or else a genericity notion for surface subgroups inside a fixed manifold. Until one of these is specified, the original question has no single formal yes-or-no interpretation.

Related open problems.

1. *Original direction.* Decide whether a fixed π_1-injective homotopy class in a closed hyperbolic 3-manifold can contain two noncongruent minimal immersions.

2. *Original direction.* Choose a precise model of "most," for example random Heegaard splittings or a complexity-ordered family of closed hyperbolic manifolds, and test uniqueness in that model.

3. *Reference-driven formulation.* Characterize the quasi-Fuchsian manifolds in which the closed incompressible minimal surface is unique, beyond the sufficient almost-Fuchsian condition [U, HW1, W, HW2].

4. *Reference-driven formulation.* For a surface subgroup of a closed hyperbolic 3-manifold, relate uniqueness in its subgroup cover to index, least-area properties, and quantitative bounds on the principal curvatures.

5. *Reference-driven formulation.* Determine how the number and Morse indices of closed incompressible minimal surfaces change when one crosses the boundary of almost-Fuchsian space [HW1, HW2].

6. *AI-generated formulation.* Develop a wall-crossing theory for minimal incompressible surfaces in quasi-Fuchsian 3-manifolds, where new minimal surfaces appear or disappear at degenerate critical points of the area functional.

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Problem 107

Let M be a complete minimal surface in $\mathbb{R}^3$. Osserman [O] has proved that the Gauss map of M cannot omit a set of positive capacity in $\mathbb{S}^2$ and he conjectured that it in fact could not omit more than four points in $\mathbb{S}^2$. Recently, Xavier [X] proved that it cannot omit more than eleven points. Based on the method of Xavier, Bombieri [B] improved the number to seven. Can one improve it to four? Can one generalize these assertions to three-dimensional minimal hypersurfaces?

Status. Partially resolved. The surface case in $\mathbb{R}^3$ is solved. Fujimoto proved that the Gauss map of a complete nonflat minimal surface in $\mathbb{R}^3$ can omit at most four points of $\mathbb{S}^2$, and this bound is sharp. Higher-dimensional and higher codimension analogues remain separate questions and are only partially understood in general.

The historical line is now quite clear. Osserman's theorem first showed that the Gauss image of a complete nonflat minimal surface cannot omit a set of positive capacity. Xavier then proved that the Gauss map cannot omit seven points; the older attribution to an unpublished Bombieri improvement should not be used as a separate reference here. Fujimoto completed the Osserman–Xavier program by proving the sharp bound four. The sharpness comes from classical examples of Voss: complete minimal surfaces whose Gauss maps omit four prescribed points.

For complete minimal surfaces with finite total curvature the situation is more rigid. Jorge–Mercuri proved the sharp bound that the Gauss map of a complete nonflat minimal surface of finite total curvature can omit at most two points [JM], with equality realized by the catenoid. This uses the meromorphic extension of the Gauss map to the compactified surface and topology of the covering, so it is a different mechanism from Fujimoto's general completeness argument.

For higher-dimensional minimal hypersurfaces the Gauss map takes values in a sphere but is no longer a meromorphic function on a Riemann surface. The complex-analytic Nevanlinna theory behind Fujimoto's proof is therefore not directly available. This is the main reason the three-dimensional hypersurface analogue remains a different and harder problem.

There are nevertheless higher-dimensional Bernstein-type substitutes. Ding proved that a complete oriented embedded minimal hypersurface in $\mathbb{R}^{n+1}$ with Euclidean volume growth is an affine hyperplane if its Gauss image avoids a neighborhood of a half-equator [D]. The proof treats the Gauss map as a harmonic map and combines image avoidance with curvature estimates and a Liouville argument. It is a strong rigidity theorem, but not a finite omitted-point theorem. Jost–Xin–Yang obtain related Bernstein conclusions when the normal Gauss image lies in suitable regions of a Grassmannian [JXY].

For two-dimensional domains, complex analysis still gives sharper analogues in higher codimension. Aiyama–Akutagawa–Imagawa–Kawakami study complete minimal surfaces in $\mathbb{R}^4$ [AAIK], while Dethloff–Ha treat ramification on annular ends in $\mathbb{R}^3$ and $\mathbb{R}^4$ [DH]. Huynh proves that a nondegenerate generalized Gauss map of a complete minimal surface in $\mathbb{R}^n$ cannot omit a generic projective hypersurface of an explicit bounded degree [Hu]. These results show which parts of Nevanlinna theory survive a change of codimension, but they do not answer Yau's higher-dimensional-domain question.

Related open problems.

1. *Original direction.* Find the optimal omitted-value theorem for the Gauss map of complete minimal hypersurfaces in $\mathbb{R}^{n+1}$, $n \geqslant 3$.
2. *Original direction.* Determine which omitted subsets of $\mathbb{S}^n$ can occur as the complement of the Gauss image of a complete nonflat minimal hypersurface.
3. *Reference-driven formulation.* Remove or weaken the Euclidean-volume-growth and half-equator-neighborhood hypotheses in Ding's higher-dimensional rigidity theorem [D].
4. *Reference-driven formulation.* Classify complete minimal surfaces in $\mathbb{R}^3$ whose Gauss map omits exactly four points.
5. *Reference-driven formulation.* Determine the optimal projective-hypersurface defect relation for generalized Gauss maps of complete minimal surfaces in $\mathbb{R}^n$ [AAIK, DH, Hu].
6. *AI-generated formulation.* Relate omitted-value bounds for higher-dimensional Gauss maps to quantitative stability or Liouville theorems for the harmonic Gauss map.

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Problem 108

Suppose H is an area-minimizing hypersurface in a manifold M. Prove that by a perturbation of the metric on M, the singularities of H may be eliminated, retaining the $N-1$ dimensional homology class represented by H. For some related problems, see B. White [W].

Status. Partially resolved. For integral codimension-one homology minimizers, the requested arbitrarily small metric perturbation is now known through ambient dimension 11. Above dimension 11, perturbation gives a strictly improved bound on the singular-set dimension, but full smoothness remains open. The PL question is also open and is logically separate.

Is it true that the support of a codimension-one area-minimizing current has a PL structure? For a general high-codimension minimal current, it is not true that the support is a real analytic variety (see Milani [M]). For the present state of the problem, see Almgren [A].

The starting point is the Federer–Fleming and De Giorgi regularity theory for mass-minimizing hypersurfaces. In codimension one, the singular set has codimension at least 7 in the hypersurface. This explains why the problem is invisible in low dimensions and why dimension 8 is the first critical case, with the Simons cone as the basic model singularity.

White's work initiated a systematic study of how singularities and tangent cones behave under perturbations. Later regularity theorems of Schoen–Simon and Wickramasekera gave powerful structural control for stable and almost minimizing hypersurfaces. In high codimension, Almgren's regularity theorem and the De Lellis–Spadaro program show that the singular set is much smaller than the whole current, but the local structure can be highly nonclassical; Milani's examples warn against expecting real analyticity of supports in general.

The metric-perturbation problem has a sharp recent chronology. Smale proved it for homologically area-minimizing hypersurfaces in ambient dimension 8 [Sma]. Chodosh–Mantoulidis–Schulze extended the homology and Plateau results to ambient dimensions 9 and 10 [CMS] and then obtained improved generic partial regularity in all higher dimensions [CMS2]. Finally, Chodosh–Mantoulidis–Schulze–Wang proved the following precise form through dimension 11: for a closed oriented $(n+1)$-manifold, a nonzero class in $H_n(M;\mathbb{Z})$, and any metric g, there are arbitrarily C^∞-small perturbations g' such that every g'-minimizing current in that class is a sum of disjoint smooth oriented hypersurfaces when $n+1 \leqslant 11$ [CMSW]. When $n+1 \geqslant 12$, the same theorem gives

$$\dim_{\mathcal{H}} \mathrm{Sing} \leqslant n - 10 - \varepsilon_n$$

for a dimensional $\varepsilon_n > 0$. Thus Yau's perturbation request is solved through ambient dimension 11, not merely supported by examples there.

Li–Wang's generic regularity and nondegeneracy theorem for locally stable and min–max hypersurfaces in ambient dimension 8 [LW] belongs to the same program, but it should not replace Smale's earlier theorem for homology minimizers. The proof mechanisms in the recent homology results combine a one-parameter family of minimizers, cone splitting, quantitative separation of leaves by positive Jacobi fields, and, in dimension 11, a second blow-up analysis using an Almgren frequency for Jacobi fields on cylindrical minimizing cones.

The codimension-one hypothesis is essential. Liu proved that in many higher-codimension ranges there are open sets of metrics for which every minimizer in a smooth-representable homology class remains singular; for dimensions $d, c \geqslant 3$ the forced lower bound is $\max\{d-5, d-c\}$ [Liu]. This negative theorem does not contradict [CMSW], which concerns hypersurfaces. It also reinforces that the PL question is a different structural problem: rectifiability and stratification of the singular set do not automatically triangulate the entire support.

Related open problems.

1. *Original direction.* Extend the full metric-perturbation theorem for homologically area-minimizing hypersurfaces from ambient dimension 11 to all dimensions [CMSW].

2. *Original direction.* Determine whether one can perturb a specified singular minimizer itself to a smooth minimizer, rather than only obtain smoothness of minimizers after perturbing the metric in the fixed homology class.

3. *Reference-driven formulation.* Decide whether the support of a codimension-one area-minimizing current always has a natural PL structure.

4. *Reference-driven formulation.* Classify which tangent cones to minimizing hypersurfaces persist under generic perturbations of the metric.

5. *Reference-driven formulation.* Improve the $n-10-\varepsilon_n$ singular-set estimate in ambient dimensions at least 12, and identify the tangent cones that obstruct another dimension jump [CMS2, CMSW].

6. *Reference-driven formulation.* Resolve the remaining higher-codimension pairs not covered by Liu's persistence theorem, especially codimension 2 and the low-dimensional cases $d = 3, 4 < c$ [Liu].

7. *AI-generated formulation.* Construct a deformation complex for singular minimizing hypersurfaces whose obstruction classes measure whether a metric perturbation can remove each singular stratum.

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Problem 109

Let Ω be a k-dimensional, compact, minimal submanifold of $\mathbb{R}^N$. Prove the isoperimetric inequality

$$\mathrm{Vol}(\Omega)^{k-1} \leqslant c_k \,\mathrm{Vol}(\partial\Omega)^k$$

where c_k is given by

$$c_k = \frac{\mathrm{Vol}(B(1))^{k-1}}{\mathrm{Vol}(\partial B(1))^k}$$

with $B(1)$ signifying the unit ball in $\mathbb{R}^k$. It is true if $k = N$, that is, domains in $\mathbb{R}^N$.

Status. Partially resolved. The sharp inequality is classical for domains in $\mathbb{R}^k$ and for several classes of minimal surfaces. Brendle proved the sharp isoperimetric inequality for compact minimal submanifolds of codimension at most 2 in Euclidean space, with equality only for a flat round ball. The fully general sharp statement in codimension at least 3 remains open.

This inequality with a constant larger than the sharp Euclidean constant above is known to be true; see [FF], [Alm], [All], [MS], and [BDG].

If $k = 2$ and Ω is simply connected, this result is classical and is due to Carleman (see Osserman [O]). If $k = 2$ and Ω is doubly connected, it is again true, and due to Osserman and Schiffer [OS] and Feinberg [F].

An approach is to show that the extremal case for the inequality can be realized as a stationary integral varifold which one might be able to show is a flat k-disk. That flat k-disks are indeed extreme among nearby nonparametric surfaces has been studied by B. White [W].

The early history begins with the classical Euclidean isoperimetric inequality and its two-dimensional minimal-surface analogues. Carleman proved the sharp inequality for simply connected minimal surfaces. Osserman–Schiffer and Feinberg proved the doubly connected case. These results show that minimality supplies enough extra structure to recover the Euclidean sharp constant in important surface cases.

For higher-dimensional submanifolds, the Michael–Simon Sobolev inequality gives a powerful isoperimetric inequality, but not with the sharp Euclidean constant. Almgren's optimal filling inequality gives sharp results for area-minimizing currents, which is stronger than mere stationarity but applies to a different class of objects.

Brendle's work introduced an ABP-type method for submanifolds and proved a sharp Sobolev inequality in arbitrary dimension and codimension. In codimension 2, and hence also in codimension 1 by embedding a hypersurface into one higher Euclidean space, this yields exactly the sharp isoperimetric inequality stated here. Equality is rigid: the extremal submanifold must be a flat round ball. The arbitrary-codimension case remains the natural remaining form of the problem, so the safest status is partial resolution rather than a complete solution of Yau's full statement.

The ABP method has continued to generate related sharp inequalities. Brendle also proved a sharp logarithmic Sobolev inequality for submanifolds in Euclidean space, including the mean-curvature term in the Michael–Simon spirit [B2]. Balogh–Kristály later obtained sharp and rigidity-enhanced logarithmic-Sobolev inequalities on noncompact Euclidean submanifolds, with codimension-free constants in the minimal case [BK]. Lu extended the ABP approach to sharp logarithmic-Sobolev inequalities in manifolds with nonnegative curvature [Lu]. These results do not settle the sharp isoperimetric constant in arbitrary codimension, but they make clear that the remaining problem is part of a larger sharp-constant program for Sobolev, logarithmic Sobolev, and isoperimetric inequalities on submanifolds.

Two recent expositions make the remaining gap especially transparent. Brendle–Eichmair give both the ABP and unequal-dimensional optimal transport proofs in the hypersurface case [BE], while Brendle's 2026 survey isolates the normal-bundle Jacobian estimate that produces the sharp codimension-2 constant [B3]. Balogh–Kristály–Mester obtain explicit L^p Sobolev inequalities for minimal submanifolds of every codimension; for $p \geqslant 2$ their constants are codimension-free and asymptotically sharp as the dimension grows, but they explicitly remain nonsharp in codimension at least 3 [BKM]. Hence these advances do not close Yau's $p = 1$ sharp-constant problem.

Related open problems.

1. *Original direction.* Prove the sharp Euclidean isoperimetric inequality for compact minimal submanifolds in arbitrary codimension.

2. *Original direction.* Determine whether Brendle's ABP method can be sharpened to give the Euclidean constant beyond codimension 2.

3. *Reference-driven formulation.* Classify equality and near-equality cases for sharp isoperimetric inequalities on minimal submanifolds.

4. *Reference-driven formulation.* Extend the sharp inequality to singular stationary integral varifolds under natural boundary hypotheses.

5. *Reference-driven formulation.* Find a codimension-independent $p \downarrow 1$ limit of the transport inequalities in [BKM] that retains the Euclidean sharp constant.

6. *AI-generated formulation.* Develop a quantitative stability version: if a compact minimal submanifold nearly attains the Euclidean isoperimetric constant, then it is close, in a suitable varifold or graphical sense, to a flat round disk.

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Problem 110

Let Σ be a compact surface, and let $f : \Sigma \to M$ be a minimal immersion into a three-dimensional manifold such that f has least area among all the maps homotopic to f. (If Σ has boundary, we consider only immersions which are embeddings on $\partial\Sigma$ and we fix the image of $f(\partial\Sigma)$ also.) If Σ is $\mathbb{S}^2$ or a planar domain, Meeks–Yau [MY] prove that f is an embedding. When Σ has higher genus, Freedman–Hass–Scott [FHS] prove that f is an embedding assuming that f is homotopic to an embedding. Without this latter assumption, f need not be an embedding. However, it is believed that f tends to minimize the complexity of the self-intersection set. It is of basic interest for the topologists to estimate the number of triple points of f.

Status. Open for effective triple-point estimates, but the motivating claim that least area should minimize the number of triple points is false in general. Gulliver–Scott constructed a least-area incompressible torus with triple points that is homotopic to a general-position immersion with no triple points [GS].

The foundational results separate the problem into two regimes. For spheres and planar domains, Meeks–Yau proved strong embeddedness results for least-area maps. For higher genus incompressible surfaces, Freedman–Hass–Scott proved that a two-sided least-area representative homotopic to an embedding is embedded, apart from the explicit possibility that it double covers a one-sided embedded surface [FHS]. Their proof is based on area swapping, cut-and-paste, and rounding: an inessential product region or removable intersection would yield an area-decreasing competitor.

Their slogan that least-area surfaces "intersect least" requires the right invariant. For a two-sided incompressible immersion f, they define a self-intersection invariant $D(f)$ that remains meaningful when the immersion is not in general position, and prove that a least-area map minimizes D in its homotopy class. For tori and Klein bottles this controls the ordinary number of double curves. For higher-genus surfaces, they give an example showing that a least-area map need not minimize the number of double curves.

Gulliver–Scott show that triple points are even less well behaved [GS]. Their main example is a Riemannian 3-manifold and a least-area incompressible map of a torus having triple points, although the same homotopy class contains a general-position immersion without triple points. Consequently neither triple-point count nor the lexicographic complexity (t, d) can be the universal invariant minimized by area. The remaining problem is to bound triple points or to relate them to $D(f)$ under additional transversality, analyticity, or topological hypotheses.

There are purely topological constraints. Banchoff proved that for a self-transverse immersion of a closed surface in $\mathbb{R}^3$, the number of triple points is congruent modulo 2 to the Euler characteristic of the surface [B]. Carter–Ko generalized the parity formula to immersions in 3-manifolds, adding the Euler characteristic of the image [CK]. These are obstructions, not upper bounds.

There are also useful existence results in noncompact hyperbolic settings. Collin–Hauswirth–Mazet–Rosenberg constructed compact embedded and least-area incompressible minimal surfaces in complete finite-volume hyperbolic 3-manifolds [CHMR]. Huang–Wang proved that closed immersed incompressible surfaces in cusped hyperbolic 3-manifolds can be deformed to closed immersed least-area surfaces in their homotopy classes [HW]. These results do not directly estimate triple points, but they broaden the supply of least-area representatives on which the refined questions can be tested.

Related open problems.

1. *Original direction.* Give an effective upper or lower bound for the number of triple points of a least-area representative in terms of topological data of the homotopy class.

2. *Original direction.* Relate the number of triple points to the Freedman–Hass–Scott invariant $D(f)$ under a generic or analytic metric [FHS, GS].

3. *Reference-driven formulation.* Determine hypotheses under which a least-area representative minimizes the lexicographic complexity (t, d), despite the Gulliver–Scott counterexample [GS].

4. *Reference-driven formulation.* Decide whether a least-area immersion for a bumpy or real-analytic metric has finitely many triple points and admits an effective bound.

5. *Reference-driven formulation.* Combine the parity formula with normal-surface or subgroup complexity to obtain nontrivial lower bounds for unavoidable triple points [B, CK].

6. *AI-generated formulation.* Define a "minimal self-intersection norm" on surface homotopy classes in a 3-manifold and compare it with the number of triple points of least-area representatives.

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Problem 111

Let $f : M_1 \to M_2$ be a diffeomorphism between two compact manifolds with negative curvature. If $h : M_1 \to M_2$ is a harmonic map which is homotopic to f, is h a univalent map? For $n = 2$, this was proved in [SY] and [S].

Status. True in dimension 2 and false in every dimension $n \geqslant 6$; the cited counterexample technology does not settle dimensions 3, 4 and 5. Schoen–Yau and Sampson proved univalence for harmonic maps between negatively curved surfaces [SY, S]. Farrell–Ontaneda–Raghunathan proved that for every $n \geqslant 6$ there are diffeomorphisms between closed negatively curved n-manifolds whose harmonic representatives are not one-to-one [FOR, FOs].

For $n > 2$, Calabi has a counterexample if we do not impose conditions on M_1 and M_2. (In Calabi's example, M_2 is a torus.)

The distinctions among non-univalence, failure to be a diffeomorphism, and non-cellularity are essential. The maps in [FOR] answer Yau's question negatively by failing to be one-to-one. Farrell–Ontaneda also constructed harmonic cellular maps which are limits of diffeomorphisms but are not diffeomorphisms [FO]. Other exotic structure constructions produce harmonic homotopy equivalences that are not cellular [FJO2, FOs]. These are related phenomena, but they are not interchangeable conclusions about one example.

The positive two-dimensional result is a consequence of the special structure of harmonic maps between surfaces: the Hopf differential, negative curvature, and degree-one topology combine to rule out folding. In higher dimensions the analytic uniqueness of harmonic representatives does not force pointwise injectivity.

The high-dimensional counterexamples are topological as much as analytic. Farrell–Jones rigidity and exotic smooth or triangulated structures make it possible to construct negatively curved manifolds which are homeomorphic or homotopy equivalent in controlled ways but whose harmonic representatives fail to be diffeomorphisms or even limits of homeomorphisms. Thus the original question, without further geometric or dimensional hypotheses, has a negative answer.

The remaining meaningful directions are therefore to settle dimensions 3–5 and to isolate hypotheses which restore univalence: curvature pinching, control of the Jacobian, or assumptions on the smooth structure and the harmonic representative.

The contrast with the general existence and uniqueness theory for harmonic maps is important. Eells–Sampson proved existence for targets of nonpositive curvature, and Hartman proved uniqueness in a homotopy class under suitable nonpositive-curvature hypotheses [ES, H]. These theorems explain why there is a canonical harmonic representative in the homotopy class; the Farrell–Ontaneda type examples show that canonicity and uniqueness of the harmonic representative are still much weaker than being a homeomorphism or a diffeomorphism.

Related open problems.

1. *Original direction.* Decide the original question in dimensions 3, 4 and 5.
2. *Original direction.* Find curvature pinching or Jacobian hypotheses which imply univalence in dimensions greater than 2.
3. *Reference-driven formulation.* Classify the possible singular or non-injective behavior of harmonic cellular maps in the Farrell–Ontaneda examples.
4. *Reference-driven formulation.* Study whether stronger target curvature assumptions, such as locally symmetric negative curvature, force better regularity or injectivity of the harmonic representative.
5. *Reference-driven formulation.* Separate the roles of existence, uniqueness, cellularity, and injectivity in the nonpositive-curvature harmonic-map theory, especially for maps homotopic to homeomorphisms or diffeomorphisms [ES, H, FO, FOR].
6. *AI-generated formulation.* Define a quantitative "harmonic folding defect" for degree-one harmonic maps and determine whether it is controlled by exotic smooth-structure invariants.

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Problem 112

Prove that $\pi_i(\mathbb{S}^N)$ can be represented by harmonic maps. What happens if we replace $\mathbb{S}^N$ by a compact manifold with finite fundamental group?

Status. Open for the standard round metrics. No theorem is known which realizes every element of every $\pi_i(\mathbb{S}^N)$ by a single harmonic map $\mathbb{S}^i \to \mathbb{S}^N$. Smith's join construction does prove that every class in $\pi_n(\mathbb{S}^n)$ has a harmonic representative for $1 \leqslant n \leqslant 7$ [S]. Peng–Tang and later Clifford-system constructions realize further specific unstable homotopy classes [PT, QTY]. These results are substantial but do not solve the uniform problem.

One should refer to the paper of R. T. Smith [S].

Peng–Tang developed additional special cases [PT]. If one allows deformation of the domain metric, more examples can be obtained [R].

The problem sits between homotopy theory and the variational theory of the energy functional. Some basic representatives, including totally geodesic maps, holomorphic maps between round 2-spheres, Hopf fibrations, and maps built from harmonic polynomials or orthogonal multiplications, are harmonic for geometric reasons. Smith's construction joins such polynomial maps by solving an equivariant nonlinear ordinary differential equation [S]. Peng–Tang use isoparametric and equivariant geometry to construct additional representatives [PT].

The difficulty is that the target sphere has positive curvature, so the direct Eells–Sampson convexity theory for nonpositively curved targets does not apply. Minimizing or flowing the energy in a prescribed homotopy class can lose energy through bubbling, and the limiting harmonic maps need not preserve the original homotopy class in a simple way. This is why the question is much subtler than merely knowing the homotopy group.

Ratto's work shows that deforming the domain metric produces additional harmonic representatives [R]. This suggests a useful separation between two problems: fixed round domain and target metrics, versus the flexible problem where the domain metric is part of the unknown.

The first Hopf case already displays the gap. For every class in $[\mathbb{S}^3, \mathbb{S}^2]$ one can choose metrics for which a harmonic representative exists, while for the standard round metrics the only presently known examples are compositions of a Hopf fibration with conformal maps of $\mathbb{S}^2$. Recent rigidity results recover this form under extra hypotheses such as horizontal weak conformality or constant energy density [Geo].

The modern analytic obstruction is best understood through bubbling. Sacks–Uhlenbeck compactness produces harmonic spheres after perturbing the energy, but in positive curvature the limiting object may be a bubble tree rather than a single harmonic representative of the original class [SU]. Parker–Wolfson gave a systematic bubble-tree compactification for harmonic maps [PW]. Thus a current form of the problem is to decide when a homotopy class has an honest smooth harmonic representative, rather than only a limiting bubble configuration whose components represent pieces of the class.

Related open problems.

1. *Original direction.* Determine exactly which elements of $\pi_i(\mathbb{S}^N)$ admit harmonic representatives for the standard round metrics.
2. *Original direction.* Determine whether every element of $\pi_i(\mathbb{S}^N)$ can be represented by a harmonic map after a suitable deformation of the domain metric.
3. *Reference-driven formulation.* Extend the representation problem from spheres to compact manifolds with finite fundamental group, using their universal covers.
4. *Reference-driven formulation.* Understand how bubbling in harmonic map heat flow changes the homotopy class of maps between spheres.
5. *Reference-driven formulation.* Classify which homotopy classes of spheres are represented by a single harmonic sphere and which are represented only after bubble-tree decomposition [SU, PW].
6. *Reference-driven formulation.* Classify harmonic maps $\mathbb{S}^3 \to \mathbb{S}^2$ for the round metrics without the extra horizontal-conformality or constant-energy hypotheses [Geo].
7. *AI-generated formulation.* Construct an "energy spectrum" for each element of $\pi_i(\mathbb{S}^N)$ recording whether the class has a smooth harmonic critical representative, only a bubble-tree representative, or no representative for the fixed metrics.

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Problem 113

(Affine geometry).

1. (Chern). Establish Bernstein's theorem for affine geometry: any convex graph over affine space which is an affine maximal hypersurface must be a paraboloid.

2. Classify 3-dimensional compact affine-flat manifolds. In general, it is not known whether any compact affine-flat manifold has zero Euler number; see Milnor [M], Kostant–Sullivan [KS], Sullivan [S1], [S2], and Wood [W].

Status. Partially resolved, with three formulations that must be kept separate. Trudinger–Wang proved the smooth classical affine Bernstein theorem in graph dimension 2 [TW]. For smooth, locally uniformly convex classical affine-maximal hypersurfaces, the higher-dimensional problem remains open already in dimension 3. Their generalized nonsmooth notion has a counterexample in dimensions $n \geqslant 10$. Counterexamples for other values of the parameter in the broader affine-maximal-type equation do not by themselves disprove the classical affine-maximal conjecture [Du1, Du2].

The complete three-dimensional affine crystallographic groups were classified by Fried–Goldman [FG], but this is not a classification of all compact, possibly incomplete, affine 3-manifolds. Chern's Euler-characteristic conjecture for arbitrary compact affine manifolds also remains open; it is proved for complete affine manifolds [KS] and for special affine manifolds, whose holonomy preserves a volume form [Kl].

Milnor formulated the obstruction problem for flat affine structures in terms of flat torsion-free connections [M]. Kostant and Sullivan proved that the Euler characteristic of a complete affine space form is zero [KS], and Sullivan and Wood developed related characteristic class obstructions [S1], [S2], [W]. Trudinger and Wang later proved Chern's affine Bernstein conjecture for smooth surfaces. Their explicit high-dimensional example is affine maximal only in the generalized nonsmooth sense, and they explicitly left the existence of smooth high-dimensional counterexamples open [TW]. In the affine-manifold direction, Fried and Goldman classified three-dimensional affine crystallographic groups [FG], but the broader Chern conjecture that every compact affine manifold has zero Euler characteristic remains open.

Later developments split the two parts of the problem even further. For affine manifolds, Klingler proved Chern's conjecture for compact special affine manifolds, i.e. the case with parallel volume form [Kl]. Bucher–Gelander proved Milnor–Wood type inequalities for manifolds locally modeled on products of hyperbolic planes, giving another important family where affine structures are obstructed [BG]. On the analytic Bernstein side, Du constructed non-quadratic, Euclidean-complete affine-maximal-type hypersurfaces for $N \geqslant 3$ and $\theta \in (1/2, (N-1)/N)$ [Du1], with further explicit examples in related ranges [Du2]. The classical affine-maximal exponent is $\theta = (N+1)/(N+2)$ and lies outside that interval. Sun–Xing–Xu study Calabi-affine and centroaffine Bernstein problems and construct complete examples in those related, but different, geometries [SXX].

Related open problems.

1. *Original direction.* Prove or disprove the smooth classical affine Bernstein statement in every graph dimension $n \geqslant 3$.

2. *Original direction.* Decide Chern's Euler-characteristic conjecture for compact affine manifolds without completeness or parallel-volume hypotheses.

3. *Reference-driven formulation.* Classify compact affine-flat manifolds beyond the known low-dimensional and complete cases.

4. *Reference-driven formulation.* Determine whether special affine, amenable-holonomy, locally symmetric, or product-hyperbolic methods can be combined to prove new cases of Chern's affine Euler-characteristic conjecture [BG, Kl].

5. *Reference-driven formulation.* Determine whether the nonsmooth $n \geqslant 10$ example of Trudinger–Wang can be regularized to a smooth, locally uniformly convex classical affine-maximal hypersurface, or prove a rigidity obstruction to such a regularization [TW].

6. *Reference-driven formulation.* Separate the classical affine-maximal exponent from affine-maximal-type, Calabi-affine and centroaffine variants, where higher-dimensional non-quadratic complete examples are known [Du1, Du2, SXX].

7. *AI-generated formulation.* Relate the analytic Bernstein problem for the affine maximal equation to holonomy and characteristic-class restrictions for compact affine manifolds.

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VII. General Relativity and the Yang–Mills Equation

Problem 114

This is the problem of "cosmic censorship", as coined by Penrose.

Let M be a three-dimensional manifold equipped with a metric g_{ij} and a symmetric tensor h_{ij}. Assume that g_{ij} and h_{ij} satisfy the compatibility requirement necessary for them to represent the induced metric and second fundamental form, respectively, that M would inherit as a spacelike hypersurface of four-dimensional asymptotically flat spacetime satisfying the (vacuum) Einstein field equations.

In the study of global solutions to the vacuum field equation with Cauchy data g_{ij} and h_{ij} on M, one wishes to know the nature of the singularities of the solution obtained. Perhaps the most important open problem in general relativity is this:

Is it true that generically a singularity will have a horizon? (Is there no "naked" singularity?)

This question amounts to asking if the future can be theoretically predicted.

One should consult the book of Hawking and Ellis [HE] for background on this problem.

Status. Open in the full generic form. Cosmic censorship is not a single theorem but a family of weak and strong formulations, and each formulation requires precise hypotheses on asymptotics, matter model, regularity and genericity. Without such hypotheses simple counterexamples occur [Ch].

Hawking and Ellis gave the classical causal and singularity-theorem background [HE]. Christodoulou and Klainerman proved the nonlinear stability of Minkowski space for small asymptotically flat vacuum data [CK], and Lindblad and Rodnianski later gave a wave-coordinate proof of the same stability theorem [LR]. Andersson and Moncrief proved future stability for certain expanding Einstein flows [AM]. Ringström established strong cosmic censorship for $\mathbb{T}^3$-Gowdy spacetimes [R]. These theorems are major confirmations in special regimes, but they do not settle Penrose's generic censorship problem. The nonlinear stability of the Kerr family for small angular momentum has also been proved through the Klainerman–Szeftel and Giorgi–Klainerman–Szeftel program [KS, GKS]. This is a major black-hole stability theorem, but it is not by itself a theorem that generic large-data gravitational collapse forms a horizon; final-state stability and weak cosmic censorship are logically distinct statements. Current literature still treats strong and weak cosmic censorship as central open problems, with different behavior in asymptotically flat, de Sitter, charged, rotating, symmetric, and matter-coupled settings.

The Kerr and de Sitter stability literatures sharpen the modern landscape. Hintz–Vasy proved nonlinear stability of the Kerr–de Sitter family for slowly rotating black holes [HV], and Häfner–Hintz–Vasy proved linear stability of slowly rotating

Kerr black holes in the asymptotically flat setting [HHV]. The Giorgi–Klainerman–Szeftel proof of nonlinear stability for slowly rotating Kerr in vacuum is therefore part of a broader program linking final-state stability and black-hole exterior dynamics [KS, GKS]. Recent work reaches the full subextremal Kerr–de Sitter range only conditionally on mode stability [HPV].

Strong cosmic censorship is sensitive to the regularity class used for extension. Dafermos–Luk prove C^0 stability of a portion of the Kerr Cauchy horizon, while Luk–Oh prove a generic C^2-inextendibility statement for the spherically symmetric Einstein–Maxwell–scalar model [DL, LO]. Thus a formulation may fail at C^0 and hold at a higher regularity threshold. On the weak side, An–Tan prove a generic spherical-symmetry result for the Einstein–Maxwell–charged-scalar system [AT]. None of these symmetry-reduced or perturbative theorems settles the full asymptotically flat vacuum conjecture.

Related open problems.

1. *Original direction.* Formulate a generic weak cosmic censorship theorem for asymptotically flat vacuum initial data and prove that singularities visible from null infinity are nongeneric.

2. *Original direction.* Prove strong cosmic censorship with the correct regularity threshold for physically natural Einstein–matter models.

3. *Reference-driven formulation.* Determine which matter perturbations preserve censorship and which produce stable naked singularities.

4. *Reference-driven formulation.* Connect nonlinear stability of slowly rotating Kerr and Kerr–de Sitter spacetimes with precise final-state and weak-censorship formulations [GKS, HHV, HV].

5. *Reference-driven formulation.* Identify the regularity thresholds at which Cauchy horizons violate or preserve strong cosmic censorship in charged, rotating, and cosmological models [DL, LO].

6. *Reference-driven formulation.* Remove spherical symmetry from the charged-scalar weak-censorship theorem, or identify the first new obstruction created by angular momentum [AT].

7. *AI-generated formulation.* Compare censorship thresholds under asymptotically flat, asymptotically de Sitter and spatially compact boundary conditions in one common function-space framework.

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Problem 115

The splitting theorem of Cheeger and Gromoll [CG] says that if a Riemannian manifold M of nonnegative Ricci curvature contains a line γ (i.e., an absolutely minimizing geodesic), then M decomposes isometrically as a Riemannian product $\mathbb{R} \times N$, the first factor being represented by γ.

It would be of interest in studying the structure of space-time to prove that a geodesically complete Lorentzian 4-manifold of nonnegative Ricci curvature in the timelike direction which contains an absolutely maximizing timelike geodesic is isometrically the cross product of that geodesic and a spacelike hypersurface.

Status. Solved under the standard global hypotheses of the Lorentzian splitting theorem. Newman's theorem proves Yau's original timelike-geodesically-complete formulation [N]. Galloway proved the complementary globally hyperbolic formulation without assuming timelike geodesic completeness [G].

Cheeger and Gromoll's Riemannian splitting theorem is the model [CG]. In the Lorentzian setting the Ricci nonnegativity condition is replaced by the timelike convergence condition, and a line is replaced by a complete timelike line. Under the strong energy condition, the existence of such a line and either timelike geodesic completeness or global hyperbolicity gives an isometric splitting $\mathbb{R} \times S$ with S complete [N, G, BGMOS]. Later work, including Galloway–Vega's achronal-limit approach, clarified variants with weaker completeness or curvature hypotheses [GV]. Recent work moves the problem in two directions. Woolgar–Wylie proved Lorentzian splitting and singularity theorems under Bakry–Emery curvature-dimension hypotheses, where the equality case can become a warped product rather than the rigid product in the classical theorem [WW]. Beran–Ohanyan–Rott–Solis proved a synthetic splitting theorem for globally hyperbolic Lorentzian length spaces with nonnegative timelike curvature and a complete timelike line [BORS]. Braun–Calisti and Braun–McCann developed timelike Ricci lower bounds through optimal transport and timelike curvature-dimension conditions in low-regularity or nonsmooth spacetimes [BC], [BM]. A unified elliptic proof now covers the smooth Galloway and Newman alternatives [BGMOS], Galloway has replaced pointwise timelike Ricci nonnegativity by an averaged condition along timelike rays [G25], and a weighted globally hyperbolic splitting theorem is available for C^1 metrics [BGMOS2]. Thus the original smooth problem is settled, but its natural descendants concern how much regularity, completeness and pointwise Ricci control can be removed while preserving a genuine product conclusion.

Related open problems.

1. *Original direction.* Determine whether Galloway's averaged Ricci condition along every timelike ray is sharp, and whether it can be replaced by a condition imposed only along the given line [G25].
2. *Original direction.* Extend the C^1 globally hyperbolic theorem to the Newman alternative based solely on timelike geodesic completeness [BGMOS2].
3. *Original direction.* Understand the borderline cases where timelike lines exist only after taking limits of maximizing segments.
4. *Reference-driven formulation.* Determine when Bakry–Emery or curvature-dimension timelike convergence hypotheses give an isometric product, and when the best possible rigidity is only a warped product [WW].
5. *Reference-driven formulation.* Push below C^1 to locally Lipschitz or distributional Lorentzian metrics using synthetic timelike Ricci lower bounds, and compare the conclusion with the C^1 theorem [BC, BM, BGMOS2].
6. *Reference-driven formulation.* Identify the exact relationship between Lorentzian length-space splitting and smooth Lorentzian splitting by approximation, regularization and stability of timelike lines [BORS].

7. *AI-generated formulation.* Classify almost-splitting phenomena: if the timelike Ricci curvature is nearly nonnegative and a very long almost-maximizing timelike line exists, quantify the closeness to a Lorentzian product on large causal diamonds.

8. *AI-generated formulation.* Develop null and timelike versions of splitting in one framework, especially for spacetimes where complete timelike lines degenerate to null lines at infinity.

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Problem 116

Prove that a static stellar model is isometric to a sphere. See Lindblom [Lin] for the case when the model has uniform density.

S. Hawking demonstrated that a static black hole is axially symmetric, but his argument is based in part on physical reasoning.

From the work of Israel, Hawking, Carter, and Robinson, one knows that a stationary, rotating black hole must be the Kerr black hole (see [Ro]). Can one make a similar statement about a charged stationary black hole? If the metric is Riemannian, there are similar questions. Lapedes (these proceedings) points out that Robinson's method does not apply, but Israel's approach still works, in the static case.

Status. Partially resolved. The stellar and black-hole parts must be separated. Static asymptotically flat perfect-fluid stars are spherical for broad physically important classes of equations of state, but the unrestricted fluid-ball conjecture still requires equation-of-state and boundary hypotheses. Stationary black-hole uniqueness likewise depends on dimension, matter model, horizon structure, regularity and asymptotic assumptions.

Robinson proved uniqueness of the Kerr black hole in the classical stationary vacuum setting [Ro]. Lindblom proved spherical symmetry for static uniform-density stars [Lin]. Beig–Simon and Masood-ul-Alam proved spherical symmetry theorems for static perfect-fluid stars under suitable conditions [BS], [M]. These results should not be read as a theorem for an arbitrary perfect fluid. Coutinho–Leandro–Reis give a Robinson-divergence proof under an explicit equation of state and state the unrestricted fluid-ball conjecture separately [CLR]. Chruściel and Tod treated static electro-vacuum black holes [CT], while Hollands–Ishibashi–Wald generalized Hawking's axisymmetry theorem to higher dimensions [HIW]. Chruściel–Costa–Heusler survey the modern uniqueness picture and emphasize that the classical "no-hair" form must be refined in broader theories [CCH]. Bunting–Masood-ul-Alam's conformal argument gives a central proof of static vacuum black-hole uniqueness [BMA]. For charged stationary black holes, the modern electrovacuum classification is close to the Kerr–Newman answer only

after strong hypotheses are imposed: analyticity, I^+-regularity, connectedness or a precise treatment of degenerate horizons. Lopes Costa obtained a classification of stationary analytic electrovacuum black holes in terms of Weinstein solutions and recovered Kerr–Newman uniqueness for connected horizons [LC]; Amsel–Horowitz–Marolf–Roberts and Chruściel–Nguyen treated important extremal or degenerate Kerr–Newman uniqueness cases [AHMR], [CN]. Thus the original charged question has substantial affirmative answers, but the sharp geometric hypotheses remain part of the modern problem. Chruściel's 2023 note completes the connected, analytic, suitably regular four-dimensional stationary vacuum statement and clarifies which near-horizon ingredients do not need analyticity; it does not remove analyticity from the global stationary theorem [C23].

Related open problems.

1. *Original direction.* Determine the sharp hypotheses under which static stellar models must be spherically symmetric.
2. *Original direction.* Classify stationary charged black holes under minimal regularity, connectedness and nondegeneracy assumptions.
3. *Original direction.* Extend black-hole uniqueness theorems to higher-dimensional and nonlinear matter models, or exhibit the precise failures.
4. *Reference-driven formulation.* Identify which uniqueness results survive if asymptotic flatness is replaced by weaker geometric falloff conditions.
5. *Reference-driven formulation.* Remove analyticity or connected-horizon assumptions from stationary electrovacuum Kerr–Newman uniqueness, or construct explicit counterexamples to the naive formulation [LC], [CN].
6. *Reference-driven formulation.* Extend the conformal positive-mass method of static black-hole uniqueness to static perfect-fluid stars with the weakest physically natural equation of state hypotheses [BMA], [BS], [M].
7. *Reference-driven formulation.* Determine the precise relation between axisymmetry theorems and uniqueness theorems in higher-dimensional Einstein–Maxwell or Einstein–matter systems [HIW].
8. *AI-generated formulation.* Build a "hypothesis map" for black-hole uniqueness: horizon connectedness, nondegeneracy, analyticity, topology, matter model, asymptotic geometry and dimension should each be marked as removable, essential or unknown.
9. *AI-generated formulation.* Formulate a quantitative near-uniqueness theorem: if a static star nearly saturates a spherical-symmetry criterion, prove closeness to a spherically symmetric Tolman–Oppenheimer–Volkoff model.
10. *Reference-driven formulation.* Replace the differential equation-of-state inequalities in known fluid-ball theorems by the weakest causality, monotonicity and free-boundary assumptions [CLR].

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Problem 117

Prove that any Yang–Mills field on $\mathbb{S}^4$ is either self-dual or anti-self-dual.

See the paper of Bourguignon and Lawson [BL] in this volume. Atiyah, Drinfeld, Hitchin, and Manin [A+] have classified the self- and anti-self-dual solutions.

Status. False. There exist Yang–Mills connections on $\mathbb{S}^4$ which are neither self-dual nor anti-self-dual.

Atiyah–Drinfeld–Hitchin–Manin classified the instanton solutions, namely the self-dual and anti-self-dual Yang–Mills fields [A+]. This does not classify all Yang–Mills fields. Sibner–Sibner– Uhlenbeck proved the existence of non-self-dual Yang–Mills solutions [SSU]. Sadun and Segert later produced explicit non-self-dual Yang–Mills connections with nonzero Chern number [SS]. Thus the original assertion fails, while the instanton sector remains the distinguished absolute-minimizer sector in four-dimensional gauge theory. Subsequent analytic work has shifted the emphasis from the false dichotomy "instantons or anti-instantons" to the structure of the full Yang–Mills critical set. Feehan proved discreteness results for the possible Yang–Mills energies over closed four-manifolds and energy-gap theorems near the ground-state strata [F]. This supports a more Morse-theoretic version of the problem: understand the non-instanton critical points, their indices, their bubbling compactifications and their relation to the instanton moduli spaces. In this form, the problem is still useful even though the original claim is false. Dayaprema–Waldron prove a parabolic gap statement: on an $SU(r)$-bundle of charge κ over $\mathbb{S}^4$, the space of connections with Yang–Mills energy below $4\pi^2(|\kappa| + 2)$ deformation-retracts under Yang–Mills flow to the instantons [DW]. Hence the first non-instanton critical level is constrained from below, but the critical set above that threshold is not classified.

Related open problems.

1. *Original direction.* Classify non-self-dual Yang–Mills critical points on $\mathbb{S}^4$ up to gauge and conformal symmetry.
2. *Original direction.* Understand the Morse-theoretic role of non-self-dual critical points for the Yang–Mills functional.
3. *Original direction.* Determine which compact four-manifolds admit explicit non-instanton Yang–Mills connections with prescribed topological charge.
4. *Reference-driven formulation.* Compute the Morse index and nullity of the Sibner–Sibner–Uhlenbeck and Sadun–Segert non-self-dual examples, and locate them inside the full Yang–Mills energy spectrum [SSU], [SS], [F].
5. *Reference-driven formulation.* Determine whether the Yang–Mills energy spectrum on $\mathbb{S}^4$ for a fixed compact structure group and topological class can be described explicitly, beyond the general discreteness theorem [F].
6. *Reference-driven formulation.* Describe bubble-tree limits of sequences of non-self-dual Yang–Mills critical points on $\mathbb{S}^4$ and compare them with Uhlenbeck compactification of instanton moduli spaces.
7. *AI-generated formulation.* Build an "atlas" of non-instanton critical points on $\mathbb{S}^4$: symmetry type, structure group, Chern number, energy level, Morse index and degeneration behavior.

8. *AI-generated formulation.* Search for min–max constructions of non-self-dual Yang–Mills connections whose critical values lie between consecutive instanton energy levels.

9. *Reference-driven formulation.* Determine whether the parabolic gap threshold is attained by a non-self-dual critical point in each rank and charge, and classify the equality and bubbling scenarios [DW].

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Problem 118

Prove that the moduli space of the self-dual fields on $\mathbb{S}^4$ with a fixed Pontrjagin number is connected. Prove that the Pontrjagin number of a L^2-integrable gauge field on $\mathbb{R}^4$ is an integer.

Both Problems 117 and 118 are very well known. See the excellent article of Atiyah [A].

Status. Solved in the standard $SU(r)$ instanton setting. The two assertions require different analytic theorems. Taubes proved path-connectedness of the instanton moduli spaces on $\mathbb{S}^4$ [T]. Uhlenbeck's 1985 theory of Chern classes of Sobolev connections proves integrality for an arbitrary connection on $\mathbb{R}^4$ whose curvature is in L^2 [UC]. Her 1982 removable singularity theorem [U] gives the stronger smooth compactification statement when the connection also satisfies the Yang–Mills equation.

Atiyah's ICM article records the gauge-theoretic formulation and the role of instantons [A]. Uhlenbeck proved that a finite-energy Yang–Mills field on a punctured four-ball extends smoothly after gauge [U]. For a general finite-energy connection no Euler–Lagrange equation is available; the integer is instead obtained by constructing the associated Sobolev bundle and identifying its Chern–Weil number with an integral Chern class [UC]. Taubes proved path-connectedness for the relevant instanton moduli spaces [T]. Later developments put these solved statements into a larger moduli-space framework. Donaldson–Kronheimer give the standard gauge-theoretic treatment of instanton compactification and four-manifold applications [DK]. Nakajima's quiver-variety theory extends the ADHM viewpoint and relates instanton moduli spaces to representation theory and singular hyperkähler geometry [N]. Recent mathematical-physics work computes Hilbert series and Coulomb-branch descriptions of instanton moduli spaces, including higher charge and nonclassical groups [HMR], [CFHM]. These results do not change the connectedness and integrality answers, but they give more explicit algebraic invariants of the same moduli spaces. Dayaprema–Waldron give a new differential-geometric proof valid for every $SU(r)$-bundle: a sharp Yang–Mills sublevel deformation-retracts to the instanton space, and Taubes' charge-subtraction/gluing induction then gives path-connectedness [DW].

Related open problems.

1. *Original direction.* Describe the singular strata and compactifications of Yang–Mills moduli spaces in a way adapted to explicit geometry.

2. *Original direction.* Compare the connectedness and compactification picture for instantons with the corresponding Seiberg–Witten moduli spaces.

3. *Original direction.* Extend removable-singularity and integrality results to higher-dimensional Yang–Mills fields under sharp analytic hypotheses.

4. *Reference-driven formulation.* Give an explicit stratified description of the Uhlenbeck compactification of instanton moduli spaces on $\mathbb{S}^4$, compatible with the ADHM and quiver-variety models [DK], [N].

5. *Reference-driven formulation.* Compute Hilbert series, intersection cohomology or other algebraic invariants of fixed-charge instanton moduli spaces and relate them to geometric compactification strata [HMR], [CFHM].

6. *Reference-driven formulation.* Identify the sharp finite-energy hypotheses under which removable singularity and integrality remain true for Yang–Mills fields in dimensions greater than four.

7. *AI-generated formulation.* Produce a table, by compact structure group and instanton number, comparing connectedness, smoothness, singular strata, known algebraic models and known cohomological invariants.

8. *AI-generated formulation.* Look for a direct path construction between two ADHM data sets of the same charge which keeps the path inside a controlled regular stratum except at explicitly described wall crossings.

9. *Reference-driven formulation.* Compare the 2026 Yang–Mills-flow deformation retraction with the algebraic Uhlenbeck compactification and determine whether it controls intersection cohomology of the boundary strata [DW], [CFHM].

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Problem 119

Physicists have a notion of asymptotically flat manifolds (see [SY], for example). The definition depends heavily on the choice of coordinate system and is not intrinsic. If one replaces the definition by requiring the curvature to decay suitably, do we obtain an equivalent condition?

Status. Partially resolved, but curvature decay alone is not equivalent to asymptotic flatness. Under standard decay and volume-growth hypotheses, Bartnik gave an intrinsic treatment of mass and asymptotic coordinates [B], Chruściel analyzed the corresponding Hamiltonian boundary conditions [C], and Bando–Kasue–Nakajima constructed ALE coordinates at infinity from fast curvature decay and maximal volume growth [BKN]. Asymptotic flatness is the trivial-group case; topology at infinity and further regularity cannot be omitted.

Schoen–Yau's positive mass theorem uses the classical coordinate definition of asymptotic flatness [SY]. Bartnik showed that the mass has a geometric meaning under appropriate falloff and coordinate change hypotheses [B]. Chruściel gave an alternative Hamiltonian analysis of boundary conditions at spatial infinity [C]. Bando–Kasue–Nakajima supplied a Riemannian coordinate construction at infinity from fast curvature decay and volume growth [BKN]. Thus the desired implication is true in important precise regimes only after one controls the finite group at infinity and the analytic hypotheses. Curvature decay by itself also contains no information about the extrinsic curvature of an Einstein initial data set or the parity needed for some asymptotic

charges. More recent work has shifted part of the question from "which coordinates may be chosen?" to "which coordinates and asymptotic charges are geometrically selected?" Nerz proved geometric characterizations of asymptotic flatness and linear momentum using CMC-type covers near infinity [N]. Cederbaum–Metzger review and announce results on geometrically defined asymptotic coordinates, asymptotic foliations and the geometric meaning of mass, energy, linear momentum, angular momentum and center of mass [CM]. Thus the coordinate-dependence problem remains active even after BKN-type coordinate existence: one wants canonical, invariantly selected asymptotic charts and charges under weak falloff hypotheses.

Related open problems.

1. *Original direction.* Determine the weakest curvature-decay and volume-growth hypotheses that force a useful asymptotically flat coordinate system.

2. *Original direction.* Extend intrinsic asymptotic-flatness criteria to lower regularity initial data used in the Einstein constraint equations.

3. *Original direction.* Compare spatial-infinity and null-infinity definitions of asymptotic flatness in a single geometric framework.

4. *Reference-driven formulation.* Determine whether CMC/STCMC or related asymptotic foliations characterize asymptotic flatness under weaker hypotheses than current geometric characterization theorems [N], [CM].

5. *Reference-driven formulation.* Compare BKN harmonic coordinates, Bartnik harmonic coordinates and geometrically selected asymptotic coordinates in the same decay scale [B], [BKN], [CM].

6. *Reference-driven formulation.* Develop intrinsic criteria that recover not only asymptotic flatness but also ADM linear momentum, angular momentum and center of mass without imposing a preferred coordinate system at the outset [N], [CM].

7. *AI-generated formulation.* Formulate a stability theorem: if curvature decay and volume growth are almost compatible with an asymptotically flat end, quantify how close the end is to admitting a canonical asymptotic chart.

8. *AI-generated formulation.* Build a comparison table of coordinate-free definitions at spatial infinity and at null infinity, recording which conserved quantities each definition controls.

9. *Reference-driven formulation.* Find intrinsic conditions that force the finite ALE group at infinity to be trivial, thereby upgrading BKN coordinates to genuinely asymptotically flat coordinates [BKN].

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Problem 120

Given an asymptotically flat space, can one give a good definition of total angular momentum? What would the relationship be with total mass? (See [Pe].)

Status. Partially resolved, with several definitions depending on the asymptotic setting. Classical ADM-type angular momentum was analyzed by Beig–Ó Murchadha [Bo], and Chen–Wang–Yau gave a quasi-local definition whose spatial-infinity limit supplies total angular momentum under weak asymptotic hypotheses [CWY]. At null infinity, Chen–Wang–Wang–Yau developed a supertranslation-invariant formulation [CWWY].

Penrose listed total angular momentum as one of the fundamental problems of classical general relativity [Pe]. Beig–Ó Murchadha identified the Poincaré symmetry and Hamiltonian charges at spatial infinity [Bo]. Szabados determined the corresponding falloff and parity conditions and the Poincaré transformation law [Sz]. Chruściel–Delay used angular momentum in the constraint gluing framework [CD]. Chen–Wang–Yau derived quasi-local conserved quantities from optimal isometric embeddings and proved that their limits give total angular momentum, center of mass and the expected dynamical formulas for isolated systems [CWY]. Later work at null infinity removes the supertranslation ambiguity in the angular momentum and center-of-mass definitions [CWWY]. Chen–Wang–Wang–Yau subsequently verified the supertranslation invariance of their angular momentum at null infinity in double-null gauge [CWWY2]. Javadinezhad–Porrati proposed supertranslation- invariant angular-momentum flux formulae for gravitational scattering and analyzed remaining covariance/frame-fixing puzzles [JP1], [JP2]. Cederbaum–Metzger's geometric-coordinate program also links angular momentum and center of mass to asymptotic foliations at spatial infinity [CM]. Javadinezhad–Porrati have since given a covariant prescription for fixing both supertranslation and translation frames and clarified the difference between invariant Lorentz charges and generators of BMS coordinate transformations [JP3]. The modern frontier is therefore not merely the existence of a definition, but compatibility among spatial infinity, null infinity, flux laws, BMS covariance, numerical relativity and black-hole final-state questions. On the inequality side, Han–Khuri–Weinstein–Xiong prove a multiple-black-hole mass–angular-momentum inequality modulo an explicit extreme no-hair alternative, still assuming axisymmetry and maximal initial data [HKWX]. A 2025 preprint of Xu claims an axisymmetric angular-momentum Penrose inequality [Xu]. Neither result removes symmetry, so no general mass–angular-momentum relation follows from these theorems.

Related open problems.

1. *Original direction.* Compare all major notions of angular momentum for general asymptotically flat initial data without parity assumptions.
2. *Original direction.* Prove sharp flux and balance laws for angular momentum through null infinity in low regularity radiating spacetimes.
3. *Original direction.* Relate total angular momentum, total mass and center of mass to black-hole uniqueness and stability problems.
4. *Reference-driven formulation.* Develop a computable version of the Chen–Wang–Yau angular momentum for numerical relativity data sets with controlled error estimates.
5. *Reference-driven formulation.* Compare spatial-infinity CWY angular momentum, null-infinity CWY angular momentum and geometrically selected asymptotic-coordinate definitions under one set of decay and radiation hypotheses [CWY], [CWWY], [CWWY2], [CM].
6. *Reference-driven formulation.* Decide which supertranslation-invariant flux formula has the correct covariance, frame-fixing and scattering behavior in nonlinear radiating spacetimes [JP1], [JP2].
7. *Reference-driven formulation.* Establish mass–angular momentum inequalities and balance laws without axial symmetry, or identify the minimal replacement for symmetry.
8. *AI-generated formulation.* Create benchmark families of asymptotically flat numerical initial data for which ADM, CWY, Bondi-Sachs and scattering-based angular momenta can all be computed and compared.
9. *AI-generated formulation.* Formulate a final-state angular-momentum theorem: prove that the total angular momentum radiated through null infinity plus the final Kerr angular momentum equals the initial angular momentum in a fully geometric definition.

10. *Reference-driven formulation.* Remove axisymmetry from mass–angular-momentum inequalities by finding a geometric replacement for the twist-potential harmonic-map reduction [HKWX], [Xu].

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